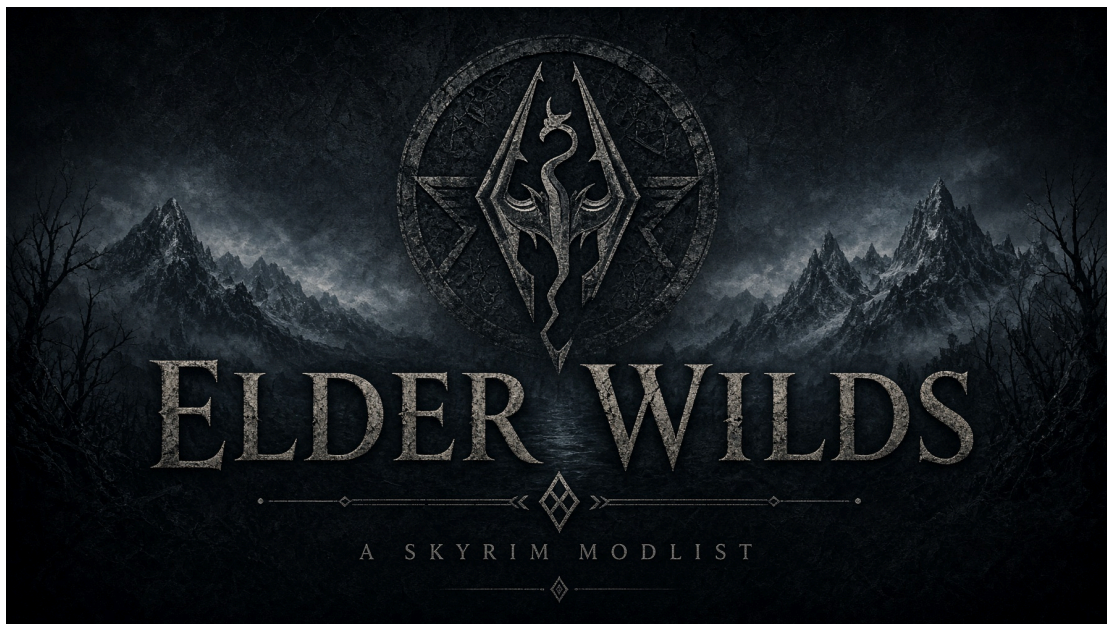




Elder Wilds

Skyrim AE 1.6.1170 Modlist Design Guide

Modern UI & Graphics
Modern Mechanics

Big, Dark, Awe-Inspiring World
Third-Person Gamepad Parity

Lots of New Content
Living the World

Version 0.2.0 · 2026-08-08

Contents

1. Elder Wilds	7
1.1. Installation	7
1.2. Research Workflow	7
1.3. Reference Docs	7
2. Installation Guide	8
2.1. Pre-Installation: Steam, Skyrim, And Creation Kit	8
2.2. Core Setup Goals	9
2.3. Recommended Base Toolchain	9
2.4. Mod Organizer 2 Setup	9
2.5. Output Mods To Create In MO2	10
2.6. SKSE64 Installation	10
2.7. Register Tools In MO2	11
3. Foundations and Compatibility	13
3.1. Core Framework Baseline	13
3.2. Official Masters Cleanup	15
3.3. Patching Technique And Strategy	16
3.4. ESP To ESL Conversion	18
3.5. Targeted Bugfix Mods	19
4. Graphics Pipeline	21
4.1. Overview	21
4.2. Pipeline Order	21
4.3. Key Principles	21
5. PGPatcher	22
5.1. PGPatcher Workflow	22
6. Community Shaders	23
6.1. Community Shaders Core Setup	23
6.2. Community Shaders Add-Ons And Required Features	23
6.3. Parallax Framework Dependencies	25
6.4. Physically Based Rendering (PBR) Support	25
6.5. Open Research	27
7. Textures & Meshes	28
7.1. Mesh Improvements And Parallax Support	28
7.2. Furniture, Clutter, And Item Replacers	31
7.3. Unique Entity Replacers	33
7.4. Blood, Decals, And Combat Visual Effects	33
7.5. Book and Paper Visual Overhauls	34
7.6. Additional Texture And Mesh Replacements	35
8. Lighting	40
8.1. Lighting Overhaul Strategy	40
9. Weather & Water	42
9.1. Weather And Atmosphere	42
9.2. Water Visuals	43
9.3. Sky, Stars, And Auroras	44
10. Terrain & Flora	46
10.1. Terrain, Roads, And Snow	46

10.2. Flora Visuals	48
10.3. Tree Overhauls For Dense Forests	50
11. Characters & Creatures	52
11.1. Character, Skin, And Creature Visuals	52
11.2. Hair, Eyes, And Beards	55
11.3. Skeleton And Bone Replacers	57
12. LOD & Distant Detail	58
12.1. LOD Generation And Distant Detail	58
13. Modernized UI	62
13.1. UI Framework Prerequisites	62
13.2. HUD Overhaul	62
13.3. Inventory And Item Card Improvements	64
13.4. Map Improvements	65
13.5. Magic And Journal Improvements	66
13.6. Dialogue UI Improvements	66
13.7. Controller-Friendly UI Support	67
13.8. UI Scaling For Ultrawide And High Resolution	67
13.9. Compass, Markers, And Minimal HUD	68
13.10. Crafting Menu Improvements	68
13.11. Loading Screens	68
13.12. Optional HUD	69
13.13. Open Research	69
14. Animations and Movement	70
14.1. Animation Framework Landscape — What Goes With What	70
14.2. Pandora Framework And Prerequisites	79
14.3. Skeleton And Behavior Prerequisites	79
14.4. Parkour, Climbing, And Free-Form Movement	79
14.5. Locomotion	81
14.6. Combat Animation Packs	82
14.7. BFCO Install Workflow Reference	84
14.8. Non-Combat Interaction Animations	85
14.9. Conditional Animation Systems	86
14.10. Camera-Aware Animation Support	87
14.11. Equipment Display Framework	87
14.12. Creature Animations	87
14.13. Animation Conflict Management	88
14.14. Open Research	88
15. Third-Person Gameplay	89
15.1. Third-Person Camera Framework	89
15.2. Camera Presets And Shoulder Switching	89
15.3. True Directional Movement	89
15.4. Target Lock And Targeting Behavior	90
15.5. Dodge, Sprint, And Movement Responsiveness	90
15.6. Hit Reactions, Stagger, And Impact Feedback	90
15.7. Third-Person Combat Systems	91
15.8. Crosshair And Aiming Behavior	91
15.9. Horse Camera And Mounted Controls	92
15.10. Dragon Mounted Controls	92
15.11. Third-Person Interactions: Lockpicking	92
15.12. Open Research	92

16. Expanded Systems	93
16.1. Overview	93
16.2. Sub-Files	93
16.3. Key Principles	93
17. Character & Progression	94
17.1. Character Creation And Starting Choices	94
17.2. Race Overhauls	95
17.3. Standing Stones	95
17.4. Character Progression Mechanics	96
17.5. Character Traits And Identity Hooks	96
17.6. Religion Or Roleplay Systems	97
17.7. Race-Specific Overhauls	97
17.8. Train And Study Systems	99
17.9. Experience and Leveling	100
18. Magic & Perks	102
18.1. Perk Overhaul Compatibility	102
18.2. Magic System Expansion	102
18.3. Spell Learning Mechanic	103
18.4. Stealth And Detection	103
19. Survival & Needs	105
19.1. Survival Systems	105
19.2. Roleplay Upkeep And Hygiene	105
19.3. Death Alternative System	108
20. Crafting & Economy	109
20.1. Alchemy Overhaul	109
20.2. Cooking Systems	109
20.3. Smithing And Crafting Expansion	110
20.4. Economy And Loot Balance	112
20.5. Weapons, Armor, And Equipment Content	113
21. Followers & Reputation	114
21.1. Crime, Reputation, And Social Systems	114
21.2. Followers And Companion Systems	114
21.3. Romance, Marriage, And Companionship	115
22. World Progression Philosophy	117
22.1. The Three Styles	117
22.2. The Interlocked Dial	117
22.3. Deleveled	118
22.4. Static Leveling	119
22.5. Vanilla With Curated Lists	120
22.6. Suggested Combinations	121
22.7. How To Use This Document	123
23. Immersive Scale and World Feel	124
23.1. Timescale And Calendar Adjustments	124
23.2. Sleep And Bedtime Life	124
23.3. Travel Pacing And Carriage Fast Travel Rules	125
23.4. Horse Travel	125
23.5. Encounter Spacing And Wilderness Feel	126
23.6. Soundscape And Ambient Audio	126
23.7. City Size, Outskirts, And Population Feel	126
23.8. Roadside Clutter, Ruins, And Landmark Density	127

23.9. Wildlife And Creature Ecology	127
23.10. Road Network And Surface Overhaul	128
23.11. Environmental Atmosphere: Wind And Dynamic Effects	128
23.12. Night Darkness And Visibility Balance	129
23.13. NPC Dialogue Immersion	130
23.14. Fire And Camp Interaction	130
23.15. Activation Text	130
23.16. Open Research	130
24. World Content	131
24.1. Landscape Overhauls	131
24.2. Cities, Towns, And Villages	131
24.3. College of Winterhold	133
24.4. Inns, Farms, And Small Settlements	134
24.5. Player Homes	135
24.6. Dungeons And Ruins	136
24.7. Points Of Interest And Roadside Content	137
24.8. Underwater	139
24.9. Side Activities: Fishing, Hunting, And Downtime	139
24.10. Weapons, Armor And Equipment Additions	141
24.11. Quest Additions	141
25. NPCs	147
25.1. NPC Appearance Overhauls	147
25.2. NPC AI & Behavior	148
25.3. Civilian And Traveler Population Additions	149
25.4. NPC Name Variety	149
25.5. NPC Dialogue Expansions	150
25.6. Custom Followers	152
25.7. Playable Race Additions	154
25.8. FacegenBaseline & NPC Stat Patcher Reference	156
25.9. Open Research	157
26. Enemies & Creatures	158
26.1. Enemy Variety	158
26.2. Creature Visual Overhauls	158
26.3. New Creature Additions	160
26.4. Creature Behavior Improvements	161
26.5. Wildlife Density And Rarity Balance	162
27. Weapons & Armor	163
27.1. Weapon Pack Additions	163
27.2. Armor Pack Additions	165
27.3. Artifact And Unique Item Overhauls	171
27.4. Clothing And Wardrobe Extensions	171
28. Audio and Feedback	174
28.1. Music Overhaul Strategy	174
28.2. Weapon, Magic, And Impact SFX	175
28.3. Creature And NPC SFX	176
28.4. Combat And Immersion Audio	176
28.5. Ambient And Environmental Audio	176
29. Survival, Difficulty, and Balance	178
29.1. Difficulty Framework	178
29.2. Combat Lethality And Pacing	178

29.3. Enemy AI Improvements	179
29.4. Injury, Stamina, And Resource Pressure	179
29.5. Enemy Resistances And Knowledge Checks	180
29.6. Survival Systems Compatibility	180
30. Legacy of the Dragonborn	182
30.1. Core Mod Baseline	182
30.2. Supported Content	182
30.3. Patch Strategy	183
30.4. Submods And Quality-Of-Life	184
31. Adult Content	185
31.1. Framework Prerequisites	185
31.2. Romance Content	186
31.3. NPC Relationship Content	186
31.4. Pregnancy & Family	187
32. Main Character Voicing	188
32.1. Framework Overview	188
32.2. Two Custom Voice Packs	189
32.3. Custom Voice Pack Pipeline	189
33. Performance and Technical Workflow	192
33.1. Overview	192
33.2. Sub-File Index	192
33.3. Key Principles	192
33.4. Related Sections	192
34. Performance Strategy	194
34.1. Performance Budgeting	194
34.2. VRAM-Heavy Mod Review	194
34.3. CPU-Heavy Script Review	195
35. Optimization & Configuration	196
35.1. Shader Performance Impact	196
35.2. Grass, Tree, And Draw-Call Optimization	196
35.3. INI Tuning And Display Settings	197
35.4. Save Safety Considerations	197
36. Tool Pipeline	198
36.1. Tool Workflow	198
36.2. 4K LOD Tool Baseline	198
36.3. Optional Diagnostics And Performance Tools	200
37. Bashed Patch & Synthesis Configuration	201
37.1. Patcher Discipline	201
37.2. Wrye Bash (Bashed Patch) Configuration	201
37.3. Synthesis Configuration	205
38. Testing & Maintenance	214
38.1. Performance Testing And Tuning	214
38.2. Maintenance And Rebuild Epilogue	216

1. Elder Wilds

Target build: Skyrim Special Edition / Anniversary Edition runtime `1.6.1170` via Steam

Scope notes:

- No paid Anniversary Edition Creation Club content
- Research categories are organized around five pillars: modern graphics, expanded systems, modernized UI, immersive world scale, and third-person-first gameplay. The entire modlist is designed to be completely played in third-person with a gamepad — no interaction should require first-person or keyboard/mouse alone.
- This modlist is also built around `Legacy of the Dragonborn` as a major collection, museum, and patch-ecosystem pillar rather than treating it as an optional side add-on

1.1. Installation

The full setup and installation guide (Steam config, MO2 setup, SKSE, tool registration, operating rules) lives in [Section 2](#).

1.2. Research Workflow

- Work through one category at a time
- For each category, define goals, constraints, candidate mods, compatibility risks, and acceptance criteria
- Do not lock final picks until foundations, graphics baseline, animation framework, and UI framework are validated together
- Prefer mods with active maintenance, clear Steam `1.6.1170` support, strong patch ecosystem, and stable load-order behavior

1.3. Reference Docs

- [Section 22](#) — Design dial connecting leveling, encounter zones, loot distribution, difficulty, XP, and perks. Adopted combination for Elder Wilds: **1 Static with Hard Threat** + “Living the World” layer (player home, family, slow questing).

2. Installation Guide

This document covers everything needed to set up the toolchain and environment for Elder Wilds. The section index and project scope live in [Section 1](#).

2.1. Pre-Installation: Steam, Skyrim, And Creation Kit

2.1.1. Steam & Skyrim Setup

1. Install Steam and Skyrim SE/AE outside `Program Files`, ideally under a shallow path such as `C:\Games\Skyrim Special Edition`.
2. Confirm the game is on runtime `1.6.1170`. The **Anniversary Upgrade** is not used.

2.1.1.1. Configure Steam

- **Disable the Steam Overlay:** In Steam, go to **Steam > Settings > In-Game** and uncheck "Enable the Steam overlay while in-game." Alternatively, disable it per-game: right-click **The Elder Scrolls V: Skyrim Special Edition** in your library, select **Properties**, and uncheck "Enable the Steam Overlay while in-game."
- **Update-proof Skyrim:** In Steam, right-click **The Elder Scrolls V: Skyrim Special Edition > Properties > Updates**, set **Automatic Updates** to **"Wait until I launch the game."** Updates will not trigger when launching through MO2 via SKSE. If an update slips through, use the [Skyrim downgrade patcher](#) to roll back.

2.1.1.2. Vanilla Skyrim First Launch

Launch Skyrim Special Edition through Steam to open the default launcher: click **Options**, select the **Ultra** preset, set **Aspect Ratio** and **Resolution** to your monitor's optimal values, set **Antialiasing** to **Off** (prepares for Community Shaders), ensure **Windowed Mode** is unchecked, click **OK** to save, and exit the launcher.

2.1.1.3. Install The Creation Kit

- **Skyrim Special Edition: Creation Kit** (https://store.steampowered.com/app/1946180/Skyrim_Special_Edition_Creation_Kit/): Download and install via Steam. Launch the Creation Kit through Steam to allow the **Scripts** folder to extract (automatic on first launch). Exit once loaded. Download the pre-made [CreationKitCustom.ini](#) and extract it to the Skyrim root folder. This configures the CK to allow multiple masters and load DLC content. If the CK crashes on first launch, add an [outbound firewall rule](#) for `CreationKit.exe`.
- **Creation Kit Platform Extended** (<https://www.nexusmods.com/skyrimspedition/mods/71371>): Download **CK Platform Extended 0.6-b550** and extract directly into the main Skyrim Special Edition folder — NOT into a sub-folder.

2.1.1.4. Create Modding Folders And Install Core Tools

Create a separate modding path with sub-folders (e.g., `C:\Modding` and `C:\Modding\Tools`) and install the tools from the toolchain list below into `C:\Modding\Tools`.

2.1.2. FOMOD Plus Installation

Install **FOMOD Plus - A Mod Installation Overhaul for Mod Organizer 2** (<https://www.nexusmods.com/skyrimspecialedition/mods/141001>) after MO2 setup: extract the archive contents directly into your MO2 installation folder, merging the `plugins` folder. Restart MO2 — FOMOD Plus activates automatically, adding a `FOMOD` column in the left pane and a `FOMOD` tab on the right pane.

2.2. Core Setup Goals

- Build and maintain `Elder Wilds` through `Mod Organizer 2` with clean separators and reproducible generated output
- Keep the setup aligned to Steam runtime `1.6.1170`
- Separate source mods, generated outputs, patches, and downloads clearly enough that rebuilds and troubleshooting stay manageable

2.3. Recommended Base Toolchain

- `Mod Organizer 2` as the main mod manager and profile system
- `SKSE64` and `Address Library for SKSE Plugins` as the runtime baseline
- `SSEEdit` for conflict review, patch inspection, cleaning guidance, and record-level troubleshooting
- `Wrye Bash` for later leveled-list and compatibility review where it still adds value
- `Synthesis` for selective patcher-based automation once the stack is stable enough to justify it
- `Cathedral Assets Optimizer` for asset cleanup and conversion tasks when a mod actually needs it
- `BethINI Pie` for baseline INI management and graphics/settings normalization
- `Skyrim Special Edition: Creation Kit` for resaving LE plugins to SE format; used alongside `Creation Kit Platform Extended` for stability
- `xLODGen`, `TexGen`, and `DynDOLOD` for terrain/object/tree distant detail workflow
- `BodySlide` and `Outfit Studio` and `Pandora Behaviour Engine Plus` are installed as regular mods — see → [Section 11](#) (BodySlide) and → [Section 14](#) (Pandora). They are still registered as MO2 executables (see tool registration below).

2.4. Mod Organizer 2 Setup

- Follow STEP's standalone-folder approach: keep the game under a shallow custom path such as `C:\Games\Steam\steamapps\common\Skyrim Special Edition` and keep modding tools under a separate path such as `C:\Modding\Tools`.
- Install `Mod Organizer 2` as a standalone application under `C:\Modding\Tools\Mod Organizer` rather than inside the game folder.
- On first launch, create a **global instance**, select the Steam *Skyrim Special Edition* install, and name the instance something obvious like *Elder Wilds 1.6.1170*.
- If MO2 prompts for profile settings during instance creation, tick every box so the profile keeps its own `INI` files and game-specific settings.
- Connect MO2 to Nexus during first launch and let MO2 handle `nxm` links.

- In *Settings > Workarounds*, tick **Enable archives parsing (experimental)** so MO2 reports asset conflicts more reliably.
- Create at least one clean working profile before major category testing, and keep separators aligned to this repo's structure in `separators.md`.
- Keep generated outputs in their own enabled mods, never mixed into source mod folders.

2.5. Output Mods To Create In MO2

Before registering tools, create these empty mods in MO2 under the `Output` separator:

- `xEdit Output`
- `SKSE Scripts`
- `SKSE Output`
- `BodySlide Output`
- `Pandora Output`
- `Grass Cache Output`
- `Terrain LOD Output`
- `TexGen Output`
- `DynDOLOD Output`
- `Occlusion Output`
- `Synthesis Output`
- `Bashed Patch Output`

2.6. SKSE64 Installation

- Follow STEP's split install pattern for `SKSE64` on Steam runtime `1.6.1170`: install the two root files into the Skyrim game folder, but manage the `Data` portion through MO2 as its own mod.
- Use the current AE `1.6.1170` build of `SKSE64`, not an older AE build and not a GOG/VR variant.
- Copy `skse64_loader.exe` and the matching `skse64_1_6_1170.dll` from the archive into the Skyrim root folder next to `SkyrimSE.exe`.
- In MO2, create an empty mod named `SKSE Scripts`, then copy the archive's `Data\Scripts` folder into that mod so the script portion stays visible and manageable in the left pane.
- If an `SKSE\skse.ini` is needed later, place it inside the `SKSE Scripts` mod rather than writing directly into the game folder. Minimal modern baseline:

```
[General]
ClearInvalidRegistrations=1

[Display]
iTintTextureResolution=2048
```

- Register `skse64_loader.exe` in MO2 as the `SKSE Skyrim Launcher` executable and launch the game through that entry after the vanilla initialization step.
- Community troubleshooting on `r/skyrimmods` still clusters around the same mistakes: wrong runtime build, root files copied into the wrong folder, scripts not installed as a mod, or launching outside MO2. Guard against those explicitly.

2.7. Register Tools In MO2

- Open MO2's executable editor from the gear icon and add each tool with **Add from file...**
- Tick **Use application's icon for desktop shortcuts** for each entry so shortcuts remain readable.
- For tools that generate files, tick **Create files in mod instead of overwrite** and point them at the correct output mod.
- Arrange the executables in a stable order so the dropdown reflects the real workflow instead of becoming random over time.

2.7.1. MO2 Executable Reference

Tool	Executable	Arguments	Output Handling	Notes
BethINI Pie	Bethini.exe	none	none	Run through MO2 so it edits the active profile INIs instead of global game INIs.
LOOT	LOOT.exe	--game="Skyrim Special Edition"	none	If LOOT fails through MO2, use <code>--single-process .</code> <code>--game="Skyrim Special Edition"</code>
SKSE Skyrim Launcher	skse64_loader.exe	none	SKSE Output	Use this to launch the game after the root files are in the game folder and the SKSE Scripts mod is enabled.
xEdit	xEdit.exe	-SSE - IKnowWhatImDoing -AllowMasterFilesEdit	xEdit Output	Main conflict review and manual patching entry.
xEditQuickAutoClean	xEditQuickAutoClean.exe	-SSE	xEdit Output	Only use for plugins LOOT explicitly flags for cleaning.
xLODGen	xLODGenx64.exe	-lodgen -SSE - o:"DriveLetter:\Moddir	Terrain LOD Gen\Output	Replace <code>DriveLetter</code> with the actual drive letter used for the modding folder.
TexGen	TexGen64.exe	-SSE	TexGen Output	Keep output isolated and pack it into the

				dedicated mod immediately after generation.
DynDOLOD	DynDOLODx64.exe	-SSE	DynDOLOD Output	Keep output isolated and enable <code>DynDOLOD.esm</code> / <code>DynDOLOD.esp</code> after generation.
Wrye Bash	Wrye Bash.exe	none	dedicated patch/output mod if used	Register only if the list still needs a Bashed Patch or leveled-list review pass.
Synthesis	Synthesis.exe	none	dedicated patch/output mod if used	Give it its own output mod once the final patcher set is known.

2.7.2. MO2 Operating Rules

- Launch the game through `SKSE`, not through Steam or the stock launcher, after the vanilla initialization step is complete.
- Run `LOOT` before gameplay tests and before any major cleaning or conflict-review pass.
- Use `xEditQuickAutoClean` only for plugins LOOT explicitly marks as needing cleaning.
- Use `xEdit` after each major category to review conflicts instead of waiting until the whole list is assembled. The `Patching Technique And Strategy` section in [Section 3](#) defines the workflow to follow from the first mod.
- Treat the `Overwrite` folder as a failure state; if files appear there, either move them into the proper output mod or fix the executable configuration that produced them.
- Rebuild generated outputs after major changes to bodies, animations, grass, LOD, or other systems that produce artifacts.
- Document manual fixes, custom arguments, and tool-specific decisions directly in this repository so the plan stays reproducible.

3. Foundations and Compatibility

3.1. Core Framework Baseline

Install all core libraries and frameworks upfront — they are non-negotiable dependencies for virtually every mod in the list. Bulk-install prevents missing-master errors and lets every subsequent section work against the real framework stack.

3.1.1. SKSE & Scripts

- **SKSE64** — Script extender; required by almost every SKSE plugin. [Nexus](#)

3.1.2. Core Libraries

Mod	Purpose
Address Library for SKSE Plugins	Universal SKSE plugin compatibility layer for 1.6.1170.
USSEP	Foundational record patch. Requires the four free CC masters — no paid AE Upgrade needed.
SSE Engine Fixes	Critical engine-level stability and bug fixes.
SSE Display Tweaks	Mandatory display and frame-pacing control.
Scrambled Bugs	Collection of low-level engine tweaks.
powerofthree's Tweaks	SKSE plugin tweak framework; hard dependency for many modern mods.
powerofthree's Papyrus Extender	Extended Papyrus scripting functions.
PapyrusUtil	Papyrus array/map utilities.
JContainers	JSON data storage for Papyrus.
Keyword Item Distributor (KID)	Distributes keywords to items without patching.
Spell Perk Item Distributor (SPID)	Distributes spells/perks to NPCs dynamically.
Base Object Swapper (BOS)	Swaps base objects via rules without direct overrides.
Crash Logger SSE AE VR	Crash dump logging for diagnosing CTDs.
SkyPatcher	SKSE-based vanilla record patcher; hard requirement for several NPC/AI mods.
Dual Casting Fix	Required by Blade and Blunt.
Andrealphus' Papyrus Functions	Required by Dragon War.
ConsoleUtilSSE NG	Required by Dragon War and several animation behavior mods.
Kris's Papyrus Extender	Required by Bathing in Skyrim - Renewed.

Animation Queue Fix	Animation-stack stability for heavy OAR/behavior loads.
Papyrus Tweaks NG	Script-engine performance optimization; widely adopted baseline.
Lexicon SKSE	Required by Actor Value Generator.
Actor Value Generator	Required by Classic Attributes.
XEMI Utility	Required by FYX - Nordic Doors and Traps Collisions.
Sonders Keyword Distribution Resources	Keyword distribution framework for item/keyword tagging.
Skyrim Save System Overhaul 3 (SSSO3)	Save-system replacement with Save Shield, rotating caps, and safe timed hardsaves. Requires <code>JContainers SE</code> + <code>NL_MCM</code> .
Unofficial Skyrim Modder's Patch - USMP SE	Modder-side patch collection fixing record-level bugs across DLC and CC content. Complements USSEP.
USMP - Patch Emporium	Centralized patch hub for USMP — third-party compatibility patches for various content mods. Install after USMP base.
Object Impact Framework (OIF)	Physics-based object impact system; prerequisite for several immersion/destruction mods.

3.1.3. SSSO3 Configuration

Vanilla autosaves, quicksaves, and manual saves all write the same `.ess` format — the instability risk is **when** they fire (cell transitions, combat, half-evaluated script state), not the format. SSSO3 replaces vanilla triggers with a Save Shield that defers saves during unsafe states.

Installation:

1. Install requirements (already in Core Libraries): `JContainers SE`, `NL\_MCM - A Modular MCM Framework`, `Regional Save Names`, `SSE Engine Fixes`
2. Install the SSSO3 main file and `SSSO3 - French` (if applicable). Place SSSO3 anywhere in load order.
3. **Disable all vanilla autosaves** in BethINI Pie: untick Autosaves, Save on Travel, Save on Rest, Save on Wait. SSSO3 force-deactivates them, but BethINI settings prevent the INI values from conflicting.
4. Disable `iAutoSaveCount` in `Skyrim.ini` (`[SaveGame]`) to prevent the engine from reserving autosave slots — SSSO3 manages its own.

In-game setup (required before the rotating system activates):

1. Create your character and exit the starting cell (alternate-start mod or vanilla Helgen sequence).
2. Make one manual save to generate a save file on disk.
3. Open the SSSO3 MCM. Go to **Player Setup**.
4. Press the **Find Player ID** button — SSSO3 reads it automatically from the save file. If it fails, use **Bypass Check** and type your character name exactly.
5. Assign a **Quick Save key** and a **Quick Load key** in the MCM.
6. **Select a preset profile.** For Elder Wilds, start with **Vanilla+** — it replicates vanilla autosave behavior with Save Shield protection and timed saves, no hardcore restrictions.

7. From the MCM's **Rotating System** page, set your preferred save cap (20-30 is reasonable for a playthrough).

SSSO3 Configuration for Elder Wilds: Starfrost (locked survival baseline) does not use Campfire files — no patch conflict expected. If a campfire-required mod is adopted later, SSSO3 includes compatibility patches.

3.1.4. SkyPatcher And SkyPatched Mods

SkyPatcher is an SKSE plugin that patches vanilla records at runtime from INI-style configuration files rather than traditional ESP plugins. No winner-loser override chain, no per-pair patches for the records it covers.

A **"SkyPatched"** variant ships record changes as SkyPatcher configs instead of an ESP. Assets (meshes, textures, scripts) still install in MO2's left pane, but the plugin slot is replaced by files under `SKSE/Plugins/SkyPatcher/`.

When to choose SkyPatched:

- Mod only changes vanilla records (book stats, weapon damage, armor ratings, ingredient values), not new records
- Load order is approaching the 254-plugin cap — each SkyPatched variant saves one ESP slot
- Multiple mods edit the same records — runtime application avoids patch debt
- Author explicitly provides a SkyPatched alternative

When to stick with traditional plugin:

- Mod adds new records (CELL, WRLD, QUST, NPC_ with facegen)
- Mod needs explicit load-order priority — Bashed Patch or xEdit patches give control that "all apply" doesn't
- No SkyPatched variant available

Mods using SkyPatcher in this list:

- `Book Covers Skyrim - SkyPatched` (→ [Section 7](#))
- `Dawnguard Arsenal SSE - SkyPatched` (→ [Section 24](#))
- `Diverse 4thUnknown Dragons` — uses SkyPatcher for combat-mod compatibility (→ [Section 26](#))
- `Patchifier` Synthesis patcher generates SkyPatcher patches (→ [Section 37](#))

3.1.5. Mesh & Texture Fixes

Mod	Purpose
Particle Patch	Particle shader/LOD corrections, CS-compatible.
Assorted Mesh Fixes	Corrections for common visual mesh errors.
Flickering Meshes Fix	Resolves distant-object mesh flicker in landscape transitions.

—

3.2. Official Masters Cleanup

Clean all six-plus vanilla masters before installing any mod. Use `xEditQuickAutoClean` on each file individually, collect cleaned `.esm` files into a `Cleaned Vanilla Masters` MO2 mod, and restore originals from `xEdit Backups` back into `Data\`.

3.2.1. Which Masters To Clean

- `Skyrim.esm`, `Update.esm`, `Dawnguard.esm`, `HearthFires.esm`, `Dragonborn.esm`
- All Creation Club `.esm` files shipped with 1.6.1170
- Do not clean `\_ResourcePack.esl` — contains no ITM or UDR records.

3.2.2. Procedure

1. Create empty mod `Cleaned Vanilla Masters` under `Output` separator. Leave disabled.
2. Run LOOT, sort, apply, close.
3. For each `.esm`: select `xEditQuickAutoClean` from MO2 executables, check ONLY that master, wait for completion, close.
4. Move cleaned masters (recent mod date) from `Data\` into `Cleaned Vanilla Masters\` mod folder. Keep only `.esm` files; delete `.backup` / `.bak`.
5. Restore originals from `xEdit Backups` back into `Data\`.
6. Enable `Cleaned Vanilla Masters` in MO2. Run LOOT again — no masters flagged.

3.2.3. Risks & Compatibility

- Cleaning `Update.esm` is standard despite some mod-author cautions; USSEP's records win over cleaned originals.
- Never clean a third-party mod's plugin unless LOOT explicitly flags it — many mods ship ITMs intentionally.
- `Cleaned Vanilla Masters` must remain enabled for the entire playthrough.

—

3.3. Patching Technique And Strategy

Adopt the canonical `xEdit` mod-by-mod patching workflow: install one mod at a time, run `xEdit` in `-veryquickshowconflicts` mode, resolve only new conflicts, use per-pair patches and `ModGroups`. Reserve `Bashed Patch` and `Synthesis` for the categories they genuinely own.

3.3.1. Wrye Bash And The Bashed Patch

The **Bashed Patch** is an automatically generated plugin merging leveled lists, applying game-setting tweaks, and resolving category-tagged record conflicts. Replaces dozens of per-pair patches for the record types it owns.

Rebuild triggers: adding/removing leveled-list mods, changing `.TweakSettings`, adding a mod with `Bash Tags`, form-ID compaction. It is step 2 in the full build order (after `SSEEdit`, before `Synthesis`; see → [Section 37](#)).

What the Bashed Patch does not do: resolve arbitrary record conflicts (→ `xEdit` / `Synthesis`), merge plugins (→ ESL flags), or iterative/rule-based patching (→ `Synthesis`).

Mods complementing or requiring Wrye Bash:

- **Leveled-list expanders** (Immersive Weapons, Immersive Armors, Hunterborn) — need `ReLev` / `DeLev` / `Invent.Add` tags
- **World loot frameworks** (Open World Loot SkyPatched — locked)

- **Economy mods** (Trade and Barter, Ars Metallica, Honed Metal, Apothecary) — need `Stats / Invent.Change`
- **Lighting/weather** (CS Light, True Light, Azurite III CS) — need `C.Light / C.Water / C.Climate`
- **NPC/follower mods** (RDO, Nether's, Serana Dialogue) — need `Actors.*` tags
- **Audio overhauls** (AOS, ISC, Sounds of Skyrim) — need `Sound / C.Acoustic`

Exact `Bash Tag` per plugin, `.TweakSettings` table, and build-order placement → [Section 37](#).

3.3.2. Cyclic Workflow (per new mod)

1. **Install and place.** Heuristic: broad world-changers early; narrow/local edits late.
2. **Run xEdit with `-veryquickshowconflicts`**. Register xEdit in MO2 with that argument. Right-click → **Hide no conflict and empty rows**.
3. **Try load order first.** Move mod up/down if that resolves most conflicts; re-run.
4. **Identify bad conflicts.** Only conflicts where the loser was supposed to win.
5. **Build per-pair patch. Copy as override into...** → `\<new file>.esp [Template] ESL`. Name explicitly (e.g., `LoreRim\_x\_RWT.esp`). Drag missing fields from loser into new record. Accept master addition. Run **Add Masters...** for every referenced mod. Save into `Patches` separator.
6. **Build per-pair `ModGroup` s.** One `ModGroup` per conflicting pair. Move `*.ModGroup` files from `MO2 overwrite` into the source mod's own MO2 folder.
7. **Re-open xEdit and confirm zero visible conflicts.**

3.3.3. Starting State

Clean load order (`Skyrim.esm`, `Update.esm`, three DLCs) plus USSEP. USSEP ships a premade `ModGroup` since xEdit 4.1.7 — keep it.

3.3.4. Cleaning Discipline

Run LOOT after new mod install. If flagged for cleaning, run **Quick Auto Clean** then **Check For Errors**. Red errors = reconsider the mod. QAC only removes ITM/UDR records.

3.3.5. ESL-Flagged Patches

Always use `\<new file>.esp [Template] ESL` for patches. Never manually re-flag ESP↔ESL unless the mod author or xEdit explicitly says to.

3.3.6. Risks & Compatibility

- Keep Bashed Patch and Synthesis scoped — don't let them absorb every decision.
- Date-stamp patches and re-run xEdit after every source-mod update.
- Patches copying from downstream plugins inherit unresolved conflicts — read the master chain.
- xEdit normal mode writes current loaded state — never use it to save. Save only in fast mode.
- Promote tiny per-pair patches into themed region/feature patches when count becomes unmanageable.
- Deliberately test that disabling one upstream source mod cleanly stops its `ModGroup` and dependent patches.

—

3.4. ESP To ESL Conversion

Converting ESP to ESL-flagged ESP frees a regular plugin slot (limit 254) by moving into the 4096 light-plugin address space.

3.4.1. What Makes A Plugin Convertible

- Fewer than 2048 new record forms
- No new CELL/WRLD/NAVM/NAVI records
- No QUST/DIAL overrides with hard-coded form ID references
- **Trivially convertible:** Bodyslide outputs, SKSE plugin dummy ESPs, texture-replacer ESPs

3.4.2. Automated Discovery

1. **Built-in script:** `Find ESP plugins which could be turned into ESL.pas` — ships with SSEEdit.
2. **Fractal's** `ESP-ESL Finder v1.1` ([Nexus](#))
3. `ESLify` ([Nexus](#)) — batch-convert wrapper. Verify candidates manually first.
4. `SSEEdit Script - Header Change to Version 1.71` ([Nexus](#)) — unlocks full 4095-record range for 1.6.1130+.

3.4.3. Safe Conversion (low risk)

1. Right-click → **Compact Form IDs for ESL**.
2. Right-click → **Set Is-Esl Flag** (ESL only, not ESM).
3. Save. Re-sort with LOOT.

3.4.4. Full Conversion (xEdit workflow)

1. Load plugin + dependencies. Apply **Hide no conflict and empty rows**.
2. Right-click → **Compact Form IDs for ESL**. If zero, skip to step 4.
3. **Check for broken references:** Run **Check for Errors**; for external refs, load full load order → **Referenced By** → **Apply as filter**; or use `List records referencing specific plugins.pas`.
4. Right-click → **Set Is-Esl Flag** (ESL only).
5. Save. Re-sort.

3.4.5. Red Lines

- **Never** convert plugins with CELL/WRLD/NAVM/NAVI records without verifying every cross-reference.
- **Never** convert mid-playthrough.
- **Never** convert a plugin another mod lists as a master without verifying.
- ESL-flagged plugins still count toward the 4096 light-plugin limit.

3.4.6. Recommended Candidates

- Bodyslide outputs, texture-replacer ESPs, SKSE plugin dummies
- Single-weapon/armor/spell mods with <50 new records and no scripts
- Crafting recipe-only ESPs
- Patches created via `[Template] ESL`

3.4.7. Never Convert

- Major quest mods (LotD, Interesting NPCs, Vigilant)
 - City/settlement overhauls with new CELL or WRLD records
 - Overhauls with new dialogue branches or follower framework quest aliases
 - Any plugin LOOT or SSEEdit flags as not ESL-safe
-

3.5. Targeted Bugfix Mods

Mod	Purpose
Fuz Ro D-oh - Silent Voice	Subtitle safety for unvoiced dialogue.
Actor Limit Fix	Crowded scene / facial animation reliability.
Equip Enchantment Fix	Enchantments dropping off while equipped.
NPC AI Process Position Fix - NG	NPC processing after waiting/fast travel.
Simple Offence Suppression	Accidental hostility from stray hits.
Bug Fixes SSE	Crafting-menu, perk-application, and water-behavior fixes.
Navigator - Navmesh Fixes	Follower pathing in complex worldspaces.
Barter Limit Fix	Merchant gold cap overflow.
Mum's the Word NG	Worthless stolen-item tagging.
Regional Save Names	Save labeling for testing hygiene.
Better Jumping SE	Consistent jumping control.
LeveledList Crash Fix	Leveled-list crash protection.
Explosion Collision Fix	Prevents explosion-force spells from launching objects into camera.
Inertia (Floating Gear Fix)	Stops dead-NPC equipment freezing mid-air by briefly keeping collision capsules active after death (configurable duration). SKSE plugin, zero records, all NPCs/creatures.
SMP-NPC crash fix	Prevents the known FSMP crash when loading NPCs with SMP hair. Required companion to the locked FSMP baseline. SKSE plugin.
Dragon Stalking Fix	Dragon corpses no longer follow the player or fall out of the sky. ESL, SPID-based corpse cleanup. Requires Po3 Tweaks (baseline).
Beard Mask Fix	Beards use slot 44 — prevents mask clipping. SKSE plugin.
Luma Utility	Utility SKSE plugin.
Description Framework Patch Hub	Centralized patch collection for the Description Framework.
Exit Sneak On Sprint	Exits sneak mode when sprinting. Lightweight SKSE plugin, no records.
Naked Dead NPC Fix	Auto-equips body/hands/feet/head slots when a dead NPC loads naked (outfit-distribution

	glitch). SKSE, zero records, "compatible with everything".
Native EditorID Fix	Bug-fix utility (Bug Fixes/SKSE) correcting engine-side native EditorID handling. SKSE plugin, requires Address Library (baseline).
Weapons Armor Clothing and Clutter Fixes (WACCF)	Broad record fix for weapon/armor/clothing/clutter records. High compatibility debt — test carefully with the crafting and artifact stack. Optional — skip if conflicts outweigh gains.

3.5.1. Not Installed

- **Alt-Tab Stuck Key Fix** ([Nexus](#)) — Prevents stuck modifier keys after alt-tabbing. Optional fix, evaluate if alt-tab issues arise during testing.
- **I'm Walkin' Here NG with Pets** ([Nexus](#)) — Ally/pet body-blocking fix.
- **Bard Instrumentals Mostly - Sing Rarely** ([Nexus](#)) — Repetitive tavern singing. Not needed — other bard mods cover this.

4. Graphics Pipeline

4.1. Overview

This section owns the full graphics rendering pipeline for `Elder Wilds`: from the post-process shader framework through textures, lighting, weather, water, terrain, characters, and distant detail. Each pillar of the pipeline lives in its own sub-file.

The pipeline is designed to be evaluated in order — each stage depends on the one before it. Do not jump ahead to textures or characters before the Community Shaders framework and its addon modules are locked.

4.2. Pipeline Order

Stage	File	Category
1	Section 5	PGPatcher — parallax/PBR mesh generation
2	Section 6	Community Shaders — post-process framework and addon modules
3	Section 7	Textures & Meshes — all surface-level asset replacement
4	Section 8	Lighting — interior, exterior, and effect lighting
5	Section 9	Weather & Water — atmospheric and liquid rendering
6	Section 10	Terrain & Flora — ground surfaces, grass, trees
7	Section 11	Characters & Creatures — body, skin, hair, creature textures
8	Section 12	LOD & Distant Detail — far-field rendering

4.3. Key Principles

- PGPatcher runs after the texture and mesh stack is stable → `Graphics - PGPatcher`
- Community Shaders is the universal post-process baseline; no ENB is used → `Graphics - Community Shaders`
- All texture and shader choices must be evaluated for third-person readability at 4K
- Performance budgeting for the pipeline is owned by `modlist-performance.md` → `Performance`
- Visual coherence rules (no mixing low-res and high-res in the same category) are owned by `modlist-curation.md` → `Testing & Curation`

5. PGPatcher

5.1. PGPatcher Workflow

PGPatcher is a post-installation patcher that generates parallax and PBR-compatible meshes from the installed texture and mesh stack. It must run after the main texture, mesh, parallax, and PBR candidate set is mostly chosen so it works against a stable asset baseline.

Mod	Type	Notes
PGPatcher	Baseline	Post-installation parallax/PBR mesh generator. Keep output in own MO2 folder.

5.1.1. Workflow Rules

1. Run after installing/changing PBR texture packs, parallax-enabled textures, or major mesh/parallax support mods.
2. Let the generated patch load after the source texture/mesh mods it supports.
3. Re-run whenever a major landscape, architecture, ruins, or clutter texture pack is replaced.
4. Only keep generated output if it produces visible improvement without new artifacts.
5. Hard requirement for mods such as [Steel Armor Redone - PBR and Complex Material](#).

5.1.2. Alternatives

- Skip route: rely on hand-authored parallax meshes from individual texture packs and skip post-processing entirely.

5.1.3. Risks & Compatibility

- Running PGPatcher before the texture stack is stable means regenerating the patch each time a texture mod changes.
- Generated output can conflict with hand-authored parallax meshes — always test before committing the generated patch.

6. Community Shaders

6.1. Community Shaders Core Setup

This section owns the graphics-framework decision for the shader-first visual stack. Build modern visuals around `Community Shaders`, then layer materials, lighting, weather, water, and distant detail on top.

6.1.1. Baseline

- **Community Shaders** ([Nexus](#)) — Core framework.
- Validate prerequisites before judging later graphics layers: `SKSE64` ([Nexus](#)), `Address Library for SKSE Plugins` ([Nexus](#)), `Luma Utility` ([Nexus](#)), and any current page-listed dependencies.
- Record baseline performance before installing add-ons so module decisions are measured against a real control point.

6.1.2. Alternatives

- Non-CS alternatives (ENB) are out of scope unless the project abandons the shader-first direction.

6.1.3. Risks & Compatibility

- Some CS features are much heavier than the core framework.
 - Older ecosystem pages may refer to legacy behavior or outdated version requirements.
 - A shader-first stack still needs discipline; enabling every feature isn't the same as building a coherent visual direction.
-

6.2. Community Shaders Add-Ons And Required Features

How aggressive the first CS feature layer should be — improve shadowing, wetness, water response, and material depth without front-loading the full performance cost.

6.2.1. Feature Tiers

Tier	Modules	When To Enable
Conservative	Screen Space Shadows, Grass Lighting, Water Effects, Wetness Effects	Start here
Balanced	Conservative + Terrain Shadows	After trees, grass, weather, water, LOD are stable

High-End	Balanced + SSGI, Subsurface Scattering	Luxury, not requirement
-----------------	--	-------------------------

Install the CS main download (v1.7+) and enable modules via FOMOD. Install optional plugins separately.

6.2.2. Plugins (separate downloads from core)

- **Skylighting** — Sky-driven ambient occlusion. Pairs cleanly with the conservative tier. Worth including for authentic grim-dark look.
- **Skyrim Upscaler - DLSS FSR2 XeSS (Nexus)** — Only if real performance testing proves it's needed.
- **Embers XD (Nexus)** — The fire-and-ember answer (140K+). Install after lighting overhauls so it inherits their campfire/hearth placements.

Optional separate plugin downloads: `Cloud Shadows`, `Hair Specular`, `HDR`, `SSGI`, `Skylighting`, `Terrain Helper`, `Terrain Blending`, `Terrain Variation`, `Upscaling`, `Wetness Effects`.

6.2.3. Related Add-Ons

Mod	Description
Dynamic Wetness	Wetness VFX triggered by swimming, rain, and water contact. SKSE plugin, complements CS Wetness Effects.

—

6.2.4. Graphics Testing Protocol

6.2.4.1. Recommended Tool Stack

- Performance capture: `PresentMon`
- Analysis and graphing: `CapFrameX`
- Record keeping: Markdown table in this repo first, spreadsheet only if comparison volume grows
- Visual comparison: fixed test saves plus matching screenshot captures

6.2.4.2. Test Scene Rules

- Use the same character save for all graphics testing.
- Same location, camera angle, FOV, and time of day for each comparison pass.
- Keep one indoor test cell and at least two outdoor test cells: forest and open landscape.
- Record weather state for every run.
- Add or remove only one graphics variable at a time.

6.2.4.3. Suggested Metrics

- Average FPS
- 1% low FPS
- Frame-time consistency notes
- VRAM pressure notes from monitoring tools
- Shader artifacts, flicker, ghosting, water issues, grass mismatch, shadow instability
- Subjective visual verdict: keep, retest, reject

6.2.4.4. Markdown Table Format

Date	Build	Test Area	Weather/ Time	Mod Changes	Avg FPS	1% Low	Visual Notes	Decision
YYYY-MM-DD	Steam 1.6.1170	Falkreath Forest	Overcast / 3 PM	Added Grass Lighting	00	00	Better depth, no flicker	Keep

6.2.5. Risks & Compatibility

- Module stacks that look good in isolation can fight later weather, water, or lighting winners.
- Measuring several shader changes at once makes later decisions impossible to trust.
- Screenshot appeal can hide long-session readability or frame-time problems.

—

6.3. Parallax Framework Dependencies

Auto Parallax and Complex Parallax Materials form the parallax rendering layer bridging CS parallax support with mesh and texture mods downstream.

6.3.1. Baseline

- Install **Auto Parallax** ([Nexus](#)) and **Complex Parallax Materials** ([Nexus](#)) as infrastructure before adding parallax-enabled textures or meshes. CPM hard-depends on Auto Parallax. Install after the texture stack is mostly chosen but before PGPatcher runs.

6.3.2. Risks & Compatibility

- Without Auto Parallax, Complex Parallax Materials will not function — both must be present.
- Installing too early means regenerating parallax data after texture changes; install after the main texture stack is locked.

—

6.4. Physically Based Rendering (PBR) Support

PBR is an adoption strategy layered on top of Community Shaders, not a separate renderer choice. The real decision is how selectively to use PBR-ready materials and conversions.

6.4.1. Baseline — Balanced PBR Route

Broad AIO base + targeted terrain and rock overrides.

- **Vanilla PBR AIO** ([Nexus](#), 11.5 GB) — Primary base. Covers all architecture, clutter, actors, dungeons with vanilla-faithful art direction.
- **Faultier's PBR Landscapes 4k** ([Nexus](#), 3.0 GB) — Fills VPBR's terrain gap. Do NOT install [Faultier's Skyrim AIO](#) alongside VPBR.
- **Enhanced Rocks and Mountains - Complex Material and PBR** ([Nexus](#)) — Modern rock/mountain overhaul. Load after VPBR to override its mountain textures. Install both main files (meshes then textures, choose PBR in FOMOD).

- **Tomato's PBR Whiterun (Nexus)** — Optional Whiterun override. Install 2K (735 MB), skip 4K (2 GB). Optional: `Improved Dragon Carvings` (beta, 2K) and `PBR Talos`.
- **Tomato's PBR Farmhouses 2.0 (Nexus)** — Optional farmhouse override. Install Grey 2K (84 MB), skip 4K. Optional: `WSU PBR Farmhouse windows patch` if Window Shadows Ultimate is in the stack.

Community Shaders ships **Dynamic Cubemaps** and **Image-Based Lighting (IBL)** as standard features. Verify the chosen PBR texture set uses both before adding extra reflection/cubemap mods.

6.4.2. Alternatives

- **Minimal PBR:** Enable support but only use a few showcase conversions.
- **Full PBR push:** Broad world coverage with higher maintenance and consistency risk.

6.4.3. Priority Coverage

Priority 1 — seen constantly in travel and third person:

- City architecture; farmhouses and rural buildings
- Nordic ruins and dungeon stonework
- High-visibility clutter and props

Priority 2 — fill in after core look settles:

- Select landscape regions with strong material identity
- City sets from settlement overhauls that survive later testing
- Broader regional packs that fill obvious gaps

Defer for now:

- Full-world landscape replacement
- Low-visibility assets adding patching/VRAM cost without changing moment-to-moment feel
- Niche regional conversions before core city/dungeon/farmhouse look is settled

6.4.4. Priority 2 Candidate Mods

- **A Cathedralist's PBR Landscape (Nexus)**
- **TomatoRim PBR Landscapes AIO (Nexus)**
- **PBR textures for The Great Cities Collection (Nexus)**
- **PBR Hub (Nexus)** — Discovery hub for later expansion.

6.4.5. Working Rollout Order

1. Install **Vanilla PBR AIO** — broad PBR base for architecture, clutter, actors, dungeons.
2. Install **Faultier's PBR Landscapes 4k** — fills VPBR's terrain gap.
3. Install **Enhanced Rocks and Mountains** after VPBR — overrides VPBR's mountain textures.
4. Optionally install **Tomato's PBR Whiterun** and **Tomato's PBR Farmhouses 2.0** after VPBR.
5. Run **PGPatcher** last — all PBR texture mods above require it.
6. Only after core stack passes review, test optional landscape additions (`Cathedralist's PBR Landscape`, `TomatoRim PBR Landscapes AIO`).

6.4.6. Risks & Compatibility

- VPBR is newer (March 2026) than Faultier's PBR Skyrim (v3.0) — single-point-of-failure risk if abandoned. Active development so far (v1.0 → v1.32 in 2 months).

- Faultier's PBR Landscapes 4k + ERM cover VPBR's landscape gap; Faultier's PBR Armors and Clothes covers the gear gap.
 - VPBR includes mountain textures that must be overridden by ERM. Load ERM after VPBR.
 - Do NOT install `Faultier's Skyrim AI0` alongside VPBR — same texture paths.
 - PBR ambition can outrun the rest of the stack if weather, trees, water, and lighting aren't held to a similar bar.
-

6.5. Open Research

Open research for the graphics pipeline is tracked in `TODO.md`.

7. Textures & Meshes

7.1. Mesh Improvements And Parallax Support

Improve shape quality and silhouette readability without creating an unmaintainable conflict swamp. The best outcome is a stable baseline of mesh improvements, targeted fixes where they matter, and parallax support only where textures benefit from it.

 **Install order:** Core foundation → doors & walls → locks → furniture & city detail → mesh replacers → environment props → alternatives. Follow the grouping below.

7.1.1. Core Foundation

Mod	Description
Static Mesh Improvement Mod (SMIM)	Core mesh foundation.
Unofficial Material Fix	Corrects material records for Community Shaders PBR. Install after SMIM.
Simplicity of Snow v0.26	Main file only (32.8 MB). Skip Parallax Meshes and patches for mods not in the list.
XEMI Utility	Prerequisite for FYX collision fixes below.
FYX - Nordic Doors and Traps Collisions	Collision geometry fix. Requires XEMI Utility.
Flame VFX Edit	Mesh-only fire spell effect improvement. CS-compatible.

7.1.2. Weapon Mesh Improvements

Mod	Description
LeanWolf's Better-Shaped Weapons	Upgraded vanilla weapon silhouettes and mesh shapes. Stacks with weapon retextures and Heavy Armory.
Imperial Gladius — Imperial Sword Replacer	Replaces vanilla Imperial Sword model. Single mesh replacer.

7.1.3. Doors, Walls & Structures

Mod	Description
Glorious Doors of Skyrim (GDOS)	Higher-detail door meshes for all architecture types. FOMOD: 2K textures , install all door categories , include Splendid Mechanized Dwemer Door (ESL).

Whiterun Has Walls Redone	3D geometry and parallax textures for Whiterun's exterior walls. Replaces flat walls.
Stockades of Skyrim 3D	Base layer for 3D stockade walls and barricades.
FYX - 3D Stockades	Overwrites Stockades of Skyrim 3D meshes. Install after the stockade base.
Born in a Barn - Door Curtains for Shacks	BOS-based door curtains for shacks and poor homes. No ESP.

7.1.4. Security Overhaul SKSE Series

Install together. Use **2K textures** for all — locks are examined up close but 4K is wasted VRAM.

Mod
Lock Variations
Regional Locks
Lock Add-ons
Some More Locks
Extra Locks
PBR

7.1.5. Furniture, Clutter & City Detail

Mod	Description
Snazzy Furniture and Clutter Overhaul (SFCO)	Clutter objects, geometry, and variety beyond VPBR's base textures.

Snazzy City series (BOS-based, city-by-city clutter improvements):

Mod
Whiterun AIO
Windhelm AIO
Solitude AIO
Markarth AIO
Riften AIO
Falkreath AIO
Dawnstar AIO
Winterhold AIO
Towns and Settlements
Misc Locations
Player Homes
Snazzy Interiors - Dawnstar Mortar and Pestle
Snazzy Interiors - Winterhold Birna's Oddments
Snazzy Interiors Patch Collection

7.1.6. Mesh Replacer Series

All mesh-only, no ESP.

Mod
V.A.E.R.
Thrones Expanded
Rustic Repose - A Common Bed Replacer
Use Those Blankets
Sleipnir Beds
The Royal Seat
Divide and Conquer
Oddments and Miscellanea
Utenlands Nordic Tents
Redoran Reverie
HFs Archery Target Remodel
Diverse Grindstones
Blary's Booksets
ElSopa Misc Ruins Redone
Snazzy Orc Chests

7.1.7. Environment & Props

Mod	Description
TB's 3D Driftwood	3D driftwood mesh replacer. No ESP. Complements coastal areas.
WiZkiD Hunter's Camp Overhaul	Visual overhaul for hunter camps (tents, campfires, pots, clutter). No ESP.
Improved Fish SE	Higher-quality fish meshes and textures.
Improved Fish PBR	PBR textures for fish. Requires Improved Fish SE as mesh base.
CC Fish PBRed	PBR textures for CC fish. Requires free AE fish addon.
JS Badges of Office	Replacer for Jarl's crown, court-mage's circlet, steward's key. Mesh-only, no ESP.

> **PGPatcher:** Use the PGPatcher workflow in → [Graphics - PGPatcher](#) for late-stage parallax generation rules.

7.1.8. Alternatives

Mod	Description
High Poly Project	Optional. Keep only if the extra geometry earns its conflict cost.
RUSTIC CLUTTER COLLECTION	Higher-res diffuse/normal replacer. May be redundant with VPBR's clutter coverage — evaluate side-by-side.

7.1.9. Mesh Section Notes

- High Poly Project adds overlap and complexity for surprisingly small on-screen gains.
- Parallax looks excellent on some surfaces and broken on others if support is inconsistent.
- Snow, rock, landscape, and water meshes should not be finalized independently of texture and weather winners.

—

7.2. Furniture, Clutter, And Item Replacers

Texture and mesh upgrades for furniture, clutter, and items beyond the Snazzy city series — Rally's, Arc's, and Kanjs author lines plus standalone BOS replacers. No cell-record edits; all are BOS-based or pure replacers.

7.2.1. Rally's, Arc's, And Snazzy Clutter

Mod	Description
Rally's Noble Furniture - BOS	
Rally's Orc Furniture	
Rally's Upper Furniture	
Rally's Barsets	
Rally's Common Furniture	
Rally's Dark Elf Furniture (High Poly - ENB Light)	
Snazzy Common Wardrobes - BOS	Choose the BOS meshes main file.
Whiterun Objects SMIMed (and fixes too)	
Arc's WispMother Redux 2K-4K	Choose 2K.
Arc's Kettle REDUX 2K-4K	Choose 2K.
Arc's Kitchen Redux 2K-4K	Choose 2K.
Arc's MeadBarrel Redux 2K-4K	Stack: install all three (base + Bloody + Distributed). Choose 2K for base.
Arc's Bloody MeadBarrel Redux	
Distributed Arc's Mead Barrels	
Arc's Tankard Redux	
ElSopa - HD Iron Tools Redone SE	
Diverse BOS Kitchenware	
HFs - Diverse Inkwell and Quill (BOS)	Diverse inkwell and quill meshes via BOS or Model Swapper.
Dynamic Things Alternative - BOS	
Nordic Stonewalls	
Nordic Stonewall Terraces	
Nordic Stonewalls Complex Material and PBR	PBR upgrade. Install after base Nordic Stonewalls.
Snazzy Wall Mounted Trophies - BOS	

Fluffy Wall Mounted Dead Animals	
Wall Mounted Dead Animals Fixes	Install after Fluffy Wall Mounted Dead Animals.
Snazzy Diverse Carriages - BOS	
Snazzy Diverse Carriages - Seasonal Patch	Required if Seasonal Landscapes is adopted.

7.2.2. Vanaheimr Series

Mod
Vanaheimr - Mines and Caves - Complex Material - PBR
Vanaheimr - Ore Veins - Complex Material - PBR

7.2.3. Animated Items (mesh replacers)

Mod	Description
Kanjs - Sanguine Rose Animated	
Kanjs - Soup and Stew AIO Animated (SkyPatched)	Choose the SkyPatched version over the base.
Kanjs - Human Heart Animated and Beating Motion	
Kanjs - Taproot Animated and Beating Motion	
Animated Filled Soul Gems	
Animated Ingredients	
Kanjs - Stalhrim Weapons Animated	Animated Stalhrim weapon mesh replacer.
Kanjs - Nordic Puzzle Door Animated	Animated Nordic puzzle door mesh replacer.
Kanjs - Ancient Falmer Tomes Animated	Animated Ancient Falmer tomes (Dawnguard lore books), remesh + 4K + animated emissive.
Kanjs - The Boethiah's Proving Book Animated	Animated Boethiah's Proving Book, remesh + 4K + animated emissive.
Kanjs - Master Spell Tomes Animated	Animated Master spell tomes (College), remesh + 4K + animated emissive.
Kanjs - The Dreamstride Book Animated	Animated Dreamstride Book, remesh + 4K + animated emissive.
Zero frost's Tamriel Alchemy	Animated alchemy ingredient and potion replacer. Conflicts with Tamriel Alchemy Replacer.

7.2.4. Risks & Compatibility

- Kanjs Soup and Stew has two variants: the base animation mod (113104) and a SkyPatched version (113191). The SkyPatched version is recommended for better compatibility.
- Nordic Stonewalls + Complex Material: install the base mod first, then the PBR upgrade on top.
- Most Rally's and Arc's mods are texture-only or mesh-only; verify each against the load order for BOS overrides.
- Snazzy Diverse Carriages Seasonal Patch hard-depends on Seasonal Landscapes (→ [Terrain & Flora](#)).

- Zerofrost's Tamriel Alchemy (185386) conflicts with Tamriel Alchemy Replacer (165730) — pick one.
- The four Kanjs unique tomes (Ancient Falmer / Boethiah's Proving / Master Spell / Dreamstride) override [Book Covers Skyrim](#)'s texture on those same records. Install them **after** the Book Covers stack so the Kanjs animated look wins for those books; drop a specific file if you'd rather keep BCS's cover for that title. (Kanjs's [KanJS Spell Tomes Animated](#) 103057 in [Additional Replacements](#) targets a different set — regular spell tome meshes — no overlap.)

7.3. Unique Entity Replacers

Model and texture upgrades for unique named entities and one-of-a-kind objects — characters, corpses, set-piece models not covered by broad replay packs. Closes visual gaps without swelling into full NPC beautification.

Mod	Description
Night Mother	High-quality model/texture replacer for the Night Mother's corpse. ESL-flagged, zero script or record conflicts.
Skyland Imperial and Nordic Tents	4K tent textures with new mesh for the large Nordic fur tent. No ESP.

7.3.1. Risks & Compatibility

- Verify Night Mother does not conflict with lighting/interior overhauls that modify the sanctuary cell.
- Do not let this subsection grow into general NPC or creature overhaul — those are owned by → [NPCs](#).

7.4. Blood, Decals, And Combat Visual Effects

In-world visual effects during combat: blood pools, persistent decals, lingering marks. Separate from lighting/weather (ambient worldstate) and from hit-reaction/stagger (→ [Third-Person Gameplay](#), → [Expanded Systems](#)).

Mod	Description
Dynamic Bloodpool Framework	Blood and decal framework. Lock in after weather/lighting baselines are stable (decals must read correctly under both daylight and your night setup).
Dragon Breath VFX Edit	Dragon breath projectile/impact VFX replacer. Mesh-only, no ESP.

7.4.1. Alternatives

Mod	Description
-----	-------------

Improved Sparks	Orange spark effects on metal impacts and grindstones. FOMOD with density options (1X-20X).
Lightning VFX Edit	Lightning bolt VFX replacer. Mesh-only, no ESP.
Light Magic VFX Edit	Candlelight/Magelight VFX glow replacer. Mesh-only, no ESP. Sibling of Dragon Breath/Lightning VFX Edit.
Sanguine Symphony	Combat impact overhaul: adaptive blood splatters, detailed decals, HD wound textures, custom SFX, slow-motion kills. Verify overlap with Dynamic Bloodpool Framework and Simplicity of Splatter — decal/impact ownership; check slow-mo kills against Precision/Valhalla/BFCO feel.
Dynamic Footprints SKSE	Persistent footprints on snow, mud, sand. Add after blood/decals baseline is stable.

7.4.2. Risks & Compatibility

- Blood/decals can interact poorly with CS weather mods handling screen-space wetness. Verify under a locked weather baseline.
- Persistent blood decals can strain frame-rate and memory in long fights. Check MCM for decal lifetime and density.
- Blood pools can look odd in snowy regions if the framework lacks surface-specific blending.
- Ownership scope: persistent decals/pools, not hit-effect layer.

7.5. Book and Paper Visual Overhauls

Visual upgrades for books, notes, and paper — the most-read world objects. Texture and mesh improvements make reading feel more tactile and library exploration more rewarding without touching gameplay records.

 **Install order:** Book Covers Skyrim → Enhanced Textures → SkyPatched → Lost Library (optional content).

7.5.1. Base + Enhancements

Mod	Description
Book Covers Skyrim	Core book-cover texture replacer. Every book gets a unique hand-crafted cover. 200K+ endorsements. Choose the Main Files download (not the USSEP Update variant).
Book Covers Skyrim Enhanced Textures	2K/4K texture upgrade. Install after base mod and overwrite. Optional but recommended for 4K displays.
Book Covers Skyrim - SkyPatched	SkyPatcher-based forward-port resolving record-level conflicts. Choose Recommended Main File .
Book Covers Skyrim - Lost Library	Library of 150+ lore books from earlier TES games. Content companion; quest-level decision owned by

	Section 24 . Add the PBR-compatible cover variant where available.
Book Covers Skyrim - PBR (155254)	PBR conversion of the Book Covers stack (book main + Enhanced Textures 178820). Requires the Community Shaders PBR pipeline (locked baseline); install after BCS + Enhanced Textures. 2K recommended.
HFs - Ruined Books	High-quality ruined/destroyed book clutter meshes. Complements Book Covers Skyrim for damaged books.

7.5.2. Risks & Compatibility

- With USSEP in the load order, use the SkyPatched variant — it handles USSEP forwarding plus broader compatibility.
- Enhanced Textures is recommended for 4K displays. At 2K the base BC7 textures are adequate.
- Verify compatibility with mods touching book inventory records (e.g. Unread Books Glow). SkyPatched resolves most record-level conflicts.
- Base mod is 600 MB for 2K. Evaluate whether unique book covers earn their VRAM and disk space cost.

—

7.6. Additional Texture And Mesh Replacements

Small texture/mesh replacers that don't fit the main topic sections above. Each is a single-author or single-object upgrade.

Mod	Notes
Visual Animated Enchants	Original VAE — superseded by V.A.E.R. in the main Mesh Replacer Series. Keep only if V.A.E.R. conflicts.
Apophysis Dragon Priest Masks SE	Dragon priest mask retexture. Optional cosmetic upgrade.
Runed Nordic Weapons	Nordic weapon mesh/texture replacer.
Elven Armor replacer glow	Elven armor retexture with glow maps.
Elsopa Iron weapons redone	Iron weapon retexture.
Ennead Banners and Shields	Banner/shield retexture.
Elsopa Quivers Redone	Quiver retexture.
Believable greatswords sheathed	Greatsword sheath mesh fix.
EISopa - HD Medieval anvil	Anvil retexture.
Elsopa Grindstones	Grindstone retexture.
Renthals workbench	Workbench retexture.
VFX Edit AIO - Spells Retexture	Spell visual effect retexture.
JFresh's Mead Stein - A Tankard Replacer	Mead stein/tankard retexture.
SD's Horn Candles	Candle retexture.

JS Dragon Claws	Dragon claw retexture.
Skyrim Remastered - Soul Gems	Soul gem retexture.
Rally's Market Stalls	Market stall retexture.
KanJS Spell Tomes Animated	Animated spell tome mesh replacer.
Alt Markarth Forge	Markarth forge retexture.
HD Unique Handmade Signs (ElSopa)	Inn/shop sign retexture.
Skyking Windhelm Bridge Overhaul	Windhelm bridge mesh/texture replacer.
Tomato's Windhelm - PBR or Complex Material	Windhelm city-wide PBR retexture. 2K main file recommended.
Distinguished Drapery	Curtain/hanging fabric retexture.
The Mists of Blackreach	Blackreach mist/fog visual effect overhaul.
Valhalla Bridges	Higher-detail bridge mesh replacers. Mesh-only, no ESP.
HFs - Imperial Jail Grates	High-quality remodel of Imperial jail grate doors/gates. Mesh-only clutter improvement.
JS Solitude Sewer Cover SE	3D remake of the Solitude sewer cover (1K/2K/4K, 4K recommended). Opening/closing animation for activators. No ESP, straight-swap replacer.
JS Dwemer Puzzle Cube SE	Dwemer puzzle cube retexture (johnskyrim). Complements the Dwemer visual stack.
JS Dwemer Artifacts SE	Dwemer artifacts (Keening, Aetherial items, etc.) retexture. Complements the Dwemer visual stack.
JS Dwemer Control Cube SE	Dwemer control cube retexture. Complements the Dwemer visual stack.
RUSTIC RELIEFS	Compilation retexture: monuments & tombstones, Nordic murals, standing stones, word walls (2K). Gamwich. Verify mural overlap vs Temples of the Ancients (38544) before locking.
HFs - War horns - remodel	New Nord war horn model + HD textures. Standalone ESL options for Torgg's war horn and Horn of Jurgen Windcaller. Choose main file; ESL add-ons independent.
HFs - Safe - remodel	New metal-safe model + textures (safeandlock01.nif). No ESP, no scripts, safe mid-game. AI-Generated tag — quality-check before locking.
Valhalla Bridges - Addons Patches and Fixes	Compatibility patches and addon coverage for Valhalla Bridges. Install after base.
Improved Closefaced Helmets	Mesh/texture overhaul for closed-face helmets.
Improved Closefaced Helmets - SkyPatched	SkyPatched variant; preferred over the base ICH for compatibility.

Ivy - Gates of Riverwood	Ivy-covered gate mesh replacer at Riverwood entrance. BOS-based, optional cosmetic.
Better Wine Labels - Vanilla and LotD	Wine bottle label retexture. PBR patch: 175588
Basic Dining Set Replacer	Higher-detail dining table and chair meshes. No ESP.
RUSTIC COOKING - Special Edition	High-res cooking pot, oven, and kitchen clutter textures.
Petroglyphs of Skyrim	Rock carving/runestone visual replacers. Mesh-only, no ESP.
Diverse Windmill Sails - BOS	BOS-based windmill sail variety.
Diverse Foods - BOS	BOS-based food item variety.
Diverse Candles - BOS	BOS-based candle variety.
Diverse Woodchopping Blocks - BOS	BOS-based woodchopping block variety. Requires Happy Little Trees.
Handcarts - BOS	BOS-based handcart variety (6 variants).
Diverse Catches - BOS Fish Racks	BOS-based fish rack variety at docks/markets.
CC Fishing Gear Diversified - BOS	BOS-based CC fishing gear model variety. Requires AE.
Kanjs - Bird Nests and Eggs - up to 4k	Bird nest and egg retexture. Required by Diverse Bird's Nests.
Diverse Bird's Nests - BOS	BOS-based bird nest variety. Requires Kanjs - Bird Nests and Eggs above.
Better Pelts and Hides	Pelt/hide texture replacer.
Fluffy Hanging rabbits	Hanging rabbit mesh/texture replacer.
FrankBlack's Honey in a Jar	Honey jar retexture.
Slightly Better Nordic Henges - BOS	Nordic henge object variety via BOS.
Skeps of Skyrim - Diverse Beehives - BOS	BOS-based beehive/skep variety. ESL-flagged. Requires BOS.
Diverse Tanning Racks - BOS	BOS-based tanning rack variety at crafting stations. Requires BOS.
HFs - Buttocks Rest (Bar Stool BOS)	BOS-based bar stool variety in inns/taverns.
Arc's Gem Holder Redux	High-quality gem holder mesh/texture replacer. No ESP.
Cabbage - A Cabbage Mod	Hyper-realistic cabbage mesh/texture replacer. Mesh-only, no ESP.
Comfy Coffins	Coffin and vampire coffin mesh replacer. Mesh-only, no ESP.
Showcases of Skyrim - Display Case Replacer	Display case mesh/texture replacer.
Vanilla Table Replacers	Higher-poly table mesh replacer for noble, common, and rustic tables. Alternative — verify no overlap with Basic Dining Set Replacer (30055).
Vanilla Table Replacers PBR Conversion	PBR textures for Vanilla Table Replacers above. CS-compatible. Install after VTR base.

Cisterns of Skyrim - Rain Barrel and Blood Barrel Replacer	Rain barrel and blood barrel mesh replacer. No ESP.
Diverse Downed Logs - BOS	BOS-based downed log variety in wilderness.
Glorious Grainmills	Grainmill mesh/texture replacer. No ESP.
Diverse Chicken Coops - BOS	BOS-based chicken coop variety at farms.
Diverse Hunter Camps - BOS	BOS-based hunter camp overhaul with new assets, diverse catches, and supplies.
Nordic Farmfield Stonewalls	Stonewall mesh variety for farm fields. Distinct from Nordic Stonewalls (57686).
Simplicity of Splatter - High-Res Blood Textures	Blood splatter/decal retexture. Verify compatibility with Dynamic Bloodpool Framework (172080).
Khajiit Has Tents	Khajiit caravan tent mesh replacer. Install before Khajiit Has Tents Animated.
Khajiit Has Tents Animated	Animated tent meshes. Install after Khajiit Has Tents base.
Rope Bridges Remastered	Rope bridge mesh/texture replacer. Broken bridge variants. No ESP.
Skyfall's Whiterun Bridges Replacer	Lore-friendly Whiterun footbridge mesh replacer. No plugin.
Common Clothing Retexture SE	Common clothes texture replacer.
One Quiver to rule them all	Quiver attachment positioning fix.
HFs - Flagon (BOS and Model Swapper)	Flagon/tankard variety via BOS.
Deep Slumber - Dwemer Bed Replacer	Dwemer bed replacer. New and Timeworn variants, optional BOS distribution.
Divergence - Skytone - Runes	Custom rune texture replacer.
Temples of the Ancients (2K - 4K)	Nordic temple/Aedric ruin retexture. 2K recommended.
Temples of the Ancients Complex Material and PBR	PBR upgrade for Temples of the Ancients. Install after base mod.
SB - Pressure Plate Trap Blending - BOS	Blends pressure plates into floor textures via BOS.
Skyland Bits and Bobs - A Clutter Overhaul	Clutter item texture/mesh overhaul. May overlap with Snazzy and Rally's coverage — evaluate side-by-side.
CleverCharff's AIO	Architecture and dungeon texture overhaul. Broad coverage — test for cohesion with VPBR.
RUSTIC MAPS	High-res world map and local map texture replacer. Redundant with FWMF paper maps — pick one map route.
Parallax Textures For Vanilla Game and Various Mods	Broad parallax texture pack for vanilla and mod-added objects. Use 2K for performance.
Detailed Rugs	Higher-detail rug texture/mesh replacer. No ESP.

Detailed Rugs 4K-2K - Cleaned and Upscaled Textures	Cleaned/upscaled rug textures. Overwrites Detailed Rugs — pick one.
JS Rumpled Rugs SE	Rumpled rug mesh/texture replacer. Alternative to Detailed Rugs.
Parallax Caves 4k-2k	Cave wall parallax texture replacer. Use 2K. Complements Vanaheimr Mines and Caves.
Stacks of Septims - 3D Coin Piles	3D coin pile meshes. Mesh-only, no ESP.
Stacks of Septims - My patches SE by Xtudo	Compatibility patch hub — C.O.I.N., LotD, JS Septims, Ennead, Skyland Bits, Maple Manor, etc. Install after base.
JS Purses and Septims SE	Higher-detail coin purse and septim meshes/textures.
Skyrim Fantasy Overhaul - Fantastic City Walls SE-AE	City wall texture overhaul with fantasy-inspired style.
Kanjs — Dragon Stone and Map of Dragon Burials	Animated dragon stone and burial map mesh replacer. No ESP.

8. Lighting

8.1. Lighting Overhaul Strategy

Build lighting as a coherent layer supporting the shader-first visual direction, stronger world scale, and third-person readability. Interior mood matters, but so do readability, compatibility cost, and coherence with the final weather route. The baseline is a Community Shaders-native stack with much lower patch burden than the LUX family.

 **Install order:** SLT → CS Light + True Light + WSU → Based Lighting Configs → Ambient Templates → Helios → Embers XD → Luminous Atronachs.

8.1.1. Core CS-Native Stack

Mod	Role	Notes
Standard Lighting Templates (SLT)	Foundation	Standardized light template records. Replaces per-cell template work.
CS Light	Core	CS-native interior/exterior lighting.
True Light	Exterior bulbs	Exterior bulb placement via Light Placer. Pin v3.4.1 — v3.5.x breaks LoS II lamppost ground illumination.
Window Shadows Ultimate (WSU)	Interior	Interior shadowed windows.
Ambient Templates for Lighting Mods	Required	Required by the CS-native stack.

> **True Light FOMOD:** Select **Exterior mode** — attaches bulbs to exterior meshes (braziers, campfires, torches, lanterns) via Light Placer. This replaces Lux Orbis's exterior lighting role.

8.1.2. Configuration & Ambient

Mod	Role	Notes
Based Lighting Configs	Presets	Standardizes tone/behavior across interiors/exterior.
Helios	Ambient	Dynamic auto-ambient adjustment for interiors and exteriors via Community Shaders.

8.1.3. Fire & Effects

Mod	Role	Notes
Embers XD	Fire	140K+ endorsements. Install after main lighting overhauls. → Graphics - Lighting
Luminous Atronachs	Effects	Self-illuminating atronachs via particle lights.
KD Realistic Fireplaces	Fire	Optional fireplace visual upgrade. Cosmetic only.

8.1.4. Alternatives

- **EmbersXD Torch Edit** ([Nexus](#)) — Torch visual replacer for Embers XD. Optional cosmetic.
- **Relighting Skyrim SE** ([Nexus](#)) + **Luminosity Lighting Overhaul** ([Nexus](#)) — Lightest vanilla-plus route. Falls back if the CS-native stack fails compatibility testing.
- **ELFX Shadows** ([Nexus](#)) — Strong alternate if CS-native proves insufficient for interior mood. Hard-requires base [Enhanced Lights and FX](#) ([Nexus](#)).
- **iLluminous Elementals** ([Nexus](#)) — Broader elemental-light approach. Alternative to Luminous Atronachs.
- **Window Shadows RT - Updated** ([Nexus](#)) / **Ambiance** ([Nexus](#)) — Fallback interior shadow mods.

8.1.5. Risks & Compatibility

- The CS-native stack has 35-55 total patches versus LUX's 80-120 across the Patch Hub.
- CS Light + True Light + WSU relies on Light Placer non-shadow-casting bulbs — test together before locking.
- WSU can hit shadow-caster limits and flicker in certain interiors — monitor during playtesting.
- Very dark interiors look impressive in screenshots and feel worse in long play sessions.
- Choosing lighting before weather is settled can produce a mismatched visual tone.
- True Light author is doing a major rework — monitor for a post-v3.5.x fix for the LoS II lamp post bug.
- SLT provides template-level standardization; verify it does not conflict with CS Light's template expectations.

9. Weather & Water

9.1. Weather And Atmosphere

Weather should be chosen as part of the Community Shaders presentation layer, not as an isolated plugin choice. The target is a grounded but modern visual tone that preserves forest, mountain, and travel readability in third person.

Primary candidates — test side-by-side:

Mod	Description
Azurite III CS	Baseline. Requires the balanced CS add-on tier or higher (SSGI, Screen Space Shadows, Grass Lighting, Wetness Effects). 1,263 endorsements, FOMOD with darker nights and reduced bloom for tunable grim-dark aesthetic.
NAT.CS III	Alternative. Works on the conservative CS stack (no SSGI required).
Sky Sync - Community Shaders	Sky colour/dome enhancement for CS. Lighter than full weather replacement.

9.1.1. Alternatives

Mod	Description
Raid Weathers CS	Gameplay-clarity route. Requires base RAID Weathers .
RAID Weathers (base)	
Real Weathers Remastered	CS-native ground-up build. DALC, moon phase lighting, physical sky approximation. 526 endorsements. Nights reported as very dark — test before adopting.
Vanilla Weathers for CS	Vanilla-plus CS-native. Has a True Storms patch.
Obsidian Weathers	Fallback benchmark.
Obsidian CS	Fallback benchmark.
Cathedral Weathers	Fallback benchmark.
Azurite Weathers III - Enhanced	Later tuning layer, not the first decision.
Skydreamus CS Presets	Later tuning layer, not the first decision.
Mists of Tamriel	Mist/fog atmospheric enhancement. CS-compatible — verify fog density in third-person readability.

9.1.2. Azurite III Addon

Mod	Description
-----	-------------

Azurite III CS - Realistic Darker Nights	Darker night preset for Azurite III CS. Requires testing — older nights affect third-person readability and combat detection. Needs Azurite III CS as base.
Morning Fogs SSE	Rolling morning fogs over lakes/ivers (dawn windows, MCM-tunable). Adds new atmospheric fog only — compatible with everything except Interior Floating Fog Remover . Verify fog density in third-person readability (same gate as Mists of Tamriel) and Azurite III CS fog overhead before locking.

9.1.3. Risks & Compatibility

- A weather setup that looks stunning in screenshots can flatten gameplay readability in forests, storms, or dusk travel.
- Overly aggressive fog, contrast, or dark nights work against third-person exploration and combat.
- Preset stacking makes the final visual chain harder to reason about if weather, CS options, and external presets all push tone differently.

9.2. Water Visuals

Treat water as a full visual stack: base water look, CS water features, mesh support, foam, waterfalls. Target: cold, readable, grounded water that holds up in motion during travel.

9.2.1. Baseline

Mod	Description
Water Effects - Community Shaders	Baseline shader feature layer. CS 1.5.2+ ships a Unified Water module.
Simplicity of Sea	Baseline. Lightweight, proven CS water. Pairs with Water Effects CS for visual depth.
Splashes Of Skyrim	Baseline inclusion, not optional. SKSE plugin for water impact effects; CS 1.4.1+ includes FPS optimizations.
Loki's Wade In Water	Water-interaction layer for wading. Mesh/anim replacer, no ESP.

9.2.2. Alternatives

Evaluate these only after the main water base is selected.

Mod	Description
Simplicity of Sea	
Color and Transparency Tweaks	
Realistic Water Two SE	Legacy baseline.
Natural Waterfalls	

Rally's Water Foam	
Water Debris	Floating debris VFX.
Underwater Bubbles - SKSE Plugin	No ESP, complements surface stack.
Refined Water Droplet Replacer for R.A.S.S.	Rain/water droplet visual replacer. Works with R.A.S.S.
Wade in Water Redux - Swimming Overhaul	Swimming overhaul with wading physics. Broader scope than Loki's Wade.
GKB Waves Reborn	High-quality animated wave shader for shorelines. Prerequisite for GKB Waves For Various Mods.
GKB Waves For Various Mods	GKB wave coverage for mod-added coasts and new-lands. Requires GKB Waves Reborn. Verify compatibility with chosen water mod.
WAVY Waterfalls Effect	Animated waterfall mesh replacer with wave motion. CS-compatible.

9.2.3. Risks & Compatibility

- Water can look strong in still shots and distracting in motion if foam, reflections, or wave response are overdone.
- Water tone can clash with the chosen weather route.
- Mesh fixes and waterfall add-ons can help one setup and hurt another.

9.3. Sky, Stars, And Auroras

The cosmic visual layer: night-sky stars, aurora rendering, sky-dome enhancements. Kept separate from weather because sky-dome work is about clear-night rendering and long-horizon reads, not precipitation and overcast coverage.

9.3.1. Baseline

Mod	Description
Praedy's Night Sky AIO	Locked. Complete night-sky overhaul: stars (9 options), nebula (16 options), galaxy (4 options), constellations, Masser + Secunda moons, auroras. FOMOD-tunable. Choose 2K. 5,795 endorsements, v2.1.
	Shooting Stars SE
Locked. Randomized shooting-star streaks overlaying the static sky. Script-free, no overlap with Praedy's textures.	

9.3.2. Alternatives

Mod	Description
-----	-------------

Rally's Celestial Canvas	Rejected — Praedy's AIO chosen for higher endorsement signal (5,795 vs Rally's) and granular FOMOD tuning (9 star options, 16 nebula options).
AURORA S.E.	Rejected — redundant. Praedy's includes aurora coverage.
Realistic Galaxy / Ultra Hi-Res Nightsky / Worlds Behind Glass	Rejected — single-texture replacers. Praedy's AIO provides coherent full-sky coverage.
	Draco's fantasy auroras / Draco's moons
Rejected — redundant with Praedy's aurora and moon coverage.	
Texture Overhaul Stars and Galaxy of Nirn	Rejected — single-texture replacer. Praedy's AIO preferred for coherent stack.
Skyrim Textures Redone - Stars	Rejected — single-texture replacer. Praedy's AIO preferred for coherent stack.

9.3.3. Risks & Compatibility

- Praedy's aurora textures can look out of place in non-northern settings; verify southern holds.
- Sky-dome edits can fight with weather mods that tweak the night sky. Give Praedy's sky-dome priority for night-sky changes.
- Strong auroras require a real night-darkness baseline from → [Performance](#) ; bright nights kill the aurora.
- Shooting Stars SE is script-free and adds dynamic streaks on top of Praedy's static textures — no overlap risk. Verify the combined visual reads cleanly under Azurite III CS dark nights.

10. Terrain & Flora

10.1. Terrain, Roads, And Snow

Treat terrain, roads, and snow as one connected presentation layer. Must stay coherent with Community Shaders, PBR support, rock meshes, weather, and later LOD generation.

10.1.1. Baseline

Mod	Description
Better Dynamic Snow SE	Snow-and-ash multistage accumulation. Works with Simplicity of Snow ; the BDSPatcher Synthesis patcher handles mod-added objects automatically.
Better Dynamic Ash SE	Solstheim ash equivalent.

10.1.2. Landscape And Terrain

Mod	Description
TomatoRim PBR Landscapes AIO	Locked. PBR-native landscape textures. Coherent with Tomato's PBR Vanilla Trees (locked tree texture baseline) for unified CS visual pipeline. Choose 2K.
Enhanced Rocks and Mountains - Complex Material and PBR	Primary mountain/rock identity. CS-native, no plugins, Complex Material support. Drop Majestic Mountains entirely (author inactive since June 2022, mesh misses snow coverage on rock piles).
Enhanced Rocks and Mountains - Fix and Addon	Extends ERM to underground spaces and Whiterun Skyforge.
Tomato's Whiterun Remake - PBR or Complex Material	Whiterun-specific terrain retexture.

10.1.3. Alternatives

Mod	Description
Skyking Fantasia Landscapes	Rejected — fantasy-leaning aesthetic conflicts with grim-dark pillar.
A Cathedralist's PBR Landscape	Rejected — TomatoRim chosen for shared-author coherence with locked tree textures.
Vanaheimr - Landscapes - AIO - Complex Material - PBR	Rejected — TomatoRim chosen for shared-author coherence with locked tree textures.
Atlantean Landscape - Majestic Edition	Rejected — lower community signal.

Atlantean Landscape - Complete - Complex Terrain Parallax	Rejected — older version; superseded by Majestic Edition.
Tomato's Complex Parallax Material Landscapes AIO	Rejected — TomatoRim PBR is the newer locked version.
Seasonal Landscapes	Rejected — excessive patch debt at this modlist's complexity level.
Tomato's Whiterun Remake PBR - Seasonal Landscapes Patch	Rejected — depends on Seasonal Landscapes which is rejected.

10.1.4. Roads

Mod	Description
Blended Roads - Light Plugin (ESL)	
HD Remastered Blended Roads	HD hand-made rework. Choose 2K for performance, 4K for visual sweet spot. Requires Blended Roads base.
Simplest Roads	Minimal road texture replacer. Alternative to Blended Roads.
Simple Complex Parallax Roads	Parallax road textures requiring CS Complex Material. Alternative to Blended Roads.

10.1.5. Snow And Support

Mod	Description
Simplicity of Snow	
Nordic Snow	
Nordic Snow - Complex Material	
Hyperborean Snow SE - 8K	Comparison candidate, not locked.
Better Dynamic Snow SE	Multistage accumulation with terrain-blended coverage.
Better Dynamic Ash SE	Solstheim ash equivalent.
Enhanced Rocks and Mountains - CM and PBR	See Landscape section above; listed here for snow coherence.
Softly Obscuring Snowfall - No Dirty Ruins Snow	Removes dirty brown vertex-color shading on snow statics for a clean fresh-snow look. Meshes based on vanilla/SMIM/Assorted Mesh Fixes. FOMOD has optional patches for Simplicity of Snow, Snozz' Resource Pack, Enhanced Rocks and Mountains, Wyrmsstooth, Deadly Dragon Lairs. Fits the CS clean-snow direction.

10.1.6. Alternatives

- **Windhelm Is Snowy - Base Object Swapper** ([Nexus](#)) — BOS-based snow coverage for Windhelm. Complements Simplicity of Snow.

- **Snowy Standing Stones for Snowy Regions** ([Nexus](#)) — BOS-based snow coverage for standing stones in snowy regions.

10.1.7. Risks & Compatibility

- Snow can look good in isolation and fail once mixed with mountains, roads, and weather-heavy scenes.
 - Heavy terrain parallax can look impressive close up while adding artifact risk or visual noise in motion.
-

10.2. Flora Visuals

Flora is the ground-level readability layer between terrain materials and tree coverage. It should improve grass density, wildland texture, and plant variety without making traversal unreadable or creating constant patch work.

10.2.1. Grass

Mod	Description
Skoglendi - A Grass Mod	Baseline grass.
Merethic Grasslands	Variety enhancer; pairs with Skoglendi for northern grassland texture.
Origins Of Forest - 3D Forest Grass	Density enhancer. Add only if forests feel too sparse after terrain and tree choices are locked.

10.2.2. Plant Replacers

Mod	Description
Mari's flora	
Real Wheat Fields	Fuller wheat field meshes/textures. 2K, ESL. Option B adds a big wheat field east of Whiterun. Requires Wheat Replacer (112074) as base.
Freak's Floral Solstheim	Solstheim ash-covered grass/flora replacer.
A Nirnroot	Nirnroot mesh/texture replacer with glowing effects. No ESP.
Flora Orientalis	Dependency for Vinland Grass Patch + mixed-route coverage.
Waterplants	Dependency for Vinland Grass Patch + mixed-route coverage.
Flora Additions - Water Plants	Optional water plant flora additions.
Reimagined Mountain Flowers - Base Object Swapper - Seasons of Skyrim	Mountain flower replacer with BOS and Seasonal Landscapes support.

10.2.3. 3D Plant Replacers (mesh-only, no ESP)

Mod	Description
Cathedral - 3D Plants (Clover, Deathbell, Dragons Tongue, Lavender, Thistle, Snow Berries)	Full Cathedral 3D series.
Cathedral - 3D Stonecrop	
DrJacopo - 3D Tundra Shrubs	
Cathedral PBR Plants	PBR texture layer for the Cathedral 3D series. Requires original Cathedral plants and PGPatcher.
PBR Waterplants	PBR textures for underwater plants.

10.2.4. Tree-Adjacent

Mod	Description
Skyfall's Sleeping Hist Tree Overhaul	Riften Sleeping Hist tree replacement. Mesh/texture only, no ESP.
TMD The Rift Leaves	3D leaf particle replacer for The Rift. No ESP.
Immersive 3D Leaves	Replaces Riften's flat 2D leaves with 3D models. By MaximusTheWizard.
Bigger trees	Optional tree size multiplier.

10.2.5. Environment-Plant Extras (visual polish, not baseline)

Mod	Description
High Quality Ivy Replacer - BOS	Replaces ivy meshes/textures. Animated/static via FOMOD. No performance cost.
Grumbledook Swordferns - Optional PBR	New swordfern meshes/textures. Hard-requires BOS; PBR variant requires CS.

> Evaluate these after grass, 3D plant replacers, and tree choices are locked. Both layer via BOS (no cell-record edits).

10.2.6. Alternatives

Mod	Description
Folkvangr - Grass and Landscape Overhaul	Main heavier alternate.
Vinland Grass Patch	Only after the final grass combination is narrow enough.
QW's Grass Patch 2	Only after the final grass combination is narrow enough.
Grassreach - Blackreach Grass	Blackreach-specific grass replacer.
Wildlands Renewal	Grass and groundcover improvement mod. Evaluate as Skoglendi alternative or layering candidate.

Freak's Floral Meadows	Full grass overhaul by TheBloodyFreak. Replaces Skoglendi — not a plant replacer. iMinGrassSize 60, requires Cathedral 3D Grass Library meshes. No other grass mods. Performance-friendly alternative to FFF.
--	--

10.2.7. Risks & Compatibility

- Grass that looks strong in screenshots can make traversal and combat readability worse.
- Layering multiple grass and plant mods creates real patch and cache maintenance cost.
- Forest-floor enhancers can become too dense once weather fog and heavy tree coverage are added.

—

10.3. Tree Overhauls For Dense Forests

Trees are the main large-scale world-shaping layer for wilderness mood, canopy silhouette, and perceived regional scale.

10.3.1. Baseline

Mod	Description
Nature of the Wild Lands	Baseline. Regional tree variety per hold — natural distribution. Fits “Living the world” with varied forest biomes.
Tomato's PBR Vanilla Trees	PBR tree texture option; can layer under whichever tree overhaul wins testing.

10.3.2. Alternatives

Mod	Description
Traverse the Ulvenwald - 3.3	Dense forest alternative. More uniform canopy.
Happy Little Trees	Safer fallback baseline.
Fabled Forests	Fantasy-leaning large trees.
Happy Little Trees PBR	Support signal.
Nature of the Wild Lands - Animations Addon	OAR-based tree interaction animations. Requires NotWL base (locked).
Nature of the Wild Lands - PBR	PBR texture layer for NotWL trees. Requires NotWL base.
Nature of the Wild Lands - Snowier Spruce Trees	Spruce tree snow coverage addon for NotWL. Requires NotWL base.

10.3.3. Risks & Compatibility

- Dense tree overhauls can make roads, structures, and combat spaces harder to read in motion.

- Heavier worldspace tree edits create clipping, blocked sightlines, floating objects, and broad patch debt.
- A tree overhaul that looks excellent nearby can still fail if its distant silhouette is weak.

11. Characters & Creatures

11.1. Character, Skin, And Creature Visuals

Baseline visual treatment for player bodies, skin textures, and common creature visuals before NPC-specific overhauls are chosen. Target: grounded modern look that holds up in third person without drifting into overly glossy, doll-like, or hyper-stylized presentation.

11.1.1. Body Bases And Skins

 **Install order:** CBBE → CBBE 3BA → HIMBO → RaceMenu → BnP skins → female preset → male preset → overlay/freckle packs.

11.1.1.1. Female Body

Mod	Notes
CBBE NSFW	Female body base.
CBBE 3BA (3BBB)	Extension and preset ecosystem.
BnP — Female Skin	Female skin textures.

11.1.1.2. Male Body

Mod	Notes
HIMBO	Male body base.
BnP — Male Skin	Male skin textures.

11.1.1.3. Character Creation

Mod	Notes
RaceMenu SE	Character creation framework.
Racemenu Enhancer	RaceMenu addon: smooth camera, player rotation, freeze (incl. blinking), idle previews, hide UI, strip equipment.
High Poly Head 1.4 SE	Google Drive mirror . Pair with Expressive Facegen Morphs + Expressive Facial Animation (Female + Male).
SlimFantasy — 3BAv2 Body Preset	Female Bodyslide preset. Slim/lean dancer physique.
HIMBO Preset — Chadborn	Male Bodyslide preset. Muscular build.
Lovely Makeup — Racemenu Overlays	Makeup overlay collection.
Lovely Makeup 2	Additional makeup overlays.
Koralina's Freckles and Moles	Freckle and mole overlays.

11.1.2. Face Mesh Branch Notes

- `High Poly Head 1.4` still has current ecosystem support — used in Wabbajack compilations with a whitelisted Google Drive source.
 - Treat it as a player-face and selective NPC-support branch, not a full beauty-overhaul stack.
 - The strongest modern alternative (`Alternate High Poly Head`) still depends on the original HPH 1.4 package.
 - Headpart mismatches, facegen conflicts, and vampire edge cases are still common — keep downstream patching documented in → `Performance` .
-

11.1.3. BodySlide And Outfit Studio Setup

- **BodySlide and Outfit Studio** ([Nexus](#))
- Install as a normal mod in MO2.
- Register `BodySlide x64.exe` in MO2 as an executable.
- Create a dedicated `BodySlide Output` mod before generating anything.
- Keep `BodySlide Output` in the `Output` separator.
- Keep CBBE, CBBE 3BA, HIMBO, the chosen preset, and body-aware armor/outfit mods installed before building meshes.

11.1.3.1. Recommended Setup Order

1. Install CBBE NSFW
2. Install CBBE 3BA (3BBB)
3. Install HIMBO
4. Install BodySlide and Outfit Studio
5. Install SlimFantasy preset
6. Install OBody NG and all body-variety preset packs (before first BodySlide build)
7. Install selected skin textures (if the skin package expects a specific layout)
8. Install armor/clothing packs needing body refits (before first BodySlide build)
9. Add and enable the empty `BodySlide Output` mod before generating anything

11.1.3.2. BodySlide Configuration

- Run BodySlide through MO2, not from the game folder.
- Set output path to the dedicated `BodySlide Output` mod location.
- **Female meshes:** Select the `SlimFantasy` preset (not plain CBBE).
- **Male meshes:** Select the `Chadborn` preset before batch building.
- Use **Batch Build** once the baseline armor/clothing set is stable enough to avoid constant rebuilds.
- If multiple body variants appear, choose the option matching active CBBE 3BA and HIMBO ecosystems.
- Rebuild after installing or replacing major armor packs, outfit conversions, or body-shape presets.

11.1.3.3. 3BA And Preset-Specific Notes

- The chosen female body path is CBBE NSFW + CBBE 3BA + SlimFantasy preset.
- Do not batch build female outfits against plain CBBE if the actual in-list preset requires 3BA/3BBB.
- When armor mods offer both CBBE and 3BA/3BBB options, choose 3BA/3BBB.
- Keep the female body stack consistent across base, preset, and outfit conversions.

11.1.3.4. Outfit Studio Guidance

- Treat Outfit Studio as the exception tool for fixing problem outfits, not a default for every armor.
 - Use when a chosen armor/clothing mod does not match the active CBBE or HIMBO setup cleanly.
 - Keep edited/converted outfit output separate from the original source mod.
 - Document manual outfit fixes so future rebuilds don't depend on memory.
-

11.1.4. Body Variety And Distribution

Distributes different BodySlide presets across NPCs so the world doesn't look like everyone shares the same body shape.

11.1.4.1. Baseline

- **OBody NG** ([Nexus](#)) — Core distribution layer.
- **Real Body - 3BA (3BBB) CBBE Presets** ([Nexus](#)) — Curated semi-realistic 3BA presets for OBody distribution.
- **Realistic Racial Body Diversity - OBody NG Config** ([Nexus](#)) — Per-race body distribution (broader Nords, leaner Elves, heavy Orcs). Covers both CBBE 3BA and HIMBO.
- **Individual presets:** [Rugged Body - 3BA](#) , [Skinny Berry - 3BA](#) , [Realistic chubby and curvy body - 3BA](#) .
- **Male variety: HIMBO SliderPresets Pack 02** ([Nexus](#)) — 10 presets.
- **Racemenu - Thigh Position Sliders** ([Nexus](#)) — Adds 3 thigh sliders (Up/Down, Forward/Back, Spacing) to the Body Scale category for finer thigh-gap customization. Requires RaceMenu + XPMSE (both baseline). Overwrites [RaceMenuPluginXPMSE.pex](#) — **verify against the locked XPMSE RaceMenu plugin before committing** (skeleton baseline accommodation gate).

11.1.4.2. Alternatives

- OBody NG is the only maintained OBody version that works with both OStim and standalone.
- Additional male presets: [The Common Guy - SerketHetyt's HIMBO Preset](#) , [Greek God -- HIMBO Bodyslide Preset](#) (adult-content page; review before adding).

11.1.4.3. Notes

- OBody NG requires building all included presets in BodySlide with [Create Morphs](#) ticked — larger batch build.
 - Every additional preset pack increases BodySlide build time. Keep pack size intentional.
 - OBody NG works with CBBE 3BA and HIMBO but may need config adjustment for other body mods.
 - SlimFantasy preset remains the player-character default; OBody distributes additional presets to NPCs only.
-

11.1.5. Clothing And Armor Texture Upgrades

Mod	Notes
Faultier's PBR Armors and Clothes (2.9 GB)	Complete PBR retexture of all vanilla armors/weapons/clothes. Requires CS + PGPatcher.
RUSTIC CLOTHING — SE	Higher-res clothing textures. Test for style consistency.

Dwemer Armors and Weapons Retexture SE	Gear-side companion to VPBR's Dwemer ruin textures.
Steel Armor Redone — PBR	PBR retexture of CC Steel Armor. Requires CC Alternative Armors — Steel Soldier, CS, PGPatcher. Optional.
Robes Retexture SE	Higher-quality robe textures for all vanilla robes.
Divine Crusader Retexture SE	Retexture of Divine Crusader armor set.
aMidianBorn Book of Silence SE	Comprehensive armor/weapon/dragon retexture. 207K endorsements.
aMidianBorn Content Addon	Extends aMidianBorn coverage to additional armors. Install after Book of Silence.
aMidianBorn Content Addon - 3BA and HIMBO Patch	Body refit patch for Content Addon.
Resurgence Armory - Artifact Weapons and Armor Overhaul	Visual overhaul for artifact weapons and armor models.
3BA and HIMBO Refits for Resurgence Armors	Body refit for Resurgence Armory armors.
HDT-SMP Vanilla Armors	Adds HDT-SMP physics to all vanilla armors.

11.1.6. Recommended Creature Baseline

Mod	Notes
Bellyaches Animal and Creature Pack SSE	Creature texture foundation.

11.1.7. General Notes

- Body-base choices create a large armor-refit and outfit-conversion workload later.
- Choosing 3BA/3BBB increases female outfit compatibility requirements versus plain CBBE.
- Letting BodySlide output overwrite source mods makes auditing and rebuilds much harder.
- Skin mods that look strong in screenshots can still appear too polished or glossy in gameplay lighting.
- High Poly Head 1.4 adds headpart and facegen complexity with vampire fixes, NPC overhauls, and custom presets.
- Do not let this section expand into full NPC beautification — that belongs in → [NPCs](#).

11.2. Hair, Eyes, And Beards

Per-character face assets: hairstyles, eye textures, and beard options. Ensures character close-ups match the grim-dark tone without pulling in full NPC face-gen overhauls.

Mod	Type	Notes
-----	------	-------

KS Hairdos SSE	Baseline	Community-canonical hair pack. 141K endorsements.
High Poly Vanilla Hair	Baseline	Higher-poly vanilla hair meshes (smooth geometry, 3D braids, UV tweaks). Replaces the vanilla styles NPCs/ player use; complements KS Hairdos (which adds its own styles, not vanilla replacements). Human races. No ESP.
Modular SMP Hairstyles	Add-on	100+ unisex SMP hairstyle combos (mix & match front/back) added to the hair slider. Requires FSMP (baseline). Physics cost on the wearer only. Verify against SMP-NPC crash fix (baseline) when used on NPC distribution. ESL.
Eyes Nouveaux	Baseline	Khisartin-style eye textures, grim-dark tone.
LDD Clean Cubemap for Eyes 4K	Baseline	Cleaner eye cubemaps. Stacks on eye-texture baseline.
Brows	Baseline	Brow shape and texture replacer for all races.
Better Argonian Horns	Baseline	Higher-quality Argonian horn meshes and textures. BOS-based.
The Eyes Of Beauty SSE	Alternative	Community-standard alternative to Eyes Nouveaux.
Argonian - Khajiit FabULook Eyes SSE	Alternative	High-res eye textures for Argonian and Khajiit.
Beards of Power	Alternative	High-fidelity beard meshes and textures. Adds variety over vanilla.
BnP - Teeth Overhaul	Alternative	Higher-quality teeth meshes/textures. Complements BnP skin stack.
New Beast Feet	Alternative	Higher-detail beast race (Khajiit/ Argonian) feet with claws/digits.
SG Female Eyebrows + Improved	Alternative	Higher-fidelity female eyebrow shapes. Use Improved variant.
Eyes of Beauty - Nouveaux	Alternative	Combines Eyes Nouveaux meshes with The Eyes of Beauty textures. Requires both as dependencies.
Discipline-first route	Alternative	Rely on vanilla hair and eye textures entirely.

Keep the beard decision minimal; vanilla beards work well with the chosen body and face mesh stack.

11.2.1. Risks & Compatibility

- High-poly hair can clash with extreme lighting/shadow setups; verify KS Hairdos works with the CS-native lighting stack.

- Eye textures that lean too stylized look out of place; pick a realistic pack matching grim-dark.
 - Hair assets are common in NPC-overhaul conflicts; check for hair-distribution patches when → `NPCs` is active.
 - Adding a beard pack without testing creates obvious neck clipping under most lighting.
-

11.3. Skeleton And Bone Replacers

Visual upgrades to in-world skeleton and bone assets: skulls, bone piles, skeleton models in dungeons, Nordic ruins, draugr crypts, dragon priest rooms. Kept separate from the rigged skeleton baseline in → `Animations`.

Mod	Type	Notes
Skeleton Replacer HD — SE	Baseline	High-res skeleton/bone models. Choose 2K main file.
Skeleton Replacer HD — Mesh Patches	Baseline	Add only if modlist includes a creature/content mod needing a mesh patch.
Discipline-first route	Alternative	Rely on vanilla skeleton assets.

11.3.1. Risks & Compatibility

- Skeleton Replacer HD can conflict with creature mods that ship their own skeleton/bone assets.
- Some vanilla draugr and dragon priest models have unique assets the mod can override; verify they're preserved.
- High-resolution bones can look out of place in a dungeon without matching texture work.

12. LOD & Distant Detail

12.1. LOD Generation And Distant Detail

Distant detail determines whether `Elder Wilds` feels large and coherent during travel instead of collapsing into obvious pop-in and flat backgrounds.

12.1.1. Baseline

- `DynDOLOD`-based distant detail is the assumed standard.
- Use the balanced route: strong object and tree LOD quality with targeted support mods.
- Add support mods only where they visibly improve the chosen tree and texture stack.
- Judge distant detail from actual travel routes, mountain overlooks, and city approaches — not static showcase shots.

12.1.2. Required Tools And Dependencies

12.1.2.1. Core Support Mods

Mod
DynDOLOD Resources SE 3
DynDOLOD DLL NG

12.1.2.2. Optional Distant Detail Support

Mod	Notes
HD LODs Textures SE	
Skyfall's Sleeping Hist Tree Overhaul — DynDOLOD Add-On	Billboard generation for replaced tree. Required for correct distant-LOD display.
Happy Little Trees DynDOLOD Optimizations	
Far Object LOD Improvement Project SSE (FOLIP)	Improves distant object LOD.

12.1.2.3. World Map Support

`Elder Wilds` uses its flat map / paper-map framework — see [guide/modlist-ui.md](#).

12.1.2.4. Core Workflow Dependencies

Mod
No Grass In Objects
Worldspaces with Grass SSEEdit Script

Grass Cache Fixes

xLODGen Resource — SSE Terrain Tamriel

12.1.3. Build Order And Configuration

12.1.3.1. Setup

- Install `DynDOLOD Resources SE 3` as a normal mod in MO2. During FOMOD installation, uncheck `Low-Res LOD Textures` and `Holy Cow`. On Skyrim SE 1.6+, also uncheck `Whiterun Exterior Grass`. Place the mod relatively high in the left pane.
- Install `DynDOLOD DLL NG` as a normal mod in MO2. Provides the large-reference bug workaround from its own side — only use if the modlist ships with it and you understand the implications; not recommended unless the list explicitly requires it.
- Install `No Grass In Objects`, `Grass Cache Fixes`, and `xLODGen Resource` as support content.
- Register **TexGen**, **DynDOLOD**, and **xLODGen** as MO2 executables, not mixed into mod folders.
- Treat occlusion as generated output, not as a separate mod pick.

12.1.3.2. Dedicated MO2 Output Mods (create before generation)

Output Folder	Purpose
<code>Grass Cache Output</code>	Generated grass cache
<code>Terrain LOD Output</code>	xLODGen terrain output
<code>TexGen Output</code>	Generated billboards
<code>DynDOLOD Output</code>	DynDOLOD output
<code>Occlusion Output</code>	Occlusion data

Keep generated outputs in the `Output` separator. Do not generate final outputs until tree, terrain, snow, flora, and large worldspace picks are mostly stable.

12.1.3.3. Run Order

1. Confirm load order and conflict resolution first.
2. Generate grass cache (No Grass In Objects workflow).
3. Install/update `Grass Cache Output`.
4. Run **xLODGen** for terrain LOD (once terrain and snow are stable).
5. Install/update `Terrain LOD Output`.
6. Run **TexGen**.
7. Install/update `TexGen Output`.
8. Run **DynDOLOD**.
9. Install/update `DynDOLOD Output`.
10. Generate **occlusion data** near the end, keep in `Occlusion Output`.

Re-run relevant generated layers when a major tree overhaul, architecture change, landscape shift, grass change, or LOD-relevant texture pack is replaced.

12.1.3.4. Recommended Starting DynDOLOD Configuration

- **First-pass preset: High** — start here even if the setup is graphics-heavy. Move above only after checking travel performance and distant coherence.
- Generate both object and tree LOD on the first real pass to judge forest silhouette quality early.
- Keep large-reference and ultra-aggressive options off for the first pass.
- First-pass questions: do distant trees match the overhaul? Do mountain/road/city approach views feel coherent? Is performance acceptable in real travel?
- If using **Happy Little Trees**, test baseline output first, then compare DynDOLOD Optimizations.
- If using heavier tree overhauls (**Ulvenwald**, **Fabled Forests**, **Nature of the Wild Lands**), prioritize clean transitions and stable horizons before longer range.

12.1.3.4.1. Tree-Specific DynDOLOD Rules

Certain tree overhauls require specialised tree rules in DynDOLOD for correct distant rendering. Example for **Happy Little Trees** (from the DynDOLOD add-on page):

LOD Level	Rule
LOD4	Level0 / Mesh
LOD8	Billboard4
LOD16	Billboard(1)
LOD32	Billboard(6)

Apply matching rules for your chosen tree overhaul if its DynDOLOD add-on or author documentation specifies them.

12.1.3.4.2. Additional Configuration Guidance

- **Ultra Trees (3D tree LOD):** Recommended once first-pass High is validated. 3D tree LOD provides substantially better visual quality for a minor performance cost and unloads properly, unlike hybrid LODs. Anything above 1024 tile size for billboards provides diminishing returns — keep at or below 1024.
- **Billboard brightness:** Adjust the R/G/B brightness values in DynDOLOD's grass/billboard settings if distant tree LOD appears too bright in game. A common starting point is 0.500 across all channels.
- **DynDOLOD DLL NG:** If using DLL NG for the large-reference bug workaround, ensure DynDOLOD settings match its expectations exactly (tick the large-reference workaround options in DynDOLOD's advanced GUI). Not recommended unless the modlist requires it.

12.1.3.5. Baseline Starting Profile For Elder Wilds

- **Preset:** High
 - **Quality target:** Balanced object and tree LOD with stable travel performance
 - **First-pass goal:** Believable distant forests, readable city approaches, clean mountain silhouettes
 - Keep optional LOD texture upgrades and tree-specific optimization mods disabled at first
-

12.1.4. Grass Cache And Occlusion

- Use No Grass In Objects only after the main grass/flora direction is narrow enough that cache generation isn't immediately obsolete.
- Keep Grass Cache Fixes as part of the serious grass-cache path.

- Use Workspaces with Grass SSEEdit Script if the final setup needs tailored workspace grass support.
- Treat grass cache as invalid whenever the winning grass combination, landscape coverage, or major workspace edits change materially.
- Regenerate occlusion near the end of a major graphics pass, after grass, terrain LOD, and DynDOLOD inputs are stable.
- Keep occlusion output separate from all other generated mods for easy identification and replacement.

12.1.5. Validation

- No missing distant objects, giant billboard errors, or broken tree lines
- Near and far tree silhouettes transition cleanly during normal travel
- City approaches, mountain overlooks, and forest horizons look coherent under selected weather/lighting
- Generated outputs are enabled in MO2 and load after source mods
- Old generated outputs are replaced when the visual stack changes, not left active by accident

12.1.6. Risks & Compatibility

- Strong near-field visuals can still look broken in motion if distant terrain and tree lines don't match.
- Optional LOD texture and tree-optimization add-ons create extra maintenance for limited payoff.
- Generating too early produces stale outputs that quietly poison later comparisons.
- Partial rebuild habits leave mismatched terrain, tree, and occlusion outputs active simultaneously.

12.1.7. Rebuild Rules

- **Grass cache:** Rebuild if grass baseline, major flora layering, or workspace edits change.
- **Terrain LOD:** Rebuild if landscape, snow, mountain, or terrain-color direction changes.
- **TexGen + DynDOLOD:** Rebuild if tree overhauls, large architecture, major LOD-support mods, or visual workspace edits change.
- **Occlusion:** Refresh when major workspace edits or the distant-detail pass changes enough to affect visibility/culling.
- If multiple major visual categories change together, prefer a clean full rebuild over partial guesswork.

—

12.1.8. Seasonal LODs

When using **Seasons of Skyrim SKSE + Turn of the Seasons**, DynDOLOD and xLODGen can generate seasonal LOD passes so distant detail matches the current season. This multiplies generation time considerably — plan accordingly.

xLODGen seasonal: Tick the **Seasons** checkbox in the bottom-left corner and select which seasons need terrain LOD. Generation takes significantly longer than a single pass.

DynDOLOD seasonal: Tick the **Seasons** checkbox, the **Snow** checkbox, and select the desired seasons in DynDOLOD's GUI. Generation time scales with the number of seasons selected.

Rebuild seasonal LOD whenever the underlying terrain, tree, or landscape seasonal-swap data changes.

13. Modernized UI

13.1. UI Framework Prerequisites

The base menu and interface framework the rest of the UI stack builds on.

13.1.1. Baseline

Mod	Description
SkyUI	Non-negotiable baseline.
Oathvein UI	Locked visual direction. Grim-dark presentation matches the project tone; cleaner dependency chain than Norden UI.
UIExtensions	Support infrastructure, not competing UI identity.
Constructible Object Custom Keyword System (COCKS)	Crafting-menu category infrastructure.
MCM Helper	Persists MCM settings across saves. Hard dependency of TDM, CNO, and others.
Stay At The System Page NG	Keeps System page open after loading a save. Most noticeable on gamepad.
ImGui Icons	Icon font resource for ImGui-based mods (Dialogue History, etc.).

13.1.2. Alternatives

Mod	Description
Norden UI	Modern-rustic by Nithog, but depends on Extended UI (no official SE port).
Vel'dun UI	Dunmer-themed by Nithog; too narrow a fit for Elder Wilds.
Prisma UI	Web-UI framework replacing SkyUI. Incompatible with Oathvein UI and UIExtensions mods.
SKSE Menu Framework	Low-level menu framework for SKSE-based UI mods.

13.2. HUD Overhaul

Health, stamina, magicka, target readability, and permanent interface structure during ordinary play.

13.2.1. Baseline

Mod	Description
SkyHUD	Baseline layout framework: widget positioning, visibility, style consistency.
TrueHUD	Combat-feedback: actor info bars, boss bars, player widgets, recent-loot logging. Same author as TDM.
Casting Bar	Cast/draw timing for spells, shouts, and bow draw.
Floating Damage	Optional combat-feedback overlay. Install only after confirming acceptable visual noise.
QuickLoot IE	Container/corpses loot window on interact. Improved controller support, third-person readability. FOMOD with placement presets.
Oxygen Meter 2	Breath meter when underwater. Configurable position/opacity.
Oxygen Meter 2 - Lung Shaped	Cosmetic lung-shaped icon.
Oxygen Meter 2 - Fixes and Additions	Bugfixes and feature tweaks.
Detection Meter	Stealth detection meter. Pairs with RAID.
Detection Meter - AE Support	Required for 1.6.1170.
Casting Bar Reskin - Edge UI Inspired	Cosmetic reskin of Casting Bar. Evaluate integration with Veldun UI's own casting bar (not in Elder Wilds stack).
STB Widgets	Lightweight modular widgets (clock, time, compass-free direction, stat bars). SKSE + MCM.
Follower Stats	MCM-based follower stat tracking. Pairs with Nether's.
Tween Menu Overhaul Improved	Main-menu and loading-screen visual overhaul. Updated version with additional scenes.
RUMINATE - Frost and Snow - Main Menu Video Pack	Frost-and-snow themed main menu cinematic.
Skeuomorphism of Skyrim - Immersive Plaque Reading Interface	3D in-world plaque model. Requires BOS. ESL-flagged.
Dynamic Location Pop-ups	Location discovery notification banner.
Complete Controller Setup	SKSE-based controller config: remapping, radial-menus, camera profiles for gamepad.

13.2.2. Alternatives

Mod	Description
Stats Tracker Menu - STM	Brand new (May 2026), very low community signal. Test before locking.

Kill feed	Kill notification overlay.
---------------------------	----------------------------

13.3. Inventory And Item Card Improvements

13.3.1. Baseline

Mod	Description
moreHUD SE	Mandatory. Broader item-information layer.
moreHUD Inventory Edition	Mandatory. Extends readability into inventory views.
Inventory Interface Information Injector	Value/weight/stat-per-weight columns. Same author as moreHUD.
I4 Weapon Icons Overhaul	30+ weapon category icons. Depends on III.
I4 Armor Icons Overhaul	
I4 Shout Icons Overhaul	
The Handy Icon Collection Collective (THICC)	Thousands of high-fidelity icons. Additive to I4 family.
Favorites Menu Effects Description	Effect descriptions in favorites menu. SKSE plugin, no ESP. Essential for gamepad.
B.O.O.B.I.E.S (aka Immersive Icons)	Icon overhaul for SkyUI. 28K+ endorsements. FOMOD: 2K recommended.
B.O.O.B.I.E.S - POTIONS	Dedicated potion bottle icons.
P.E.N.I.S. for B.O.O.B.I.E.S.	Ingredient icon pack.
A.S.S. for B.O.O.B.I.E.S.	Food & drink icon pack.
Rotols More Icons	Additional coverage for items B.O.O.B.I.E.S doesn't cover.

13.3.2. Alternatives

Mod	Description
SkyUI Item Card Fixes	Cleaner item-card presentation.
Aura's Inventory Tweaks	Stronger sorting and icon structure.
Weapon Stat Viewer V2	Weapon-stat overlay. Lock only if rendered cleanly.
Enchanted Icon Overhaul	Depends on Dynamic Inventory Icon Injector.
Dynamic Inventory Icon Injector	
Book 'Em	Book reading/collecting tracking UI.

13.3.3. Item Description Cards

Add descriptive text to inventory item cards for better readability. All by the same author, share a consistent style. Install as a set.

Mod
Weapons Have Description
Ingredients Have Description
Scrolls Have Descriptions
Soul Gems Have Description
Food and Drink Have Descriptions

13.4. Map Improvements

13.4.1. Baseline

Mod	Description
Flat World Map Framework (FWMF)	Non-negotiable map foundation.
Local Map Upgrade	Local-map for interior/close-range navigation.
HD Local Map	HD textures for the local map upgrade.
Atlas Map Markers SE - Updated with MCM	Expanded map markers with MCM configuration.
Traditional (8K) Skyrim and Solstheim Paper Maps for FWMF (AE and SE)	High-fidelity paper-map replacer for FWMF.
Custom Map Markers for Traditional Skyrim and Solstheim Paper Maps	Cosmetic map marker styling for the Traditional map set.
Updates to various maps for FWMF	Coverage patches for additional FWMF worldspaces.

13.4.2. Alternatives

Mod
The Nightmare Paper Map for FWMF by Limon
Skyrim and Solstheim Paper Maps by Mirhayasu for FWMF
Immersive Paper Map (3rd Edition)
Skyrim Paper Map by FreelanceCartography for FWMF
Legendary Map
Pastel Map Markers
Wurmstooth Paper Map for FWMF
Apocrypha Paper Map for FWMF
Beyond Reach Paper Map for FWMF
Blackreach Paper Map for FWMF
Soul Cairn Paper Map for FWMF

13.5. Magic And Journal Improvements

13.5.1. Baseline

Mod	Description
Convenient Reading UI - SE	Reading comfort for normal play.
Wider MCM Menu for SkyUI	Support fix for cramped config menus.
Modern Wait Menu	Clock-face interface showing time, weather preview. Gamepad-friendly. SKSE plugin, no ESP.
Completionist - Skyrim Completion Tracker (NG)	MCM-based quest/item/location/book/shout completion tracker. Run the Automated GUI Patcher.
Completionist Addons	Mod-added content tracking.
Quest Journal Overhaul	Journal UI redesign with quest tracking improvements.
Quest Journal Overhaul - Unofficial Mods Support	Extended mod support patch. Install after the main QJO.
Character Menu SE	Character stats/sheet UI replacement. Alternative.
	Character Menu SE - More Factions
Additional faction tracking for Character Menu SE.	
Even Better Quest Objectives SE	Quest objective text clarifications and journal improvements. Alternative.
	Knotwork
Custom quest journal knotwork art for non-vanilla questlines. Complements Quest Journal Overhaul. SKSE plugin. Alternative.	

13.6. Dialogue UI Improvements

13.6.1. Baseline

Mod	Description
Better Dialogue Controls	First-pass dialogue friction fix.
Better MessageBox Controls	Same for message boxes.
Dialogue History	Scrollable dialogue log. Essential for gamepad.
Improved Alternate Conversation Camera	Dialogue camera baseline. Test for CS letterbox conflict.

13.6.2. Alternatives

Mod	Description
Dialogue Interface ReShaped	Visual dialogue flow.
Convenient Dialogue UI - SE	
Smart Talk (Dialogue Menu Enhancer)	Highlights quest options, reorders choices, gamepad-friendly skip. 5.5K endorsements.
Various Dialogue Tags	Adds contextual tags to dialogue options (e.g., [Lie], [Persuade], [Ask]). ESL-flagged.
Various Book Tags	Adds contextual tags to book-reading prompts. Same author as Various Dialogue Tags.
Skyrim Autocorrect - Dialogue Grammar Fixes	Fixes grammar, spelling, and punctuation in vanilla dialogue. Text-only, no records.

13.7. Controller-Friendly UI Support

13.7.1. Baseline

Mod	Description
Skyrim SE Controller Interface Fix	
Show Player In Menus	Renders player character in menus for gear previews. Essential for third-person parity.

13.7.2. Alternatives

Mod	Description
Gamepad Plus Plus	Broader controller-input answer.
Wheeler - Quick Action Wheel	Radial interaction layer.
WHEELER - Refined	Updated visuals, better controller support.
Auto Input Switch	Automatically switches input device.
Show Player In Inventory	Renders player model in inventory screen.

13.8. UI Scaling For Ultrawide And High Resolution

Mod	Description
Sovngarde - Mist's Font Replacer	Text readability for 16:9 4K HDR. Prefer Bold V8.9 first.
Complete Widescreen Fix	Secondary for ultrawide layouts.

13.9. Compass, Markers, And Minimal HUD

13.9.1. Baseline

Mod	Description
Compass Navigation Overhaul	First-pass navigational layer.
CoMAP - Common Marker Addon Project	Marker consistency across the FWMF ecosystem.

13.9.2. Alternatives

Mod	Description
Immersive HUD - iHUD SE	Minimal-HUD branch.
MiniMap	Very optional. Last updated May 2021; occasional crash reports on 1.6.1170.

—

13.10. Crafting Menu Improvements

Mod	Description
COCKS	See Framework Prerequisites above. Directly improves the Constructible Object Menu.
SkyUI - Vanilla Crafting menu	Fallback for vanilla-style split categories.

—

13.11. Loading Screens

13.11.1. Baseline

Mod	Description	Notes
Lore-Friendly Load Screen Compendium (Loading Screens)	Locked baseline. Lore-friendly loading screens with in-world art and text. Requires Improved Loading Screen Colors (61712) and SSE Display Tweaks (34705) for the load screen filter — both already in the stack.	By ChildofDragons. Install at 100% frequency to suppress vanilla tips.

13.11.2. Alternatives

Mod	Description	Notes
-----	-------------	-------

The Elder Scrolls Legends - Loading Screens	110 Legends card-art loading screens. Install at 100% frequency in 16:9 with Fixed Height border.	Previously locked baseline. Keep as known-good fallback.
---	---	--

—

13.12. Optional HUD

Mod	Description
Skyrim Party Sheet - Follower and Player HUD	Persistent overlay for player attributes and follower info. v2.7 (June 2026). Research before installing — verify no conflicts with TrueHUD, Oathvein, or Nether's.

—

13.13. Open Research

Open research for the UI stack is tracked in `TODO.md`.

14. Animations and Movement

14.1. Animation Framework Landscape — What Goes With What

Skyrim's animation stack has four distinct layers. Each layer has exactly one active owner. Understanding what replaces what, what depends on what, and what is mutually exclusive is the single most important prerequisite to building a stable animation load order.

14.1.1. The Four Layers

Layer	Role	One Active	Elder Wilds Pick
Behavior Engine	Generates behavior files from animation data. Runs as an external tool (MO2 executable).	Yes	Pandora
Conditional Replacer	Selects which <code>.hkx</code> file plays based on runtime conditions (weapon type, location, weather, NPC identity, etc.).	Yes	OAR (Open Animation Replacer)
Combat Animation Framework	Overhauls attack behavior: replaces vanilla attack chain logic with modern combo systems, enables motion-data-driven attacks, adds jumping/swimming/charged attacks.	Yes	BFCO
NPC Combat AI	Teaches NPCs to use combat animation movesets intelligently — combo selection, distance management, attack commitment.	None required	SCAR (Baseline after Precision is proven)

14.1.2. Behavior Engines

Behavior engines are the foundation. They read your installed animation mods, resolve conflicts between their behavior patches, and write out the final `behaviors` directory. You run them once after changing any animation mod that includes behavior data, and the output lives in a dedicated MO2 mod.

Engine	Status	Creature Support	Reads Nemesis Patches	Reads FNIS Formats	Notes
FNIS	Deprecated	Limited (Add-on)	No	Natively	Closed source. Last meaningful update 2016. No AE support. Do not use.
Nemesis	Superseded	Partial (never completed)	Natively	Partial	Open source. Replaced FNIS around 2020. Creature support was a promised feature that never shipped. Still works on AE 1.6.1170 but Pandora is the direct upgrade.
Pandora	Current	Full	Yes	Yes	Open source. Cross-platform. Error-tolerant (isolates illegal edits so one broken mod doesn't poison the whole patch). Faster generation. Reads both Nemesis patch format and legacy FNIS XML. Elder Wilds baseline.

Key compatibility rule: Pandora replaces both FNIS and Nemesis. You do NOT install FNIS or Nemesis alongside Pandora. Most mods that say “Requires Nemesis” work under Pandora without modification — the Nemesis patch format is read natively. The rare exception is a hypothetical mod using a Nemesis-only code plugin that Pandora hasn’t implemented, but this is essentially nonexistent in a modern AE load order.

Nemesis page lists “Project New Reign — Nemesis Unlimited Behavior Engine” at mod ID 60033 (the real page), NOT the various fork/translation pages that share the Nemesis name.

14.1.3. Conditional Animation Replacers

Conditional replacers sit between the behavior engine output and the game engine. They don't generate behavior files — they decide which animation file to play at runtime based on configurable conditions.

Replacer	Status	Backward Compat	Open Source	Notes
DAR (Dynamic Animation Replacer)	Maintenance-only	—	No (closed)	Author inactive. Works on AE 1.6.1170 but no new features expected. If DAR ever breaks from a future Skyrim update, nobody can fix it except the original author.
OAR (Open Animation Replacer)	Current	Full DAR backward compatibility	Yes	Implements every DAR condition. Adds: animation variants (random/sequential), presets (reusable condition blocks), in-game editor, constant polling, paired animation support, graph variable conditions. Elder Wilds baseline.

Key compatibility rule: Any mod packaged for DAR works in OAR without modification. OAR reads DAR's folder structure and condition format natively. You do NOT need DAR installed alongside OAR — OAR is a complete replacement.

DAR-based mods use folder paths like `meshes\actors\character\animations\DynamicAnimationReplacer\CustomConditions\...`. OAR reads these same folders with no conversion needed. Mod authors increasingly ship OAR-native configs (JSON-based, richer conditions) but DAR-format mods remain fully functional.

14.1.4. Combat Animation Frameworks

Combat frameworks replace Skyrim's vanilla attack behavior — directional power attacks, combo chains, attack commitment, and motion-data-driven movement during swings. They are the layer that makes combat feel modern.

Framework	Requires	Built-In Features	Moveset Format	Notes
MCO / ADXP (Modern Movement Combat Overhaul / Attack - Distar Experience)	AMR + Payload Interpreter + Pandora/Nemesis + OAR/DAR	Directional power attacks (via separate mod), combo chains, motion-driven attacks	<code>MCO_Attack*.hkx</code> naming	Established standard. Vast moveset library. Active community. The "Distar ecosystem" includes MCO, DMCO (dodge), and related mods. Distar's mods have moved to Nexus (mod ID 117115 for the <code>.esp</code> ; main files and movesets still reference the off-site download).
BFCO (Attack Behavior Framework)	AMR + Payload Interpreter + DMK + Pandora/Nemesis + OAR	Directional power attacks (built-in, single hotkey), combo chains, motion-driven attacks, jump attacks , swim attacks , charge attacks , NPC combo AI (bfcoAI), gamepad MCM hotkeys	<code>BFCO_Attack*.hkx</code> naming	Newer alternative. More features packed into one framework. Has an MCO→BFCO converter tool for movesets. Mutually exclusive with MCO/SkySA/ABR.

Key compatibility rule: MCO and BFCO are mutually exclusive. They perform the same function (attack behavior overhaul) and conflict on behavior files, power attack handling, and animation event processing. Pick one. Do not install both.

Both MCO and BFCO require **AMR** and **Payload Interpreter** as hard dependencies — these are not optional. BFCO does NOT eliminate the need for AMR; it lists AMR as a "must" requirement on its Nexus page. The shared dependency chain for either path is:

```
Pandora (behavior generation)
+ AMR (motion data in attacks)
+ Payload Interpreter (animation payload processing)
+ OAR (conditional animation selection)
+ XPMSE (skeleton)
+ [MCO OR BFCO] (attack behavior framework)
```

BFCO also requires **Directional Movement Keys (DMK)** for directional power attacks to work correctly — this is unique to BFCO. MCO handles directional input through its own `.esp` and optional companion mods like “Separate Power Attacks.”

Moveset portability: MCO movesets (.hmx files using `MCO\_Attack*` naming) can be converted to BFCO format using the community “MCO To BFCO Converter” tool. The reverse (BFCO → MCO) is less common since BFCO has additional attack types (jump, swim) that have no MCO equivalent.

Companion mods that BFCO replaces internally (do NOT install alongside BFCO):

- One Click Power Attack NG / Elden Power Attack / For Honor Power Attack (power attack hotkey — BFCO has its own in MCM)
- Dual Wield Parrying (built into BFCO)
- UCBO — Unarmed Combat Behavior Overhaul (built into BFCO)
- One Handed Crossbow Framework (built into BFCO)
- CGO (conflicts — jump attacks cause stuck-in-falling state)

14.1.5. NPC Combat AI

NPC combat AI teaches enemies to use combat animation movesets the way a player would — picking appropriate attacks based on range, committing to combos, and varying their patterns.

Mod	Requires	Works With	Notes
SCAR (Skyrim Combos AI Revolution)	Pandora/Nemesis + Address Library. AMR recommended (not mandatory).	MCO, BFCO, or vanilla	NPCs use SCAR-annotated animations via scarAI. With BFCO: SCAR annotations take priority; animations without SCAR annotations fall back to bfcoAI. With MCO: SCAR handles all NPC combo logic. Elder Wilds baseline (after Precision is confirmed stable).

SCAR does NOT require AMR as a hard dependency (not listed in its Nexus requirements), but the two are designed to work together — AMR’s motion data makes SCAR-driven NPC attacks feel grounded and weighty rather than ice-skating.

14.1.6. Practical Compatibility Q&A

Q: I have a DAR mod from 2021. Will it work with OAR? Yes. OAR implements every DAR condition and reads DAR folder structures natively. Install OAR, don’t install DAR, and the DAR-packaged mod will work.

Q: I have a mod that says “Requires Nemesis.” Can I use Pandora instead? Yes, in virtually all cases. Pandora reads Nemesis patch format natively. The mod’s Nemesis patch checkbox will appear in Pandora’s UI just as it would in Nemesis.

Q: Do I need FNIS for creature animations (e.g., werewolf, vampire lord)? No. Pandora has full creature support. FNIS is completely unnecessary in a modern AE load order.

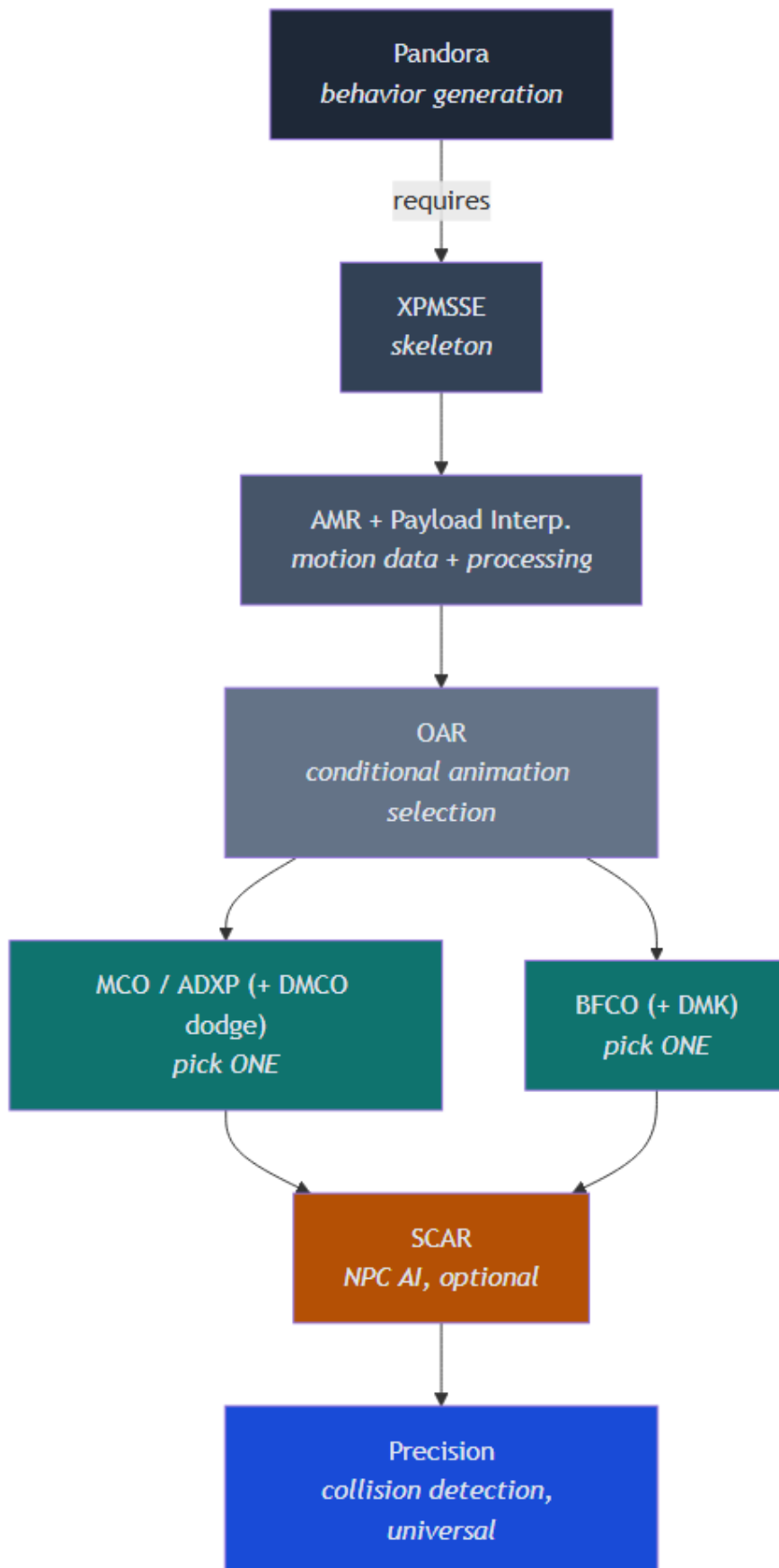
Q: I want to try BFCO instead of MCO. What do I need to change? Remove MCO and any MCO-specific movesets. Install BFCO, DMK, and BFCO-format movesets (or convert MCO movesets with the converter tool). Re-run Pandora, ticking BFCO’s patch instead of MCO’s. AMR, Payload Interpreter, OAR, and XPMSSE all stay — they’re shared requirements.

Q: Can I install SCAR without MCO or BFCO? Technically yes — SCAR has no hard dependency on either. But SCAR is designed to work with MCO/BFCO movesets, and without them NPCs can only use vanilla attack animations, which defeats the purpose. Always pair SCAR with a combat animation framework.

Q: What about SkySA and ABR? Both are predecessors to MCO from the Distar ecosystem. SkySA was the original attack behavior mod; ABR was a fork. MCO superseded both. BFCO lists both as incompatible. Do not use either in a modern load order.

Q: Does Precision (accurate melee collisions) work with all of this? Yes. Precision runs at the collision-detection layer, independent of which combat framework you choose. It works with MCO, BFCO, and vanilla. No special compatibility configuration needed.

14.1.7. The Framework Dependency Map



14.1.8. Research Findings (July 2026)

All four tasks researched. Sources: Nexus mod pages, Nexus community posts tabs, official patch compatibility statements.

Task 1 — Pandora compatibility with modlist behavior mods

Pandora v4.3.1 is confirmed compatible with all behavior-requiring mods currently in the Elder Wilds modlist. The BFCO author (BF001) explicitly tested and confirmed: *"I tested in Pandora v4.3.1 + Skeleton Auto Patch, everything works perfectly"* (BFCO sticky post, May 2024, updated for v3.100+). Pandora reads both Nemesis patch format and legacy FNIS XML natively — the only edge case identified is SCAR (see Task 3 below).

Behavior-requiring mods in the current modlist and their Pandora status:

Mod	Has Pandora Patch	Notes
MCO/ADXP	Yes	Tick in Pandora UI. BFCO author confirmed Pandora works.
BFCO	Yes	Tick in Pandora UI. Author-tested with v4.3.1.
SCAR	Yes (requires fix)	Tick in Pandora. Also needs SCAR - Pandora - Fix loaded after SCAR. Without the fix: WARN : Display defaultfemale~1hm_b #2521/event/Element6
Animated Armoury OAR	Yes	Tick in Pandora for new weapon-type behaviors.
EVG Animated Traversal	Yes	Tick in Pandora.
SkyParkour	Yes	Tick in Pandora.
SkyClimb	Yes (if chosen)	Tick in Pandora.
XPMSSE	Yes (OR skip)	Do NOT tick the XPMSSE patch checkbox if using Universal Behaviour Runtime — Auto Skeleton Patch (mod 176724). The Auto Skeleton Patch replaces the old Pandora XPMSSE checkbox.
Precision	No patch needed	SKSE plugin only; no behavior generation.
AMR	No patch needed	SKSE plugin only; no behavior generation.
Payload Interpreter	No patch needed	SKSE plugin only; no behavior generation.
OAR	No patch needed	SKSE plugin only; no behavior generation.
IFrame Generator RE	No patch needed	SKSE plugin only; install only when a pack explicitly lists it.

Task 2 — MCO vs BFCO community signal on AE 1.6.1170 with Pandora

Both frameworks are actively maintained and Pandora-compatible. The community signal divides along a clear line:

- **MCO** remains the established standard with the larger moveset library (hundreds of MCO-format animation packs). It is the proven path with the most community tutorials and troubleshooting resources. Distar's ecosystem (MCO, DMCO, related mods) is mature.
- **BFCO** is the rapidly growing alternative. Key advantages: built-in jump attacks, swim attacks, charge attacks, vanilla attack speed support, and gamepad MCM hotkey support — all features that MCO requires separate companion mods (or can't do) to achieve. BFCO v3.100+ is described by the author as "almost done with my idea," indicating maturity.

The **MCO→BFCO Converter** ([mod 119926](#), v1.2.2) is actively maintained and handles: file renaming, annotation conversion (attack speed, power windows, recovery, next-attack chaining, multi-window annotations), and batch processing. It converts `MCO\_powerattackloop#emph[.hkx]` and `MCO\_powerattackoutro[.hkx]` files (supported since converter v1.2.1 / BFCO >= 3.3). The converter has gone through 10+ bugfix releases, with progressively better annotation fidelity.

BFCO also has a **BFCO NG** companion ([mod 160505](#)) for flexible hotkey assignment. BFCO's FOMOD offers pre-input behavior choices: "Vanilla Like" (can only input next attack after hit frame) vs "MCO Like" (can input next attack almost immediately — same feel as MCO).

Recommendation: BFCO is locked for Elder Wilds. Built-in gamepad support, fewer companion-mods-required, and the MCO→BFCO converter making the MCO moveset library accessible make it the right long-term fit. MCO is rejected.

Task 3 — SCAR + Pandora interaction confirmed

SCAR works with Pandora but needs the **SCAR - Pandora - Fix** ([mod 164638](#)). Requirements: Pandora Behaviour Engine Plus + SCAR AE Support. Install order: SCAR → SCAR AE Support → SCAR Pandora Fix. Then re-run Pandora.

Additional SCAR compatibility notes from community posts:

- If using **SCAR AE Support** ([mod 77285](#)) with BFCO: do NOT install the default animations in the SCAR AE Support FOMOD — they cause compatibility issues with BFCO.
- SCAR version 2.0 from GitHub (not the Nexus page) reportedly has issues with BFCO — stick to the Nexus version (v1.06b) for now.
- SCAR does not hard-require MCO; it works with MCO, BFCO, or SkySA/ABR. The SCAR Nexus comments explicitly state: *"You don't need MCO if you are using ABR or SkySA for this mod to work."*

Task 4 — MCO→BFCO converter quality assessment

The converter tool (v1.2.2) is actively maintained with good annotation fidelity but is not lossless. The changelog reveals the conversion has gone through multiple rounds of bugfixing:

- **v1.1.3:** Multi-window annotations fixed (animations with more than one `MCO\_nextattack` or `MCO\_PowerWinOpen` now convert correctly).
- **v1.1.4:** Added `MCO\_powerWinOpen` / `MCO\_powerWinClose` → BFCO annotation recognition.
- **v1.1.5:** Added attack speed annotation conversion (`MCO\_AttackSpeed` → `BFCO\_AttackSpeed`).
- **v1.1.6:** Old BFCO annotations are removed on re-conversion.
- **v1.2.1:** Power attack loop and outro files supported (`MCO\_powerattackloop#emph[.hkx]`, `MCO\_powerattackoutro[.hkx]`).

Known limitations: the converter handles standard MCO annotations. MCO movesets with heavily custom or non-standard annotations may not convert perfectly. Complex movesets should be tested individually after conversion for: animation timing drift, ice-skating (lost motion data), missing combo-chain windows, and power attack trigger reliability.

Risk assessment for the converter: For the vast majority of MCO movesets with standard Distar-ecosystem annotations, the converter should produce clean BFCO output. Edge-case movesets with hand-tuned custom annotations are the primary risk. Given that Elder Wilds hasn't locked specific movesets yet, this is a manageable risk — verify each adopted moveset post-conversion rather than counting on batch-convert perfection.

—

14.2. Pandora Framework And Prerequisites

Mod	Type	Notes
Pandora Behaviour Engine Plus	Baseline	Single behavior-generation owner. Register as MO2 executable; output to dedicated <code>Pandora Output</code> mod.
Universal Behaviour Runtime — Auto Skeleton Patch	Baseline	Runtime skeleton patching for XPMSSSE. Do NOT tick Pandora XPMSSSE patch checkbox.
A-Pose Bug Fix — Universal Behavior Runtime	Baseline	Runtime A-pose interception and LE animation backward compatibility.

14.2.1. Risks & Compatibility

- Validate current Pandora install guide and requirements tab during setup.
- Leaving old generated output active or mixing generators makes debugging much harder.
- **SCAR compatibility:** SCAR triggers a known Pandora warning (`defaultfemale~1hm\_behavior > Replace > Element > #2521`). Install [SCAR - Pandora - Fix](#) after SCAR + SCAR AE Support to resolve it. Re-run Pandora after adding.
- Do NOT tick the Pandora XPMSSSE patch checkbox if **Universal Behaviour Runtime — Auto Skeleton Patch** is installed — they are mutually exclusive. The Auto Skeleton Patch is the preferred route.

14.3. Skeleton And Behavior Prerequisites

Mod	Type	Notes
XP32 Maximum Skeleton Special Extended — XPMSSSE	Baseline	Single skeleton baseline.
CBPC — Physics with Collisions	Baseline	Default first-pass physics for CBBE 3BA.
FSMP — Faster HDT-SMP	Baseline	SMP coverage alongside CBPC. Required by OStim and some outfits.
XPMSSSE Fixed Scripts	Alternative	Companion script fix over XPMSSSE.
ConsoleUtilSSE NG	Alternative	Keep available for script-dependent pieces.

14.4. Parkour, Climbing, And Free-Form Movement

Mod	Type	Notes
SkyParkour v3 - Procedural Parkour and Climbing Framework (SPPF)	Baseline	Vault, climb, traverse environmental geometry. 10,112 endorsements.

EVG CLAMBER - Slope Animations	Baseline	Character posture adjusts dynamically on slopes and stairs. Complements SkyParkour.
Feminine EVG Clamber Stair Animations	Baseline	Female-specific stair animations for EVG CLAMBER.
SkyClimb	Alternative	Climbing-first alternative built around EVGAT. Pick one (not cumulative with SkyParkour).
Discipline-first route	Alternative	Vanilla climbing + TDM + sprint/jump only.

14.4.1. Horse Animations

Mod	Type	Notes
Thundertrot Horse Animations	Baseline	Locked. OAR-based horse movement/idle replacer (walk, trot, gallop). 763 endorsements, v2.1.
Horse Animation Overhaul (WIP - OAR)	Baseline	Locked. OAR-based rider animation replacement with directional movement support. Requires Thundertrot for directional movement patch. WIP status — monitor for stable release.

14.4.2. Alternatives

Mod	Type	Notes
Riding Animation Overhaul - RAO (OAR)	Alternative	Rejected — HAO is the newer successor by the same author (mxnedula) with directional movement support. RAO v2.0 (3,804 endorsements) remains the fallback if HAO is abandoned.

All horse animation mods are OAR-based and work under Pandora. Thundertrot (horse layer) + HAO (rider layer) are designed to stack — HAO's directional movement patch requires Thundertrot as a dependency.

14.4.3. Companion Candidates (evaluate after baseline is locked)

- **Dova Jump** ([Nexus](#)) — Stamina-based jump height/distance. Complements parkour.
- **Walk Speed Tuner** ([Nexus](#)) — Configurable walk speed hotkey.
- **Beam Walking Assist** ([Nexus](#))
- **RaySense - Jumping over obstacles** ([Nexus](#))
- **RaySense - Edge Lookdown** ([Nexus](#))
- **Inertia - Physical Movement Response System** ([Nexus](#))

14.4.4. Risks & Compatibility

- SkyClimb and SkyParkour are competing — not harmless companions.

- Parkour can expose navmesh gaps in older worldspaces.
- This subsection owns vertical/lateral movement; dodge, sprint, and camera belong in → `Third-Person`.

14.5. Locomotion

CATA (Conditional Armor Type Animations) is the framework — choose the CATA Addon version for each pack below so animations switch dynamically based on equipped armor type.

Mod	Type	Notes
Heavy armor		
Leviathan Animations II - Male Idle Walk And Run	Baseline	CATA Addon. Male heavy-armor locomotion.
Leviathan Animations II - Female Idle Walk And Run	Baseline	CATA Addon. Female heavy-armor locomotion.
Light armor		
Vanargand Animations II - Male Idle Walk And Run	Baseline	CATA Addon. Male light-armor locomotion.
Vanargand Animations II - Female Idle Walk And Run	Baseline	CATA Addon. Female light-armor locomotion.
Mage (non-armored)		
Goetia Animations - Female Idle Walk And Run	Baseline	CATA Addon. Female mage locomotion.
Goetia Animations - Male Idle Walk And Run	Baseline	CATA Addon. Male mage locomotion.
NPC Animation Remix (OAR)	Alternative	NPC-specific movement and idle animation remix.
Arm Movement Animations (OAR)	Alternative	Hand and arm idle animation variations.
Dynamic Female Weather Idles	Alternative	OAR-based weather-aware idles. Complements survival/weather stack.
Dynamic Female Swimming Idles	Alternative	Swimming idle replacements. Complements Random Swimming Animations.
Dynamic Female Rail Leaning	Alternative	Rail-leaning idles at NPC idle markers. Player use needs Dovahkiin Can Lean (32748).
Dynamic Female Ledge Sitting	Alternative	Ledge-sitting idles at NPC idle markers. Player use needs Dovahkiin Can Lean (32748).
Dynamic Female Hand Warming	Alternative	Cold-weather hand-warming idles. Complements survival/weather stack.
Seamless Varied Masculine Idles (OAR)	Alternative	Male non-combat idle variety via OAR.
Seamless Varied Feminine Idles (OAR)	Alternative	Female non-combat idle variety via OAR.

SD's Seamless Sneak Transition and Idles (OAR)	Alternative	Sneak idles and stand-to-sneak transitions. Complements Vanargand Sneak locomotion.
EVG Animated Traversal	Alternative	Belongs in interaction/traversal bucket.
Poser Hotkeys Plus SSE	Alternative	Hotkey-based pose/idle system.
More Tavern Idles - SSE Port	Alternative	Tavern-specific idle animation variety.
Lightweight Headtracking and Emotions	Alternative	NPC/PCP headtracking and expression support.
Smooth Random Jump Animation - Rework	Alternative	Randomized jump animation replacer.
Smooth Weapon Jump Animation	Alternative	Weapon-drawn jump animation replacer.
Random Swimming Animations	Alternative	Randomized swimming animation replacer.
Simple Diving System	Alternative	OAR-based diving detection and animations for water entry from various heights. Requires Behavior Data Injector or Pandora.
Dynamic Sprint	Alternative	SKSE-based sprint animation with motion-matched lean.
Dynamic Sprint Stop	Alternative	Sprint-stop animation with deceleration. Companion to Dynamic Sprint.
Vanargand Animations - Sneak idle walk and run	Alternative	Sneak locomotion animation pack.

14.6. Combat Animation Packs

Mod	Type	Notes
Precision - Accurate Melee Collisions	Baseline	Accurate melee collision detection.
SCAR - Skyrim Combos AI Revolution	Baseline	Add after Precision is proven. Companion mods: SCAR AE Support + SCAR Pandora Fix . If using BFCO, skip default animations in SCAR AE FOMOD.
Animated Armoury - OAR	Baseline	12 new weapon types. Requires DAR Version for meshes/collision/leveled lists. Tick in Pandora for new weapon behaviors.
No Spinning Death Animation LITE	Baseline	Prevents spinning death animations.
Animation Motion Revolution + Payload Interpreter	Baseline	Required by BFCO (attack behavior framework). AMR enables motion

		data in .hkx files; Payload Interpreter handles attack event processing.
BFCO - Attack Behavior Framework	Baseline	Locked. Attack behavior framework. Requires DMK for directional power attacks. Companion: BFCO NG for flexible hotkeys. Confirmed working with Pandora v4.3.1.
Iframe Generator RE AE Support	Support	Install only when a pack explicitly lists it.
Elden Ring DLC Light Greatsword Moveset	Alternative	Requires MCO or BFCO framework. 1H and 2H moveset.
For Honor in Skyrim	Prerequisite	Asset base only. Required by For Honor Reforged for animation files. Do not run standalone — overwrite with 172176.
For Honor Reforged	Alternative	Full For Honor combat moveset rebuilt with BFCO support, NPC AI (OAR), parry/guard-break/knockdown mechanics. Requires 151478 as asset base + DMK, Chocolate Poise, BFCO.
Vindictus Fiona Moveset BFCO	Alternative	BFCO-specific moveset. Requires BFCO framework.
Vindictus Delia Animation Remake	Alternative	Combat animation pack.
MCO / BFCO / SCAR WoLong QuarterStuffs	Alternative	Works with MCO, BFCO, or SCAR. Quarterstaff moveset.
Dynamic Killmove - Pike	Alternative	Killmove animation for pike/spear weapons.
KG Animations - One-handers and Dual Wield	Alternative	One-handed and dual-wield combat animation pack.
KG Animations - Two-handers	Alternative	Two-handed combat animation pack.
Smooth Random Blocking Animation 3.0	Alternative	Randomized blocking animation replacer.
Precision - Enchanted Weapon Trails	Alternative	Weapon trail VFX requiring Precision.
Killmove Fixes	Alternative	Killmove trigger and camera fixes.
Goetia Animations - Magic Spell Casting	Alternative	Magic spell casting animation replacer.
Diverse NPC Movesets	Alternative	Varied NPC combat stances via SCAR/OAR.
Vanargand Animations - Sneak Strike Attacks	Alternative	Sneak power-attack animation replacer.
Vanargand Animations - Sneak Thrust Attacks	Alternative	Sneak thrust-attack animation replacer.
Vanargand Animations - Archery	Alternative	Archery draw-and-release animation replacer.

Vanargand Animations - Sneak Archery	Alternative	Sneak archery stance and release animation replacer.
Vanargand Animations - Dual Wield Sneak Strikes	Alternative	Dual-wield sneak attack animation replacer.
Vanargand Animations - Crossbows	Alternative	Crossbow draw-and-fire animation replacer.
Vanargand Animations - Sneak Crossbows	Alternative	Sneak crossbow stance and fire animation replacer.
Goetia Animations - Sneak Magic	Alternative	Sneak magic-casting animation replacer.
Goetia Animations - Conditional Shouts	Alternative	Conditional shout animation replacer.
Dynamic Dodge Animation	Alternative	SKSE-based dodge animation with i-frames. Evaluate as potential TK Dodge RE companion.

14.7. BFCO Install Workflow Reference

External tutorial baseline: BFCO is the locked combat animation framework for Elder Wilds (Pandora for behaviour generation, OAR for conditional selection).

14.7.1. Core Install — BFCO Stack

Mod	Type	Notes
Animation Motion Revolution (AMR)	Baseline	Required by BFCO. Motion data support for .hmx files.
Payload Interpreter	Baseline	Required by BFCO. Attack event processing.
BFCO — Attack Behavior Framework (117052)	Baseline	Locked attack behavior framework.
Directional Movement Keys (DMK, 174499)	Baseline	Required by BFCO for directional power attacks.
BFCO NG (160505)	Optional	Flexible hotkey assignment companion.
MCO→BFCO Converter (119926)	Tool	Converts MCO-format movesets to BFCO naming/annotations. Run before Pandora.

14.7.2. Install Flow (Pandora/OAR)

1. Install AMR, Payload Interpreter, DMK, and BFCO as separate MO2 mods under `Animations - Combat`.
2. Optionally install BFCO NG for flexible hotkey assignment.
3. Download BFCO-format movesets or convert MCO movesets using the MCO→BFCO Converter tool before placing in MO2.
4. Create a dedicated empty MO2 mod named `Pandora Output` to receive generated behaviour files.

5. In Pandora, tick BFCO, Payload Interpreter, and any other behaviour-requiring mods present in the load order.
6. Run Pandora: Update Engine → Launch. Keep antivirus off during generation; re-enable afterwards.
7. Launch Skyrim and verify: directional power-attack input, weapon-sheathe behaviour, and attack combo chains work end-to-end in third-person gamepad.

14.7.3. Risks & Compatibility

- DO NOT install MCO alongside BFCO — they are mutually exclusive and will conflict on behavior files.
- BFCO includes internally: One Click Power Attack, Dual Wield Parrying, Unarmed Combat Behavior Overhaul, One Handed Crossbow Framework. Do NOT install these separately.
- Third-person gamepad parity depends on BFCO's built-in MCM hotkey support. Verify directional power attack input with the bound gamepad key.

14.7.4. Research Tasks

- Verify BFCO-format moveset availability for all weapon types used in the modlist (one-handed, two-handed, dual wield, unarmed, ranged, magic).
- Cross-check [r/skyrimmods](#) signal for recommended BFCO movesets and SCAR-compatible NPC animation packs.
- Test MCO→BFCO Converter output quality with candidate movesets before adoption.

—

14.8. Non-Combat Interaction Animations

Mod	Type	Notes
Immersive Interactions - Animated Actions	Baseline	Core interaction animation framework.
Go to bed	Baseline	Bed interaction animations.
Animated Interactions SKSE	Alternative	Can coexist only if overlapping actions are deliberately disabled.
Take a Seat	Alternative	Sitting interaction animations.
Improved Barstool Exit Animation	Alternative	OAR-based barstool exit animation fix. Requires SKDR (→ Foundations).
Immersive Hunting Animations	Alternative	Hunting-related animations.
Immersive Carcass Carrying	Alternative	Carcass carrying animations.
Flute Animation Fix	Alternative	Flute playing animation fix.
Witcher Flute	Alternative	Witcher-style flute animation.
EVG Animated Traversal	Alternative	Environment traversal animations.
Beginner Bard Animations	Alternative	For Skyrim's Got Talent.
Immersive Interactions - Eating ingredients and apply poison animations	Add-on	Eating and poison-apply animations for II.

Dynamic Crafting Animations	Add-on	Crafting-station interaction animations.
Dynamic Looting and Harvesting Animations	Add-on	Looting and harvesting interaction animations.
Dynamic Horse Petting Animations for Immersive Interactions	Add-on	Horse interaction animations for II.
Tools Not Weapons (Pickaxe and Woodcutter Axe) DAR Animations	Prerequisite	DAR/OAR animations for mining/chopping tools. Prerequisite for Chop Chop and related woodcutting mods.
HSF Male Furniture Idles	Alternative	Male idle animations for furniture interactions.
Modern Female Sitting Animations Overhaul	Alternative	Female sitting animation replacements.
Improved Idle Laydown Animations	Alternative	Ground/sleep laydown idle improvements.
Improved Table Sit Transition Animations	Alternative	Prevents plate/drink knockover when sitting at tables.
Paired Animation Improvements	Alternative	Improved paired NPC interaction animations.
Ultimate Animated Potions NG	Alternative	Potion-use animation replacer.
JellyFish Ultimate Animated Potions NG	Alternative	Expanded potion animation pack for Ultimate Animated Potions NG.
Simple Wall Lean (RaySense)	Alternative	Contextual wall-leaning idle animation.
Simple Wall Lean - More feminine Female animations	Add-on	Female-specific variant for Simple Wall Lean.
Divines Prayer Animations	Alternative	Conditional prayer-idle animation at shrine activations.
Gesture Animation Remix (OAR)	Alternative	Expanded gesture/idle animation variety via OAR.
Lively Children Animations (OAR)	Alternative	Child NPC animation variety for play/idle/run.
Lively cart driver animation replacer or OAR	Alternative	Cart driver animation replacer for carriage rides.

14.9. Conditional Animation Systems

Mod	Type	Notes
Open Animation Replacer	Baseline	Single condition framework owner.
Conditional Armor Type Animations	Baseline	Armor-type-based animation switching. Pairs with Leviathan II (heavy), Vanargand II (light), and Goetia (mage) CATA Addons in → <code>Locomotion</code> .

EVG Conditional Idles	Alternative	Idle animation conditional framework.
Unique Jarl Throne Sitting Animation (OAR)	Alternative	Throne sitting animation for Jarls.
Malignis Animations - Conditions	Alternative	OAR condition pack for animation variety. Personal favourite.

14.10. Camera-Aware Animation Support

Mod	Notes
Improved Camera SE	Only if hybrid perspective is a real playstyle.
Comprehensive First Person Animation Overhaul - CFPAO	Optional first-person polish.

14.11. Equipment Display Framework

Equipment visibility, sheathing positions, and draw-sheathe animations. Builds on XPMSSSE for third-person gamepad parity.

Mod	Type	Notes
Immersive Equipment Displays (IED)	Baseline	Equipment visibility and positioning framework.
Simple Dual Sheath	Baseline	Dual-sheathed weapon support. Requires IED.
Weapon Styles - Draw-Sheathe animations for IED	Addon	Conditional draw/sheathe animations per weapon type. Requires IED.
Walking Stick - Walk with staves or polearms - IED-OAR	Addon	Staff/polearm walking animation support. Requires IED.
Ready to Play IED	Alternative	Pre-configured IED preset. Evaluate only if manual IED configuration proves too time-consuming.
Weapons On Back	Alternative	Weapon positioning on back rather than hip. Complements IED/SDS.
Guards Pose With Weapons — IED-OAR	Addon	Guard and faction guards hold-weapon poses via IED + OAR. Requires both baselines. Third-person (gamepad) flavor.

14.12. Creature Animations

Mod	Notes
-----	-------

New Creature Animation - Giant	Giant animation replacer.
New Creature Animation - Werewolf	Werewolf animation replacer.
New Creature Animation - Falmer	Falmer animation replacer.
DCA - Dragon Combat Animations	Dragon combat animation replacer.
Draugr Greatsword Animation	Greatsword-wielding draugr animation replacer.

14.12.1. Risks & Compatibility

- Validate per-pack combat-animation support individually. Keep layer small and intentional.
-

14.13. Animation Conflict Management

Strict ownership: one clear owner per layer.

14.13.1. Owners

- **Behavior generation:** Pandora only
- **Skeleton:** XPMSSE only
- **Conditional selection:** Open Animation Replacer only

14.13.2. Workflow

- Keep Pandora output isolated in dedicated MO2 output mod.
- Record which mods rely on generation, which on OAR, and which are pure presentation.
- Test in repeatable scenarios: town walking, idle downtime, dungeon corridors, uneven outdoor combat, interaction-heavy interiors, creature encounters.

14.13.3. Risks & Compatibility

- Animation conflicts look like camera, combat, or skeleton bugs until ownership is checked.
 - Mixed generated output hides whether a problem is from load order or stale behavior files.
-

14.14. Open Research

Open research for the animations stack is tracked in `TODO.md`.

15. Third-Person Gameplay

15.1. Third-Person Camera Framework

15.1.1. Baseline

- **SmoothCam** ([Nexus](#)) — Main third-person camera framework.

15.1.2. Alternatives

- **Customizable Camera** ([Nexus](#)) — Leaner fallback.
- **True Directional Movement** ([Nexus](#)) — Governs movement/targeting, not the camera framework itself.
- **Improved Camera SE** ([Nexus](#)) — Perspective support, not main third-person framework.
- **Modern Combat Gamepad Overhaul** ([Nexus](#)) — Gamepad-centric combat controls integration. Research before adding — evaluate overlap with TDM and the combat stack.

15.1.3. Risks & Compatibility

- A good preset can hide weak camera behavior — judge the framework, not the preset.
 - Over-tuned cameras look impressive in clips but become tiring.
-

15.2. Camera Presets And Shoulder Switching

Restrained centered-to-light-offset baseline. Shoulder switching is a support feature for visibility problems, not the main camera identity.

15.2.1. Risks & Compatibility

- Aggressive over-the-shoulder framing creates visibility issues in interiors and dungeons.
 - Shoulder switching that feels mandatory too often = poor baseline framing.
-

15.3. True Directional Movement

- **True Directional Movement - Modernized Third Person Gameplay** ([Nexus](#)) — Modern 360-degree character-relative movement.
 - Core value: freer facing, cleaner traversal, better animation-stack alignment.
 - May feel awkward in tight interiors, stairs, or precision positioning.
-

15.4. Target Lock And Targeting Behavior

- **TDM** as the targeting-behavior owner. Keep lock-on restrained — optional combat-readability tool, not permanent default.
- **TrueHUD** ([Nexus](#)) — Optional feedback companion. Add only if testing shows material improvement.

15.4.1. Risks & Compatibility

- Weak target selection/switching makes even good camera and movement feel unreliable.
 - TrueHUD can create illusion of a solved targeting system when the underlying behavior is still awkward.
-

15.5. Dodge, Sprint, And Movement Responsiveness

15.5.1. Baseline

- **TK Dodge RE - Script Free** ([Nexus](#)) — **Locked baseline**. Script-free dodge with configurable step/distance. Maps cleanly to gamepad. Tactical repositioning, not combat pacing center.
- **S.M.C. — Skyrim Motion Control** ([Nexus](#)) — **Baseline**. Script-free SKSE plugin: acceleration ramp-up, momentum on stop, configurable jump boost, slope speed scaling, and per-state speed/accel multipliers (bow-drawn, magic-cast, swimming, walking, auto-walk, etc.). Works on gamepad. Fully tunable INI, reloadable in-game with Enter. Skyrim Souls compat toggle (v2.4+). S.M.C.'s sprint-jump boost works only with Better Jumping v1.7.1 or below — disable in INI if using newer Better Jumping. Evaluated alongside Inertia (172783, → [Animations](#) candidate): S.M.C. handles speed/acceleration at the SKSE physics-param layer; Inertia handles physical body-response. Compatible, not redundant. Tune INI values to match Static with Hard Threat — particularly `fAcceleration`, `fStoppingForce`, and `fJumpBoost`.

15.5.2. Risks & Compatibility

- Incompatible with Scrolling Speed 2, Wounds (17581), MBO (38950) — none present in this list.
- Pace (27268): works but disable Pace's movement speed settings in MCM.
- Jump boost may fail intermittently (known issue); low `fStoppingForce` can cause momentum to not decay when another speed buff/nerf is active.

15.5.3. Alternatives

- **TK Dodge SE** ([Nexus](#)) — Legacy fallback reference.
 - **Dodge for all** ([Nexus](#)) — Newer (May 2026). Too early for baseline — low community signal.
-

15.6. Hit Reactions, Stagger, And Impact Feedback

Mod	Notes
Valhalla Combat	Main gameplay-side comparison for hit feedback and stagger.

Precision (from → Animations)	Adjacent support for contact feel.
--------------------------------	------------------------------------

15.6.1. Risks & Compatibility

- Too much stagger flattens combat; too little leaves animations unsatisfying.
- Camera, animation timing, collision, and gameplay rules all affect impact feel — ownership blurs quickly.

—

15.7. Third-Person Combat Systems

Mod	Type	Notes
Valhalla Combat	Baseline	Gameplay-rule framework.
SCAR (from → Animations)	Baseline	NPC-side attack variety.
Knockback SKSE	Support	Compatible with BFCO. Evaluate for adoption with BFCO setup.
RaySense - Cover Animation	Optional	Cover animation system.
Fire Ignites Arrows	Optional	Arrow ignition from fire sources.
Poisoned Arrows and Bolts	Optional	Poison crafting for ranged ammo.
Variadic Collision Dynamics	Optional	Runtime collision-capsule adjustments.
Variadic Collision Dynamics - Resources	Optional	Ready-to-use JSON presets for VCD.
Next-Gen Decapitations	Optional	Decapitation system replacement with improved gore visuals.
Death Drop Overhaul	Optional	Changes how NPC bodies ragdoll and drop on death.
Dismembering Framework	Optional	Dismemberment system framework.
DF - Official Humanoid Asset Pack	Add-on	Required asset pack for Dismembering Framework (humanoid).
DF - Official Creature Asset Pack	Add-on	Creature asset pack for Dismembering Framework (optional).
Eviscerate	Optional	Combat gore / dismemberment alternative.
Cinema - Dynamic Combat Letterbox	Optional	Letterbox overlay during combat.
Parry for All	Optional	Parrying support for non-equipped actors.
Deadly spell impact	Optional	Spell impact visual + audio upgrade.

—

15.8. Crosshair And Aiming Behavior

Mod	Notes
Better Third Person Selection - BTPS	Object-selection reliability.

TrueHUD	Optional aiming readability aid.
Ricochet - Arrow Physics Framework	Lock only if archery is a meaningful combat style.

15.9. Horse Camera And Mounted Controls

Mod	Type	Notes
Press H to Horse	Baseline	Actively maintained (May 2026). Few vanilla record edits.
Horses Simply Turn Better	Baseline	Free add-on. Sharpens turning radius for TDM.
Convenient Horses	Alternative	Last updated Jan 2022 (stale).
Simplest Horses	Alternative	Lightweight SPID+MCM drop-in.
HorsePower - Modernized horse riding	Baseline	Modernized horse riding overhaul. Requires SKSE, Address Library, Animation Motion Revolution, Pandora, TDM.

15.10. Dragon Mounted Controls

Mod	Type	Notes
Intuitive Dragon Ride Control	Baseline	Full dragon flight control — camera-based movement, combat, fast travel, perching. Native gamepad support, no dTry needed. Integrates with TDM (patch 160556) and TrueHUD. Requires SKSE, Po3 Tweaks, Address Library (all → Foundations).

15.11. Third-Person Interactions: Lockpicking

Gamepad-accessible lockpicking that works entirely in third-person without mouse input. Required by the design directive — every interaction must work in third-person with a gamepad.

Mod	Notes
Break The Lock - OIF	Physics-based lockpicking with breakable locks via OIF. Gamepad-compatible, third-person usable. Requires OIF framework.

15.12. Open Research

Open research for the third-person stack is tracked in [TODO.md](#) .

16. Expanded Systems

16.1. Overview

This section owns the gameplay systems that expand Skyrim beyond its 2011 baseline: character creation, progression, magic, perks, survival, crafting, economy, followers, and social systems. Each pillar of the Expanded Systems section lives in its own sub-file.

The system is designed to be evaluated in order — progression choices (races, standing stones, perks) come before magic, survival, crafting, and social systems that build on them.

16.2. Sub-Files

Area	File	Separator	Topics
Character & Progression	Section 17	Expanded Systems - Character & Progression	Alternate starts, races, standing stones, traits, religion, vampire/werewolf/shouts
Magic & Perks	Section 18	Expanded Systems - Magic & Perks	Perk overhauls, magic expansion, stealth/detection
Survival & Needs	Section 19	Expanded Systems - Survival & Needs	Survival frameworks, hygiene, death alternative
Crafting & Economy	Section 20	Expanded Systems - Crafting & Economy	Alchemy, cooking, smithing, enchanting, economy
Followers & Reputation	Section 21	Expanded Systems - Followers & Reputation	Crime/reputation, follower frameworks, romance/marriage

16.3. Key Principles

- Simonrim progression family (`Adamant` , `Mysticism` , `Thaumaturgy` , etc.) is the locked baseline. EnaiRim (`Ordinator` , `Apocalypse` , etc.) is rejected.
- Survival, difficulty, and balance interactions across the full load order are owned by `modlist-survival-combat.md` → `Survival & Combat`
- All system choices must be evaluated for third-person gamepad usability and compatibility with the `Legacy of the Dragonborn` museum pillar → `Legacy of the Dragonborn`

17. Character & Progression

17.1. Character Creation And Starting Choices

How [Elder Wilds](#) begins: a grounded alternate start, a utility-first start, or a configurable route. The decision should improve replayability and roleplay expression without pre-answering later race, progression, religion, survival, or follower choices.

Mod	Type	Dial	Notes
Skyrim Unbound Reborn	Baseline	all	Locked. Configurable start with full Dragonborn-quest control.
Why I Came to Skyrim - Origin Stories	Baseline	all	Background-driven starting scenario with narrative hook. Companion to Skyrim Unbound Reborn.
Why I Came to Skyrim - Skyrim Unbound Reborn patch	Add-on	all	Patch for Origin Stories + Skyrim Unbound Reborn.
Skyrim Unbound Reborn - New started room	Add-on	all	Start room improvement for Skyrim Unbound Reborn.
Dealing with Backstories	Alternative	1, 2, 3, 4	Origins companion — background-driven starting skills, bonuses, drawbacks, gear.
Take Notes — Journal of the Dragonborn SSE	Alternative	1, 2, 3, 4	Persistent in-game journal. Low-risk roleplay companion.
Roleplaying In Skyrim — Origins	Alternative	1, 2, 3, 4	Origin-driven flavor and bonuses. Requires SKSE + Papyrus Ini Manipulator; soft-requires Actor Value Generator + SkyUI.
Choose Your Starting Skills and Stats	Add-on	all	New-game dialogue to pick starting skills/perks before the run begins. Complements the Skyrim Unbound Reborn + Why I Came to Skyrim start; sets starting state only, does not fight the locked Experience/Static Skill Leveling dial.
Choose Your Starting Skills — Oathvein UI	Add-on	all	UI skin for the above matching the locked Oathvein UI baseline (→ Modernized UI). Install with 25630.

17.1.1. Risks & Compatibility

- Start frameworks carry world-state assumptions that ripple into later quest, city, survival, or follower decisions.
 - [Dealing with Backstories](#) must be judged against the chosen race, standing-stone, perk, and survival stack.
 - [Take Notes](#) only earns a slot if the list expects real roleplay or long-form journaling.
 - A more immersive start loses value if it adds onboarding friction without materially improving the early-game tone.
 - This subsection gets muddy if race, perk, religion, or survival decisions leak into it too early.
-

17.2. Race Overhauls

How race traits shape long-term character identity. Choose before perks, magic, and religion build on top of it.

Mod	Type	Dial	Notes
Aetherius — Race Overhaul	Baseline	1, 2, 4	Locked. Pairs with Evenstar. Simonrim progression family.

17.2.1. Roleplaying in Skyrim Companion Series: Race

> All RiS mods require SKSE. Check individual mod pages for specific dependencies.

Mod	Purpose	Dial	Requirements
RiS — Races	Biological race overhaul	all	KID, SPID
RiS — Lineages	4 subraces per vanilla race	all	KID, SPID, Papyrus Ini Manipulator

17.2.2. Risks & Compatibility

- Race overhauls can quietly pre-decide too much of the character-building experience.
 - A very expressive route can push optimization-heavy play even if the project wants immersive tone.
 - A very light route can undershoot if later systems assume stronger identity hooks.
-

17.3. Standing Stones

How standing stones and birthsigns provide ongoing passive identity. Judge together with race pick so the progression baseline stays coherent.

Mod	Type	Dial	Notes
Evenstar — Standing Stones	Baseline	1, 2, 4	Locked. Pairs with Aetherius. Simonrim progression family.

17.3.1. Roleplaying in Skyrim Companion Series: Standing Stones

> All RiS mods require SKSE. Check individual mod pages for specific dependencies.

Mod	Purpose	Dial	Requirements
RiS — Standing Stones	13 stones overhaul	all	aTweaks and Utilities, KID, Perk Entry Point Extender, SkyUI
RiS — Birthsigns	Monthly waxing/waning birthsign effects	all	KID, Papyrus Ini Manipulator

17.3.2. Risks & Compatibility

- Stone and birthsign overhauls can quietly pre-decide too much of the character-building experience.
- Judge race and stone picks as a pair so the progression baseline stays coherent.

—

17.4. Character Progression Mechanics

Supporting progression layers that govern how stats and skills grow over time.

> All RiS mods require SKSE. Check individual mod pages for specific dependencies.

Mod	Purpose	Dial	Requirements
RiS — Classic Attributes	10 classic RPG attributes	1, 4	Actor Value Generator
RiS — Useful MiscStats	Misc stat progression layer	all	Actor Value Generator
RiS — Skill Based Stat Growth	Health/Magicka/Stamina grow with skills	1, 4	Papyrus Ini Manipulator

—

17.5. Character Traits And Identity Hooks

Opt-in character-defining quirks shaping a run from the start without replacing the broader progression stack. Goal: stronger roleplay identity and replayability, not a giant second character-build framework.

Mod	Type	Dial	Notes
Biggie Traits	Baseline	1, 4	Run-defining tradeoffs. Fallout-style traits. No hard dependencies.
Pumping Iron — Dynamic Muscle Growth	Alternative	1, 2, 3, 4	Cosmetic muscle growth from combat + sleep. No gameplay impact.
Minimalist route	Alternative	2, 3	Leave identity to race, standing stones, perks, and self-imposed roleplay.

17.5.1. Risks & Compatibility

- Trait systems can duplicate what race, perk, religion, or survival choices are already doing.
 - If the best traits are too obviously optimal, the whole idea collapses into routine min-maxing.
 - Traits should sharpen character identity and replayability, not replace the race-and-perk decisions above.
-

17.6. Religion Or Roleplay Systems

How much explicit spiritual identity and day-to-day roleplay texture `Elder Wilds` wants.

Mod	Type	Dial	Notes
Pilgrim — A Religion Overhaul	Baseline	1, 2, 4	Locked. Fits Simonrim progression family.

17.6.1. Risks & Compatibility

- A very light religion route can undershoot and leave roleplay identity too close to vanilla.
 - A heavy deity-centered framework can make religion dominate character identity.
 - Survival, economy, and follower expectations can drift into this section if faith routines aren't scoped carefully.
 - Judge by ordinary play rhythm: prayer habits, shrine relevance, exploration texture.
-

17.7. Race-Specific Overhauls

Dedicated overhauls for vampire, werewolf, and shout gameplay — three character-identity systems vanilla leaves shallow. All three are by EnaiSiaion and require no cross-patching.

Mod	Type	Dial	Notes
Sacrosanct — Vampires of Skyrim	Baseline	1, 4	Vampirism progression, 4 feeding tiers, blood-power by age. 133K endorsements.
Dark Envoy — Vampire Powers	Add-on	1, 4	Small questline + new VL powers (Bat Form flight, Bat Teleport, Gate of Coldharbour). 2,157 endorsements. Author-tested with Sacrosanct.
Vampire Lord Renewed (VL Framework) ESL	Add-on	1, 4	Humanoid VL form, inventory/spell menu access in VL form, loot/activate enabled. Explicitly supports Sacrosanct. Fully ESL-flagged. v6.4.11 actively maintained.
Vampire Lord Overhaul	Add-on	all	Visual-only VL replacer — 8K skin, armors, bigger wings, glowing eyes. Compatible with all gameplay overhauls.

Just Bite — DAK Patch for Sacrosanct	Add-on	1, 4	Replaces Sacrosanct feeding pop-up menus with Dynamic Activation Key single-button feed/drain. Better gamepad/third-person UX. Requires DAK (96273).
Vampire Feed Decals	Add-on	1, 4	Blood decals on body/face when feeding as a vampire. BodySlot-based (race-independent). Requires Vampire Animations (100349), Behavior Data Injector. Optional SPID for NPC decals.
Vampires Cast No Shadow 2	Add-on	1, 4	SKSE plugin (powerofthree): vampires cast no shadow in sunlight (humanoid + Lord forms). Zero records, compatible with every mesh/vampire mod. Visual immersion fix — aligns sun behavior with vampiric lore. v1.2.2 supports AE 1.6.1170.
Dark Destiny — Become a Vampire Unwillingly	Add-on	all	Short voiced quest scene where the player is captured and turned into a vampire against their will in Pinemoon Cave. AI-voiced (ElevenLabs). No hard dependencies. Compatible with any vampire overhaul.
Seeking The Cure — A Rising At Dawn Quest Overhaul	Add-on	all	Makes vampirism incurable — Falion's ritual fails. Rising At Dawn becomes a non-repeatable story quest. 100+ lines AI-voiced dialogue. Pairs with Dark Destiny for a full unwilling-vampire arc.
Ancient Mausoleum SSE	Add-on	all	Small vampire-themed player home with custom storage and artifact displays (dragon claws, masks).
Vampire Clothing Expansion	Add-on	all	New vampire-themed clothing items. Author: Hornclonic.
Growl — Werebeasts of Skyrim	Baseline	1, 4	Beast-form progression, perks, 5 werebear varieties.
Thunderchild — Epic Shouts	Baseline	1, 2, 4	19 new shouts (57 words), meditation mechanics, boss encounters.
Shout Progression	Add-on	1, 2, 4	Perk-tree and progression for vanilla + Thunderchild shouts. XP-for-unlock loop replaces

			meditation. Gamepad-friendly MCM.
Discipline-first route	Alternative	3	Vanilla vampire/werewolf/shout systems.

17.7.1. Risks & Compatibility

- All three core overhauls are by EnaiSiaion and operate independently of the progression stack. Death-alternative (discipline-first) does not apply — verify Sacrosanct death scenarios work with vanilla death/reload.
- [Sacrosanct](#) and [Growl](#) interact with [Heart of the Beast](#) (werewolf transformation audio) in section 10. Load transformation-sfx after transformation-system mods.
- [Thunderchild](#) 's new shouts interact with shout-audio mods ([Airgetlam - Shouts SFX](#)) from section 10. Verify sfx replacements apply to Thunderchild's shouts, not just vanilla.
- Vampire feeding interacts with the crime-and-reputation baseline ([Skyrim Reputation](#) from → [Followers & Reputation](#)). Test feeding in cities.
- [Thunderchild](#) 's boss encounters may overlap with world-content mods from → [World Content](#) .
- [Dark Envoy](#) is visual- and gameplay-only; verified by author against Sacrosanct with no conflicts. Powers stand on their own as a complementary layer.
- [Vampire Lord Renewed](#) touches VL transformation scripts and race records — install after Sacrosanct so its patches detect the correct overhaul. Pack 01 (MCM) is required; Pack 02 (NPC humanoid VL) is optional. Requires MCM Helper, PO3 Papyrus Extender, PapyrusUtil.
- [Vampire Lord Overhaul](#) is mesh/texture only — no ESP records to conflict. Safe to install at any point.
- [Just Bite](#) modifies Sacrosanct's feeding perk — install after Sacrosanct and DAK. Safe to update mid-save per author.
- [Vampire Feed Decals](#) requires [Vampire Animations](#) (100349), [Behavior Data Injector](#) , and optionally [Payload Interpreter](#) for blood spray. Install Vampire Feed Proxy after any vampire overhaul mod. Test via MCM ("Vampire Blood Decals" → "Main" → "Apply Blood Decals").
- [Dark Destiny](#) + [Seeking The Cure](#) form a coherent unwilling-vampire narrative arc — Dark Destiny triggers the turning, Seeking The Cure makes it permanent. Both are AI-voiced and compatible with any vampire overhaul (Sacrosanct, Sacrilege, Better Vampires).
- [Ancient Mausoleum](#) is a standalone player home with no gameplay record conflicts. Safe to install at any point.
- [Vampire Clothing Expansion](#) adds items via leveled lists. Verify no overlap with [Armor and Clothing Extension](#) or other wardrobe mods.

—

17.8. Train And Study Systems

Training and study utilities for off-combat skill progression.

Mod	Type	Dial	Notes
Train and Study	Baseline	1, 2, 3, 4	Training overhaul: study skill books, practice at training dummies, learn from NPCs.
Train and Study - Base Object Swapper	Baseline	1, 2, 3, 4	BOS addon for Train and Study — enables training dummy

			interaction via BOS. Install both together.
--	--	--	---

17.9. Experience and Leveling

How the player gains levels and skill points. Instead of vanilla sleep-to-level, use Experience (discovery-based leveling) paired with Static Skill Leveling Rewritten (cooldown-gated skill advancement) for progression earned through exploration and deliberate practice.

Mod	Type	Dial	Notes
Experience	Baseline	1, 4	Core leveling — exploration, quests, combat, discovery. Not skill grinding.
Experience — MCM	Baseline	1, 4	MCM config for XP rates and per-source toggles.
Static Skill Leveling Rewritten	Baseline	1, 4	Cooldown-based skill advancement. ESL-flagged.
No Startup Quest XP	Baseline	1, 4	Prevents XP from startup quests with alternate start.
Skyshards	Baseline	1, 4	60+ glowing shards granting XP on collection. ESL-flagged.
Discipline-first route	Alternative	2, 3	Vanilla sleep-to-level. Avoids patcher requirement.
Static Skill Leveling — Custom Preset	Alternative	all	Manual tuning instead of Simpler Preset.

17.9.1. Risks & Compatibility

- Experience's Synthesis Patcher must be re-run whenever the load order changes.
- Static Skill Leveling's cooldown may feel restrictive; the Simple Preset mitigates this. Tune via .ini if needed.
- Skyshards placement may clip with city overhauls, new-lands mods, or landscape edits.
- All four mods are ESL-flagged or lightweight, adding minimal plugin pressure.

17.9.2. Experience Configuration For 1 (Static With Hard Threat)

Settings tuned for the adopted combination: power fantasy XP pace, high level cap, rewarding discovery and quest completion, no skill grinding.

Setting	Value	Notes
<code>iMaxPlayerLevel</code>	250	Very high cap for content-heavy list — quest mods, discovery, clearing, Skyshards. Room to keep growing through all major quest content.
<code>fSkillCapBase</code>	18	Default — skills cap relative to player level.

<code>fSkillCapMult</code>	2	Default — +2 skill cap per player level. Reaches skill 100 at level 41.
<code>bEnableKilling</code>	true (default)	Kill XP enabled. Use the Synthesis Patcher for compatibility.
<code>iXPQuestMain</code>	75 (default)	Main quests feel rewarding.
<code>iXPQuestSide</code>	75	Slightly boosted from 50 — side content is where “living the world” lives.
<code>iXPQuestDaedric</code>	75 (default)	Daedric quests feel meaningful.
<code>iXPQuestObjectives</code>	10 (default)	Partial completion still awards progress.
All discovery XP	Default	Location discovery awards modest XP.
All clear XP	Default	Location clearing awards meaningful XP.
<code>fXPLevelUpBase</code> / <code>fXPLevelUpMult</code>	Vanilla	Uses the default Skyrim leveling curve (no change needed).
<code>bUseRacialCaps</code>	true (default)	Racial skill bonuses matter.
<code>bEnableSkillXP</code>	false (default)	No skill grinding — XP comes from doing, not training.

18. Magic & Perks

18.1. Perk Overhaul Compatibility

How much rules density and specialization pressure the list actually wants once race and standing-stone choices are in place.

Mod	Type	Dial	Notes
Adamant — A Perk Overhaul	Baseline	1, 2, 4	Locked. Simonrim progression family.

18.1.1. Risks & Compatibility

- A dense perk framework can dominate the character-building experience and make later systems feel secondary.
- A restrained framework can make the list feel under-committed if other systems also land on the conservative side.
- Compare perk candidates against the already-preferred race-and-stone direction so the full progression stack stays coherent.

18.2. Magic System Expansion

How broad, readable, and build-defining magic should feel once race, standing-stone, and perk baseline are in place.

Mod	Type	Dial	Notes
Mysticism — A Magic Overhaul	Baseline	1, 2, 4	Locked. Primary magic overhaul.
Triumvirate — Mage Archetypes	Baseline	1, 2, 4	Locked. Identity-driven archetype spells (Cleric, Druid, Shadow Mage, Shaman, Warlock). Complements Mysticism.
Strange Runes	Alternative	1, 2, 3, 4	Floating Nordic runes during spell charging. Visual only, no records.
Elemental Mastery Magic	Alternative	1, 4	Elemental specialization and tiered mastery bonuses.
Immersive Illusion Spells	Alternative	1, 4	New illusion types beyond fear/calm/frenzy.
360 Ward	Alternative	all	Wards block from any direction. Third-person improvement.

Revealing Rune	Candidate	all	Makes rune traps visible. Evaluate after primary overhaul locked.
Cancel Spells - Proper Spell Sheathing	Add-on	all	Cancel a cast to recover magicka instead of lowering hands. QoL, no records.

18.2.1. Risks & Compatibility

- A very large spell ecosystem can dominate progression and make race, perk, and religion feel secondary.
- Layering multiple spell packages casually turns this into a compatibility problem.
- [Mysticism](#) is the locked primary magic overhaul. [Odin](#) is rejected.
- Choose one primary magic overhaul first, then decide whether extra spell volume is actually needed.

—

18.3. Spell Learning Mechanic

Replaces the vanilla “eat-tome-learn” system with time-based study, supporting the slow leveling and immersive pacing pillars.

Mod	Type	Dial	Notes
Immersive Spell Learning - DESTified	Baseline	1, 2	Time-based spell study via DEST framework. One spell at a time.
Immersive Spell Learning	Parent	1, 2	Original mod this is based on. DESTified version preferred.

18.3.1. Risks & Compatibility

- Works with any spell mod that uses standard spell tomes. No patches needed.
- Compatible with Survival Mode and Starfrost (studying induces Fatigue).
- Has MCM for configuration. Only one spell can be studied at a time; switching loses progress.

—

18.4. Stealth And Detection

How Skyrim’s detection system should be recalibrated for modern stealth mechanics without replacing the existing perk, magic, and combat layers.

Mod	Type	Dial	Notes
Realistic AI Detection (RAID)	Baseline	1, 4	Script-free detection recalibration. 23.8K endorsements.
Hide in Bushes	Add-on	1, 4	Bush cover reduces detection. No scripts, third-person friendly.

Take a Peek	Add-on	1, 4	Peek through keyholes before opening doors. JaySerpa. Hard dependency: <code>Dynamic Activation Key</code> (currently only an Alternative in → <code>World Content</code> — promote to a shared framework if adopted). DAR/OAR optional for the third-person peek animation; SkyUI (baseline). Activation via hold-sneak (gamepad-friendly) or the Shift+E contextual prompt. No detection-record edits, so RAID interplay is nil.
Throwing Things Makes Noise - OIF	Add-on	1, 4	Thrown objects make noise and attract enemies. Requires OIF. Verify sound-triggered detection against RAID.
Discipline-first route	Alternative	2, 3	Rely on perk overhaul + Foundations bugfix mods.
Deferred high-commitment branch	Alternative	all	Add stealth toolkit only if playtesting requires it.

18.4.1. Risks & Compatibility

- Detection recalibration can make encounter zones feel too easy or hard if it interacts with difficulty mods from → `Survival & Combat`.
- A detection overhaul can make stealth-only play too dominant if it stacks with a perk overhaul that already buffs sneak.
- Detection changes can affect follower behavior; verify follower behavior under RAID with the chosen follower framework.
- `RAID` last updated April 2022 — script-free means less breakage risk, but verify detection feels right against the chosen perk overhaul.

19. Survival & Needs

19.1. Survival Systems

How much day-to-day bodily upkeep the list wants: enough hunger, fatigue, cold, weather, disease, and wilderness friction for roleplay and travel texture, or a heavier survival framework as a central system. This section owns the baseline survival direction; → `Survival & Combat` owns later balance and difficulty interactions.

Mod	Type	Dial	Notes
Starfrost — A Survival Overhaul	Baseline	1, 2, 4	Locked. Simonrim progression family.
Immersive Diseases 2.0	Add-on	1, 4	Disease overhaul: OAR animations, RaceMenu overlays, blood/dirt compat. Reqs OAR/RaceMenu/SkyUI/SPID (all in stack). Fills the disease axis Starfrost doesn't own. Test.

19.1.1. Risks & Compatibility

- Confirm Starfrost (Simonrim) ships no same-named disease system so Immersive Diseases owns the affliction axis cleanly (its scope is cold/hunger/fatigue). Versions 1.1+ feature per-disease OAR animations and RaceMenu overlays.
- 9 open bug reports on Nexus — load-test disease acquisition/progression before locking.
- A heavy survival system can dominate pacing and crowd out other expanded systems.
- A very light setup can undershoot the immersive-travel pillar if later downtime decisions expect more texture.
- Cold, disease, food, sleep, and camping can blur together between survival, cooking, roleplay, and religion ownership.
- Judge survival primarily on travel texture, downtime rhythm, and world feel rather than maximum hardship.

19.2. Roleplay Upkeep And Hygiene

Lighter day-to-day body-maintenance layer between full survival needs and pure visual flavor. Adds believable downtime rituals without turning hygiene into a second dominant survival framework.

Mod	Type	Dial	Notes
Bathing in Skyrim — Renewed	Baseline	1, 2, 3, 4	SKSE + KPE + PO3 Papyrus Extender. Auto freecam, custom

			bathing animations, washable blood decal removal, bundled dirt overlays, unique soap with SkyPatcher distribution.
Saunas of Skyrim - The Bathing Mare (External)	Add-on	1, 2, 3, 4	Bathhouse outside Whiterun with custom NPCs, vendor, sauna buffs, brandy kegs, rentable bed. Compatible with JK's Whiterun and BiSR.
Real Water in Buckets and Barrels	Add-on	all	Visible water in containers (buckets, barrels, pots, pans). iNeed supported. BOS-based. Enhances survival-immersion visual detail.
Dynamic Dialogue Replacer — DDR	Framework	all	SKSE runtime dialogue-replacement framework (v1.4.1). Needed by the Simple Inn Bath dialogue expansion below; also reusable for future non-AI dialogue mods.
Simple Inn Bath	Add-on	all	Bathhouse access at inns via innkeeper dialogue — minimal cell edits, high compatibility. Confirmed working with the locked Candlehearth inn baseline (→ modlist-world-content.md , Inns) — no patches needed (acts via dialogue/faction, not cell edits).
Simple Inn Bath - Dialogue Expansion - DDR	Add-on	all	DDR-based voiced dialogue lines for Simple Inn Bath (80 lines: spliced innkeeper + bathing dialogue). Requires DDR (135618) + Simple Inn Bath . Carries an AI-Generated Content tag — quality-eval per policy; user-approved (same gate precedent as Autumngate's xVASynth add-on).
Simple Inn Bath for New Lands — SPID	Add-on	all	SPID patches granting Simple Inn Bath dialogue to Bruma, Falskaar, Skyrim - Obscure's College of Winterhold, Beyond Reach, Wyrmstooth, and Midwood Isle innkeepers so you bathe at those new-land inns. Requires SPID (baseline).
Minimalist route	Alternative	all	Leave hygiene implied through inns, rivers, roleplay.

19.2.1. Eating & Drinking Animations

Visible eating and drinking as a downtime ritual — animated consumption with sound effects, NPC autonomous eating, MCM-configurable. Complements survival needs without adding hunger mechanics.

Mod	Type	Dial	Notes
Animated Eating Redux SE	Baseline	all	Third-person eating/drinking animations with sound. NPCs eat autonomously based on time of day. FOMOD with optional animated potions, coffee/tea. Requires SKSE + SkyUI + Pandora (behavior engine).
Animated Eating Redux - Settings Loader	Companion	all	MCM settings loader/preset. Install after Animated Eating Redux.
Eating Animations and Sounds	Alternative	all	Lighter eating/drinking animation replacement. v1.9.4, 11,210 endorsements. Competes with the locked Animated Eating Redux baseline on the same animation events — choose one.

19.2.2. Key Features

- **Washable Blood** — Uses a KPE function (not Dirt & Blood's overlay-based system) to remove blood decals directly. Compatible with Enhanced Blood Textures, Just Blood, Sanguine Symphony.
- **Dirtiful Dirt** — Bundles Zaki's 2K dirt overlays. Optional: Zaki's higher-res variants or MoonMods alternative textures.
- **Custom Animations** — Packaged male animations (Tweens) and female animations (BakaFactory, Krzpz). Auto-detects JVRaven's Malignis Bathing animations. Falls back to vanilla bathing idles — no FNIS/Pandora requirement for basic use.
- **Unique Soap** — Craftable at cooking pots from vanilla materials, or buy from general merchants (SkyPatcher). Multiple soap types with different benefits. Linen wash rag as a soap-free fallback.
- **Automatic Freecam** — Switches to third-person freecam during bathing, matching the pattern of other animation frameworks.
- **Follower Support** — Tracks and manages follower hygiene alongside the player.

19.2.3. Hard Dependencies

Already in the → [Foundations](#) baseline: Kris's Papyrus Extender, PapyrusUtil, PO3's Papyrus Extender, RaceMenu. No additional dependency installs needed.

19.2.4. JSON Configuration

BiSR supports auto-start via JSON in [ModOrganizer.ini](#):

```
[CustomConverter]
\path\data\mods\Bathing in Skyrim Renewed\SKSE\plugins\BathingInSkyrimRenewed.json
```

Use `"!!doautoLoad" : 1` and `"!!doautostart" : 1` to skip MCM setup on new games.

19.2.5. Configuration Discipline

- Hygiene busywork risk. Configure overlay intervals and NPC tracking counts to taste.
- Equipment handling, overlays, and optional integrations need stricter discipline than a simple immersion add-on.
- Only hard runtime requirements should be treated as mandatory.
- For Steam runtime `1.6.1170`, Backported Extended ESL Support requirement does not apply.

—

19.3. Death Alternative System

What happens when the player falls in combat. Goal: add consequence and narrative texture to defeat without making every lost fight permanent.

Mod	Type	Dial	Notes
Discipline-first route	Baseline	all	Vanilla death-and-reload loop. Chosen over Shadow of Skyrim for zero compatibility risk.
Shadow of Skyrim	Alternative	1, 4	Nemesis system — respawn with debuffs, enemies gain name/buffs/presence. Rejected due to script debt at this list's complexity.

19.3.1. Risks & Compatibility

- Death-alternative systems interact with every combat mod. Verify compatibility with `Valhalla Combat`, `Precision`, `TK Dodge`, and combat-animation mods from → `Animations`.
- The nemesis system adds persistent NPCs and effects; verify save file size remains reasonable over long playthroughs.
- Death-alternative mods can create edge cases with scripted-death sequences in quests. Check compatibility with `Skyrim Unbound Reborn`, `Wyrms tooth`, `The Forgotten City`, and major quest mods from → `World Content`.
- Confirm compatibility with Steam `1.6.1170` and the SKSE plugin stack from → `Foundations`.

20. Crafting & Economy

20.1. Alchemy Overhaul

Whether alchemy is a disciplined support system for potions/poisons/ingredients or a larger rework reshaping cooking and survival-adjacent loops.

Mod	Type	Dial	Notes
Apothecary — An Alchemy Overhaul	Baseline	1, 2, 4	Balanced modern route.
Complete Alchemy and Cooking Overhaul	Alternative	1, 2, 4	Large integrated alchemy + cooking pillar.
Alchemy Potions and Food Adjustments	Alternative	all	Lighter fallback.

20.1.1. Risks & Compatibility

- A very large alchemy package can quietly absorb cooking and food design.
- A restrained overhaul can undershoot if too many adjacent systems also land on conservative choices.
- Treat large food-and-cooking coverage as a deliberate expansion choice, not something hitchhiking into the alchemy baseline.

20.1.1.1. Flavor Additions

Mod	Dial	Notes
Alchemy Requires Bottles	all	Empty potions at alchemy lab. Adds bottle-crafting step. Tentative — evaluate gameplay friction.

20.2. Cooking Systems

Whether food and cooking are mostly ambient support for travel and downtime or a deliberate progression-and-preparation layer.

Mod	Type	Dial	Notes
Gourmet — A Cooking Overhaul	Baseline	1, 2, 4	Balanced modern route.
Cooking with Fire - OIF	Add-on	1, 2, 4	Cook raw ingredients with fire spells/staves/shouts via OIF. Scriptless, ESPFE. Verify cooked-

			item records match Gourmet's food edits.
Complete Alchemy and Cooking Overhaul	Alternative	1, 2, 4	Integrated alchemy + cooking.
Alchemy Potions and Food Adjustments	Alternative	all	Lighter fallback.
Mealtime	Alternative	all	Recipe-density companion if Gourmet's scope feels too narrow.
More Plants and Recipes SE	Alternative	1, 2, 4	Adds new plants and recipes to cooking/alchemy. Complements Gourmet.

20.2.1. Visual Companions

Mod	Dial	Notes
Hyper Realistic Cooked Beef + Hyper Realistic Bread	all	Mesh/texture-only replacers. Zero plugin overhead.
Skyrim 3D Cooking	all	Higher-detail cooking pot, oven, food-prep meshes. Mesh-only, no plugin.
Realistic HD Food Remastered	all	Photorealistic food textures.
Steaming Hot Soups and Stews	all	Animated soup/stew bowl mesh + texture replacer with subtle steam effect. No ESP; compatible with soup/stew retextures. Choose the SMIM bowl version (SMIM baseline).

20.2.2. Risks & Compatibility

- Cooking can become too dependent on the survival framework and stop making sense as its own system.
- A very large cooking package can quietly absorb alchemy and economy ownership.
- Judge cooking on inns, camps, home bases, travel prep, and downtime rhythm rather than recipe volume alone.

—

20.3. Smithing And Crafting Expansion

What crafting is supposed to do: lightly improve vanilla smithing, become a broader rules-and-materials framework, or shift responsibility toward NPC services and roleplay texture.

Mod	Type	Dial	Notes
Ars Metallica	Baseline	1, 2, 3, 4	Smithing XP, tanning, mining improvements. No heavy rules restructuring.
Glowing Ore Veins - Light Placer	Add-on	all	Ore veins emit light in the dark via Light Placer (CS-native bulb

			framework). Pairs with Glowing Ore Veins 300 SSE - Optimized.
Honed Metal	Baseline	1, 2, 3, 4	NPC crafting/enchanting services. Makes gold economy matter.
Thaumaturgy — Enchanting Overhaul	Baseline	1, 2, 4	Simonrim-aligned. Pairs with Mysticism + Adamant. Prerequisite for artifact overhauls.
Complete Crafting Overhaul Remastered	Alternative	1, 2, 4	High-complexity material logic and crafting rules.
Summermyst — Enchantments	Alternative	1, 2, 4	140 new enchantment effects. Verify compatibility with Thaumaturgy.
Honed Thaumaturgy (FLM)	Patch	1, 2, 4	Lets Honed Metal NPC enchanters use Thaumaturgy enchantments. Requires FLM (optional).

20.3.1. Risks & Compatibility

- A very large crafting framework can absorb economy and loot-balance ownership.
- Mixing player-centric crafting with strong NPC-service assumptions can produce an incoherent progression loop.
- `Ars Metallica` and `Honed Metal` serve different purposes and should be locked together as complementary.

20.3.2. Woodcutting Support

Prerequisite chain for Chop Chop woodcutting gameplay:

Mod	Type	Dial	Notes
Woodcutting Tweaks	Prerequisite	all	Woodcutting behavior improvements.
KN Cutting Trees	Prerequisite	all	Cutting Trees framework for wood-chopping. Required by Chop Chop.
Chop Chop	Baseline	1, 2, 3, 4	Interactive wood-chopping gameplay with choppable trees and resource gathering.

Also requires `Tools Not Weapons (Pickaxe and Woodcutter Axe)` `DAR Animations` from → [Section 14](#) for mining/chopping weapon animations.

20.3.3. Farming & Gardening

Mod	Dial	Notes
Simplicity of Seeding - Better Hearthfires and Farming CC Planter Scripts	all	Improves Hearthfires planter scripts for more reliable farming. ESL-flagged.
Cut The Wheat - OIF	all	Harvest wheat by cutting farm-field stalks. Requires OIF.

20.4. Economy And Loot Balance

What economy and loot should feel like: lightly rebalanced merchant-and-pricing, deliberate loot distribution, or harsher scarcity-driven progression.

Mod	Type	Dial	Notes
Mod	Type	Dial	Notes
Trade and Barter	Baseline	1, 2, 4	Merchant-pricing route.
C.O.I.N. — Coins of Interesting Nature	Baseline	1, 2, 4	Regional coin variety. No price/loot changes.
C.O.I.N. — Treasury Exchange	Baseline	1, 2, 4	Currency exchange for coin type conversion.
Open World Loot - SkyPatched	Baseline	1, 4	Locked. Curated loot progression. SkyPatched — no ESP slot, runtime patching. Pairs with Arena encounter zones and the Static with Hard Threat dial.
Open World Loot	Alternative	1, 4	Original ESP version. Use SkyPatched variant above instead.

20.4.1. RiS Economy Series (optional)

Mod	Purpose	Dial	Requirements
RiS — Encounter Zones	Reworks encounter zones with randomizer. ESL-flagged.	1, 4	None
RiS — Evolving Economy	Dynamic pricing by resources, season, Civil War, reputation, distance.	1, 4	Papyrus Ini Manipulator, Po3 Papyrus Extender
RiS — Loot Overhaul	Rarity-driven loot, equipment breakage, delevated artifacts.	1	Papyrus Ini Manipulator, SkyUI

20.4.2. Flavor & Content

Mod	Dial	Notes
Narrative Loot	all	1,629 lore-friendly misc items. No economy changes. Stacks cleanly.
SUDs — Skyrim's Unique Drinks	all	50+ unique drink items with custom meshes, textures, effects.
Amulets of Skyrim SSE	all	Adds wearable amulets for every deity, new quest amulets, and improved visuals.

Amulets of Skyrim - Tweaks and Fixes	all	Bugfix and compatibility patch for Amulets of Skyrim. Install after base.
Immersive Jewelry SSE	all	800+ earrings, rings, necklaces, bracelets. Heavy leveled-list integration.
Immersive Jewelry - Lower Value	all	Value rebalance for Immersive Jewelry items. Optional — review for economy fit.

20.4.3. Risks & Compatibility

- A heavy loot-distribution framework can absorb crafting and progression ownership.
- Harsh scarcity can make exploration, crafting, and reward pacing feel punitive rather than immersive.
- `Barter Limit Fix` belongs in → `Foundations` (Targeted Bugfix Mods), not as the economy-framework answer.

—

20.5. Weapons, Armor, And Equipment Content

Weapon packs, armor packs, artifacts, and clothing content have been consolidated into their own dedicated section. See → [Section 27](#).

21. Followers & Reputation

21.1. Crime, Reputation, And Social Systems

How much the world should notice what the player does socially — reputation, crime tools, city suspicion, and guard behavior.

Mod	Type	Dial	Notes
Skyrim Reputation	Baseline	1, 2, 3, 4	Main reputation baseline.
The Choice is Yours	Baseline	1, 2, 3, 4	Quest auto-start control for content-heavy runs.
To Your Face SE	Baseline	1, 2, 3, 4	Social polish for city/tavern greetings.
Steal off Scoundrels	Baseline	1, 2, 3, 4	SKSE plugin — pickpocketing criminals not flagged as stealing. No ESP.
Suspicious City Guards	Alternative	all	Urban tension around suspicious behavior.
Book Of Shadows	Alternative	all	High-expression crime and stealth pillar.

21.1.1. Risks & Compatibility

- A very light social-response layer can leave crime and reputation too close to vanilla.
- A stealth-heavy crime toolkit can make one playstyle dominate a subsection meant for broader roleplay.
- City suspicion systems can become annoying without enough payoff.
- Targeted fixes like [Simple Offence Suppression](#) belong in → [Foundations](#) (Targeted Bugfix Mods), not as the social-system answer.
- Keep stealth-expression tooling separate from broader world-reaction goals.

21.2. Followers And Companion Systems

How companion management should function at the systems level.

Mod	Type	Dial	Notes
Nether's Follower Framework	Baseline	1, 2, 3, 4	Modern flexible route.
EFF — Extensible Follower Framework	Alternative	all	Simpler established management.
Amazing Follower Tweaks SE	Alternative	all	Heavy-management branch.

21.2.1. Risks & Compatibility

- A very light or dated framework can leave companions awkward to manage once the modlist grows.
- A heavy follower-control framework can make party management dominate ordinary play.
- Generic frameworks can conflict with custom companions better left outside them.
- This subsection owns party-management philosophy, not named companion selection (→ [NPCs](#)).

—

21.3. Romance, Marriage, And Companionship

The SFW romance and marriage layer: expanded marriage mechanics, modern dialogue with spouses and romance interests, and opt-in quest-driven romance arcs. Baseline is fade-to-black, not explicit content. Custom named followers belong in → [NPCs](#).

Mod	Type	Dial	Notes
Relationship Dialogue Overhaul — RDO	Baseline	1, 2, 3, 4	5,000+ voiced lines for NPCs/followers/spouses. 137K endorsements, fully SFW.
Amorous Adventures (clean variant)	Baseline	1, 2, 3, 4	Opt-in romance quests, fade-to-black. Lock only clean build.
Marriage Mod — To Have And To Hold	Baseline	1, 2, 3, 4	Up to 11 spouses, divorce, enhanced wedding, dynamic events.
Serana Dialogue Add-On	Baseline	1, 2, 3, 4	7,000+ voiced lines, new story arcs, romance. 51K endorsements.
I'm Glad You're Here	Add-on	all	Fully-voiced follower/spouse/adopted-kids appreciation via dialogue + hug animation. v3.6.0, 10K endorsements. Reqs FLM, KID, SPID, OAR, MCM Helper, SKSE — all baseline. "Living the world" homecoming layer.
Discipline-first route	Alternative	all	Vanilla marriage mechanics.

21.3.1. Risks & Compatibility

- The wrong Amorous Adventures variant breaks the SFW rule. Download must be the clean build.
- Do not stack [To Have And To Hold](#) with another multi-spouse mod.
- Romance dialogue can conflict with custom-voiced follower mods from → [NPCs](#) that replace the same NPC's voice.
- Romance quest mods add non-trivial content — review against → [World Content](#) to avoid doubling up on a single NPC's quest line.
- Marriage events triggered by [To Have And To Hold](#) can fire during unintended moments; check MCM for event timing/cooldown settings.

- `I'm Glad You're Here` dialogue sits in the same spouse/follower topic space as the `RD0` baseline — verify in testing that topics coexist (expected: different topics, no overwrite). Hug animation depends on OAR (baseline); the load-game “welcome back” hug can be disabled in MCM.

22. World Progression Philosophy

> **Adopted Combination:** `#1` — **Static with Hard Threat** (see [Suggested Combinations] (suggested-combinations)). The static-with-threat dial is the base; the “Living the World” pillar is layered on top — the home is a real anchor (survival warmth, food, rest, family), and the player quests when they want, not on the main quest’s clock.

This document owns the design dial that connects four concerns: **leveling**, **encounter zones**, **loot distribution**, and **difficulty**. They are not independent picks — change one and the other three shift. The modlist’s section files cite this document when they need to justify a concrete pick.

No concrete mod picks live here. The owning sections (notably → [Section 20](#) for loot distribution, → [Section 29](#) for encounter zones and difficulty) make the final call. This document presents the three styles, analyzes them across shared axes, and ends with **suggested combinations** for different list identities — the curator picks the combination that matches the list’s intent.

Tagging convention: Mods in the section files are tagged in a `Dial` column with one or more combination codes (`#1` , `#2` , `#3` , `#4` , `all` , `none`). `#1` is the adopted combination for Elder Wilds; mods tagged `#1` (or `all`) are on-dial, others are off-dial for the list’s intent and need explicit justification.

—

22.1. The Three Styles

Deleveled — World no longer scales to the player. A Bandit Chief is dangerous at level 1 and level 50; loot tables are decoupled from player level so a high-tier weapon can be found in a low-level dungeon. `Open World Loot` (SkyPatched) is the locked implementation.

Static Leveling — World is fixed but the loot tables are hand-curated rather than the messy vanilla lists. Top-tier gear is rare, but it isn’t gated behind specific zones — it’s gated behind a curated drop table that the player learns over time. `Open World Loot` (default philosophy) is the closest popular implementation.

Vanilla With Curated Lists — Keep leveled lists essentially as Bethesda shipped them, but curate content additions (new weapons, armors, jewelry) to fit the existing economy rather than inflating it. The base game is the default; new items plug in cleanly. `Trade and Barter` style interventions, content packs that match vanilla tier-spread, light Rarity-style interference at the edges.

—

22.2. The Interlocked Dial

The four concerns are coupled, and the dial really has **six knobs** — the four named plus the **experience mod** (how fast the player levels and what skill-ups do) and the **perks overhaul** (what perks do and how perk points are spent). All six need to sing together or the list feels incoherent:

- **Leveling style** (deleveled / static / vanilla) — what gear the world offers at what level.
- **Encounter zones** — what threat the world offers at what location.

- **Loot distribution** — what the player can actually pick up off that threat.
- **Difficulty** — how hard any given encounter feels.
- **Experience mod** — how fast the player accumulates levels and skill-ups relative to those four knobs.
- **Perks overhaul** — what the player can build into, which determines whether a given dial feels like a wall or a puzzle.

Common failure patterns when the six knobs don't sing:

- **Deleveled loot + encounter-zone overhaul** stacks gating. The world closes in fast; progression has to come from skill, knowledge, and LoTD-style meta-progression, not from better gear.
- **Vanilla loot + no encounter overhaul** produces the canonical problem: by level 30, glass and ebony are everywhere and nothing feels like a reward.
- **Static loot + encounter-zone overhaul** is the most common "modern list" dial — it gates the *threat* without gating the *gear*, so the player feels strong against a curated threat ladder.
- **Deleveled loot + survival framework** can overshoot into punitive early-game, especially in cold biomes where food and warmth are themselves scarce.
- **Heavy perks overhaul + fast XP** floods the player with build options before they've explored the world — the list feels like a build simulator rather than a world.
- **Vanilla perks + slow XP + deleveled loot** starves the player of both gear and build options; every fight is a coin-flip.
- **Heavy perks overhaul + deleveled loot** is actually a *natural* pairing: the player can build into a counter for any encounter the world throws at them, which softens delevel's flat-late-game problem and gives the perks tree real purpose.

A list that locks a strong position on one or two knobs without thinking through the other four will feel incoherent. This document is the place to think it through.

22.3. Deleveled

22.3.1. Power Fantasy

Late-game power fantasy is **weaker**: the world never feels "below you" because the world is the world. Early-game power fantasy is **stronger in a punishing way**: nothing hands you gear, you have to earn every step. For a list that wants the player to feel small and threatened, this is the right dial. For a list that wants the player to feel like they're growing into a legend, it can feel like the dial never moves.

22.3.2. Longterm Play

Excellent for replayability — knowing where the best gear is, or where the hardest fights are, doesn't trivialize the playthrough because the player level doesn't unlock anything. A new character in six months has the same discovery space. **Weak for in-list long-term progression** — if the player plans to spend 200+ hours in a single character, the lack of a power curve can feel flat late.

22.3.3. Exploration & Discovery

Excellent incentive structure. "I should go everywhere, because anything I find could be a game-changer." Pairs well with [Legacy of the Dragonborn](#) (museum-as-reward) and big-world lists. A high-tier weapon found at level 5 is a story, not a balance violation.

22.3.4. NPC & Encounter Balance

NPCs stay dangerous; the curve is flat. Pairs cleanly with encounter-zone overhauls that *add* a curve (so the world still has a sense of progression through the map even if the player's gear doesn't track). The risk is the **over-gated trap**: delevel + encounter overhaul + harsh survival + low early-game loot = the first ten levels feel like a wall. `Survival & Combat` already flags this combination as the "overshoot from moody into over-gated" pattern.

22.3.5. Modlist Synergy

The hardest style to balance with content-heavy modlists. Every new leveled-list mod (every new weapon, armor, jewelry, alchemy ingredient) has to be tested against the de leveled tables — does this new sword break the world at level 5? Does this new jewelry inflate the merchant economy? The more mods in the list, the more curation work. Lists that are content-rich (like this one, with Heavy Armory, Immersive Armors, Immersive Jewelry, Narrative Loot, C.O.I.N., etc.) pay a high cost for running de leveled.

22.3.6. Risks

- Punitive early-game, especially combined with survival frameworks.
- Content-mod curation overhead scales with modlist size.
- Late-game power-fantasy curve can feel flat.
- Easy to over-gate when combined with encounter-zone overhauls.

22.3.7. Experience & Perks

Pairs naturally with a heavy perks overhaul and slow-to-moderate XP. Without leveled lists to gate gear, the player's power growth comes almost entirely from perks and skill-ups, so a rich perks tree carries real weight and a fast-XP setup would front-load that growth. A heavy-perks / slow-XP / de leveled-loot triple is the cleanest expression of the "earn every step" fantasy. Vanilla perks with fast XP undermines this — the player levels up faster than they can spend points and the world still doesn't give them anything.

—

22.4. Static Leveling

22.4.1. Power Fantasy

Balanced. The player grows into a legend against a threat that also has structure. Late-game still feels earned because curated late-game loot is meaningfully better than curated early-game loot, but a high-level player isn't drowning in game-breaking gear. For a list that wants both the growth fantasy and the threat, this is the natural dial.

22.4.2. Longterm Play

Strong for in-list long-term progression — the player has a 100-hour power curve, and curated loot tables mean the curve is well-paced. **Acceptable for replayability** — a new character will find roughly the same gear at the same general progression points, but the discovery of *where* is still open. LoTD-style meta-progression carries replayability.

22.4.3. Exploration & Discovery

Incentive structure is moderate. “Go everywhere to find good gear” is still true, but the curve is shallower than delevelled because the player knows tier-1 zones won’t drop tier-5 items. Pairs well with maps and markers, with the museum, and with curated content packs that respect the tier spread.

22.4.4. NPC & Encounter Balance

The cleanest combination with encounter-zone overhauls. Encounter zones add a *spatial* progression (Whiterun hold is easier than the Rift, the Rift is easier than Skrim wide); static loot adds a *temporal* progression (early-game player has starter gear, late-game player has curated late-tier gear). The two dials don’t fight.

22.4.5. Modlist Synergy

The most content-mod-friendly style. New weapons, armors, jewelry, and ingredients slot into curated tier slots. A new weapon mod that adds ten glass-tier swords integrates naturally; the same mod in a delevelled list has to be tested for “does this break level-5 progression.” Lists that are content-rich benefit from static’s lower curation overhead.

22.4.6. Risks

- The vanilla-curated-lists approach can feel like “vanilla with extra steps” if the curation isn’t visibly better than Bethesda’s default tables.
- Late-game still risks the glass/ebony-everywhere problem if the curator isn’t careful with the tier-4 and tier-5 entries.
- Less distinctive than delevelled — the dial is “more polished” not “different feel.”

22.4.7. Experience & Perks

Pairs naturally with a heavy perks overhaul and a power-fantasy XP curve (high level cap, rewarding quest/discovery, no skill grinding). The player levels up faster than vanilla through quests and exploration while the static threat world keeps every level meaningful.

—

22.5. Vanilla With Curated Lists

22.5.1. Power Fantasy

Closest to Bethesda’s intent. The player grows into a legend against a world that grows with them. Late-game the world hands you legendary glass and ebony because you earned it. For a list that wants to honor the original game and add modern systems on top, this is the right dial.

22.5.2. Longterm Play

Strong for in-list long-term progression (same as static). **Weaker for replayability** — a new character finds the same gear in the same places at the same levels. The strongest replayability lever is [Legacy of the Dragonborn](#) (different museum-fill order, different artifact quest chains, different pacing) rather than the loot dial itself.

22.5.3. Exploration & Discovery

Weakest incentive structure of the three styles. The player learns the loot ladder quickly and the motivation to “go everywhere” weakens. Lists that want exploration incentive need to add it through other levers: `Legacy of the Dragonborn` museum-fill, Evolving Locations content, hand-placed world encounters, `Interesting NPCs` style content, etc.

22.5.4. NPC & Encounter Balance

Pairs cleanly with the least intervention. Vanilla encounter zones + vanilla loot + light difficulty tuning produces a coherent, low-curation list. Adding heavy encounter-zone overhauls to vanilla loot often overshoots (the world gates but the gear doesn’t, producing late-game “I’m level 50 in glass armor and a Bandit Chief one-shots me” frustration).

22.5.5. Modlist Synergy

The lowest curation overhead. New content packs that respect vanilla tier-spread integrate trivially. The cost is that the list feels like “vanilla plus mods” rather than “a designed list.”

22.5.6. Risks

- Late-game loot inflation is the canonical failure mode of this style.
- Exploration incentive is weak; other systems have to carry it.
- Distinctiveness depends entirely on what you put on top of vanilla (art, content, mechanics) — the loot dial alone doesn’t sell a list.
- Easy to under-tune and end up with the unmodded game’s pacing problems.

22.5.7. Experience & Perks

Pairs naturally with vanilla or light perks and a moderate-to-fast XP curve. The Bethesda-style growth fantasy works best when the player is unlocking perks at the rate Bethesda intended — fast enough to feel the build opening up, slow enough that the unlock matters. A heavy perks overhaul here can work, but the dial is doing less of the lifting, so the perks tree has to be the thing that makes the playthrough feel different. Slow XP with vanilla perks and vanilla loot produces a list that feels underpowered even at level 30 — the most common “is this modlist even on?” failure mode for this style.

—

22.6. Suggested Combinations

These are **suggestions for different list identities**, not a ranking. The owning sections still pick the concrete mods; the combinations just steer those picks in compatible directions. The curator picks the combination that matches the list’s intent.

22.6.1. 1. “Big, Dark, Awe-Inspiring World” — Deleveled Or Static With Hard Threat

Suggested dial: Static leveling (preferred) or deleveled (committed), encounter-zone overhaul (`Arena` baseline or RiS Encounter Zones), harsher difficulty, deliberate loot scarcity at tier 1-2, **heavy perks overhaul (Simonrim-class depth) with slow-to-moderate XP** so the player’s power growth comes from build choices against a flat world, not from gear inflation. Survival framework (cold, hunger, fatigue, disease) is on-dial — it makes the home a real anchor.

What it serves: Exploration that makes Skyrim feel vast and threatening. The player feels small, the world feels dangerous, gear feels earned, and the perks tree is the primary expression of “I’m getting stronger.” Pairs naturally with [Legacy of the Dragonborn](#) because the museum becomes the player’s long-term reward ladder. The “Living the World” layer sits on top: a player home with a family, choosing when to quest, living a life in between the dragon-born moments.

Trade-offs: High curation overhead for content mods if deleveled. Risk of over-gating early-game if survival and encounter overhaul stack. Late-game power curve depends heavily on the perks tree carrying it; a shallow perks overhaul breaks the fantasy.

Owned by: → [Section 20](#) for the loot-distribution mod, → [Section 29](#) for encounter-zone and difficulty mods.

22.6.2. 2. “Living The World / Museum-And-Collection” — Vanilla With Curated Lists

Suggested dial: Vanilla with curated lists, light or no encounter-zone intervention, balanced difficulty (Simply Balanced tier), LoTD-anchored progression carrying the long-term reward ladder, **moderate perks overhaul (Path of the Berserker or Adamant tier) with vanilla-or-slightly-slow XP** to preserve Bethesda’s growth pacing while giving the build a little more shape.

What it serves: A list that wants to add modern systems and lots of new content on top of a Skyrim that still feels like Skyrim. The museum is the meta-progression; the loot dial is “vanilla with polish, not overhaul.” Honors Bethesda’s pacing intent while modernizing everything around it.

Trade-offs: Late-game loot inflation is the canonical failure mode — mitigation is required curation of any new gear mods. Exploration incentive is weak; LoTD and content mods have to carry it. Distinctiveness depends on what you put on top. A heavy perks overhaul (Simonrim) here can overpower the “Skyrim that still feels like Skyrim” intent — the perks tree starts to feel like it’s from a different game.

Owned by: → [Section 20](#) for trade-and-barter-style interventions and content-mod curation rules, → [Section 29](#) for minimal encounter-zone intervention and balanced difficulty.

22.6.3. 3. “Power Fantasy” — Vanilla Or Static, Lenient

Suggested dial: Vanilla with curated lists (preferred) or static, no encounter-zone overhaul, lenient difficulty, **vanilla or light perks with vanilla-to-fast XP** so the growth fantasy matches Bethesda’s intent. Late-game gear arrives as Bethesda intended; the player becomes a legend.

What it serves: A list that wants the player to feel like a growing hero, not a survivor. Pairs well with the “Dragonborn quest as the spine” playstyle — if the user is going to play the main quest, the loot dial should support that arc.

Trade-offs: Same late-game inflation risk as combination 2. Risk of feeling like an unmodded game in the first 20 levels. Exploration incentive is the weakest of the three styles; needs other systems to compensate. A heavy perks overhaul contradicts the “Bethesda’s growth pacing” intent and tends to overshoot into a build-simulator feel by mid-game.

Owned by: → [Section 20](#) for trade-and-barter and minimal loot intervention, → [Section 29](#) for the lenient difficulty pick and minimal encounter-zone changes.

22.6.4. 4. “Longterm Replayability” — Static, Moderate Gating, Tunable Difficulty

Suggested dial: Static leveling, moderate encounter-zone intervention (Arena tier, not harsher), difficulty mod with tuning knobs (Simply Balanced, not Blade and Blunt), curated content packs that

respect the tier spread, **moderate-to-heavy perks overhaul (Path of the Berserker or Simonrim tier) with moderate XP** so a new character has a fresh build-puzzle to work through every restart.

What it serves: A list the curator wants to come back to in six months and start fresh. Static gives the new character a fresh-but-familiar power curve; moderate encounter-zone intervention gives the world structure without punishing early exploration; a tunable difficulty mod lets the new character be tuned up or down as the player prefers; a moderate-to-heavy perks overhaul means each new character has a different build to commit to.

Trade-offs: Static-with-curated-lists is the most balanced but the least distinctive dial — the list has to differentiate through art, content, and mechanics. The “vanilla-with-extra-steps” risk is real.

Owned by: → [Section 20](#) for the static-with-curated-lists implementation (Open World Loot default or hand-rolled curated leveled lists), → [Section 29](#) for moderate encounter-zone changes and tunable difficulty.

—

22.7. How To Use This Document

When a section file picks a mod that touches the loot / encounter / difficulty / XP / perks dial, it should briefly note which **suggested combination** the pick serves. Example: “Picking `Open World Loot` (curated default) supports combination 1 (Big, Dark, World) or 4 (Longterm Replayability) and is not the right pick for combination 3 (Power Fantasy / vanilla-loot).”

When two sections make picks that would push the list toward different combinations, the conflict is a signal to revisit this document and pick the combination explicitly. The document does not own the final answer — it owns the framework for making the answer coherent. The owning sections for the six knobs are: → [Section 20](#) (loot distribution), → [Section 29](#) (encounter zones + difficulty), → [Section 18](#) (perks overhaul + XP mod), with → [Section 17](#) holding the cross-cutting growth philosophy.

23. Immersive Scale and World Feel

23.1. Timescale And Calendar Adjustments

How quickly days pass, whether routine actions consume believable chunks of time, and season/calendar support.

23.1.1. Baseline

- **Time Flies SE** ([Nexus](#)) — Grounded play loops without turning into abstract clock tuning.
- **Seasons of Skyrim SKSE** ([Nexus](#)) + **Turn of the Seasons** ([Nexus](#)) — **Locked seasonal baseline**. Full summer/autumn/winter/spring asset swaps. v1.0.8 (Jun 2026, actively maintained). Requires Seasons of Skyrim SKSE (62861); **Shrubs of Snow - Seasons of Skyrim SKSE** ([Nexus](#)) is highly recommended to cover winter assets missed by SoS. **Operational cost is owned by the seasonal LOD workflow** (4× DynDOLOD passes, 4× grass caches, seasonal terrain/object LOD — see → [Performance / Graphics - LOD](#)). Azurite III (locked weather baseline) is the season-driving layer.
- **Seasonal Calendar** ([Nexus](#)) — Companion for calendar readability now that the seasonal branch is baseline.

23.1.2. Alternatives

- **Extended Stay - Longer inn** — Optional longer-stay inn mechanic. Adopt only if the project leans into “living the world.”
- **Sleep Tight SE** — NPC sleep schedule enforcement and sleepwear.
- **Sleep Tight - NPCs wear underwear** — NPC sleepwear variant without sleep schedule changes.

23.2. Sleep And Bedtime Life

NPC-side sleep realism and downtime interaction — sleeping without dead-mannequin stillness, context-aware bed routines, and believable reactions for the “living the world” layer.

Mod	Type	Notes
Sleeping Expanded - Animations and NPC reactions	Add-on	JaySerpa. NPC breathing/sleeping animations, 100 fully-voiced woken-up reactions, context-aware bed enter/exit speeds (fast combat exit = no more dying while slowly getting up). v1.23 (Aug 2023), 15,444 endorsements. ESL, no vanilla edits. OAR baseline satisfies its DAR-based animations (OAR loads DAR folders). Choose main file (DAR-less only if OAR absent).

Sleep Tight SE	Alternative	NPC sleep-schedule enforcement + sleepwear (see ## Timescale Alternatives).
--------------------------------	-------------	---

23.2.1. Risks & Compatibility

- No cell or record conflicts (vanilla-free edits); verify the OAR-animation interplay with any other bed/keep animation mods in → [Animations](#).
- Pairing with [Snore of Skyrim](#) (→ [Audio](#)) layers sleep ambience cleanly — no overlap.

23.3. Travel Pacing And Carriage Fast Travel Rules

How much friction, structure, and world texture sits between major locations.

23.3.1. Baseline

- **Carriage and Ferry Travel Overhaul** ([Nexus](#)) — Improves vanilla public-travel layer. Verify 1.6.1170 compatibility (predates AE).
- **Carriage Drivers Are Alive** ([Nexus](#)) — Names, schedules, dialogue for carriage drivers.
- **Carriages and Stables Dialogue Bundle** ([Nexus](#)) — Bribes, rumors, destination chat. Stacks cleanly with above.

23.3.2. Alternatives

- **Carriage Stops of Skyrim** ([Nexus](#)) — Broader route coverage.
- **Fast Travel Cost** ([Nexus](#)) — Economic pressure on map fast travel.
- **Journeyman - A Fast Travel Overhaul** ([Nexus](#)) — Restricts fast travel unless a Travel Pack is crafted/purchased. SKSE plugin + ESP. Fits "living the world" without forcing no-fast-travel.
- **Detailed Carriages 2.0** ([Nexus](#)) — Unique redesigned look for every carriage, Hearthfire support, SMIM patch. Conflicts with the [Snazzy Diverse Carriages - BOS](#) baseline (→ [Textures & Meshes](#)) — choose one carriage visual identity. Compatible with Carriages and Stables Dialogue Bundle + Carriage Drivers Are Alive (baseline above).

23.4. Horse Travel

Horse riding overhaul and mounted NPC transport — making horse travel feel modern and functional rather than the vanilla stiff-silhouette experience.

Mod	Description	Notes
HorsePower - Modernized Horse Riding	Baseline. Total horse riding overhaul — directional animations, in-place turning, follow-up attacks, sliding fix, ragdoll.	Requires SKSE, Address Library, Animation Motion Revolution, Pandora/Nemesis, TDM. Compatible with Convenient Horses (load HorsePower below).

RSE - Shoulder Or Saddle	Pick up NPCs/corpses, carry on shoulder, secure to horse saddle for transport. RDR2-style dead-or-alive bounty hauling.	Requires OAR, Pandora/Nemesis. Compatible with all horse mods. Custom followers supported.
--------------------------	---	--

23.5. Encounter Spacing And Wilderness Feel

How busy, calm, or deliberately spaced wilderness travel feels between destinations.

23.5.1. Baseline

- **Extended Encounters** ([Nexus](#)) — Broad wilderness and road activity, lightweight.

23.5.2. Alternatives

- **Radiance Encounters** ([Nexus](#)) — World reactivity and consequences.
- **Immersive Patrols SE AE** ([Nexus](#)) — Faction-patrol route.

23.6. Soundscape And Ambient Audio

What roads, settlements, wilderness, and interiors sound like during ordinary exploration.

23.6.1. Baseline

- **Sounds of Skyrim Complete SE** ([Nexus](#)) — Roads, wilderness, dungeons, settlements in one exploration-facing project.

23.6.2. Alternatives

- **Acoustic Space Improvement Fixes** ([Nexus](#)) — Interior ambience and reverb treatment.
- **Discipline-first** — Keep current sound stack if playtesting shows world already sounds readable.

23.7. City Size, Outskirts, And Population Feel

How large, inhabited, and spatially convincing cities feel during ordinary play.

23.7.1. Baseline

- **Spaghetti's Cities - AIO** ([Nexus](#)) — Denser, more inhabited without giant patch-management.

23.7.2. Alternatives

- **Dawn of Skyrim (Director's Cut)** ([Nexus](#)) — Stronger city-core expansion. Last updated June 2020.

- **JK's Skyrim** ([Nexus](#)) — Heavier all-in-one. Only if city transformation should become a loud identity pillar.

—

23.8. Roadside Clutter, Ruins, And Landmark Density

Small-to-mid-scale worldspace texture between major destinations.

23.8.1. Baseline

- **Ancient Land** ([Nexus](#)) — Static worldspace flavor and distant landmark presence.
- **Point The Way** ([Nexus](#)) — Road guidance and signposting.
- **Ryn's Standing Stones** ([Nexus](#)) — Locked landmark baseline. Each stone becomes a unique environmental setpiece. World geometry only, no gameplay effects.
- **Haystacks - Mihail's Shards of Immersion** ([Nexus](#)) — Haystack clutter variants with retextured shovels/pitchforks. ESP (verify ESL flag).

23.8.2. Alternatives

- **Interesting Roads** ([Nexus](#)) — Lighter roadside variety. Asset-only, low compatibility risk.
- **Giant Crab Shells - Mihail's Shards of Immersion** ([Nexus](#)) — Coastal scenery. ESL-flagged.
- **Nordic Runestones** — Optional Nordic runestone world-object additions.
- **Place of Power** — Optional standing-stone mesh/texture enhancement.
- **Whiterun Horse Statues** — Cosmetic Whiterun landmark additions.
- **Solitude Statuary** — Cosmetic Solitude landmark additions.
- **s6o6t LORE - Mundus Stones** — Mundus standing-stone lore expansion.
- **Immersive Laundry** — Hanging laundry lines in cities and towns. Immersion staple.
- **Immersive Laundry - Animated** — Adds animation to the laundry lines. Install after Immersive Laundry.
- **Immersive Laundry Improvement ILIM** — High-quality laundry texture replacer by Pfuscher. Install after Immersive Laundry.
- **Immersive Laundry - Patches for Multiple City Mods** — Compatibility patches for city overhaul mods. Install after Immersive Laundry and city mods.
- **Shores of Skyrim — BOS** — Shoreline overhaul adding rocks, driftwood, and debris to lake, river, and ocean shores. BOS-based, no cell-record edits.

—

23.9. Wildlife And Creature Ecology

How varied and biologically alive wilderness travel feels.

23.9.1. Baseline

- **Animallica SE - Skyrim Wildlife Overhaul** ([Nexus](#)) — More biologically varied during travel.
- **Butterflies Land True** ([Nexus](#)) — Insects land on surfaces. No scripts, ESL-flagged, 1 KB.

23.9.2. Alternatives

- **One With Nature** ([Nexus](#)) — Creature relationship tuning. Script-heavy — verify AE compatibility.

- **Savage Skyrim** ([Nexus](#)) — Heavier danger-and-behavior.
- **Frogs - Mihail** ([Nexus](#)) — 4 frog variants. ESL-flagged.
- **Wearable Lanterns** — Optional wearable lantern mechanic for night exploration.

—

23.10. Road Network And Surface Overhaul

Roads as built infrastructure at mesh/geometry level (separate from surface texture ownership in → [Graphics](#)).

23.10.1. Baseline

- **Blended Roads - Light Plugin (ESL)** ([Nexus](#)) — Locked baseline. Seamless road-to-terrain blending. No worldspace edits, no patch requirements.
- **Interesting Roads** ([Nexus](#)) — Roadside character (barrels, carts, signs). Asset-only.
- **Point The Way** ([Nexus](#)) — Navigational signposting at junctions.

23.10.2. Risks & Compatibility

- **Northern Roads** was evaluated and rejected due to patch burden at this modlist's complexity level. **Northern Roads - Clutters Only** ([Nexus](#)) is a lighter variant with only roadside clutter — re-evaluate if the full Northern Roads patch burden remains too high.

—

23.11. Environmental Atmosphere: Wind And Dynamic Effects

Weather-aware wind physics, dynamic cloth, and airborne particles for physical world-layer motion.

Mod	Notes
Dynamic Wind Framework - SKSE Plugin	Infrastructure baseline.
Dynamic Wind - Skyrim	Required data companion with weather/location profiles.
Particle Wind - SKSE Plugin	Dust, leaves, snow, ash respond to wind direction.
Realistic Boat Bobbing SE + BOS Edition	Boat bobbing with BOS mesh swap support.
R.A.S.S. - Rain Ash And Snow Shaders	Visible rain droplets, snow accumulation, ash/frost shaders on player/NPCs. 33K+ endorsements.
Animated Icebergs	Drifting iceberg meshes in northern waters.
Animated Ice Floes	Animated ice floe meshes in cold regions.
Nocturnal Moths	Moths flutter around exterior lanterns at night (10 PM–4 AM, MCM-tunable). Requires BOS (baseline) + Lanterns of Skyrim II patch (road/settlement lantern layer is LoS II + CS Lantern Lights).
Ships On The Horizon	Ships visible offshore in the Sea of Ghosts near Solitude, Raven Rock, and Tel Mithryn. Animated version. Fits the living-coast world-feel; animated ships and dock layer are in → World Content POI.

23.11.1. OIF Ecosystem — Physics-Based Interactions & Destruction

Mod	Notes
Dynamic Inn Fireplace - OIF	Physics-enabled inn fireplace embers, smoke, and debris via OIF. Adds ambient fire life to interiors.
No Fire During The Rain	Exterior campfires and braziers are extinguished during rain/storms. Requires OIF.
Release The Soul - OIF	Break filled soul gems to release the trapped soul. Requires OIF. No conflict with the soul-gem retexture/description stack.
Burn Those Insects - OIF	Burn frostbite spider and chaurus egg sacs to ash. Requires OIF.
Stalactite Falls - OIF	Shoot stalactites to drop them on enemies below. Requires OIF. Dungeon-hazard interaction.

23.11.2. Risks & Compatibility

- **R.A.S.S. + CS coexistence:** CS Wetness Effects handles terrain/objects; R.A.S.S. handles actor surfaces. Should coexist without conflicts — verify in testing.
- OIF-based interactions depend on the OIF framework being installed and stable in the core library stack.

—

23.12. Night Darkness And Visibility Balance

Mod	Notes
Darkness baseline from the weather mod and CS pair selected in → Graphics	Base darkness level.
Simply Darker Nights	Simple image-space overlay. Verify it's not redundant before adding.
Lanterns of Skyrim II	Road and settlement exterior lantern placement. FOMOD patch hub covers 30-50 patches.
CS Lantern Lights - Lanterns of Skyrim II	CS particle lights for LoS II. Install after LoS II.
Simple Wearable Lanterns - Remastered	Preferred carried-visibility branch. Modern, lightweight, no SKSE.

Mod	Notes
Torches Candlelight and Lanterns	Physics-equipped lanterns, NPC distribution. Requires BOS, MCM Helper.
Realistic Usable Lanterns	Finite candles as consumable resource.
Handheld Lanterns - HDT-SMP	Physics-equipped with sway. Requires FSMP.
Dynamic Torches NG - OIF	Torch fire reacts to movement, collisions, and wind via OIF. Requires OIF framework.

—

23.13. NPC Dialogue Immersion

Market trader and ambient NPC dialogue variety — making recurring NPC interactions feel less repetitive.

Mod	Notes
Shouts of Stallholders	Adds varied voiced lines to vanilla market stall sellers (and mod-added NPCs with vanilla voice types). Uses SPID for NPC exclusion. AI-Generated Content tagged. Safe to install/uninstall mid-game. May be suppressed by idle-comment-suppressing mods.
NPCs React To Fire	NPCs comment/react when fires start or spread nearby (fire-related ambient lines). Script-free; requires SKSE64 + Address Library. Safe mid-save.

—

23.14. Fire And Camp Interaction

Interactive campfire behavior for survival and immersion — sitting, cooking, resting at fires.

Mod	Notes
Usable Campfires	BOS swaps worldspaces into interactive campfire furniture (cook, light, rest). Requires BOS (baseline). Optional Unlit / Rest at Campfire integration.
Campfire Animations	OAR-based animations for campfire actions (add fuel, cook, eat at fire). Requires OAR, FormList Manipulator, MCM Helper. Campfire (Chesko) no longer hard requirement — evaluate against Starfrost-frame cooking.

—

23.15. Activation Text

23.15.1. Baseline

- **Roleplaying In Skyrim - Immersive Activators** ([Nexus](#)) — Descriptive activation text (e.g., "Read Book" instead of "Activate"), hides NPC names until interaction (toggleable). 8KB, no scripts, no SKSE.

—

23.16. Open Research

Open research for the world-feel stack is tracked in `TODO.md`.

24. World Content

24.1. Landscape Overhauls

Large worldspace-content changes making regions feel meaningfully different to cross.

24.1.1. Baseline

First experiment:

Mod	Description
The Marshlands	Gives Hjaalmarch strong identity without redefining a central travel space.

24.1.2. Alternatives

Mod	Description
Whiterun Forest Borealis	Heavier compatibility pressure.
Aspens Ablaze	Rift identity layer. Keep only if → Terrain & Flora can absorb it.

24.2. Cities, Towns, And Villages

Authored settlement content — real places, not just denser vibes.

24.2.1. Baseline

Mod	Description
The Great Cities - Minor Cities and Towns SSE	First-pass baseline giving multiple underbuilt settlements stronger structure.
The Great Cities- Resources	Prerequisite resource mod for all The Great Cities series mods. Install first.
Cities of the North - Dawnstar	Default modular lock for four smaller hold capitals. Designed to stack.
Cities of the North - Morthal	
Cities of the North - Falkreath	
Cities of the North - Winterhold	
Settlements Expanded	Makes seven underbuilt settlements feel inhabited (no city geometry changes). Must load late.

The Great City of Solitude SSE	Port expansion with new homes, shops, warehouses, NPCs.
JK's Whiterun Outskirts	Whiterun-area expansion. Independent of JK's Skyrim AIO; stack without patching.
JK's Riften Outskirts	Riften stables, farmhouses, and gate-to-south-road connection. Same author/series as Whiterun Outskirts. Requires JK's Riften Outskirts Patch Collection by Czasior. Verify vs The City of Crossed Daggers - Riften Expansion (168629) cell overlap.
JK's Windhelm Outskirts	Windhelm Talos Shrine, stables, and three farmhouses. Requires JK's Windhelm Outskirts Patch Collection by Czasior. Windhelm is already heavy with edits (Tomato's Windhelm, Spaghetti's Cities) — verify patch coverage before locking.
JK's Markarth Outskirts	Markarth stables, Salvius farm, Left Hand Mine. ESLified v2.0. Requires JK's Markarth Outskirts Patch Collection by Czasior.
JK's Raven Rock	Solstheim settlement. ESL-flagged.
JK's Tel Mithryn	Telvanni tower. ESL-flagged.
FYX - Eastern Empire Company Building	
FYX - Riften - Window Consistency	
The City of Crossed Daggers - Riften Expansion	Riften market and canal expansion with new shops, NPCs, and quests.
Grand Solitude - The Walls of High King Erling	Solitude wall and exterior expansion adding new areas and detail.
Bards College Rework	Solitude Bard College architecture/interior overhaul (mesh-only, no gameplay changes). Verify mesh stack vs Grand Solitude / Riton Solitude (choose one) + CS/PBR mesh compat.

24.2.2. Locked Faction-HQ Additions

Mod	Description
JK's Fort Dawnguard	ESL-flagged.
JK's Castle Volkihar	ESL-flagged.
JK's Dark Brotherhood Sanctuaries	ESL-flagged.

24.2.3. Minor Settlement Overhauls

Mod	Description
Gonzeh — Stonehills ReRe-imagined	Unique architecture, new interiors, mini-tavern with barmaid, reworked quarry center. ESL-flagged. Has built-in patches for Embers XD (baseline), LoTD

(baseline), AI Overhaul, Cutting Room Floor, Northern Roads, NSUTR, Spaghetti Towns.
--

24.2.4. Alternatives

| [Capital Whiterun Expansion](#) | If JK's patch missing, mutually exclusive with JK's Whiterun edits. | | [Skyfall's Fortified Morthal](#) | Morthal city overhaul. Overrides COTN Morthal — choose one. | | [The Great Town of Ivarstead SSE](#) | | | [Thuldor's Ivarstead](#) | Ivarstead town overhaul. Alternative to The Great Town of Ivarstead — choose one. AI-Generated Content tagged. | | [The Great Village of Kynesgrove](#) | | | [The Great Village of Mixwater Mill SSE](#) | | | [The Great Town of Shor's Stone SSE](#) | | | [The Great City Of Winterhold SSE Edition](#) | Full Winterhold city overhaul. Incompatible with COTN Winterhold — choose one. | | [Spaghetti's Towns - Riverwood](#) | Lightweight Riverwood tweaks (clutter, fishing spot, mill supplies). ESL-flagged. | | [Fleiwas Riverwood Addition](#) | Riverwood object additions. Verify no overlap with Spaghetti's Towns - Riverwood or JK's Whiterun Outskirts. | | [Riverwood Edge](#) | Decorations, small bridge, and spring at the edge of Riverwood. Same overlap verify as above. | | [Fleiwas White River Crossing](#) | White River crossing near Embershard Mine. Verify vs Rope Bridges Remastered (149446) + JK's Whiterun Outskirts. | | [Spaghetti's Towns - Rorikstead](#) | Lightweight Rorikstead tweaks (food shipments, crops, farm animals). ESL-flagged. | | [COTN Winterhold Blacksmith](#) | Adds a blacksmith NPC and shop to COTN Winterhold. Requires COTN Winterhold. | | [Riton Solitude](#) | Solitude architecture overhaul. Likely conflicts with Grand Solitude — choose one. |

—

24.3. College of Winterhold

Coordinated three-mod sub-stack designed to work together without inter-mod patches.

24.3.1. Baseline

Mod	Description	Separator
Obscure's College of Winterhold	Architecture baseline. FOMOD with 50 built-in patches including LotD.	World Content - Overhauls
Immersive College NPCs	ESL-flagged, vanilla assets, no cell edits. Integration patch in Obscure's FOMOD.	World Content - Overhauls
College of Winterhold - Quest Expansion	Skill-gated entry, 7 starter lessons before Saarthal. ESL-flagged, by jayserpa.	World Content - Quests
JK's College of Winterhold	Exterior and interior overhaul. Stack on Obscure's via combo patch below.	World Content - Overhauls

24.3.2. Alternatives

Mod	Description
-----	-------------

Magical College of Winterhold	Cleaner fantasy aesthetic but weaker grim-dark fit.
Immersive College of Winterhold	Architecture+NPCs+crafting. Unmaintained since Dec 2020.
College of Winterhold - Quest Expansion (alone)	Questline-only option — lowest patch overhead.

24.3.3. Patch Support

Mod	Description
JK's College of Winterhold (Immersive or Obscure) Combo Patches	Compatibility patch enabling JK's College to stack over Obscure's or Immersive College.

24.3.4. Compatibility Notes

- Obscure's ↔ COTN Winterhold: Compatible without patch. Load COTN before OCW.
- Obscure's ↔ CS-native lighting: No official patch. Verify CellSettings.esp forwarding for CS Light + True Light bulbs. Install **OCW Meshes Optimized and Merged for CS patch** ([Nexus](#)).
- Obscure's ↔ LOTD: Patch in OCW's own FOMOD. ESP-FE format.
- ICN ↔ NPC appearance: Load ICN after broad NPC overhauls to avoid blackface.
- Quest Expansion ↔ FDE Brelyna Maryon: Both touch her dialogue tree — test for overlap.

24.4. Inns, Farms, And Small Settlements

Travel-stop identity, small community presence, memorable rural pauses.

24.4.1. Baseline

Mod	Description
The Great Village of Old Hroldan SSE	
Candlehearth - An Inn Overhaul	Locked inn-overhaul baseline. Every inn in Skyrim.
Ask Innkeepers To Show Room SE	Preview room before renting. 5,351 endorsements.
Inns Can Be Closed	Inns close at night, must knock for entry.

24.4.2. Alternatives

Mod
Granite Hill
Lund's Hamlet - An Overhaul
The Great Town of Karthwasten SSE
Oakwood

24.4.3. Tavern Life And Inn Economy

Inn-side downtime flavor layered on the Candlehearth baseline: tavern economy, daily menus, and innkeeper interactions. Kept separate from the locked economy/loot dials (→ [Crafting & Economy](#) owns OWL / Trade and Barter).

Mod	Type	Notes
Tavern Games - Mini Games in Skyrim	Add-on	Card ("Battle of Heroes"), dice, and lottery mini-games at any tavern + tavern vouchers (3 free nights). v1.2.1, SKSE-free. Top-down camera (Witcher/Fable-like); Xbox One port exists. Verify gamepad inputs in testing (2016-era SkyUILib UI).
Tavern Games - Ink-art Card Retexture	Companion	Card visuals for the above. Only with Tavern Games.
Today's Menu	Add-on	Buy the innkeeper's daily menu (20 gold): randomized main dish + drink + side/dessert, locality-aware, once/day/inn. Distributions via editable leveled lists. Complements Candlehearth.
Coherent Inns Prices	Add-on	Location-based inn prices (cheap 40 / normal 70 / expensive 100; main-city + Old Hroldan expensive). Contained per-inn dial — no effect on the locked economy/loot stack. Use the provided Candlehearth patch. Compatible with Inns Can Be Closed / Extended Stay.
Coherent Inns Prices - More Expensive	Add-on	Scales the price globals upward for a harder-sticker-shock economy (user-approved). Install alongside 55207; values remain configurable via the base mod's globals.

24.5. Player Homes

Owned "Living the world" home layer — a believable home the player lives in between quests, anchored by the LoTD museum. Named quest-content player homes keep their owning sections.

Mod	Notes
Autumngate - A Believable Player Home	Small believable home southeast of Ivarstead (Rift), ESL. Four FOMOD versions (Standard/Forge/Reduced Clutter/RC Forge). Install the Purchasable House add-on (purchasable from Wilhem at Vilemyr Inn for 12,000 gold; dialogue appears after 24 in-game hours or a quicksave/reload). Optional

Enchanting Table add-on acceptable. Purchasable voice lines are xVASynth (AI-voiced, single NPC) — quality-eval per policy, approved. Verify navmesh against Ivarstead-area mods (Northern Roads patch available).
--

24.5.1. Risks & Compatibility

- Autumngate checks in on the Ivarstead navmesh; verify against adopted town/interior mods for the southeast of Ivarstead.
- Tavern Life stack must not creep into the locked economy/loot dials ([Crafting & Economy](#)).
- Tavern Games' forced top-down camera needs a third-person/gamepad sense-check before locking.
- Candlehearth (locked inn baseline) owns the inn cells — Tavern Life mods act via dialogue/factions/leveled lists, not cell edits.

—

24.6. Dungeons And Ruins

24.6.1. Baseline

Mod	Description
Forgotten Dungeons (SSE)	
Caves of Morrowind	Hand-authored new-cave/dungeon pack in the Velothi Mountains (Dunmer smuggling operation). V3.1, self-contained cells, vanilla-asset-only (PS4-origin), no scripts/quests, navmesh + LOD generated. Level 15+ recommended — a somewhat challenging explorer experience.
Dungeons - Revisited	Improves existing ruins rather than adding new destinations.
Iconic's Crown of Barenziah	High-quality crown/gems/display stand.

24.6.2. Alternatives

Mod	Description
Hammet's Dungeon Pack 1 SE	Curated hand-authored dungeons.
Children of the North Wind	Bespoke Nordic architecture at 9 locations.
Skyrim Underground SSE	Larger subterranean layer.
Icy Dungeons Pack	Icy retexture of nordic crypt dungeons. Visual-only, no record edits.
Ancient Falmer Ruins (Snow Elf Dungeons)	7+ Ancient Falmer dungeon/ruin locations — temples, cities, vaults across Skyrim. Hand-authored, lore-friendly.
Embershard Mine - Revamped	Vanilla-plus overhaul of Embershard Mine — new paths, clutter, bandit sandbox routines, more lived-in feel. No radiant quest conflicts.

24.6.3. Navmesh Support

Mod	Description
Navigator - Navmesh Fixes - Patch Collection	FOMOD covering Dungeons Revisited, LotD, Helgen Reborn, JK's interiors, etc. Install after content mods and Navigator are in place.

24.6.4. Dungeon Visual Overhauls

Visual-only overhauls for specific dungeons. No record or navmesh edits, but confirm compatibility on a per-dungeon basis if world-content mods touch the same cell.

Mod
Bleak Falls Tower
Northern Scenery - Bleak Falls Barrow
Northern Scenery - Ansilvund
Northern Scenery - Angarvunde
Northern Scenery - IronBind Barrow
Bleak Falls Barrow Overhaul Version 2

Note: Bleak Falls Tower and Bleak Falls Barrow Overhaul touch the same location (Bleak Falls Barrow). They are likely incompatible — pick one. Northern Scenery series covers separate dungeons and can be installed together.

Mod
Immersive Dungeon's Fire
Enhanced fire VFX and ambient fire behavior in dungeons. Requires OIF. Tentative — evaluate visual and gameplay impact.

—

24.7. Points Of Interest And Roadside Content

Smaller discoverable places between headline destinations.

24.7.1. Baseline

Mod	Description
Hidden Hideouts of Skyrim SE	
Daedric Shrines - All in One	
Daedric Shrines - Patch Collection	
Environs - Abandoned Abodes	ESL-flagged, zero script overhead.
Evolving Locations - Riverside Shack	
Evolving Locations - Lucky Shack	
Evolving Locations - Cliffside Retreat	
Reinforced Civil War Camps	
Hagraven Houses Animated	BOS mesh replacer.

Ivarstead Source	Landscape fix. ESL-flagged.
Rorikstead Basalt Cliffs	Adds basalt cliff meshes around Rorikstead for visual improvement. No ESP.
More Wooden Bridges	Adds wooden bridge meshes to locations across Skyrim. Replacer + new placements.
Winterhold Docks	Adds dock structures to Winterhold. ESL-flagged light plugin. Complements COTN Winterhold.
Animated Ships	
Finer Sails	
Dynamic Boats at Docks - Official Patch Hub	
Snowy Ships for Snowy Regions	BOS-based.
Better Docks	Dock mesh overhaul for Skyrim and Solstheim docks. Lore-friendly.
Unmarked Locations Pack - All In One	1. addon packs for Bruma, Wyrmsmouth, Gray Cowl if those worldspaces are adopted.
Ascend - Hidden Peaks of Skyrim	Climbing challenge locations across Skyrim. Complements parkour movement.
Graveyards of the Marshlands	Atmospheric graveyard locations in Hjaalmarch. ESL-flagged.
The Wreck of the Argent Raptor	Adds the Argent Raptor shipwreck north of Windhelm, matching Xander's story at Solitude Docks. ESL-flagged, adds only, no vanilla record edits.
Corrupted Cradlestone Tower - FuzzBeed's Forsworn Camps	Overhauls Cradlestone Tower at the summit of Lost Valley Redoubt with aqueducts and fortified walls. Part of a Forsworn camp improvement series.
Interesting Locations	Environmental-storytelling additions to existing vanilla locations (hunting/fishing/lore details). No new quests or scripts; mostly added static/NPC-facing clutter. Verify per-cell against adopted city/tavern overhauls.
Regional Food Barrels - BOS	BOS swaps food-barrel contents by region for variety. No leveled-list or quest edits. Safe mid-save.
Classic Paintings Compilation for Skyrim interiors	Broad distribution of classic public-domain paintings across interior locations and DLC cells. No new scripts/quests. Verify against adopted clutter/art replacers per-cell.

24.7.2. Alternatives

Mod	Description
Hold Border Guards - Definitive Edition	

Hold Border Banners	Visual hold-border signposts and banners. Mesh-only, no ESP.
Immersive Hold Borders SSE Fixed	Hold-border checkpoint improvements with structures and guards.
Nordic Ruins of Skyrim SSE	
Hidden Hideouts City Edition SE	
7000 Steps of Sanctum	
Gildergreen Regrown	
The Gildergreen Grows	Incompatible with Regrown.
HAG Occult Orphan Rock	
Lore Friendly Roadside Ruins	
HAG - Occult Witchmist Grove	Witch-themed point of interest. Same series as HAG Occult Orphan Rock.
HAG - Occult Cradle Stone Tower	Occult tower point of interest. Same series.
Haunted Shipwrecks	Haunted shipwreck encounter locations. BOS-based, no cell edits.

—

24.8. Underwater

24.8.1. Baseline

Mod	Description
Depths of Skyrim - An Underwater Overhaul SSE	New grass, coral, kelp, 1000+ fish, unmarked treasures.
Depths of Skyrim - Mesh fixes	
More Underwater Treasures - AIO	56 treasures (Skyrim) + 35 (Solstheim). Marked compatible with Depths of Skyrim.

24.8.2. Setup Requirements

- Use `SSE-Terrain-Tamriel-Extend.esm` for LOD generation (→ [Performance](#)).
- Requires `iMaxGrassTypesPerTexture=7` in `Skyrim.ini` .
- Depths is script-free but adds 1000+ fish references — verify no unexpected Papyrus load.

—

24.9. Side Activities: Fishing, Hunting, And Downtime

Non-combat downtime making Skyrim feel lived-in when the player isn't dungeon-delving.

24.9.1. Baseline

Mod	Description
-----	-------------

Simple Fishing Overhaul	200+ lines improved dialogue, NPC reactions, MCM. ESL-flagged, by jayserpa.
Hunterborn SE	Dressing, skinning, harvesting, butchering. 20K endorsements. Use Hunterborn Extender via Synthesis.
Headhunter - Bounties Redone	Locked. Transforms vanilla bounties. By jayserpa, 28K+ endorsements.
Skyrim's Got Talent	Locked. Instrument practice, skill progression, inn performances. By jayserpa.
SGT - Player Reactions Addon	
SGT - Show in UI	
SGT - Visual Improvements	
Immersive Fishing	New spots, rods, fish types, radiant quests.
Fishermen Fish	No ESP, mesh replacer only.
Garbage Loot - Barrels and Sacks	ESL-flagged.
Brawling - No Hitting Bystanders	
Skyshards - Tweaks and ESLed	Collectible shard system tweaks and ESL conversion.
Skyshards and Immersive interactions FLM	First/Last mod patch for Skyshards + Immersive Interactions.
The Dragonborn's Bestiary - Lively's Alchemy Addon	Alchemy recipe data addon for the Bestiary book.
The Dragonborn's Fishiary - Bestiary Addon	Fishing data addon for the Fishiary book.

24.9.2. Alternatives

Mod	Description
Simple Hunting Overhaul	Simpler alternative by jayserpa. If chosen, replaces Hunterborn. Requires Dynamic Activation Key .
Dynamic Activation Key	Required by SHO.
Immersive Hunting Animations	SHO companion.
Immersive Carcass Carrying	SHO companion.
Hunters Loot	SHO companion.
Streamlined Fishing	
More Pond Fish	Adds more pond fish variety. Complements Simple Fishing Overhaul + Fishiary.
	Become a Bard / Bards Reborn
Not compatible with SGT without patch.	Fishing Extension Framework
Required for new-lands fishing.	

24.10. Weapons, Armor And Equipment Additions

Weapons, armor, artifacts, and clothing content has been consolidated into its own dedicated section. See → [Section 27](#).

24.11. Quest Additions

24.11.1. Everyday & Radiant

Mod	Description
Sidequests of Skyrim	
Missives	Dual everyday-quest baseline with Sidequests.
Missives - Voice and Quest Expansion	20 new types, voiced.
Missives - Notes Retexture	
Missives - Notes Retexture Darkened	
Diverse Witcher Missives Boards	BOS-based.
Missives - Unique Missive Boards	
Missives Quest Conversion	
Missives - Worldspace Additions	Required for new-lands.

24.11.2. Faction Expansions

Mod	Description
A Chance Arrangement Plus Plus	Thieves Guild quest overhaul.
Opulent Thieves Guild	Visual-progression companion.
Improved Companions - Questline Tweaks	Removes forced radiant-quest spam between Companions missions.
The Companions Extended Collection	Patches for several Companions mods. Integrates quest/dialogue/visual overhauls.
Penitus Oculatus	Locked. "Destroy the Dark Brotherhood!" replacement faction with radiant quests. 11.5K endorsements.
The Brotherhood of Old - Dark Brotherhood Continued	Post-DB-questline continuation with new sanctuary, quests, and radiant content. Complements Penitus Oculatus alternative route.

Optional alternative: Destroy The Dark Brotherhood - Quest Expansion ([Nexus](#)) — jayserpa's quest-expansion approach to destroying the DB. *Incompatible with Penitus Oculatus — choose one.*

24.11.3. Quest Pacing

Mod	Description
-----	-------------

At Your Own Pace	Quest pacing mod — delays main quest, faction quests, and College entry via optional skill-gating and trigger delays. ESL-flagged.
----------------------------------	--

24.11.4. Civil War Content

Mod	Description
Open Civil War SSE	Expands civil war into a dynamic campaign with multiple front-lines, siege battles, and faction maneuvering. High-commitment — significant world interaction and patch footprint.
Serious Civil War Defense for OCW	Adds settlement defense scenarios triggered by OCW campaign events. Requires Open Civil War.
After the Civil War - Siege Damage Repairs	Post-war world repair: damaged cities and fortifications are restored over time. Complements OCW.

Alternatives: Civil War Lines Expansion (Nexus) — jayserpa's dialogue expansion for civil war soldiers. Also listed in → [Section 25](#) (Dialogue Expansions). Check for overlap with the NPCs entry.

24.11.5. Quest Expansions (Vanilla+)

Mod	Description
House of Horrors Quest Expansion	
Paarthurnax Quest Expansion	
The Only Cure Quest Expansion	
The Whispering Door Quest Expansion	
Quest Expansions Patches Hub	
Saints and Seducers Extended Cut	Transforms thin vanilla Creation into proper Shivering Isles return.
Infiltration - Quest Expansion	Thieves Guild / Skyrim quest expansion.
Caught Red Handed - Quest Expansion	Riften jail quest expansion.
The Heart of Dibella - Quest Expansion	
Innocence Lost - Quest Expansion	
The Taste of Death - Quest Addon	Namira quest expansion.
More to do in the Soul Cairn	Soul Cairn quest and content addon.
Boethiah's Calling - Alternate Questline	Boethiah daedric quest alternative.
Mephala's Curse - Whispering Door Addon	Whispering Door quest addon. Separable from the Quest Expansion — evaluate for overlap.

Alternative: Boethiah for Good Guys (Nexus) — older alternative that allows completing Boethiah's Calling without sacrificing a follower. Incompatible with Boethiah's Calling - Alternate Questline — choose one. | [Dungeon Quests are not Miscellaneous](#) | Converts 7 misc dungeon quests to full side quests with journal entries. |

24.11.6. Standalone Adventures

Mod	Description
The Forgotten City	Locked. Most decorated Skyrim quest mod. Self-contained murder-mystery, LoTD integration.
Mysteries of the Dwemer	Dwemer-theme quest mod.
Frozen in Time - Definitely Not Another Snow Elf Waifu Mod	Quest to reanimate an ancient Snow Elf warrior. 2-part quest, 2 dungeons, custom assets, portable player home. 2,833 endorsements.
AI-voiced (ElevenLabs). Vanilla/UUNP/CBBE body options. No AFT during quest.	

24.11.7. Alternatives

24.11.7.1. New Lands

Mod
Wurmstooth
Beyond Skyrim - Bruma SE
Beyond Reach
Falskaar
The Gray Cowl of Nocturnal - 10th Anniversary
The Gray Cowl of Nocturnal - Tweaks
Gameplay tweaks for Gray Cowl of Nocturnal. Install alongside the base mod.
The Isle Of Valefrost SE

24.11.8. New Lands (High-Commitment)

Mod	Description
Land of Vominheim SE	Expansive new worldspace — Valenwood-themed region, quests, dungeons, followers. High patch and compatibility debt. Evaluate against existing new-lands commitments.
Snowpoint	New snow-covered worldspace with quest content. Lighter footprint than Vominheim.
Darkend	Dark Souls-inspired island of Pharos — no quest, no hand-holding, level 15+ recommended. New enemies + an Ancient weapon set (Trinity of Chaos). LoTD patch available. 12,166 endorsements.

24.11.9. Vicn Trilogy (GLEN-MORIL + Unslaad + VIGILANT)

The Vicn trilogy covers three connected quest mods. VIGILANT is already in the alternatives list above; GLEN-MORIL and Unslaad complete the set.

Mod

GLEN-MORIL - English Translation
GLEN-MORIL - ElevenLabs Voiced
GLEN-MORIL - Delayed Start
Unslaad SE
UNSLAAD - ElevenLabs Voiced
VIGILANT SE
VIGILANT - English Translation (Plus Voiced Addon)
VIGILANT (missing lines) - ElevenLabs Voiced
VIGILANT - GLENMORIL - UNSLAAD - Book Covers
Animated Armoury Unslaad Patch

All three have LoTD integration (→ [Section 30](#)). Voiced addons replace subtitle-only dialogue. Delayed Start postpones GLEN-MORIL's initiation until the player reaches a minimum level. The VIGILANT voiced addon covers all missing lines for v1.8.0 and requires the English Translation.

24.11.9.1. Standalone Quest Mods

Mod	Description
Moon and Star	
The Wheels of Lull	
Whispers of the Depths	
Before the End	
Val Serano - Pirate Custom Voiced Follower and Quest Adventure	
	The Frozen Heart
	Midnight Sun
	Echoes of Oblivion
AI-Generated Content.	It Came from Akavir
Short fully-voiced quest (pro VO) about a Kamal washed ashore; expands Kamal/lore, adds a new armor set + weapon. ESL-flagged, safe mid-game, level 15+. Facegen CTD issue fixed in v1.0.6 — take v1.0.6 (not 1.0.3, which needed the face-discoloration fix).	

24.11.9.2. Undeath — Lichdom Quest Mod

Undeath adds a necromancy-themed questline that lets the player become a Lich. New workspace, transformation mechanic, perk tree.

Core mod:

Mod	Description
Undeath Remastered	Base quest mod. Requires Dawnguard + Dragonborn.

Lich gameplay overhaul (pick ONE):

Mod	Description
-----	-------------

Undeath - Classical Lichdom	Recommended. Full lich perk tree and progression. 157K UDLs.
	Undeath - The Ascension
Alternative. Lighter overhaul (bugfixes + balance).	
Undeath Immersive Lichdom SSE	Alternative. Different lich mechanical approach.

Required fixes and patches:

Mod	Description
Undeath Camera Fix SE	Fixes camera during lich transformation.
Required.	Undeath Conjuration Absorb Fix
Fixes conjuration absorption interaction.	Required.
Undeath Remastered - Facegen Data Patch	Facegen for Undeath NPCs.
Required if NPC overhauls used.	Undeath GDOS Door Patch
GDOS compatibility.	REQUIRED — GDOS is baseline.
Moon and Star - Undeath Remastered Patch	Moon and Star compat. Xtudo.
Install if Moon and Star adopted.	

MO2 install order (left pane): Undeath Remastered → lich overhaul → Camera Fix → Absorb Fix → Facegen Patch (if NPC overhauls) → GDOS Door Patch → Moon and Star Patch (if adopted).

24.11.9.3. Quest Timing & Pacing

Mod	Description
Timing is Everything SE	Adjust starting conditions/level requirements for 19 quests and 9 world encounters (Dawnguard, Dragonborn, Hearthfire, Daedric, etc.). MCM-configurable. By kryptopyr.
Essential for “Living the World” pillar.	

24.11.9.4. Lightweight Additions

Mod	Description
Leaps of Faith	Lightweight pilgrimage. ESL-flagged.
Rescue Missions	Radiant rescue. ESL-flagged.
Sleepwalking Into A Nightmare - New Daedric Prince Quest	New Daedric quest mod. Evaluate for LoTD integration.
SIRENROOT - Deluge of Deceit	Quest adventure with new dungeon, voice acting. LoTD patch status: needs verification.
Legends of Aetherium - New Dungeon - Quest - Armour - Weapons	Quest mod expanding the Aetherium storyline.
Restoring the Aretino Residence SE	Short quest adding Aretino family home restoration. ESL-flagged.

Better Skyrim Parties - Weddings - Funerals - Crowd Events Overhaul	Overhauls crowd events and ceremonies. Lightweight, ESL-flagged.
SpiderWIP	Morthal-themed quest mod. Evaluate for coverage overlap.
Better Courier	Courier improvements — better delivery dialogue and behavior.
Quests Are In Skyrim	Restricts radiant quest destinations to Skyrim province only. ESL-flagged.
Andrealletius' Harder Quests	Higher requirements for quest objectives — thaneships, nirnroot, solstheim quests, etc.
Finding Derkeethus	Quest expansion to find and rescue the Argonian follower.
Finding Derkeethus Duplicate FormID Fix	Fix for duplicate FormID conflict.
Jiub's Opus	Quest to find Jiub's journal pages across Solstheim.
Informed Mail Delivery	Quest-aware mail delivery with timed courier dispatches.
A Friend in Mead	Encounter-based quest. Argonian courier Walks-Distant-Sands appears in taverns across Skyrim, shares stories, player responds based on completed quests (16 responses). 5 unique conversations, 9 tavern interactions. Fully voiced. Starts at Dead Man's Drink in Falkreath.
The Surviving Falmer - An Alternate Start	Alternate start addon for Snow Elf player characters + small player home. Supports Skyrim Unbound Reborn (locked).
Requires a Snow Elf race mod (The Ancient Falmer — locked).	

25. NPCs

25.1. NPC Appearance Overhauls

Face, hair, and presentation direction for Skyrim's named NPCs. Does not re-decide the body/skin/baseline character-visual stack already owned by → [Characters & Creatures](#).

Mod	Type	Notes
Northbourne NPCs of Whiterun Hold	Baseline	Modern, regionally textured look. Grounded route.
Northbourne NPCs of Winterhold	Baseline	Same author/series. Regionally textured look for Winterhold.
The Men of Winter SSE	Baseline	Dedicated male coverage. Last updated Aug 2020 — verify 1.6.1170.
Children of the First — Altmer NPC Overhaul	Baseline	Race-specific modular overlay for Altmer.
Children of the Ash — Dunmer NPC Overhaul	Baseline	Race-specific modular overlay for Dunmer.
High Poly True to Vanilla NPC Overhaul	Alternative	Vanilla-faithful fallback. Broad NPC coverage.
Pandorable's NPCs (AIO)	Alternative	Beauty-forward branch. Broad female coverage.
Dlbella's Blessing	Alternative	233 female characters.
Males Of Skyrim by zzjay	Alternative	Broad male coverage.
The Woven Widow — SerketHetyt's Elisif Overhaul	Alternative	Single-NPC overhaul (Elisif).
Val Serano - Another Replacer	Alternative	Single-NPC replacer for Val Serano (Pirate Quest Follower).
Fura Bloodmouth DF Edit SkyPatched	Alternative	Single-NPC replacer for Fura Bloodmouth (CC Dead Man's Dread). SkyPatcher-based. Requires AE content.
Ysolda DF Edit SkyPatched	Alternative	Single-NPC replacer for Ysolda. SkyPatcher-based.
Pride of Skyrim — AIO Male HPH Overhaul	Alternative	400 male NPCs with High Poly Head. Broad coverage.

BeastHHBB	Alternative	Khajiit and Argonian NPC/PC replacer — player character, NPCs, adoptable children, Interesting NPCs integration, fangs.
	Bijin Wives SE	Alternative
Female NPC replacer for wives and selected NPCs.		Skyrim is Cold
Add-on	SPID distribution of warm outfits to 50 named Dunmer/ Nord NPCs in Windhelm, Morthal, and Dawnstar — “living the world” cold-region dressing (outfit/presentation, not face). Requires SPID (baseline). Frostfall/survival-warm keywords — verify Starfrost hooks (Frostfall not in list).	

25.1.1. Overwrite Order

Stack broad-coverage mods first, then region/race-specific, then single-NPC overhauls last. The later mod wins for any NPC it covers.

1. **Broad base layer** — Choose one male and one female branch: `High Poly True to Vanilla`, `Pandorable's`, `Dibella's Blessing`, `Males of Skyrim`, OR `Pride of Skyrim`.
2. **Region or gender-specific** — `Northbourne NPCs of Whiterun Hold` (Whiterun only), `The Men of Winter SSE` (males broadly). Override specific holds or gender groups on top of the base layer.
3. **Race-specific overlays** — `Children of the First` (Altmer), `Children of the Ash` (Dunmer) — load after general overhauls so race-specific facegen wins for their target NPCs.
4. **Single-NPC finishers** — `The Woven Widow` (Elisif) — load last so named individuals use the intended facegen.

After the appearance stack is final, run `FacegenBaseline` via Synthesis (→ [Patcher Reference] (facegenbaseline-npc-stat-patcher-reference)).

—

25.2. NPC AI & Behavior

NPC schedule, behavior, and AI overhauls that affect how NPCs act rather than how they look.

Mod	Type	Notes
AI Overhaul SSE	Baseline	NPC schedule and behavior overhaul. Pairs with any appearance route.

25.2.1. Risks & Compatibility

- AI Overhaul SSE is compatible with all listed appearance overhauls. Load AI Overhaul after appearance mods so schedule records win.
- Quest mods that move NPC schedules may need AI Overhaul patches — check the mod page for supported mods.
- Do not install `Immersive Citizens – AI Overhaul SE` alongside AI Overhaul SSE (→ [Population](civilian-and-traveler-population-additions)).

25.3. Civilian And Traveler Population Additions

Extra background people making towns, inns, and roads feel used — without re-deciding city feel and roadside atmosphere.

Mod	Type	Notes
Crowded Streets	Baseline	Compatibility-first crowd: zero cell edits, dynamic cleanup, configurable density.
Travellers of Skyrim	Alternative	Road-traveler route. LE-era port — verify 1.6.1170.
Populated Skyrim Reborn SSE	Alternative	Heavier legacy route.
MINPCs	Alternative	Adventurer-style incidental content.
Immersive Citizens — AI Overhaul SE	Alternative	Known conflict with AI Overhaul SSE.
Interesting NPCs SE (3DNPC)	Alternative	Authored content NPCs.
Jesters of Skyrim	Alternative	Adds Jester NPCs to select Jarl's residences with idle dialogue and antics. ESL-flagged additions only. Verify placement against adopted palace/court interior overhauls.

25.4. NPC Name Variety

Making named NPCs feel more regionally distinct without changing appearance or dialogue.

Mod	Coverage	Notes
NPCs Names Distributor	Framework	Distributes name variations by race, faction, location.
Tamrielic Names	Human NPCs	Nord, Imperial, Redguard, Breton.
Reachmen Tribes Names	Forsworn / Reach	
Ashlander Nomads Names	Dunmer / Solstheim	
Dovah Names	Dragons	
Daedric Names	Daedra	
Player Name Randomizer	Player	Randomizes display name on new game.

25.5. NPC Dialogue Expansions

Targeted dialogue additions for existing NPCs — expanding what vanilla characters say via spliced vanilla voice assets.

Mod	Description	Notes
Dialogue Expansion — Windhelm	180+ lines for citizens/guards/court	Discontinued — installed copies work, no future updates.
Dialogue Expansion — Imperial Soldiers	Imperial soldier dialogue	Discontinued — same status.
Vigilants Expanded Dialogue	Expands Keeper Carcette (new voice/dialogue) + adds voiced Vigilant Yngvir with unique story. Makes the Hall of the Vigilant a meaningful place to visit before the Dawnguard attack.	Fully voiced original characters. No quest — dialogue depth only.
Bandit Lines Expansion	Spliced vanilla bandit assets	
Extended Bandit Dialogue	Additional bandit lines	
Falmer Servant Lines Expansion	483 lines, pro voice talent	ESL-flagged. Optional plugins: More Spawns, More Unique Servants.
Civil War Lines Expansion	Civil war faction dialogue	
Forsworn and Thalmor Lines Expansion	Forsworn / Thalmor dialogue	jayserpa. Spliced lines.
Vampire Lines Expansion	New spliced lines for vampires	Same series as Forsworn/Thalmor. Dialogue-only — compatible with Sacrosanct.
Guard Dialogue Overhaul SE	600+ lines for guards reacting to player deeds/gear	26.2K endorsements. Older mod — verify 1.6.1170.
Shouts of Stallholders	AI-Generated Content	Evaluate voice quality.
Naked Comments Overhaul	NPC comments on player nudity.	Alternative
Collision Dialogue Overhaul	Dialogue triggered by NPC collision events.	Alternative
Chattier Hunters and Fishermen	Expanded ambient dialogue for hunters and fishermen.	Alternative
Chatty NPCs and Followers	Varied idle/contextual dialogue for all NPCs and followers. No behavior changes.	Baseline — complements existing faction-specific dialogue expansions.
Dialogue Expansion - Indaryn-Ingun-Asgeir-Alessandra	140 lines + 4 scenes for Riften NPCs (Elgrim's Elixirs, Black-Briar Meadery, Bee and Barb). Dialogue-only, ESP-	AI-voiced (ElevenLabs) — accepted per AI gate decision (2026-08-07). AdamDunmer. Verify vs Riften-area mods

	FE, no package/worldspace/navmesh edits.	(City of Crossed Daggers - Riften, JK's Riften Outskirts).
Dialogue Expansion - Shor's Stone	250 lines + 9 scenes for Odfel, Sylgja, Grogmar, Filnjar. Dialogue-only, ESP-FE.	AI-voiced (11labs) — accepted per AI gate decision (2026-08-07). AdamDunmer. The Great Town of Shor's Stone (town geometry) has no overlap expected.

25.5.1. Follower Dialogue Expansion (FDE) series by Anbeegod

> AI-voiced, each ESL-flagged. Core entries listed below; the [official FDE collection](#) includes 26 mods total. All claim 100% compatibility and safe mid-save add.

Follower	Lines	Notes
FDE — Aela the Huntress	2700+	Romance arc, werewolf mechanics.
FDE — Brelyna Maryon	400+	
FDE — Lydia	400+	
FDE — Roggi Knot-Beard		
FDE — Aranea lenith	495	
FDE — Borgakh the Steel Heart	478	
FDE — Uthgerd the Unbroken	233	
FDE — Ysolda	180	
FDE — Jenassa	500+	
FDE — Olfina Gray-Mane	578	
FDE — Mjoll the Lioness	451	Quest and location awareness.
FDE — Faralda	441	Self-contained follower, personal quest. Verify overlap with College Quest Expansion.
FDE — Eola	233	Namira cultist dialogue.
FDE — Ralof	551	[INFERENCE] AI-voiced, ESL, 100% compatible, safe mid-game.
FDE — Senna	243	[INFERENCE] AI-voiced, ESL, safe mid-game.
FDE — Lisette	243	[INFERENCE] AI-voiced, ESL, safe mid-game.
FDE — Marcurio	210	[INFERENCE] AI-voiced, ESL, safe mid-game.
FDE — Saadia	231	[INFERENCE] AI-voiced, ESL, safe mid-game.
FDE — Elisif the Fair	942	Romance + marriage arc, approval system, Civil War awareness. v2.0. ESL.
FDE — Fura Bloodmouth	614	[INFERENCE] 2 personal quests, Volkihar, banter with Aranea/Sapphire/Serana. ESL.
FDE — Sapphire	208	Thieves Guild follower. ESL.
FDE — Illia	918	Repeatable v3.0; witch redemption story, bad-weather awareness, banter with Aela/Jenassa. ESL.
FDE — Rayya	740	Housecarl of Falkreath. ESL. Banter with Erik.

FDE — Faendal	240	Bosmer huntsman. ESL.
FDE — Camilla Valerius	581	Riverwood. ESL. v2.0.1 — includes marriage/romance content.
FDE — Jordis the Sword-Maiden	191	Housecarl of Solitude. ESL.
FDE — Erik the Slayer	222	Rorikstead. ESL status not stated on page — verify.

Mod	Description	Notes
Extended NPC Dialogue Hub	Central dialogue framework	
Small Talk	NPC-to-NPC ambient dialogue	SKSE plugin, no ESP.
Companions Dialogue Bundle	Expanded Companion faction dialogue bundle.	Alternative
Snow Elf Dialogue	Race-aware dialogue for Snow Elf PCs across quests — Helgen, TG, DB, Dawnguard.	AI-voiced (ElevenLabs). Supports The Ancient Falmer, Lost Races of Nirn, True Snow Elf Race. Requires SKSE, Address Library, KID.
Custom Race Dialogue	Consolidated race-dialogue mod combining Snow Elf, Dwemer, Ayleid, Maormer, Tsaesci.	Same author as Snow Elf Dialogue. Incompatible with Race Compatibility Dialogue SSE.
Skyrim Revoiced	Revoiced AI-overhaul — NPC voice diversity across all holds and factions. Compatible with existing dialogue expansions.	Large-scale voice replacer. Test for overlap with FDE series and custom follower dialogue.

—

25.6. Custom Followers

Custom fully-voiced follower additions with authored dialogue, quest commentary, and personal content. The follower framework decision (NFF vs EFF vs AFT) is owned by → [Section 21](#).

Follower	Lines	Notes	Dependencies / Extras
Lucien — Immersive Fully Voiced Male Follower	5000+	Personal quest, LoTD-aware. 30K+ endorsements.	See framework notes.
Remiel — Custom Voiced Dwemer Specialist	5000	Custom framework — do not assign via NFF/AFT.	Banter with Inigo, Auri, Kaidan 2.
INIGO	7000+	100K+ endorsements.	Inigo Official Patch SE required for 1.6.1170. Visual replacer: Lulu's INIGO 2.0 .
Khajiit Will Follow		4 Khajiit followers. 8K endorsements.	See framework notes.

Song of the Green (Auri Follower)	1000+	Romance option, Inigo banter. Custom framework — do not assign via NFF/AFT 2.0+.	Visual replacer: Majestic Auri .
Sofia — The Funny Fully Voiced Follower		Comic relief. May be cut for tonal mismatch.	See framework notes.
Kaidan 2	1200+	Romance arc. Custom AI — do not assign via NFF/AFT.	Use Immersive Features AIO (Nexus Extended Edition deprecated). Extras: A Makeover for Kaidan , Immersive Kaidan Start .
Miraak Follower Expansion	3200+	Post-Dragonborn-quest Miraak follower: quest/world awareness, spouse dialogue, NPC banter.	Full ESP slot (not ESL since v1.5). Verify NFF integration and quest-stage conditions against Dragonborn DLC quests.
Caesia Follower — Borne of Magic (Revamped)	730+	Anbeegod's custom mage follower (Morthal inn). Disposition system, romanceable regardless of sex/race, Shezarrine-lore hints.	Older (v2.0, 2017) — verify 1.6.1170; uses RaceMenu, KS Hairdos, Brows, Ordinary Women/SG textures.
Livia Salvian Revamped — Custom-Voiced Follower	1500+	Imperial knight in Dragonsreach; personal romance quest, custom armor, singing, banter with Onmund. Anbeegod's "Fate of Tamriel"/Shezarrine universe.	v2.2.1 (2020). Original v1 (mod 20143) superseded — install Revamped only. Verify 1.6.1170.

25.6.1. Tone & Genre Fit (anbeegod profile)

anbeegod's profile also hosts Final-Fantasy-styled custom followers (Aerith/Tifa-style characters). These are **not adopted** and flagged as tone mismatches: they contradict the grim-dark, Elder Scrolls-native direction of the list. If the user wants them, they belong as clearly-labeled Optional/off-dial entries, not in the core list.

25.6.2. Framework Notes

NFF compatibility per follower:

Follower	NFF-Safe?	Notes
Lucien	Yes	Tested with NFF, also works standalone.
Remiel	No	Custom system — banter with Inigo, Auri, Kaidan 2.
INIGO	Yes	Use Official Patch for 1.6.1170.
Khajiit Will Follow	Yes	Designed for NFF.
Auri	No	Custom system — has Inigo banter.

Sofia	Yes	Works under NFF or standalone.
Kaidan 2	No	Custom AI — use built-in track/summon powers.
Caesia	Verify	Original uses EFF/AFT-friendly dispos system; Revamped dispositions may work under NFF — verify.
Livia Salvian	Verify	No own follower system; original v1 explicitly expects EFF or AFT. Verify NFF driver lines.

- Verify cross-follower banter: Remiel + Auri have Inigo patches; Kaidan 2 has extended banter with Remiel.
- [Backported Extended ESL Support](#) listed for Majestic Auri does not apply to AE 1.6.1170 — do not install.

25.6.3. Additional Mods

- [Keep Up — Follower Locomotion Fix](#) — Alternative. Prevents followers from getting stuck behind the player. Evaluate after the follower framework baseline is locked.

—

25.7. Playable Race Additions

Playable race mods enabling new character options at game start. Snow Elf (Ancient Falmer) race mods are the primary focus — adding the option to play as the last surviving Snow Elf in Skyrim.

25.7.1. Race Framework

Mod	Notes
RaceCompatibility for Skyrim Special Edition	Required by several custom race mods.
Race Compatibility SKSE (RCS)	Modern replacement for RaceCompatibility — no ESP/ESM file. Required by Lost Races of Nirn. Do not use alongside RaceCompatibility (ESP).

25.7.2. Snow Elf Race Options

Mod	Type	Notes
The Ancient Falmer - A Snow Elf Race SE	Baseline	Most comprehensive snow elf race mod. Adds Snow Elf race, Gelebor/Vyrthur overhaul, optional followers (Valrysa, Ashian), craftable Ancient Falmer weapons. 2,752 endorsements. Requires RaceCompatibility. Last updated Oct 2023.
Lost Races of Nirn	Alternative	Modular race pack: Ayleid, Dwemer, Maormer, Snow Elf, Tsaesci. Separate plugins per race via FOMOD. Requires RCS (no-ESP framework). HPH Addon available (mod 147619).
True Snow Elf Race	Alternative	Vanilla-faithful standalone Snow Elf race. Last updated 2018. Does

		not require RaceCompatibility — standalone BSA.
--	--	---

25.7.3. Recommended Companions

Mod	For	Notes
Frosty Snow Elf Skin Renewal	The Ancient Falmer	Hyperborean BnP skin for Ancient Falmer. Download from the Ancient Falmer Patch Center. Core mod per Ancient Falmer requirements.
High Poly Head For Custom Races	The Ancient Falmer	Patches High Poly Head to work with 30+ custom races including The Ancient Falmer. 2,306 endorsements. Requires High Poly Head (off-site, vectorplexis) and RaceMenu.
Lost Races of Nirn - High Poly Head Addon	Lost Races of Nirn	HPH patches for Lost Races of Nirn 3.0+. ESP-FE. Includes Dwemer, Maormer, Snow Elf, Tsaesci.
Lost Races of Nirn - Female Skin Textures	Lost Races of Nirn	High-quality female textures based on Reverie Skin. UNP/CBBE, 4K-2K, ESP-FE. Fixes neck-seam issues with custom body textures.

25.7.4. Risks & Compatibility

- **Ancient Falmer vs Lost Races of Nirn:** These are competing race frameworks. The Ancient Falmer is more comprehensive (followers, spells, leveled-list weapons) but has a larger patch footprint (RaceCompatibility, Patch Center). Lost Races of Nirn is modular, uses the lighter RCS framework, and has better HPH support. **Evaluate which to adopt as baseline** — running both is risky.
- **RaceCompatibility vs RCS:** Do not use both simultaneously. RaceCompatibility (ESP/ESM) is required by The Ancient Falmer. RCS (no ESP) is required by Lost Races of Nirn. This is a hard fork — pick one race framework.
- **Race-aware dialogue:** Both race mods support [Snow Elf Dialogue](#) — see NPC Dialogue Expansions section.
- **Alternate start synergy:** The Surviving Falmer ([mod 131974](#)) and alternate-start frameworks (Skyrim Unbound Reborn — locked) provide Snow Elf-specific starting scenarios.
- **Third-person/gamepad:** Race mods are character-creation content — no third-person impact beyond face/body visuals at character gen.
- **Perk/Combat overhauls:** The Ancient Falmer adds custom sunfire spells and racial abilities. Verify compatibility with the chosen perk overhaul (Simonrim or other) in → [Magic & Perks](#).
- **BodySlide:** Followers from The Ancient Falmer (Valrysa, Ashian) require BodySlide builds for the chosen body mod (CBBE/3BA). Factor into → [Performance](#) BodySlide rebuild cycles.

25.8. FacegenBaseline & NPC Stat Patcher Reference

These Synthesis patchers resolve facegen and stat issues introduced by NPC appearance and behavior overhauls. Both run as part of the Synthesis pipeline documented in → [Section 37](#).

25.8.1. FacegenBaseline

What it does: Generates missing or corrected facegen data (face morphology `.nif` files and tint layers `.dds`) for NPCs added or changed by appearance mods. Without it, NPCs can display the black-face bug (dark head on light body), mismatched facial features, or incorrect tint layers.

When to run:

- After every change to the NPC appearance stack — adding, removing, or reordering appearance overhauls
- After installing any mod that adds new NPCs (quest mods, world content, new lands)
- After running any Synthesis patcher that changes NPC head parts or face morphs
- Every time the load order of NPC-related plugins changes

How to set up in Synthesis:

1. Add the `FacegenBaseline` patcher to your Synthesis pipeline (Stage 1 in → [performance-patches.md](modlist-performance-patches.mdstage-1-npc-ai-and-facegen)).
2. Configure the **Source Mods** list — select every NPC appearance overhaul and any mod that adds NPCs with custom facegen. The patcher reads facegen from these source mods and copies it into the output.
3. Set **Output Mod** to a dedicated mod folder (e.g., `FacegenBaseline Output` in the `Output` MO2 separator).
4. Run Synthesis with the full pipeline — FacegenBaseline runs after KS Hairs Bald Helmets Fixer and HP_NPC_WIGS_TO_HEADPART.

Configuration options:

Option	Recommendation	Notes
Source Mods	Select all NPC appearance overhauls + any mod listed in this file that ships facegen	Missed source mods are the 1 cause of black-face
Output Mod	Dedicated mod in <code>Output</code> separator	Do not output into an existing NPC overhaul mod
Overwrite Mode	Replace (default)	Keeps output clean on each rebuild
Include Tint Layers	Yes	Required for correct NPC skin/makeup in dialog

Troubleshooting:

Symptom	Likely Cause	Fix
Black face (dark head, light body)	NPC's facegen <code>.nif</code> missing from output, or a later mod overwrites it	Add the NPC's source mod to FacegenBaseline Source Mods; verify load order places FacegenBaseline output after all NPC overhauls

Face doesn't match appearance mod (wrong morph, wrong hair)	A different mod's facegen is winning the conflict	Reorder mods so the intended appearance mod loads after conflicting overhauls, then re-run FacegenBaseline
Grey face (no tint layers)	Tint layers not generated or not found	Verify "Include Tint Layers" is enabled; check that <code>Textures\actors\character\FacegenTints</code> exists in the output mod
CTD when approaching an NPC	Corrupted facegen data — mismatched head part count or missing assets	Delete FacegenBaseline output, re-verify all source mods are selected, re-run Synthesis from scratch
FacegenBaseline fails with "no facegen found"	Source mods list is empty or the patcher can't find compatible face NIFs	Check that at least one source mod is selected and contains <code>meshes\actors\character\facegendata\facegeom\</code>

25.8.2. NPCStatRescaler

What it does: Rescales NPC stats (health, magicka, stamina, level, skills) to match the list's encounter balance. Ensures NPCs added by appearance or content mods don't fall outside the intended power curve.

When to run: After any NPC appearance overhaul, new NPC mod, or encounter-zone change. Runs in the same Synthesis stage as FacegenBaseline.

Configuration: Set target stat ranges to match the encounter design in → [Section 29](#). Default Synthesis profile applies reasonable vanilla+-derived values.

Key relationship: NPCStatRescaler runs after FacegenBaseline in the Synthesis pipeline. If both are active, FacegenBaseline runs first so that NPC records have complete facegen data before stats are rescaled.

—

25.9. Open Research

Open research for the NPCs stack is tracked in `TODO.md`.

26. Enemies & Creatures

26.1. Enemy Variety

How hostile enemy groups feel less repetitive across ordinary play.

Mod	Type	Notes
Lawless — A Bandit Overhaul	Baseline	Cleanest vanilla-plus bandit variety. Bandits are the most overexposed enemy family.
OBIS SE	Alternative	Heavier legacy bandit route.
Haugbui — A Draugr Overhaul	Alternative	Lighter undead route.
Skyrim Revamped — Complete Enemy Overhaul	Alternative	List-wide enemy rebalance.
Skyrim Bandit Expansion	Alternative	New bandit variants, gear, and encounters.

26.2. Creature Visual Overhauls

Creature-specific visual direction layered on top of the Bellyaches texture foundation from →

[Characters & Creatures](#).

Mod	Type	Notes
Fluffworks	Baseline	Broad furred-animal presentation layer.
Bears of the North	Baseline	Species standout for common large wildlife threat.
Iconic's Dragon Retexture	Baseline	Locked. Clean dragon presentation upgrade. Compatible with Dragon War and behavior mods.
Draugrs — New models and textures	Baseline	Most-seen undead enemy.
Spider SE	Baseline	
Deer Reworked — Ultimate	Baseline	
Edmond's Nature Series — DEER	Alternative	Deer/elk model replacer. Alternative to Deer Reworked — Ultimate.
Fox Replacer — Mihail	Alternative	Fox model replacer. ESL-flagged.
Spriggans SE	Alternative	Spriggan model/texture replacer.

Riv's Wolves of the Wild — Diverse Wolf Replacer	Alternative	BOS-based wolf model variety. Verify compatibility with Diverse True Wolves and Dogs (38058).
Canidae - A Wolf Replacer	Alternative	Mesh/texture replacer for red/black/ice wolves + spectral/fiery summons. Pick one wolf replacer vs Riv's Wolves of the Wild.
Diverse Dragon Priests — Pixie's Creature Series	Alternative	BOS-based dragon priest mask/clothing variety. Verify vs Apophysis Dragon Priest Masks (5800).
Diverse Dragon Priests — Xtudo Optimized Textures	Add-on	Optimized textures for Diverse Dragon Priests. Install after base.
Skeletons SE	Baseline	Locked. Skeleton model replacer by Xtudo. Consistent undead visual direction with locked Draugrs SE and Spider SE baselines. 2,002 endorsements.
Skeletons SE - Expanded	Baseline	Locked. BOS-based skeleton variety addon. Install after Skeletons SE.
Skeleton Replacer HD (52845)	Alternative	Rejected — single-model replacer, no variety. Xtudo consistency across undead stack (Draugrs, Spiders, Skeletons) preferred.
TNT - True Nordic Trolls - Revamped	Baseline	Troll model/texture overhaul.
Cannibal Draugr on Solstheim	Baseline	Bloodmoon-flavored identity.
Hagraven — New models and Textures	Baseline	
Butterfly Improved by zzjay	Baseline	
Fusa Fusa Project	Alternative	Don't casually stack with Fluffworks.
Splendor — Dragon Variants	Alternative	Last updated Sept 2017.
Next Gen Dragon VFX	Alternative	Dragon breath/VFX visual upgrade.
Diverse 4thUnknown Dragons	Alternative	Dragon model variety via SkyPatcher. Requires Dragons SE (rejected). Verify compatibility with Iconic's Dragon Retexture before adopting.
Sleep of the Dead — Draugr Leather and Shroud Replacer	Alternative	Draugr shroud/leather retexture.
Frozen Electrocuted Combustion	Alternative	Elemental creature death effects.
Better Butterflies	Alternative	Butterfly insect visual overhaul.
Bear Replacer - Chakra's Creatures	Alternative	Bear model replacer. Alternative to Bears of the North.
Sabre Cat Replacer - Chakra's Creatures	Alternative	Sabre cat model replacer.
Sabrecats of Skyrim - CS ENB Fur Parallax	Alternative	Sabre cat fur parallax texture upgrade.

All the Sabre Cats - Model Swapper	Alternative	Sabre cat model variety via Model Swapper.
--	-------------	--

26.3. New Creature Additions

26.3.1. Mihail Creature Additions (all ESL-flagged)

Mod	Notes
Farm Geese	
Sewer Rats	
Giant Bats and Rats	
Imps	
Zombie Dogs and Skinned Hounds	Undead canine variety.
Giant Centipedes	
Ogrim	
Ruinachs	
Pigs	
Mammoth Expansion	Multiple mammoth variants (woolly, steppe, etc.). ESL-flagged.

26.3.2. Non-Mihail Creature Additions

Mod	Notes
Swamp Skeletons - Chakra's Creatures	Swamp-themed skeleton variants. ESL-flagged.
Edmond's Nature Series — BIRDS	Species: Ravens, Pheasants, Seagulls, flocking birds and bats, Bone Hawks (unique audio), Ghost Ravens (Soul Cairn), plus additional placed hawks and hawk nests. ESL-flagged. Raven tris from Mihail (with permission), Seagulls from Mr. Siika community resource. Do not stack with Seagulls — Mihail. Pair with Xtudo optimized textures — pick 2K main file.
Morrowind Creatures SE	
Ogres SE	
Sea Turtles — Mihail	
Wraiths SE	New wraith enemy type. 4thUnknown quality. ESL-flagged.

26.3.3. Additional Species Follow-Ups

Mod	Notes
Diverse True Wolves and Dogs	

Diverse Dog Textures - SPID	9 dog breeds (shepherd, retrievers, labs, mastiff, great dane, husky) via SPID skin shells. SPID-based, plays well with any dog-adding mod. Fluffworks patch exists (130419). Verify overlap with Diverse True Wolves and Dogs (38058) — pick one dog-variety layer.
Real Rabbits HD	
Rally's Shaggy Cows of Skyrim	
Realistic Horse Breeds by Krittakitty	Verify 1.6.1170 compatibility.

26.4. Creature Behavior Improvements

How creatures attack, react, and express identity once combat starts.

Mod	Type	Notes
Dragons Use Thu'um	Baseline	Smarter dragon shout usage and encounter AI.
The Restless Dead	Baseline	Undead-behavior baseline. Replaces abandoned Draugr Upgrades and Improvements. Requires Skeletons SE patch for compatibility with locked Skeletons SE (124394) baseline.
JS Shout Apart Skeletons SE	Add-on	Unrelenting Force (player or dragons) instantly destroys weaker skeletons and pushes their bones apart. No vanilla record edits; configurable word level via ini. Requires PapyrusUtil + Po3 Extender + Po3 Tweaks + SPID (all baseline). Fits the locked Skeletons SE direction.
Epic Dragon Combat REDONE	Alternative	Full dragon combat overhaul — 20+ new types, unique abilities, custom VFX. Self-contained; replaces baseline dragon visual direction.
Dragon War	Alternative	Heavier dragon route for fight structure/tempo.
Dragons actually fall down	Alternative	Dragons collapse on the ground when staggered/low health.
Epic Dwarven Centurion Combat	Alternative	Dwarven Centurion combat/animation overhaul.
New Creature Animation - Dwarven Centurion - SCAR2.0 - MCO - Nemesis or Pandora	Alternative	Dwarven Centurion animation pack. Works with SCAR (Baseline). MCO-compatible component is unused. Mutually exclusive with Epic Dwarven Centurion Combat — pick one.

SkyTEST Lite ESL / Animal Tweaks / Pelagius's Wildlife AI	Alternative	Animal-side mods drifting into ecology territory.
World Eater Beater	Alternative	Alduin boss fight overhaul — phase-based encounter, new attacks, arena improvements. Replaces vanilla final-boss design.

26.5. Wildlife Density And Rarity Balance

Mod	Type	Notes
True Hunter — fewer animals per square meter	Baseline	Spawn-chance reduction for more deliberate wildlife.
No Predators on roads	Alternative	Abandoned (Aug 2017). Coexists with True Hunter.

Open research for this section is tracked in [TODO.md](#).

27. Weapons & Armor

27.1. Weapon Pack Additions

Weapon content packs adding new named and leveled-list weapons fitting vanilla's medieval-Nordic aesthetic. More variety without diluting the loot ladder.

27.1.1. Baseline

Mod	Type	Dial	Notes
Heavy Armory	Baseline	1, 2, 4	150+ lore-friendly weapons, 11 new types. OAR-compatible polearm animations.
Dawnguard Arsenal SSE - SkyPatched	Baseline	1, 4	Dawnguard-themed weapons, enhanced crossbows. SkyPatched variant.

27.1.2. Alternatives

Mod	Type	Dial	Notes
Legendary Skyrim Crossbows and Bows	Alternative	1, 4	Specialty named/unique crossbows and bows. Lock if ranged is a viable build path.
Immersive Weapons	Alternative	1, 4	230+ weapons. Unmaintained since Apr 2018. Prefer the SkyPatched variant below.
Immersive Weapons SkyPatched	Alternative	1, 4	SkyPatched forward-port resolving record-level conflicts. Preferred over original.
Moonblade	Alternative	all	Unique moon-themed weapon with quest content.
Sorcerer Staff	Alternative	all	Unique staff / Staff of Magnus replacer with expanded mechanics.
Occiglacies - Ancient Falmer Blade	Alternative	1, 4	Ancient Falmer greatsword/sword. Switchable 1H/2H at Skyforge. 1,471 endorsements. Found in Forgotten Vale area. LoTD display patch available.
JaySuS Swords SE	Alternative	1, 2, 4	50 hand-crafted weapons (craftable/lootable/unique finds)

			from the classic pack. Older era (Dec 2020) — check vs established weapon packs; modern alternatives (Billyro packs, Reforging - To the Masses) fill the same niche. LoTD display patch exists.
Lost Relics of Tamriel - Echos of the Past	Alternative	1, 2, 4	Legendary relic/artifact collection by Noctis3DArts. Adds multiple unique weapons and artifacts.
Bloodskal Blade 4	Alternative	1, 2, 4	Remodels the Bloodskal Blade. PBR conversion available (165893).
Bloodskal Blade 4 — PBR Conversion	Add-on	1, 2	PBR textures for Bloodskal Blade 4. Requires 120399 + CS + PGPatcher.
Nordic Battle Axes	Alternative	all	3 new 2H war axes. Craftable at forge under Steel perk. Simple, lore-friendly.
Volkihar Relic Sword	Alternative	1, 4	Unique 1H vampire sword. Animated textures, bleed mechanics, Vampire Lord +30% dmg. Fits grim-dark.
Ivory Bow SE	Alternative	1, 4	New bow by johnskyrim.
Steel Battleaxe Of Fiery Souls — Truly Unique	Alternative	1, 4	Unique weapon rework with animated textures, VFX, 4K/2K. Fits grim-dark.
PBR Hub	Framework	1, 2	Collection of PBR add-ons for various weapon/armor mods. 36 files. Install only sub-files matching weapons already adopted. Requires CS + PGPatcher.
Radiance	Alternative	1, 2, 4	Glowing light sword/mace, acts as torch replacement. ESL, SKSE, follower-compatible. Strong dark-fantasy fit.
Runecarvers Edge	Alternative	1, 4	Lore-friendly sword + greatsword. Vanilla/Complex Material/PBR support. ESL, leveled lists.
Chitin Bow SE	Alternative	1, 4	Dunmer-themed bow. SPID distribution companion (162877).
Chitin Bow SPID	Add-on	all	SPID distribution for Chitin Bow SE.
Relentless	Alternative	1, 4	New sword by johnskyrim. SkyPatcher vendor injection available (114641).

Relentless — LL Injection SkyPatcher	Add-on	all	Adds Relentless to Dawnguard vendors (Gunmar/Hestla). No plugin, no conflicts.
Debaser SE	Alternative	1, 4	Lore-friendly weapon by johnskyrim. Xtudo patches available (94481).
Debaser — My patches SE by Xtudo	Add-on	all	Xtudo patches for Debaser. 13 files covering SPID, crafting, visuals.
Elexorien	Alternative	1, 4	Weapon by johnskyrim. Lore-friendly, fair/balanced, craftable, ESL.
Singularity Blades	Alternative	1, 4	Weapon mod by volodavkaa.
Discipline-first route	Alternative	3	Vanilla weapons only.

27.1.3. Risks & Compatibility

- [Heavy Armory](#) uses standard leveled-list integration; doesn't require Synthesis patcher intervention like [Immersive Weapons](#). A [Heavy Armory - Adamant Addon](#) (mod 144316) provides Simonrim-aligned perk integration.
- [LeanWolf's Better-Shaped Weapons](#) can conflict with [Unique Items Tweaks](#) and [Reliquary of Myth](#) on unique-weapon records.
- [Legendary Skyrim Crossbows and Bows](#) makes crossbows a meaningful archetype — → [Survival & Combat](#) difficulty decisions need to account for ranged output.
- All weapon mods add new models — [Pandora](#) and [BodySlide](#) rebuilds in → [Performance](#) fire when any of them is updated.

27.2. Armor Pack Additions

Armor content packs adding new named and leveled-list armors fitting the medieval-Nordic aesthetic. More variety without diluting the loot ladder.

27.2.1. Baseline

Mod	Description
Immersive Armors	55 armor sets, 396 shields. 206K endorsements.
Immersive Armors Retexture and Mesh Fixes SE	Brings visuals in line with 4K PBR stack.

27.2.2. Alternatives

Mod	Description
NordwarUA Total Armor Compilation	Visual model overhaul — 6 NordwarUA mods. Cubemap standardization.

Common Clothes and Armors	Clothing/light-armor variants for bandits, sellswords, travelers.
Bandolier — Bags and Pouches	Carry-capacity utility with MCM config.
Discipline-first route	Vanilla armor sets only.

27.2.3. Immersive Armors Visual Pipeline

> **Stack order:** Immersive Armors → Retexture and Mesh Fixes → (optional) Realistic Armors replace by Xtudo → (choose one female path) CBBE 3BA BodySlide OR HIMBO's CBBE patch → (if male) HIMBO Conversion V2.

Mod	Role	Dial	Notes
IA — SSE CBBE 3BA BodySlide	Female path	all	BodySlide for all female IA outfits. Do not stack with Retexture/Mesh Fixes' bundled CBBE meshes.
IA HIMBO Conversion V2	Male path	all	BodySlide for all IA armors for HIMBO. Stacks on Retexture/Mesh Fixes. Includes CBBE patch.
IA — Realistic Armors replace by Xtudo	Optional	all	Swaps 10 IA sets for NordWarUA Realistic Armors. Requires Realistic Armor (36151).

27.2.4. SkyPatcher-Powered Equipment Expansions

Mod	Description	Notes
Sentinel — An Equipment Overhaul	300 armor/weapon/clothing variants added via SkyPatcher. Modular — choose individual sets in FOMOD. Functionally identical to originals (cosmetic-only variants). Includes guard armor sets, faction sets, priest robes, AE content integration.	No record conflicts — SkyPatcher-based. No patches needed for NPC appearance mods. Detects Adamant automatically. Compatible with Immersive Armors (different scope — IA adds new sets, Sentinel variants existing ones). Compatible with Xavbio retextures (modlist baseline). Requires AE content for some modules.

27.2.5. Armor Set Additions

Mod	Description	Notes
Armor Variants Expansion	Visual variants for every vanilla armor set.	Has CBBE patch. Fits medieval-Nordic aesthetic.

Ufok's Reinforced Leather Armour	D&D-inspired light armor.	3BA, CBBE, UBE, HIMBO, Vanilla. ESL, craftable, Skypatcher integration.
Tera Armors Collection	Armor collection ported from TERA MMO.	CBBE. Evaluate visual fit.
Lustmord Vampire Armor - SSE CBBE Bodyslide	CBBE 3BA vampire armor with physics.	14K endorsements.
Wilderness Witch Outfit	CBBE witch/mage outfit.	Fits grim-dark aesthetic.
Dark Mage of Plegia - hdt SMP (CBBE 3BA)	CBBE 3BA SMP dark mage outfit.	9K endorsements.
Dremora Markynaz Armor SE	Daedric-themed armor set.	Requires HDT-SMP. HIMBO refit (125150), CBBE (79968).
Silver Armor SE	Silver-themed armor set.	CBBE (79319), HIMBO (87675), HDT-SMP patch (79255).
Colovian Prince Set	Colovian-themed armor set.	
Lifsworn Vestiges – Champion of Arkay Armor	Arkay-themed armor with optional SMP.	3BA/HIMBO (175271).
Traveling Mage HDT-SMP Armor	Mage travel outfit with SMP.	CBBE 3BA/HIMBO (69407).
JS Helm of Yngol SE	Unique helm mesh/texture replacer.	
Wind Ruler Armor SE	Wind-themed armor set.	HDT-SMP (63036), HIMBO (106036).
Imperial Guard Centurion Armor SE	Imperial guard armor set.	3BA/HIMBO (178841), SPID (123487).
Fur-lined Steel Armor	Fur-trimmed steel armor.	
Travelling Priest Robes	Priest robe outfit. HIMBO/CBBE.	SPID distribution (118464).
Nibenean Armors and Outfit SE	Nibenese-style armor/outfits.	SPID (123477), male HIMBO refit.
Templar Assassin Armor	Assassin-themed armor. 3BA/BHUNP/CBBE/UNP.	HDT-SMP patch (135910).
Visage of Mzund	Unique Dwemer face visor. Xtudo version.	
Legendary Alpha Shields	Shield variety pack.	SPID for NPCs (80041), Xtudo patches (79895).
Legendary Alpha Shields 2	Additional shield variety pack.	SPID for NPCs (80110), Xtudo patches (79617).
JS Unique Utopia SE - Rings	Unique ring mesh/texture replacer.	
Land of Vominheim - Unique Rewards by Xtudo	Unique rewards patch for Land of Vominheim.	Install if Vominheim is adopted.

Callisto HDT-SMP Armor	Female HDT-SMP heavy armor. Craftable, ESL-flagged.	Requires FSMP. Xtudo patches (134900) add SPID, Ancient Falmer replacer, LOTD, beast-race support. HIMBO male addon (134789).
Armors of the Velothi Pt. I	Dunmer-themed armor packs (multiple sets).	Install Pt. I then Pt. II.
Armors of the Velothi Pt. II	Additional Velothi armor sets.	Install after Pt. I.
Ancient Imperial Armor SE	Imperial-themed armor set.	
Ancient Imperial Armor for Vigilant SE	VIGILANT replacer for Ancient Imperial Armor.	Install if VIGILANT is adopted.
BS Bruma - Imperial Armors Replacer - Ancient Imperial	Replaces Bruma Imperial armors with Ancient Imperial. Xtudo.	Install if Beyond Skyrim - Bruma is adopted.
Dwemer Armor SE	Dwarven armor model/texture replacer. 4thUnknown.	CBBE 3BA (81602), HIMBO (95191), SkyPatch (159254).
Ancient Argonian Armor SE	Argonian-themed armor set.	HIMBO refit (139314).
Thalmor Ceykynd Armors	Thalmor-themed armor sets.	CBBE 3BA (140647), HIMBO (140823), HDT-SMP (147509).
Ysmir Armor SE	Nordic-themed armor set.	
The Southbound Scouts - HDT-SMP	Ranger/hunter armor with physics.	Requires FSMP.
Gorm HDT-SMP Armor	Armor set with physics.	Requires FSMP.
Arcane Mage Armor	Mage armor with multiple body options (3BA/BHUNP/CBBE/UNP/Vanilla).	
Steel Vanguard Armor (CBBE HDT-SMP)	Steel-tier armor with physics.	3BA/HIMBO refits (149974).
Freehand Armour - BHUNP - 3BA - HIMBO - SMP	Steel-tier heavy armor set (cuirass, gauntlets, greaves, armet, scarf). Nomad/murderhobo aesthetic. SMP physics. Craftable (Advanced Armors), leveled-list. Beast-race support. SPID companion (150508).	Requires FSMP. 2K textures (4K optional).
Buoyant Armiger Glass Armory	Glass-tier Dunmer/Morrowind armor set.	PBR (182690), HIMBO/CBBE bodyslides (174215).
Dark Apprentice (3BA SMP)	Mage armor with physics.	PBR (171845), SPID (172938).
Bosmeri Swallowtail Armour	Bosmer/wood elf-themed armor.	SPID distribution (176955).
Wolf Leather Armor	Light armor set.	BHUNP/HIMBO/CBBE/3BA bodyslides (136273).

Elven Chainmail (SSE Port)	Female-only elven light armor set + scimitar. Found in Northwatch Keep.	LE port by JWGaming75. Requires CBBE. Last updated Jan 2018 — verify 1.6.1170.
HD Falmer Elven Chainmail Retexture	HD retexture for Elven Chainmail.	By trion77/Vindition. Install over Elven Chainmail base mod.
Kozakowy's Female Vampire Armor Replacer	Replaces vanilla female vampire armor (3 color variants). CBBE 3BA, 811 endorsements.	Female-only, no weightslider compatibility. Pure mesh/texture replacer — no records conflict. 3BA SMP conversion available (143980).
Faction Armors and Weapons Retexture SE	All-in-one retexture of Dawnguard, Vampire, Thieves Guild, Nightingale, Dark Brotherhood, Forsworn, Wolf armors/weapons. Optional mesh additions, color variants (Shrouded).	By Xavbio. Replaces hidden mod 96855. Respects original designs with refined detail and grounded look. Covers both armor and weapons.
Xavbio Retexture Collection - Glass (87580)	Retexture of vanilla Glass armor/weapons, matching the Xavbio faction/ebony direction.	Same author as the locked Faction (169281) + Ebony (83654) stack — keeps the Glass material tier in the same visual family. Choose 2K.
Armor of Nocturnal	Dark fantasy light armor set, standalone craftable. Dwarven-tier.	ChakraSSE. 4K/2K textures, custom male/female meshes. No HDT-SMP. HIMBO refit below.
Armor of Nocturnal — HIMBO Refit — Heavy and Light Armor Patches	HIMBO bodyslide + optional heavy/light ESP patches for Armor of Nocturnal.	Requires 129809, BodySlide, HIMBO. Fixes original's mixed armor-type issue. Patches usable without HIMBO.
Armor of Alessia	Fantasy heavy armor set + Alessian Sword, standalone craftable. Ebony-tier.	ChakraSSE. 4K textures, custom male/female meshes. HIMBO via ChakraSSE HIMBO bundle (159950).
Nightstalker Nazir Armor Replacer	Armor replacer, standalone craftable.	ChakraSSE. HIMBO via ChakraSSE HIMBO bundle (159950).
Nordic Brute Armor (CBBE HDT-SMP)	Heavy armor with HDT-SMP cloth physics.	ChakraSSE. Requires FSMP. HIMBO via ChakraSSE HIMBO bundle (159950).
Drake Knight Armor	Armor set, glow variant available.	ChakraSSE. HIMBO via ChakraSSE HIMBO bundle (159950).
Armor Of Hermaeus Mora	Daedric-themed armor set.	ChakraSSE. HIMBO via ChakraSSE HIMBO bundle (159950).

More ChakraSSE Armor HIMBO Refits	HIMBO/BodySlide refits for Armor of Alessia, Nightstalker Nazir, Nordic Brute, Drake Knight, Armor of Hermaeus Mora.	Install per-armor, overwrite originals. Requires each base mod + BodySlide + HIMBO. 55 endorsements.
Master Thief Armor 3BA-BHUNP-UNP-CBBE-HIMBO-Vanilla	Armor set with CBBE 3BA / HIMBO support.	Evaluate for visual fit with the rest of the armor stack. Alternative — not baseline.
Fluted Armor SE	New heavy armor set.	4thUnknown. Esp-fe. HDT-SMP patch (106791) for physics cloth.
Fluted Armor - SPID	SPID distribution of Fluted Armor to NPCs.	Requires base 106381 + SPID (both baseline).
(Upgrade) HDT-SMP Fluted Armor Patch	Adds HDT-SMP cloth physics to the Fluted set.	Requires base 106381 + FSMP (baseline).
HDT-SMP arm hit Fluted HIMBO V5	HIMBO V5 male fit for Fluted with SMP.	Requires HIMBO (body baseline), FSMP.
Ash Pilgrim's Garments	Dunmer female light armor set (craftable).	4thUnknown family. Requires the Revived ports overlay below.
Ash Pilgrim's Garments Revived	Body ports (Vanilla/3BA/BH/HIMBO/Beast races) + plugin fixes for base 53683.	Install over base set. 3BA + HIMBO paths fit the locked body baseline.

27.2.6. Risks & Compatibility

- **Item Equip Restrictor** (109510) — SKSE plugin restricting armor/gender mismatches. Install in Foundations. Prevents NPCs equipping incompatible gender/race armors.
- **TangyTweaks' HIMBO Conversions** (135362) — Bulk HIMBO bodyslide conversions for various armor mods. Install if HIMBO is male body baseline.
- **NordwarUA Total Armor** standardizes cubemaps across all six sub-mods — any mod depending on original cubemaps needs re-testing.
- **Common Clothes and Armors** adds clothing to NPC leveled lists. Verify NPCs look right if **Armor and Clothing Extension** is also active.
- **Bandolier** changes inventory — carry-weight design must account for additional slots.
- The Retexture/Mesh Fixes mod and CBBE 3BA BodySlide are **not stackable** for female meshes. HIMBO Conversion V2 avoids this conflict (designed to stack on top of Retexture/Mesh Fixes).
- All IA companion mods trigger **Pandora** and **BodySlide** rebuilds in → **Performance**.
- **SynthEBD** (GitHub) — Synthesis patcher for NPC equipment/outfit distribution with complex rule-based config. Use as a SPID alternative when a mod has no SPID config or needs multi-condition NPC filtering. Included in the Synthesis pipeline (→ **Performance**, Stage 1). Add SynthEBD configs per armor mod when SPID distribution is unavailable or insufficient.

—

27.3. Artifact And Unique Item Overhauls

Unique-item content packs that add depth to Daedric artifacts, dragon priest masks, unique weapons, and quest items. Treats artifact overhauls as a separate layer from crafting — they change what named items are, not how the player makes generic items.

27.3.1. Baseline

Mod	Type	Dial	Notes
Reliquary of Myth	Baseline	1, 4	Artifact stats and effects. 12K endorsements.

27.3.2. Alternatives

Mod	Type	Dial	Notes
ArteFakes	Alternative	1, 2, 3, 4	Higher-detail 3D models for unique items. Stacks with Reliquary of Myth.
Konahrik's Accoutrements	Alternative	1, 4	Dragon-priest gear, masks, harder fights. LoTD museum integration.
Dragon Priest Hood Variants	Alternative	all	Hooded/unhooded variants for Dragon Priest masks. Verify vs Apophysis (5800) and Diverse Dragon Priests (105519).
Unique Items Tweaks	Alternative	1, 2, 3, 4	Gap-filler for artifacts Reliquary of Myth doesn't cover.
Artificer	Alternative	1, 2, 4	Simonrim-aligned alternative to Reliquary of Myth. Requires Thaumaturgy + Mysticism.
Discipline-first route	Alternative	3	Vanilla artifact stats.

27.3.3. Risks & Compatibility

- [Reliquary of Myth](#) + [ArteFakes](#) + [Unique Items Tweaks](#) touches every unique-item record. Verify the Synthesis patcher chain in → [Performance](#) doesn't re-balance the same records.
- [Konahrik's Accoutrements](#) adds new Dragon Priest encounters — check the [LoTD Patches \(Official\)](#) hub in → [Legacy of the Dragonborn](#) for an integration patch.
- Artifact overhauls that rebalance too aggressively can break encounter expectations.
- [Unique Items Tweaks](#) overlaps lightly with [Reliquary of Myth](#) on a small number of records — verify with a Bashed Patch.

—

27.4. Clothing And Wardrobe Extensions

Clothing content packs adding new clothing, cloaks, and wearable non-armor items. More wardrobe variety without making every NPC a fashion show.

27.4.1. Baseline

Mod	Type	Dial	Notes
Armor and Clothing Extension	Baseline	1, 2, 3, 4	Clothing diversity. 35K+ endorsements. Requires RUSTIC CLOTHING + SkyUI.

27.4.2. Alternatives

Mod	Type	Dial	Notes
Common Clothes and Armors	Alternative	all	Common-cloth variety for bandit/sellsword archetypes.
Common Clothing Expansion	Alternative	all	New lore-friendly clothing items. Verify leveled-list overlap vs ACE + Common Clothes and Armors.
CCE - CBBE 3BA	Add-on	all	CBBE 3BA rebody for Common Clothing Expansion. Requires base CCE + CBBE 3BA.
CCE - Vanilla Clothes Replacer	Add-on	all	SkyPatcher replacer of vanilla farm clothes using CCE models. Verify vs Common Clothing Retexture SE.
Sforzinda Imitations	Alternative	all	Curated boutique clothing. Cut if it pulls toward boutique-fantasy wardrobe.
Immersive Fur Garments	Alternative	all	New fur garments to leveled lists. SPID addon below.
Immersive Fur Garments — SPID	Add-on	all	SPID distribution addon (ESL-flagged). Install base mod (delete .esp) + SPID addon.
Nirn Necessities — SMP Accessories	Alternative	all	SMP-physics clothing accessories. PBR patch (135807) if using CS PBR. Requires FSMP.
Cloaks of Skyrim	Alternative	all	100+ cloaks via SPID. Highest third-person visual impact.
Colovian Noble Clothes	Alternative	all	Imperial-themed high-status clothing. Pairs with Beyond Skyrim.
Knapsack Backpacks SE	Alternative	all	Carry-and-wearable utility with MCM. Cosmetic companion to Bandolier.
Discipline-first route	Alternative	3	Vanilla clothing only.
RMB SPCH - Cloaks of Skyrim	Alternative	all	SkyPatched version. Replaces original Cloaks of Skyrim (6369). Preferred over base.
Cloaks Of Skyrim Retextured SE	Add-on	all	Retexture for Cloaks of Skyrim or RMB SPCH variant.
Cloaks Of Skyrim Retextured SE Female Mesh Patch	Add-on	all	Female mesh fix for the retexture.

Artesian Cloaks of Skyrim	Alternative	all	Higher-detail cloak mesh/texture replacer. Alternative to Cloaks of Skyrim.
Dynamic Artisan Cloaks	Alternative	all	Dynamic cloak animations via OAR. Cloak physics and movement.
RMB SPCH - Winter is Coming Cloaks	Alternative	all	SkyPatched version of Winter is Coming Cloaks.
Dynamic Armor Variants	Alternative	all	Visual armor/clothing variants based on conditions. No stat changes.
Dynamic Lowered Hoods	Alternative	all	Hoods lower dynamically when not in combat. OAR-based.

27.4.3. Risks & Compatibility

- [Armor and Clothing Extension](#) requires [RUSTIC CLOTHING](#) and [SkyUI](#) — verify load order keeps [RUSTIC CLOTHING](#) above ACE.
- [Cloaks of Skyrim](#) adds 100+ cloak items. If → [Third-Person Gameplay](#) adds a custom cloak-slot overhaul, verify stacking. RMB SPCH (116030) is the preferred SkyPatched variant over the original.
- [Knapsack Backpacks SE](#) (cosmetic) and [Bandolier](#) (inventory) can stack, but verify inventory UI doesn't show duplicate bag icons.
- Verify [Colovian Noble Clothes](#) for known issues before locking.

28. Audio and Feedback

28.1. Music Overhaul Strategy

The soundtrack direction: how much to expand Skyrim's music while staying close to the Jeremy Soule baseline.

28.1.1. Baseline

Mod	Notes
Chapter II - Jeremy Soule Inspired Music	Darker, more emotional melancholy without breaking from Soule's shape.
The Northerner Diaries - Immersive Edition (music by Jeremy Soule)	Authentic Jeremy Soule compositions: 8 curated exploration tracks (Skyrim + Solstheim exploration/sunset/night categories). HD 192kbps, lossless-sourced, ESL option. Link fixed 2026-08-07 (was pointing at the unrelated texture mod 33903).
Hun Lovaas - Skyrim Fan-Made Combat Music	Adds entirely new, lore-friendly combat tracks written for Skyrim, fitting the vanilla vibe. Non-replacer version — introduces new music rather than overwriting vanilla. Blend via Synthesis music merge; ESL option available. Pre-2020 (v1.0, 2018) — verify native AE support before locking.
The Southerner Diaries	22 Soule-inspired exploration and town tracks.
Melodies of Civilization	20+ town, tavern, and settlement tracks.
Nygthfall - Dark Fantasy Music	7 hours, 98 dark-fantasy tracks. Evaluate after baseline is in place.
Nordenhamr - Viking Music	Viking identity reinforcement.
BA Bard Songs	Bard song replacer. Compatible with Skyrim's Got Talent.

28.1.2. Alternatives

- **Yggdrasil Music and SoundFX Overhaul** ([Nexus](#)) — Harsher, more ritualistic direction. Last updated April 2021.
- **Silent Combat Music** ([Nexus](#)) — Support option if combat music becomes intrusive.
- **Minimalist route** — Keep vanilla soundtrack.

28.1.3. Installation Notes

- If combining multiple music mods, use the Synthesis music-merge plugin for deliberate playlist merging.
- Keep generated music-merge output in its own dedicated MO2 mod. Rebuild when the music stack changes.
- See → [Performance](#) for Synthesis pipeline configuration.

28.1.4. Risks & Compatibility

- A very large combined soundtrack can feel incoherent if tracks don't share tonal identity.
 - Multiple music mods must be merged via Synthesis to avoid overlapping [MUSTT](#) records.
 - Any pre-2020 music mod may lack native AE support — verify before locking.
-

28.2. Weapon, Magic, And Impact SFX

The sound-effects layer for combat, magic, items, UI, and impact. The bar: modern, weighty, clear audio readable in third-person without visual confirmation.

Mod	Notes
Immersive Sounds - Compendium	Weapon-and-armor SFX baseline. Broadest patch ecosystem.
Audio Overhaul for Skyrim SE	Master reverb-and-attenuation layer. 27K endorsements, actively maintained (v4.1.4, April 2026).
Airgetlam - Shouts SFX	Shout-audio companion. Replaces raw .wav files, no record conflicts.
Strange Runes	Magic audio layer (cross-referenced from → Magic & Perks).

28.2.1. Load Order

[ISC](#) and [AOS](#) both edit large numbers of sound records. Load [AOS](#) after [ISC](#) so AOS's reverb/attenuation wins over ISC's raw effect swaps. [Airgetlam](#) loads after both.

28.2.2. Risks & Compatibility

- ISC is one of the oldest actively used Nexus mods (first uploaded 2013). Touches so many records that any update creates unexpected conflict pressure.
 - Airgetlam replaces raw audio files — should be loaded after any mod touching shout records.
-

28.3. Creature And NPC SFX

28.3.1. Baseline

- Minimalist route — rely on ISC + AOS for creature sounds. Add dedicated replacers only if testing reveals weak vocalisations.
- **Talkative Dragons** ([Nexus](#)) — Dragons speak during combat via reused vanilla voice lines. ESL-flagged, only overwrites one record (`CreatureDialogueDragon`). 15,478 endorsements.

28.3.2. Risks & Compatibility

- Creature sound `.wav` replacers are generally safe (no record conflicts). Sound descriptor editors must load after ISC and AOS.
- Talkative Dragons edits a dialogue record, not raw audio. Verify with dragon dialogue from Epic Dragon Combat REDONE.

—

28.4. Combat And Immersion Audio

Combat-specific and immersion-audio mods layering on top of the weapon/magic SFX baseline.

Mod	Notes
Precision	Already locked in → <code>Third-Person</code> ; audio impact is secondary.
Valhalla Combat	Already locked; parry/stagger/timed-block audio feedback.
Dynamic Impacts	Weapon-type-vs-armor impact sounds and sparks. Requires MIF - Mu Impact Framework (95624). ISC/AOS patches available.
Heart of the Beast - Werewolf Sound and Camera Overhaul	Werewolf audio baseline if <code>Growl</code> is adopted.
Heart of the Beast - Feral	Alternative werewolf sounds. Choose one variant, not both.

28.4.1. Risks & Compatibility

- Only one Heart of the Beast variant active at a time. Both installed = file conflicts.
- If `Growl` adds new werewolf forms not covered by the sound replacer, forms fall back to Growl defaults.
- Dynamic Impacts requires **MIF - Mu Impact Framework (95624)**. Use its ISC/AOS patches if those audio mods are active.

—

28.5. Ambient And Environmental Audio

Environmental soundscapes, creature ambient sounds, and weather audio to reinforce world immersion outside of combat.

Mod	Notes
Wildwood Echoes	Forest and woodland soundscape expansion.
Blackreach Eerie Ambience	Blackreach-specific ambient atmosphere.
Murder of Songbirds	Ambient bird audio in wilderness areas.
Whistling in Skyrim	Selected voice types whistle tunes outdoors. ESL-flagged, lightweight.
Nordic Winds	Weather-specific wind sound effects for colder regions.
Whispering Tomes of Apocrypha	Apocrypha ambient soundscape.
SnORES of Skyrim	Sleeping NPC snoring for most vanilla voice types. Script-free, ESL-flagged. New sound records — safe against ISC/AOS.
Falmer 'Dialogue' Overhaul - Echolocation	Falmer echolocation vocalisation replacer. Creature-audio companion layer.
Potema Revoiced	Revoiced Potema dialogue. Overwrites vanilla voice files rather than dialogue records.
Mephala Revoiced	Revoiced Mephala dialogue (Daedric prince). Same raw-voice-replacer pattern as Potema Revoiced; verify against Daedric-quest expansions.
Whispers of the Daedric Princes	Ambient Daedric prince whispers in dungeons and Oblivion-related locations.

28.5.1. Risks & Compatibility

- All entries above are raw `.wav` / `.xwm` replacers or sound-descriptor mods - no record-level conflicts with ISC or AOS.
- Load sound-descriptor edits after ISC and AOS in left pane.
- Potema Revoiced replaces voice files only; verify compatibility with quest-expansion mods touching the Potema quest.

29. Survival, Difficulty, and Balance

29.1. Difficulty Framework

Cross-load-order pressure: how quickly the world pushes back, how clearly dungeon progression reads, and whether difficulty comes from better structure instead of blunt stat inflation.

29.1.1. Baseline

- **Arena** - **An** **Encounter** **Zone** **Overhaul**
[#link\("https://www.nexusmods.com/skyrimspecialedition/mods/33487"\)](https://www.nexusmods.com/skyrimspecialedition/mods/33487) [\[#1, #4\]](#) ([Nexus]) — Cleanest middle ground: more progression pressure than vanilla, still readable and open-world friendly.

29.1.2. Alternatives

- **Minimal intervention** [\[#3\]](#) — Leave encounter zones close to vanilla.
- **Heavy world-structure** [\[#1\]](#) — Deleveled loot and stronger world gating (owned by → [Crafting & Economy](#)).

29.1.3. Risks & Compatibility

- Encounter-zone structure can overlap with loot/economy decisions if this section pretends it owns world progression.
- If [Arena](#) is combined with much harsher loot distribution, the result can overshoot from moody into over-gated.

29.2. Combat Lethality And Pacing

Tunes encounter harshness and time-to-kill on top of the combat framework from → [Third-Person](#), not replacing it.

29.2.1. Baseline

- **Simply** **Balanced** [#link\("https://www.nexusmods.com/skyrimspecialedition/mods/15541"\)](https://www.nexusmods.com/skyrimspecialedition/mods/15541) [\[#1, #2, #4\]](#) ([Nexus]) — Preferred pacing tool. Final damage/progression tuning without re-owning stamina, injury, or combat behavior.

29.2.2. Alternatives

- **Blade** **and** **Blunt** - **A** **Combat** **Overhaul**
[#link\("https://www.nexusmods.com/skyrimspecialedition/mods/34549"\)](https://www.nexusmods.com/skyrimspecialedition/mods/34549) [\[#1\]](#) ([Nexus]) — Deliberate branch — it reaches far beyond simple pacing.

- **Wildcat - Combat of Skyrim** [#link\("https://www.nexusmods.com/skyrimspecialedition/mods/1368"\)](https://www.nexusmods.com/skyrimspecialedition/mods/1368) [#1, #4] ([Nexus]) — Injury system, combat AI, and lethality tweaks. Legacy baseline.
- **Minimalist route** [#3] — Keep lethality in the section-04 combat stack.

29.2.3. Candidate

- **Relentless** [#link\("https://www.nexusmods.com/skyrimspecialedition/mods/114022"\)](https://www.nexusmods.com/skyrimspecialedition/mods/114022) [#1] ([Nexus]) — Difficulty and pacing mod. Research before adding; evaluate how it interacts with the Section-04 combat stack and Arena encounter zones.

29.3. Enemy AI Improvements

29.3.1. Baseline

- Restrained route — rely on **Arena**, the section-04 combat stack, and targeted creature/faction additions.
- **Less Sniperlike NPCs** [#link\("https://www.nexusmods.com/skyrimspecialedition/mods/34199"\)](https://www.nexusmods.com/skyrimspecialedition/mods/34199) [all] ([Nexus]) — Reduces NPC projectile accuracy. Prefer applying via Wrye Bash (70.0) to avoid an extra plugin.

29.3.2. Alternatives

- **Skyrim** **Revamped** - **Complete** **Enemy** **Overhaul**
[#link\("https://www.nexusmods.com/skyrimspecialedition/mods/14598"\)](https://www.nexusmods.com/skyrimspecialedition/mods/14598) [#1, #4] ([Nexus]) — High-commitment. Add only if ordinary testing shows enemy behavior still too flat.
- **SkyTactics** - **Dynamic**
Combat Styles [#link\("https://www.nexusmods.com/skyrimspecialedition/mods/131148"\)](https://www.nexusmods.com/skyrimspecialedition/mods/131148) [#1, #4] ([Nexus]) — SkyPatcher-based NPC combat style variety.
- **NPCs** **Take** **Cover** - **Smarter** **Anti-Cheese** **AI**
[#link\("https://www.nexusmods.com/skyrimspecialedition/mods/111890"\)](https://www.nexusmods.com/skyrimspecialedition/mods/111890) [all] ([Nexus]) — NPCs take cover during combat instead of standing in the open. Lightweight, no new records.
- **Seeking** **Out** **Sneaks** - **Improved** **AI** **Search** **Behavior**
[#link\("https://www.nexusmods.com/skyrimspecialedition/mods/80416"\)](https://www.nexusmods.com/skyrimspecialedition/mods/80416) [#1, #4] ([Nexus]) — Enemies search further and hunt sneaking players in dungeons. Anti-cheese without touching lethality — dial-compatible. Verify interplay with RAID (→ [Expanded Systems](#)) and the AI Stealth Overhaul Synthesis patch. Unchanged since Dec 2022 — verify 1.6.1170.

29.4. Injury, Stamina, And Resource Pressure

Attrition that lingers beyond a single swing: drained stamina, enforced downtime, the feeling that repeated bad fights accumulate a cost.

29.4.1. Baseline

- **Stress** **and** **Fear** - **A** **Dynamic** **Sanity** **System**
[#link\("https://www.nexusmods.com/skyrimspecialedition/mods/116522"\)](https://www.nexusmods.com/skyrimspecialedition/mods/116522) [#1, #2, #4] ([Nexus]) — Emotional/

pacing consequences encouraging rest, food, inns, and quieter recovery. Design is light enough to create stories without shouting over the rest of the list.

- **Light the Way - A Stress and Fear Add-On** [#link\("https://www.nexusmods.com/skyrimspecialedition/mods/171532"\)](https://www.nexusmods.com/skyrimspecialedition/mods/171532) [#1, #2, #4] [[Nexus]] — Light-flicker effect during high-stress moments.
- **Bathing in Skyrim - Renewed - Stress and Fear patch** [#link\("https://www.nexusmods.com/skyrimspecialedition/mods/141564"\)](https://www.nexusmods.com/skyrimspecialedition/mods/141564) [#1, #2, #4] [[Nexus]] — Cleanliness affects sanity. Required if both mods installed.

29.4.2. Alternatives

- **Blade and Blunt** [#1] — Overlapping alternative for harsher stamina-and-combat pressure.
- **Minimalist route** [#3] — Keep resource pressure inside existing combat/survival layers.

29.5. Enemy Resistances And Knowledge Checks

29.5.1. Baseline

- **Know Your Enemy 2** [#link\("https://www.nexusmods.com/skyrimspecialedition/mods/93258"\)](https://www.nexusmods.com/skyrimspecialedition/mods/93258) [#1, #4] [[Nexus]] — Different creatures call for different tools; inspect system makes knowledge legible. Scope: tactical differentiation, not full enemy-ecosystem ownership.
- **Know Your Enemy 2** — **Armors** [#link\("https://www.nexusmods.com/skyrimspecialedition/mods/94067"\)](https://www.nexusmods.com/skyrimspecialedition/mods/94067) [#1, #4] [[Nexus]] — Armor-based resistance module. Adds resistances to NPC armor materials.
- **Know Your Enemy 2** — **Integration Patch** [#link\("https://www.nexusmods.com/skyrimspecialedition/mods/94078"\)](https://www.nexusmods.com/skyrimspecialedition/mods/94078) [#1, #4] [[Nexus]] — Unifies creature + armor inspect spell. Removes duplicate features between modules.
- **The Dragonborn's Bestiary — KYE2 Patch** [#link\("https://www.nexusmods.com/skyrimspecialedition/mods/124145"\)](https://www.nexusmods.com/skyrimspecialedition/mods/124145) [#1, #4] [[Nexus]] — Bridges Bestiary and KYE2 systems.

Install order: KYE2 → Armors → Integration Patch → Bestiary Patch.

29.5.2. Risks & Compatibility

- Resistance systems can feel punishing if the player doesn't have enough ways to adapt mid-run.
- For Steam 1.6.1170, the listed [Backported Extended ESL Support](#) requirement does not apply.
- If the inspect flow or resistance spread feels too noisy, the concept may be better than the lived experience.

29.6. Survival Systems Compatibility

How difficulty and pressure choices interact with the chosen survival framework.

29.6.1. Baseline

- Additive route — use pressure layers that reward existing survival behaviors (eating, resting, warming up, returning to town).
- `Stress and Fear` naturally plugs into meals, inns, sleep, and calmer downtime.
- If `Starfrost` remains the baseline, this section should amplify recovery need without forcing menu babysitting.
- **Cold Breath - NG** `#link("https://www.nexusmods.com/skyrimspcialedition/mods/174838")[all]` `[[Nexus]]` (v1.8, May 2026) — Higher-fidelity fogged breath in cold environments. Mesh/particle replacer, no ESP.

29.6.2. Alternatives

- **Navigation Restrictions - NG** `#link("https://www.nexusmods.com/skyrimspcialedition/mods/129086")[#1]` `[[Nexus]]` — Fast travel and survival restrictions. Adds survival-oriented travel pressure.

29.6.3. Environmental Hazards

Environmental damage layers that make fire, traps, and dungeon hazards genuinely threatening.

Mod	Notes
Fires Hurt - OIF	Fire sources deal damage over time when stood in. Requires OIF framework. Adds meaningful hazard pressure to dungeons and campfires.

30. Legacy of the Dragonborn

30.1. Core Mod Baseline

Treated as a foundational content pillar for the final list, not a late optional extra.

Mod	Type	Notes
Legacy of the Dragonborn v6.10.1 (May 2026)	Baseline	Non-negotiable centerpiece.
The Curator's Companion v7.0.8	Baseline	In-game tracking for museum displays and collection progress.
Legacy of the Dragonborn Patches (Official) v6.10.7 (June 2026)	Baseline	Required integration infrastructure.
Paintings Replacer for Legacy of the Dragonborn SSE (Nexus 13279, v1.1, Nov 2017)	Baseline	Texture replacer, low risk.
Fate Cards Remade - Legacy of The Dragonborn (Nexus 76418)	Baseline	130 unique card artworks, remade meshes and display assets.
Placeable Display Cases (Nexus 131107, v3.1, Jan 2025)	Baseline	Placeable museum display cases.
Achievement Injector + LoTD Achievement Injector	Baseline	Museum achievement content.

30.2. Supported Content

30.2.1. Quest Mods With Strong LoTD Integration

- **VIGILANT** — Multiple display cases in Hall of Oddities.
- **Unslaad** — Official patch hub support.
- **Glenmoril** — Patch pipeline still maturing; check before locking.
- **Project AHO** — Full display room in Hall of Secrets.
- **Carved Brink** — Displays for Faceted Stones artifacts.
- **The Forgotten City** — Unique items and choice reward display.
- **Beyond Skyrim - Bruma** — Count-flation and display-base integration.
- **Beyond Reach** — Third-party patch.
- **The Wheels of Lull** — Hall of Secrets display slots.
- **Moon and Star** — Unique armor/weapon display.
- **Wurmstooth** — Displays for unique items.
- **Falskaar** — Base museum count supports.
- **Sirenroot - A Verdant Tale** — Check patch status.

- **Midwood Isle** — Official patch support.
- **The Gray Cowl of Nocturnal** — Legendary mask display.
- **The Breton Paladin** — Official patch support.
- **Clockwork** — Unique item display slots.
- **Konahrik's Accoutrements** — Deep integration with dedicated display room.
- **Artifacts - The Ice Blade of the Monarch** — Standalone artifact with display.
- **Artifacts - The Tournament of the Ten Bloods** — New artifact quest with displays.
- **Obscure's College of Winterhold** — LoTD patch in OCW's own FOMOD. Covers private collection and unique College items.

30.2.2. Display Expansions And Room Mods

- **LoTD Display Patch Repos** — Third-party display patches not covered by official hub.
- **Museum Display Room Expansion** — Evaluate if base museum feels cramped.
- **Hall of Forgotten Guild** — Curator's Companion feature for additional collectible space.

30.2.3. Collectible Additions

- **Stones of Barenziah Quest Markers** — Dedicated display for completed crown.
- **Jewels of the Reach** — Additional gemstone collectible.
- **Dawnguard Treasure Map** — Feeds collection-driven exploration loop.
- **Legacy of the Dragonborn BadGremlins Collection** ([Nexus](#)) — Gremlin-themed collectible display items for the museum.
- **Fate Cards New Style - 2023 Remaster** ([Nexus](#)) — Alternate fate card artwork and design. Alternative to Fate Cards Remade (76418).

30.2.4. Artifact Enhancement

- **Zim's Immersive Artifacts** ([Nexus](#)) — 50+ vanilla artifact rebalances. If no official patch, load LoTD after ZIA so museum display conditions use LoTD records.

—

30.3. Patch Strategy

- Use **LoTD Patches (Official)** as default patch source.
- Add third-party patches only when official collection doesn't cover a finalized mod.
- Do not pre-install broad patch bundles for hypothetical mods.
- Re-check museum sorting behavior after each major content addition.

30.3.1. Mod Selection Questions

1. Does this mod have official or high-quality third-party LoTD patch support?
2. Does it add unique items that feel like they *should* be displayed?
3. Does it conflict with LoTD workspace edits?
4. Does the play order fit the museum's unlock progression?

—

30.4. Submods And Quality-Of-Life

30.4.1. Sorting And Inventory

- **Legacy Safehouse Plus** — Expanded safehouse with additional crafting stations. Check v6.10+ compatibility.
- **LOTD - Explorer's Room** — Explorer-themed storage/display.
- **Quick Light** — Not LoTD-specific but essential for museum navigation.
- **Map Markers for LotD** — Fast-travel marker after quest conditions met.

30.4.2. Museum-Specific UI

- **LOTD Museum Exterior Fixes** — Navmesh and door-activation fixes.
- **LOTD Hall of Secrets Door Fix** — Door activator fix.
- **Object Manipulation Overhaul** — Display arrangement without console commands.

30.4.3. Additional QoL

- **Legacy of The Dragonborn - SMIM Chests** ([Nexus](#)) — SMIM-quality chest meshes for the museum displays.
- **LOTD - Museum Exterior Music** — Low priority; evaluate after → `Audio` is locked.
- **Dynamic Activation Key** — Reduces E-key scroll-conflict with crowded displays.

30.4.4. Relic System

- **Relic Hunter** — Evaluate compatibility with LoTD v6's own relic system.
- **LOTD - Explorer's League** — Guild-like progression. Check overlap with Curator's Companion.
- **Shovels - Dig Up Treasure** — Check for conflict with LoTD dig-site expeditions.

30.4.5. Risks & Compatibility

- Legacy of the Dragonborn can drag the whole load order toward completionist design.
- Museum-first progression changes perceived value of loot and unique-item mods.
- Quest mods with strong LoTD integration may co-determine the quest mod stack.
- Third-party display patches for pre-2023 quest mods may not be maintained against LoTD v6.

31. Adult Content

31.1. Framework Prerequisites

OStim Standalone — a modern, standalone adult animation framework that does not require OSA or SexLab. The OStim Romance and Amorous Adventures patch layer depend on this.

Mod	Type	Notes
OStim Standalone	Baseline	Modern adult animation framework. Does not require OSA.
OStim Community Resource	Baseline	Shared framework for OStim add-ons.
Night-blooming Violets for OStim Standalone	Baseline	Scene framework addon.
OARE - Open Animations Romance and Erotica	Baseline	Recommended animation pack baseline.
Additional Leito's Animations for Ostim Standalone	Baseline	150 animations.
Anub's animations for Ostim Standalone	Baseline	685 animations.
PPA - Procedural Penis Animations	Baseline	CBPC collision physics addon.
OSmp - Automatic SMP physics toggle for OStim	Baseline	ESL-flagged, supports CBBE 3BA and HIMBO.
Female Moan Voicesets For Ostim Standalone	Baseline	38 voicesets keyed to voice types.
Ostim SA-Sound Replacer-Blowjob	Baseline	Sound replacer.
Male Orgasm Sounds For Ostim Standalone	Baseline	Sound replacer for male animations.
OCum Ascended	Baseline	Cum and orgasm mechanics with visual layers.
Clean Navigation Icons for OStim Standalone	Baseline	62 navigation icons.
Animation Pack Icons for OStim Standalone	Baseline	18 icons.
OTooTiredToMove	Baseline	Temporary immobilisation after climax.
OStim Prism	Baseline	OStim framework visual/settings enhancement.
OSquirt - OCum Expansion	Baseline	Expands OCum Ascended mechanics.
Ostim Standalone Sound Overhaul	Baseline	Audio overhaul for OStim animations.

31.1.1. Risks & Compatibility

- OStim Standalone must not be loaded alongside legacy OSA or SexLab.
 - OStim Romance and AA OStim patch must match the same OStim Standalone version.
 - Keep auto-start settings conservative to avoid script load.
-

31.2. Romance Content

Opt-in quest-driven romance content running on top of the OStim framework.

31.2.1. Baseline

Mod	Notes
OStim Romance	Primary romance-content baseline. Active maintenance.
OComfort - OStim Romance Fork	Romance fork adding comfort/affection mechanics. Alternative to OStim Romance.
Amorous Adventures OStim Standalone - OStim Sequences	Ports classic Amorous Adventures to OStim with sequence scenes.
Amorous Adventures OStim Standalone - Rewrite and MCM Patch with Fixes - Sequences Edition	Updated rewrite with MCM patch and fixes. Supersedes the base patch.
Amorous Adventures MCM	MCM configuration for Amorous Adventures.
Amorous Adventures - Player Dialogue Rewritten	Reworks player dialogue options for Amorous Adventures.
Caught Red Handed - Quest Expansion - OStim Patch	OStim integration for the Caught Red Handed quest expansion.

31.2.2. Risks & Compatibility

- Lock the framework version first, then verify romance mod compatibility.
 - Romance quests add dialogue/quest records that may conflict with → [World Content](#) or → [NPCs](#) overhauls.
 - Verify load order with [RDO](#) and [Serana Dialogue Add-On](#) for overlapping NPCs.
 - [Amorous Adventures OStim Standalone - Rewrite](#) (150577) is the preferred version over the base OStim Sequences patch (106773).
-

31.3. NPC Relationship Content

NPC relationship simulation — schedules, pairings, and partner-swapping running on OStim.

31.3.1. Baseline

Mod	Notes
-----	-------

[OStim NPCs - NPC Sex Lives Improved](#)

NPC relationship schedules and partner variety.
Requires OStim Standalone.

31.3.2. Risks & Compatibility

- Verify NPC schedules don't conflict with AI Overhaul SSE or other NPC behavior mods from → [NPCs](#).
- Heavy script load — test for Papyrus performance on longer playthroughs.

—

31.4. Pregnancy & Family

Pregnancy, childbirth, family systems, and child NPC overhauls. These mods form an interconnected ecosystem — evaluate as a set.

31.4.1. Baseline

Mod	Notes
SkyChild	Child NPC appearance overhaul. Replaces vanilla child models.
Wild Side - A SkyChild Add-On	Adds wild/feral child variants. Companion add-on for SkyChild.

31.4.2. Alternatives & Tentative

Mod	Notes
Beeing Female NG	Tentative — evaluate gameplay and compatibility impact. Pregnancy and female reproductive system framework.
Beeing Female - FM - FMR - Skychild Patch	Tentative — consider alongside Beeing Female NG. Compatibility patch for Beeing Female + SkyChild.
Seeds of Skyrim - Lore Pregnant NPCs	Lore-friendly pregnant NPCs. Evaluate for overlap with Beeing Female pregnancy system.

31.4.3. Risks & Compatibility

- Pregnancy frameworks add persistent actor data and scripts — monitor save file growth.
- SkyChild replaces child NPC models; verify compatibility with any mod that adds or modifies child NPCs.
- These mods interact with each other; install patches where available.
- Evaluate all pregnancy/child mods as a set before locking any individual pick.

32. Main Character Voicing

32.1. Framework Overview

Dragonborn Voice Over 2 (DBVO2) is the runtime framework that intercepts player dialogue and plays `.wav` / `.fuz` files via an SKSE plugin. Combined with the xEdit dialogue export tool and AI voice synthesis (ElevenLabs or xVASynth), the player character speaks every dialogue line.

32.1.1. Baseline

Mod / Resource	Type	Notes
Dragonborn Voice Over 2	Framework	SKSE plugin. Intercepts player dialogue and plays <code>.wav</code> / <code>.fuz</code> files. Settings via SKSE Menu Framework (F1). Per-character pack switching, per-save presets.
Bella Voice DBVO	Voice Pack	Most popular female pack, broadest mod coverage.
DBVO - Allison Voice Pack	Voice Pack	British female alternative.
Pre-made male packs	Voice Pack	Less common pre-made options available.
Laura Bailey	Recommended Actress	ElevenLabs Voice Design target for custom female pack via → [Custom Voice Pack Pipeline] (custom-voice-pack-pipeline).
Graham McTavish (The Saint / Geralt)	Recommended Actor	ElevenLabs custom target for male pack via → [Custom Voice Pack Pipeline] (custom-voice-pack-pipeline).

32.1.2. Alternatives

Mod / Resource	Type	Notes
xVASynth	Alternative	Free audio generation. Quality is noticeably synthetic. Used for male default (Geralt voice model).
ElevenLabs	Alternative	Premium audio generation. Higher quality, more natural output. Used for Laura Bailey custom target.

DBVO Dialogue Export for xEdit	Alternative	Replaces MCVO Generator for dialogue export. xEdit script with optional ESPFE fix patch.
MCVO Generator	Alternative	Legacy dialogue export. Still works with DBVO2 legacy mode. Last updated March 2023.

32.1.3. Setup Requirements

- DBVO2 requires SKSE64, Address Library for SKSE Plugins, and SKSE Menu Framework (all in → `Foundations` and `UI` stacks).
- No `dialoguemenu.swf` patching needed — works with most UI mods without modification.
- Windows 255-char path limit can crash on startup — keep install path short.
- Pre-made packs may not cover every mod. Uncovered lines fall back to subtitles.
- xVASynth batch processing can take 2-3 hours for 50,000+ lines.

—

32.2. Two Custom Voice Packs

Two custom-generated packs (Laura Bailey female, Graham McTavish male) for character flexibility. Install as separate MO2 mods, swap via MO2 profile or DBVO2 voice pack dropdown in SKSE Menu Framework (F1).

Run → [Custom Voice Pack Pipeline](custom-voice-pack-pipeline) twice with two ElevenLabs voice IDs (Laura Bailey female, Graham McTavish male). Package as separate MO2 mods and enable one at a time.

—

32.3. Custom Voice Pack Pipeline

Unified workflow covering dialogue export → audio generation → packaging into DBVO2 format. Replaces the legacy MCVO Generator + standalone ElevenLabs sections.

32.3.1. Prerequisites

- Install DBVO2 and dependencies (SKSE64, Address Library, SKSE Menu Framework). DBVO2 must be enabled in MO2 before exporting.
- Build the full modlist in MO2 with final load order.
- Download [DBVO Dialogue Export for xEdit](#) — modern xEdit script that exports player dialogue lines. Replaces MCVO Generator.
- [ElevenLabs](#) paid account — required for both custom targets (Laura Bailey and Graham McTavish).
- Download DBVO Voice Packer 2 V2.5.1 (Google Drive link) — packages generated audio into DBVO2 format with `.fuz` files and `voice-pack.json` manifest.

32.3.2. Step 1: Export Dialogue

1. Open SSEEdit/xEdit with all load-order mods active.
2. Right-click any plugin → **Apply Script** → select `DBVO\_Dialogue\_Export.pas`.
3. The script scans all dialogue in the load order and outputs one `TopicList#emph[.txt]` file per dialogue-containing mod, plus a `BrokenTopicList.txt` listing lines that need manual editing.

4. Optionally create the ESPFE fix patch (trims whitespace, deduplicates lines).
5. Save the output to a working directory outside MO2's `mods` folder.

32.3.3. Step 2: Generate Audio

ElevenLabs (recommended for Laura Bailey or Graham McTavish custom pack):

1. Create an [ElevenLabs](#) account. A paid subscription is required for full load orders (Creator \$22/mo. or Pro \$99/mo.).
2. Get your API key: Profile → **Profile Settings** → **API Key** → copy the key.
3. Choose a voice:
 - **Voice Library** — browse pre-made voices. For a Laura Bailey target, look for voices tagged with similar qualities (warm alto, American female). For Graham McTavish, look for deep, gravelly male voices (Scottish-accented where possible).
 - **Voice Design** — create a custom voice from scratch with fine-grained control over accent, age, gender, and style.
 - **Voice Cloning** — if you have a clean audio sample of the target actor, use Instant Voice Cloning.
4. Get the voice ID: every voice has a unique ID in its URL (`/voice/<voice-id>`) or via `GET https://api.elevenlabs.io/v1/voices`.
5. Download the DBVO Voice Pack Creator tool from the DBVO author's [Discord](#), `tool-download`.
6. Point the Voice Pack Creator at your exported `TopicList*.txt` files, enter your API key and target voice ID, then run.
7. The tool batch-generates `.wav` files covering every dialogue line in your load order.

32.3.4. Step 3: Package into DBVO2 Format

1. Run DBVO Voice Packer 2 V2.5.1.
2. Point it at your generated `.wav` files and the exported `TopicList*.txt` files.
3. The tool creates the correct DBVO2 folder structure and files:
 - `Sound\Voice\Dragonborn Voice Over\<pack-name>\*.fuz`
 - `voice-pack.json` manifest
4. Create an MO2 mod from the output folder. Name it descriptively (e.g., `DBVO - Laura Bailey Custom`).

32.3.5. Step 4: Install & Activate

1. Enable the mod in MO2. Place after DBVO2 in the left pane.
2. Launch the game through SKSE.
3. Open SKSE Menu Framework (Press **F1**) → Dragonborn Voice Over → General.
4. Ensure **Use legacy voice over** is **Off** (V2 packs only).
5. Select your voice pack from the dropdown.
6. Test dialogue lines with any NPC — uncovered lines fall back to subtitles.

32.3.6. Character Quota Planning

ElevenLabs Creator plan (100,000 chars/month) covers 15,000–20,000 lines. Full 50,000+ line load order may need Pro plan (\$99/month, 500,000 chars) or multiple months. Use free tier (10,000 chars/month) for testing only.

32.3.7. Generating The Second Set (Male/Female)

Repeat Steps 2–4 with the other voice target and ElevenLabs voice ID (Laura Bailey → female, Graham McTavish → male). Create a separate MO2 mod with a distinct pack name. Enable one at a time via the DBVO2 dropdown in SKSE Menu Framework.

33. Performance and Technical Workflow

33.1. Overview

This section owns the performance budgeting, optimization discipline, generated-patch pipeline, and long-term maintenance workflow that keeps `Elder Wilds` stable as the load order grows. The section is split into five sub-files that each own a distinct concern:

- **Performance Strategy:** budgeted performance analysis, VRAM-heavy mod review, CPU-heavy script review
- **Optimization & Configuration:** shader impact, grass/tree/draw-call optimization, INI tuning (BethINI Pie baseline), save safety
- **Tool Pipeline:** xEdit/Pandora/DynDOLOD/Synthesis workflow discipline, 4K LOD baseline (xLODGen/TexGen/DynDOLOD STEP config), optional diagnostics (Recursion Monitor, S.L.A.C.K.)
- **Bashed Patch & Synthesis Configuration:** exact Wrye Bash tweak settings, Bash Tag per plugin, full Synthesis patcher pipeline (11 stages), build order and rebuild triggers
- **Testing & Maintenance:** benchmark methodology and scenarios, tuning knobs, patcher revisit triggers, generator output order, change tracking, MCM recording, mod update workflow

33.2. Sub-File Index

File	Section
Section 34	Performance Strategy
Section 35	Optimization & Configuration
Section 36	Tool Pipeline
Section 37	Bashed Patch & Synthesis Configuration
Section 38	Testing & Maintenance

33.3. Key Principles

- Treat performance as four separate problems: shader cost, VRAM pressure, CPU/script load, and exterior draw-call pressure. A fix that targets the wrong category makes things worse.
- Use the disciplined route for all generated work: keep every output in its own dedicated MO2 mod, rebuild only when the owning category changes, and run tools in a repeatable order.
- The Bashed Patch and Synthesis pipeline is the single source of truth for conflict resolution. A tag or patcher exists only when a specific mod in the list requires it.
- The list's long-term health depends on disciplined rebuild habits, saved MCM profiles, and a documented changelog — not on any single mod pick.

33.4. Related Sections

- `modlist-foundations.md` — engine baseline, SSE Display Tweaks config, core stability fixes

- `modlist-graphics-shaders.md` — Community Shaders visual framework (performance envelope owned by `modlist-performance-optimization.md`)
- `modlist-curation.md` — curation rules and testing methodology

34. Performance Strategy

34.1. Performance Budgeting

How `Elder Wilds` measures performance, identifies bottlenecks, and keeps generated workflow sane once the list gets heavy.

34.1.1. Baseline

- **Budgeted route** — Separate GPU, VRAM, CPU/script, and draw-call problems so fixes target the real cause.
- Treat performance as four different problems that can look similar: shader cost, VRAM pressure, CPU/script load, and exterior draw-call pressure.
- Keep one repeatable forest scene, one city scene, and one dungeon/interior scene for comparisons.
- Record baseline and post-change captures with `PresentMon` and review in `CapFrameX`.
- Prefer a stable frame-time target over chasing the highest uncapped number.

34.1.2. Alternatives

- **Intuition-first route** — Change settings/mods by feel.
- **Average-FPS route** — Judge by headline FPS even if frame pacing and bottleneck type stay unclear.

34.1.3. Risks & Compatibility

- A single bad scene can be CPU-bound, GPU-bound, or content-bound in different directions.
 - Average FPS can improve while hitching and traversal smoothness get worse.
-

34.2. VRAM-Heavy Mod Review

Texture resolution, PBR/parallax adoption, heavy shader features, and world coverage drive VRAM pressure. Spend texture budget where visible in normal play.

34.2.1. Baseline

- **Selective route** — High resolution only for hero assets, creatures, architecture, or landscape layers dominating the screen.
- Start below maximum resolution when a visual pack offers multiple sizes; scale up only if testing proves the asset earns it.
- Treat PBR/parallax conversions as a VRAM and maintenance choice, not just a screenshot upgrade.
- If a scene stutters, verify actual memory pressure before blaming scripts or AI.

34.2.2. Alternatives

- **Max-detail route** — Broad 4K coverage, frequent material upgrades.
- **Restraint-first route** — Mostly 2K and below with few exceptions.

34.2.3. Risks & Compatibility

- High VRAM usage can look like generic stutter, but lowering unrelated gameplay mods won't fix it.
- Some landscape/architecture/material upgrades cost more in aggregate than character textures ever will.

—

34.3. CPU-Heavy Script Review

CPU and Papyrus pressure comes from layered systems, NPC density, polling scripts, and constant background updates.

34.3.1. Baseline

- **Moderated route** — Keep strong feature mods, but avoid stacking several background-heavy systems solving similar jobs.
- Treat NPC expansion, survival layers, follower frameworks, reputation systems, defeat systems, and ambient event mods as cumulative load.
- Prefer one good system per gameplay problem instead of three lighter systems all ticking in the background.
- Keep diagnostic logging off during normal play unless a bug needs it.
- Judge script health through sleeping, waiting, cell changes, combat cleanup, and long travel.

34.3.2. Alternatives

- **Permissive route** — Allow multiple overlapping always-on systems.
- **Strict route** — Aggressively cut scripted features even when they materially help list identity.

34.3.3. Risks & Compatibility

- Script delay often misdiagnosed as low FPS or "Skyrim being Skyrim."
- Mid-save uninstalls can create save instability looking like performance decay.
- A list can feel fine in short tests and still degrade over longer sessions.

35. Optimization & Configuration

35.1. Shader Performance Impact

35.1.1. Baseline

- **Conservative to balanced route** — Accept a few heavier depth/lighting features once the rest of the visual stack is stable.
- Remove or downgrade luxury modules before tearing apart the rest of the graphics stack.
- `Skyrim Upscaler` is a fallback tool from → `Graphics`, not the first answer to poor optimization.
- Use `SSE Display Tweaks` from → `Foundations` to enforce stable frame cap.
- Judge shader cost in motion, weather transitions, and dense foliage scenes.

35.1.2. Alternatives

- **Conservative route** — Keep CS close to the section-02 baseline.
 - **Luxury route** — Keep piling on expensive modules, then try to recover cost elsewhere.
-

35.2. Grass, Tree, And Draw-Call Optimization

35.2.1. Baseline

- **Content restraint first, then low-maintenance optimization.**
- **Lightened Skyrim** — Leading optimization branch. Almost invisible cleanup, low drama.
- **eFPS - Anniversary Edition** ([Nexus](#)) — Optional only if exterior draw-call pressure still needs help after main world stack is stable. Plan for patch coverage.
- Re-evaluate exterior optimization after major city, roadside, tree, or world-content choices.

35.2.2. Risks & Compatibility

- `eFPS` has strong upside but real compatibility/patch overhead for lists with heavy exterior edits.
 - Heavy cities, outskirts, roads, trees, and landmark mods can erase optimization gains faster than tweak mods can recover them.
-

35.3. INI Tuning And Display Settings

35.3.1. Baseline

- **Tool-first route** — Use `BethINI Pie` as the baseline editor.
- Keep MO2 INI path behavior simple and documented.
- After baseline is set, keep only a short list of manual overrides justified by display/shadow/frame-cap targets.
- Let `SSE Display Tweaks` handle display behavior and frame pacing.

35.3.2. BethINI Pie Configuration Baseline

Run through MO2 targeting the active profile INIs.

1. **Basic:** Reset to Defaults → High preset → Apply Recommended Tweaks. Window Mode: Borderless. FOV: First Person 75, Third Person 85.
2. **General → Saved Games:** Untick Autosaves, Save on Travel, Save on Rest, Save on Wait (survival mods handle save restrictions; manual saves are safer).
3. **Gameplay:** Difficulty Adept. Untick Disable Kill Cam. Tick Remove Borders. Untick Tutorials.
4. **Interface:** Tick Dialogue Subtitles and General Subtitles. Leave Compass and Crosshair ticked. Untick Floating Quest Markers. Leave Controller Vibration ticked.
5. **Environment:** Grass Density 60. Untick Reflect Sky.
6. **View Distance:** Leave Tree LOD Distance at High default for now. Set to `0` via INI Editor before DynDOLOD generation later.
7. **Visuals:** Max Particle Render Count 7500. Untick Lens Flare (CS handles post-processing).
8. Click Save and Exit, then confirm values in *MO2 → Tools → INI Editor*.

—

35.4. Save Safety Considerations

35.4.1. Baseline

- **Disposable-test route** — Keep clean labeled saves for pre-LOD, post-LOD, city-performance, forest-travel, and long-session stability.
- Start new game when adding major worldspace optimization layers (eFPS mod page recommends fresh-start caution).
- Don't treat old generated outputs, removed scripted mods, or stale occlusion data as safe just because the game still loads.
- Preserve one or two mature long-session saves only for validation.

35.4.2. Risks & Compatibility

- Save corruption can present as performance degradation long before it looks like hard failure.
- Reusing one heavily mutated test save makes optimization conclusions far less trustworthy.

36. Tool Pipeline

36.1. Tool Workflow

A heavy Skyrim list usually becomes unstable through stale output and sloppy rebuild habits before failing through any one mod choice.

36.1.1. Baseline — Disciplined Route

- Run `xEdit` conflict review after each major category change.
- Re-run `Pandora` when behavior, animation, or skeleton-relevant content changes.
- Re-run `Synthesis` whenever a chosen patcher-based system changes (including music-merge from → `Audio`).
- Rebuild `TexGen`, `xLODGen`, `DynDOLOD`, grass cache, and occlusion output when their inputs materially change.
- Keep every generated output in its own dedicated MO2 mod.

36.1.2. Alternatives

- **Ad hoc route** — Rebuild whenever something looks broken.
 - **Late-everything route** — Postpone most generated work until the end.
-

36.2. 4K LOD Tool Baseline

36.2.1. General Rules

- Keep tools (`xLODGen`, `TexGen`, `DynDOLOD`) outside game folder and outside MO2 mod folders.
- Keep output folders outside Program Files, Steam, Documents, Desktop, and MO2 instance.
- Install finished output back into MO2 as separate mods: `Terrain LOD Output`, `TexGen Output`, `DynDOLOD Output`.
- Generate in this order: `xLODGen` → `TexGen` → `DynDOLOD`.
- Use `x64` versions of tools.

36.2.2. xLODGen 4K STEP Baseline

- MO2 executable: `xLODGenx64.exe -lodgen -SSE -o:"\<output path>"`.
- Use `xLODGen` only for terrain LOD.
- Use **SSE-Terrain-Tamriel-Extend.esm** as terrain-boundary resource (from STEP xLODGen forum; Mega mirrors available). **Activate it only during xLODGen generation** — place it below the last worldspace ESM in the right pane, then deactivate it afterward and reactivate `Terrain LOD Output`.

- In MO2, ensure any landscape/tree mods that ship LOD-generation resources (e.g., Majestic Mountains, Cathedral Landscapes) are active during xLODGen.
- In the xLODGen UI: right-click the Worldspace area → **Select All**. Tick only **Terrain LOD**. Apply LOD settings per STEP recommendations.
- 4K users: double diffuse and normal sizes for LOD4/LOD8/LOD16/LOD32.
- Initial LOD4: keep **Optimize Unseen** off. For map/LOD32: use higher **Quality** (0-10) and **Optimize Unseen** 550.
- After generation, run **ACMOS Road Generator**:
 1. Navigate to the ACMOS Road Generator tool folder and open the executable.
 2. Set the roads dropdown to **Path Only**.
 3. Browse to your xLODGen output folder and select it.
 4. Click **Generate**. If prompted to overwrite, click **Yes**.
 5. When complete, zip the output folder (the tool offers this automatically) and install it as a new mod via MO2's **Add mod from file**.

36.2.3. TexGen 4K STEP Baseline

- MO2 executable: `TexGen64.exe -SSE`.
- Run after `xLODGen`, before `DynDOLOD`.
- Preset matching rendered resolution: **2160p = 4K**.
- Grass options: tick **Grass** if no complex grass; tick **HD grass** if using complex grass / CS grass features.
- `TexGen\SSE.ini`: `GrassModelHeightMultiplier=1.15`, `TreeMSAlphaThreshold=144`, `ObjectMSAlphaThreshold=96`.
- `ForceComplexGrass`: `0` (default) or `1` (STEP complex-grass branch).
- Move output to `TexGen Output` mod, enable before running `DynDOLOD`.

36.2.4. DynDOLOD 4K STEP Baseline

- MO2 executable: `DynDOLODx64.exe -SSE`.
- `DynDOLOD\SSE.ini`: `Expert=1`, `Level32=1` `ALLHDL0D32=1`.
- Grass brightness values: `0.500` for all R/G/B Top and Bottom channels.
- `DoubleSidedTextureMask=mountain,mtn`, `DoubleSidedMeshMask=mountain,mtn`.
- Complex grass branch: `ComplexGrassBillboard=5`, matching brightness values, `ComplexGrassBacklightMask=25`.
- GUI: right-click worldspace list → **Select All**. Tick **Candles** and **FXGlow**. Load the **High** rules preset.
- **Occlusion** only on first run. **Grass LOD** only if generating grass LOD.
- **Ultra Trees**: Enable 3D tree LOD for substantially better distant tree quality. Keep tile size at or below `1024` (diminishing returns beyond this).
- Target: **4K** preset with **Optimal** tree/catch-all rules + recommended **LOD32** rules.
- Move output to `DynDOLOD Output` mod. In the right pane: place `DynDOLOD.esm` as the last ESM after all worldspace ESMs, `DynDOLOD.esp` second-to-last, and `Occlusion.esp` last (unless using Synthesis, which goes last).
- Apply ACMOS-required rule edits in DynDOLOD's advanced rule editor: **Tree** → LOD32 set to **Billboard(6)**; \ → LOD32 set to **Level0**.

36.2.5. 4K Validation

- Benchmark before and after LOD from clean Whiterun save. Also check the Rift (`cow tamriel 40 -24`) for aspens.

- If performance is already below STEP's pre-LOD target, lower the broader graphics baseline before blaming LOD.
 - If travel performance still too expensive after LOD, step DynDOLOD from High → Medium/Low, skip Grass LOD.
 - Document emergency tweaks next to the MO2 executable or output mod.
-

36.3. Optional Diagnostics And Performance Tools

Neither requires ongoing configuration or conflict maintenance once installed.

36.3.1. Recursion Monitor

- **Nexus:** [Recursion Monitor](#) — Detects broken Papyrus scripts stuck in recursive loops. Requires SKSE, Address Library.
- **Updated fork:** [recursion-fix-updated](#) (May 2026) — Removes blocking debug popup; writes to SKSE log instead. Prefer for production list.

36.3.2. S.L.A.C.K. (Save And Load Accelerator For SKSE Cosaves)

- **Nexus:** [S.L.A.C.K.](#) — Cosaves save up to 150× faster, load up to 15× faster. Error-friendly mode catches exceptions from other SKSE plugins' cosave handlers.
- Requires: SKSE 1.6.1170, Address Library, SSE Engine Fixes (SKSE64 Preloader).
- Safe to install or uninstall at any time.
- Complements [Seamless Saving](#) (targets main save serialisation, not cosaves). Safe to run both.

36.3.3. Log Watcher

- **Nexus:** [Log Watcher - Real-time Analysis of SKSE Logs](#) — Monitors SKSE logs in real-time for errors, warnings, and crash indicators. Diagnostic aid for testing and load-order debugging.

36.3.4. Risks & Compatibility

- Neither mod touches worldspace records, leveled lists, or asset files.
- Recursion Monitor is read-only in normal operation. Updated fork removes the only known compatibility concern.
- S.L.A.C.K. hooks SKSE's internal save/load API; error-friendly mode catches exceptions instead of crashing.
- Both are safe to remove from an existing save.

37. Bashed Patch & Synthesis Configuration

37.1. Patcher Discipline

The generic category maps only become useful when tied to the actual mods `Elder Wilds` ships with. This section is that binding: exact `Wrye Bash` tweaks, `Bash Tags` per plugin, and `Synthesis` patcher pipeline in exact order. Default is “do not touch” — add a `Synthesis` patcher or `Bash Tag` only when a specific mod in the list makes it necessary.

37.1.1. Pre-Build Checklist

Before every Bashed Patch or Synthesis rebuild, verify:

- ☐ **MO2 backup profile created** — Right-click profile → Copy. Name with date and build stage (e.g., `Elder Wilds - 2026-07-18 pre-patch`).
- ☐ **All plugins active** in the right pane — disabled plugins cause missing-master errors.
- ☐ **xEdit conflict review** completed on the last category added — quick auto-conflict scan (`Apply Filter to show Conflicts`).
- ☐ **No form ID compaction** happened since the last build — compacting form IDs breaks Synthesis patchers that reference original FormIDs. If compaction was done, full rebuild is mandatory.
- ☐ **LOOT sorted** — sort before building so load order is stable. Tags auto-detected from LOOT masterlist. Only manually set tags not already present in LOOT.
- ☐ **Synthesis patcher list verified** — disable any patcher whose prerequisite mod is not installed (e.g., `ELE Patcher` with CS-only stack).

37.2. Wrye Bash (Bashed Patch) Configuration

37.2.1. Bashed Patch Dialogue Settings

When building the Bashed Patch, configure these checkboxes at the top of the Build Patch dialogue:

Option	Setting	Notes
Merge Patches	OFF	Synthesis handles merge-level conflict resolution. Do not let Wrye Bash merge plugins — it produces merge plugins that are fragile to load-order changes and hard to debug.
Import [Categories]	ON	Required for Bash Tags to function. Each category below corresponds to a Bash Tag type — only categories with tagged plugins are active.
Leveled Lists	ON	Core function — merges leveled-list changes from Delev/Relev/Invent.Add tagged plugins. Without this, only the last plugin's leveled-list edits win.

Tweak Settings	ON	Applies the <code>.TweakSettings</code> configured below.
Alias Mod Names	ON	Human-readable names in the patch header. Aids debugging when examining the Bashed Patch in xEdit.

37.2.2. Tweak Settings

Enable in *Wrye Bash* → *Bashed Patch* → *Tweakers* only the items below.

Tweak Actors:

- **Opposite Gender Anims: Female and Male** — needed for CBBE 3BA, HIMBO, KS Hairdos, EVG Conditional Idles, Goetia, Leviathan II packs.

Tweak Assorted:

- **All Armor Playable** — required by Immersive Armors (60+ sets with non-playable flag).
- **No Light Fade Value Fix** — CS Light, True Light, ELFX, Relighting Skyrim, Luminosity all interact with fade values.
- **No Light Flicker** — RAID Weathers, Cathedral, Obsidian, Azurite III CS weather variants.
- **Remove Load Screen Models** — SkyParkour v3, RaySense, Ricochet add 3D geometry that can leak into load screens.
- **Save Sorting Fix** — real issue with this much plugin weight.
- **Set Light Radii** — use 100% multiplier to leave lighting radii alone.
- **Set Sound Attenuation Levels: Nirnroots Only** — leave default.
- **Uniform Groundcover** — Skoglendi, Origins of Forest, Mari's flora need matching density variance.

Tweak Names:

- **Body Part Codes** — drive Sort: Armor/Clothes correctly with CBBE 3BA.
- **Sort: Armor/Clothes** — with Immersive Armors, Bruma, Spaghetti's Cities, unsorted inventory is unreadable.
- **Sort: Scrolls, Sort: Spells, Sort: Weapons/Ammunition** — Apocalypse, Mysticism, Odin, Adamant, Vokrii, Ordinator, Ars Metallica all add to these lists.
- **Lore Friendly Text: Dwarven → Dwemer** — PBR textures use Dwemer terminology.

Tweak Races:

- **Races Have All Head Parts** — KS Hairdos adds hundreds; set types code to 45.
- **Playable Head Parts**
- Leave **Force Behavior Graph Gender** off — Elder Wilds does not standardize that flag.

Tweak Settings:

- **AI: Bump Reaction Delay** — raise to Adult Content to mute spammy follower reactions.
- **AI: Conversation Chance** — leave default.
- **AI: Max Active Actors** — raise to 80.
- **AI: Max Dead Actors** — raise to 120.
- **Arrow: Max Arrows Attached to NPC** — raise to 20.
- **Combat: Max Actors** — raise to 60.
- **Combat: Stealth Damage Bonus** — leave default (RAID is canonical authority).
- **Msg: Cannot Equip Item Fix** — **must** be enabled (Campfire, Frostfall, SunHelm/Starfrost/Last Seed rely on it).
- **Msg: No Fast Travel** — leave default.
- **Player: Max Draggable Weight** — set to 1000.
- **Player: Underwater Breath Control** — leave default.

- **Soul Trap: Common/Grand/Greater/Lesser Soul Level** — set to vanilla values.
- **Visuals: Masser/Secunda Size/Speed** — set to 1.0.
- **Warning: Exterior/Interior Distance To Hostiles** — leave default.
- **World: Cell Respawn Time / Respawn Time (Cleared)** — leave default.
- **World: Timescale** — set to 12 and never change. **Save-baking warning:** this setting bakes into SSE saves.

37.2.3. Bash Tags

Bash Tags tell Wrye Bash which record types to import from which plugins into the Bashed Patch. LOOT auto-detects most tags; only set manual tags for plugins not covered by LOOT's masterlist.

Tag priority (lowest → highest):

1. Plugin header description (set by mod author)
2. LOOT masterlist.yaml (`%LOCALAPPDATA%\LOOT\Skyrim Special Edition\`)
3. LOOT userlist.yaml
4. Manual tags in `Data\BashTags\<PluginName>.txt`
5. Manual tags set in Wrye Bash UI (overrides all above)

When to manually set tags:

- Plugin is new and not yet catalogued in LOOT masterlist
- Plugin's LOOT entry has wrong or missing tags
- Plugin ships with a `BashTags` folder containing a `.txt` file — copy those tags exactly

How to verify: After building the Bashed Patch, open it in xEdit. If a plugin's leveled-list edits, actor stats, or item stats are missing from the patch when they should be present, the plugin is missing tags.

Set the following `Bash Tags` on the listed plugins (one file per plugin under `Data\Bash Patches\`):

Leveled List / Economy:

- `Immersive Weapons.esp` — `Relev`
- `Immersive Armors.esp` — `Relev`
- `Immersive Armors Retexture.esp` — `Graphics` (retexture, not content add)
- `Hunterborn.esp` — `Invent.Add`, `Delev`, `Relev`
- `Simple Fishing Overhaul.esp` — `Invent.Add`, `C.MiscFlags`
- `Streamlined Fishing.esp` — `Invent.Add`
- `Open World Loot.esp` — `Delev`, `Relev`, `C.Encounter`
- `MorrowLoot Ultimate.esp` — `Delev`, `Relev`
- `Trade and Barter.esp` — `Stats`, `Invent.Change`
- `Ars Metallica.esp` — `Stats`, `Invent.Change`
- `Complete Crafting Overhaul Remastered.esp` — `Stats`, `Invent.Change`
- `Honed Metal.esp` — `Invent.Change`
- `Apothecary.esp` — `Invent.Add`, `Stats`, `Keywords`
- `CACO.esp` — `Invent.Add`, `Stats`, `Keywords`
- `Alchemy Potions and Food Adjustments.esp` — `Stats`, `Keywords`

Race / Perk / Magic:

- `Aetherius.esp` / `Morningstar.esp` / `Imperious.esp` — `R.AddSpells` (whichever wins)
- `Adamant.esp` / `Vokrii.esp` / `Ordinator.esp` — `NPC.Perks.Change`, `NPC.Perks.Add`
- `Mysticism.esp` / `Odin.esp` / `Apocalypse.esp` — `SpellStats`, `EffectStats`

Lighting / Weather / Water:

- CS Light.esp / True Light.esp — C.Light , C.ImageSpace
- ELFX.esp / ELFX - Shadows.esp — C.Light , C.Fog
- Relighting Skyrim.esp / Luminosity.esp — C.Light
- RAID Weathers.esp / Raid Weathers CS.esp — C.Climate , C.Music (if pack changes music)
- Cathedral Weathers.esp / Obsidian Weathers.esp / Obsidian - CS.esp — C.Climate
- Realistic Water Two.esp — C.Water
- A Water Made For CS in mind.esp — C.Water
- Simplicity of Sea.esp — C.Water
- Enhanced Rocks and Mountains.esp / Fix and Addon.esp — Graphics , ObjectBounds

City / Workspace:

- Beyond Skyrim - Bruma SE.esp — C.Climate , C.Music , C.Light , C.Water , C.ImageSpace , C.Location , C.Regions
- Spaghetti's Cities - AIO.esp — C.Light , C.MiscFlags
- The Great Cities - Minor Cities and Towns SSE.esp / Dawn of Skyrim (Director's Cut) SE.esp / JK's Skyrim.esp — C.Light , C.MiscFlags

NPC / Dialogue / Audio:

- RUSTIC CLUTTER COLLECTION.esp / RUSTIC CLOTHING.esp — Names , Stats
- High Poly Project.esp — Graphics , ObjectBounds
- KS Hairdos SSE.esp — Graphics , Names
- High Poly True to Vanilla NPC Overhaul.esp — Graphics , Names , Actors.Stats
- RDO.esp — Actors.AIPackages , Actors.Spells
- Amorous Adventures.esp (clean) — Actors.AIPackagesForceAdd , Actors.SpellsForceAdd , Invent.Add
- Marriage Mod - To Have And To Hold.esp — Actors.AIPackagesForceAdd , Actors.Factions
- Serana Dialogue Add-On.esp — Actors.AIPackages , Actors.Spells
- Pilgrim.esp / Trua.esp / Wintersun.esp — Actors.Spells (whichever wins)
- Book of Shadows.esp — Actors.Spells
- Skyrim Reputation.esp — Actors.Factions , Relations.Add
- Suspicious City Guards.esp — Actors.AIPackages
- Audio Overhaul for Skyrim SE.esp — Sound
- Immersive Sounds - Compendium.esp — Sound
- Sounds of Skyrim Complete SE.esp — Sound , C.Acoustic

Survival / Follower:

- Campfire.esp — C.MiscFlags , Invent.Add
- Frostfall.esp — C.Climate , C.ImageSpace
- Starfrost.esp / SunHelm.esp / Last Seed.esp — Stats , Keywords , Invent.Add (whichever wins)
- Nether's Follower Framework.esp / EFF.esp / Amazing Follower Tweaks SE.esp — Actors.Factions , Invent.Add

No Tags: TrueHUD.esp , Sovngarde - Mist's Font Replacer.esp

—

37.3. Synthesis Configuration

The `Synthesis` patcher pipeline runs as a single batch via **Run Mutagen** in MO2, output to `Patches\Synthesis Patch.esp`. Patchers listed in run order — do not reorder.

37.3.1. Stage 1 — NPC, AI, And Facegen

- `KS Hairs Bald Helmets Fixer`
- `High Poly Head Vampire Fix`
- `HP\_NPC\_WIGS\_TO\_HEADPART`
- `FacegenBaseline` — after all NPC overhauls installed; fallback to zEdit FaceGen patcher if needed. See `### FacegenBaseline Configuration` below.
- `SynthEBD` — NPC equipment/outfit distribution via rule-based config. Use as a SPID alternative for armor/clothing distribution when SPID can't cover the target (e.g., complex outfit rules, multi-condition NPC filtering). Requires `SynthEBD` config files per armor/outfit mod. See `### SynthEBD Configuration` below.
- `NPCStatRescaler`

37.3.2. FacegenBaseline Configuration

Generates missing or corrected facegen data (face morphology `.nif` files and tint layers `.dds`) for NPCs added or changed by appearance mods. Prevents the black-face bug, mismatched facial features, and incorrect tint layers.

When to run:

- After every change to the NPC appearance stack — adding, removing, or reordering appearance overhauls
- After installing any mod that adds new NPCs (quest mods, world content, new lands)
- After running any Synthesis patcher that changes NPC head parts or face morphs
- Every time the load order of NPC-related plugins changes

Setup in Synthesis:

1. Add `FacegenBaseline` to your Synthesis pipeline (already listed in Stage 1 above).
2. Configure the **Source Mods** list — select every NPC appearance overhaul and any mod that adds NPCs with custom facegen. The patcher reads facegen from these source mods and copies it into the output.
3. Set **Output Mod** to a dedicated mod folder (e.g., `FacegenBaseline Output` in the `Output` MO2 separator). Do not output into an existing NPC overhaul mod.
4. Run Synthesis with the full pipeline — `FacegenBaseline` runs after `KS Hairs Bald Helmets Fixer` and `HP\_NPC\_WIGS\_TO\_HEADPART`.

Configuration options:

Option	Recommendation	Notes
Source Mods	Select all NPC appearance overhauls + any mod listed in the NPCs section that ships facegen	Missed source mods are the 1 cause of black-face
Output Mod	Dedicated mod in <code>Output</code> separator	Do not output into an existing NPC overhaul mod
Overwrite Mode	Replace (default)	Keeps output clean on each rebuild

Include Tint Layers	Yes	Required for correct NPC skin/makeup in dialogue
---------------------	-----	--

Troubleshooting:

Symptom	Likely Cause	Fix
Black face (dark head, light body)	NPC's facegen <code>.nif</code> missing from output, or a later mod overwrites it	Add the NPC's source mod to Source Mods; verify load order places FacegenBaseline output after all NPC overhauls
Face doesn't match appearance mod (wrong morph, wrong hair)	A different mod's facegen is winning the conflict	Reorder mods so the intended appearance mod loads after conflicting overhauls, then re-run FacegenBaseline
Grey face (no tint layers)	Tint layers not generated or not found	Verify "Include Tint Layers" is enabled; check that <code>Textures\actors\character\FacegenTints</code> exists in output
CTD when approaching an NPC	Corrupted facegen data — mismatched head part count or missing assets	Delete FacegenBaseline output, re-verify all source mods are selected, re-run Synthesis from scratch
FacegenBaseline fails with "no facegen found"	Source mods list is empty or the patcher can't find compatible face NIFs	Check that at least one source mod is selected and contains <code>meshes\actors\character\facegendata\facegeom\</code>

37.3.3. SynthEBD Configuration

Port of zEBD to Synthesis/Mutagen. Distributes equipment (armor/clothing/weapons), body shapes, textures, and heights to NPCs via rule-based config files. Treat as the primary SPID alternative when an armor/outfit mod has no SPID config or needs multi-condition NPC filtering.

Download: [GitHub Releases](#) or [Nexus](#). Extract the `SynthEBD` folder to your mod utilities directory. Launch `SynthEBD.exe` through MO2.

Setup in Synthesis:

1. Add `SynthEBD` to your Synthesis pipeline (already listed in Stage 1 above, after FacegenBaseline).
2. Run **SynthEBD.exe** (standalone GUI) before Synthesis to configure settings and install config files. The Synthesis patcher reads the configuration produced by the standalone tool.
3. Set the **Output Data Folder** to a dedicated MO2 mod (e.g., `C:\MO2\mods\SynthEBD Output`) via the `General Settings` tab.
4. Install config files via the `Textures and Meshes` tab using the **Install Config From Archive** button — select each config file you want active.
5. Run SynthEBD standalone to build the configuration database, then run Synthesis with the full pipeline.

General Settings:

Option	Recommendation	Notes
--------	----------------	-------

Output Name	SynthEBD.esp	Generated plugin name
Output Data Folder	Dedicated MO2 mod folder	All generated files (ESP + assets) land here
Apply Textures and Meshes	Yes	Enables asset distribution from config files
Apply Body Shape Using	None (OFF)	Elder Wilds uses OBody NG for body distribution — let OBody handle it
Apply Height Changes	OFF	Leave height control to other patchers in Stage 1 (<code>RacialHeights</code> / <code>HarmonizedRaceHeights-Patcher</code>)
Enable Consistency	Yes	Keeps the same body shape on NPCs across save loads
Verbose Mode for Conflict NPCs	ON (debugging)	Logs decision-making for NPCs with assignment conflicts; disable once configs are stable

Textures and Meshes settings:

Option	Recommendation	Notes
Allow config files to change NPC textures	Yes	Core distribution function
Allow config files to change NPC meshes	Yes	Core distribution function
Patch NPCs with custom bodies	ON	Ensures NPCs from mods with custom WNAM get textures
Patch NPCs with custom faces	ON	Ensures NPCs from appearance overhauls get assigned face textures
Force Vanilla Body Mesh Paths	OFF	Elder Wilds uses OBody NG — body mesh paths managed separately
Generate combination assignment log	ON (debugging)	Disable once configs are stable

Config File Types:

Config Type	Purpose	Elder Wilds Usage
Asset Config	Distributes armor/clothing/weapon textures and meshes to NPCs	Primary use case — install when an armor mod lacks SPID config
BodyGen Config	Distributes BodyGen morphs	OFF — OBody NG handles body distribution
BodySlide Settings	Distributes BodySlide presets via OBody/AutoBody	OFF — OBody NG handles body distribution
Height Config	Distributes NPC heights	OFF — use Stage 1 height patchers instead

Attribute Groups: The core controller of SynthEBD's distribution system. Config files reference Attribute Groups to control which NPCs get which assets. Attribute Groups define filters by race, gender, faction, class, voice type, unique NPC, etc. Check **Supercede Plugin Group Definitions with Main** in General Settings to control distribution from all config files from a single centralized menu — recommended for Elder Wilds.

When to run: After adding or updating any armor/clothing mod that uses a SynthEBD config for NPC distribution (typically as a SPID fallback). Run the SynthEBD standalone GUI first to re-process configs, then run the full Synthesis pipeline.

Troubleshooting:

Symptom	Likely Cause	Fix
NPCs not getting assigned equipment	Config file not installed or Attribute Group doesn't match	Verify config installed via <code>Textures and Meshes</code> tab; check Attribute Group filters match the target NPC
Body shapes won't assign	Body shape distribution enabled but no BodySlide/BodyGen config installed	Set <code>Apply Body Shape Using</code> to None — OBody NG handles this separately
Duplicate NPC entries	Config file has redundant Attribute Group rules	Enable <code>Verbose Mode</code> , check <code>SynthEBD\Logs</code> for conflicting assignments

- `AIOverhaulPatcher` / `ICAI0 AI for Mods`
- `AI Stealth Overhaul` — with RAID.
- `Followers-are-Sneaky`
- `FollowerFavorCarryLimitPatcher`
- `NpcProtector` — flags essential quest NPCs.
- `RaceCompatibilityDialogue`
- `RacialHeights` / `HarmonizedRaceHeights-Patcher`
- `SynBanditWarForwarder`
- `SynOppositeGenderAnimsTweak` — **do not enable** (Bashed Patch handles this).

37.3.4. Stage 2 — Encounter Zones

- `EZLeveler` — configure to unleveled-band target for OWL/MLU.
- `EnemyReleveler` — delevel Bruma/Falskaar/Wyrmstooth enemies within new band.
- `No More Easy Enemies`
- `EncounterZoneNames` / `CellEncounterLevelsInName`
- `MissingEncounterZonesPatcher`
- `TrueUnleveledSkyrim` — only if dropping OWL and MLU.

37.3.5. Automated Leveled List Addition Configuration

A Synthesis patcher that automatically adds weapons and armor to leveled lists, reducing manual patching for standalone weapon/armor mods. Runs in Stage 3 before `OWLLeveledListAddition`.

What it does: Scans all loaded plugins for weapons and armor records, then injects them into appropriate vanilla leveled lists based on material tier, type, and stat comparison against existing list entries.

Setup:

1. Add `Automated Levelled List Addition` to your Synthesis pipeline (already listed in Stage 3 above).
2. In the patcher settings, configure three **Output Plugins** — one each for weapons, armor, and misc items. Use a dedicated output plugin (e.g., `ALLA Output.esp`).
3. Set default output plugin names in the script's Settings section to avoid re-typing on each run.
4. Run after adding new weapon/armor mods and before `OWLLeveledListAddition`.

Configuration options:

Option	Recommendation	Notes
Output Plugin (Weapons)	<code>ALLA Weapons.esp</code>	Separate output for weapon leveled-list entries
Output Plugin (Armor)	<code>ALLA Armor.esp</code>	Separate output for armor leveled-list entries
Output Plugin (Misc)	<code>ALLA Misc.esp</code>	For jewelry, clothing, misc items
Smart Level Assignment	Yes	Matches item tier to list level based on stats
Skip Already Levelled Items	Yes	Avoids duplicating items already in leveled lists
Skip Unique/Quest Items	Yes	Prevents unique artifacts from appearing in random loot

When to run: After adding any standalone weapon or armor mod to the load order. Since Elder Wilds now has 16+ standalone weapon mods, this patcher is essential — run it every time the Weapons & Armor section changes.

Key relationships: Runs before `OWLLeveledListAddition` and `LeveledListresolver`. Does not replace OWL integration — it handles inject-only distribution while OWL controls tier-gating and deleveling logic.

37.3.6. Stage 3 — Levelled Lists And Loot

- `Automated Levelled List Addition` ([Nexus](#)) — Automatically adds weapons/armor to leveled lists. Configure output plugins per section (weapons, armor, misc). Set default output plugins in the script's Settings section to avoid re-typing plugin names. Run after new weapon/armor mods are added and before `OWLLeveledListAddition`. See [### Automated Levelled List Addition Configuration](#) below.
- `OWLLeveledListAddition` — adds IA/IW to OWL lists.
- `OWLPatcher`
- `Skyrim-LevelledLoot`
- `HalgarisConsistentRPGLOot`
- `Container and Loot Patcher`
- `LootableCrates`, `MakeFirewoodPilesIntoContainers`, `Lootable Things & Training Dummies Patcher`
- `GeneralStoresIngredientsPatcher`
- `ImmersiveMusicPatcher` / `MusicTypePatcher-Revised-2024`
- `LeveledListresolver` — after OWL and MLU.
- `Feminized Levelled Lists` / `Masculinized Levelled Lists`

37.3.7. Stage 4 — Items, Crafting, Alchemy

- `ApothecaryAutoPatcher` — pair with `CACO Patcher` if CACO in stack.
- `BreakdownRecipeGenerator` / `CreateTemper`

- `HonedMetalAdditionalIngredients`
- `Heim Recipe Patcher`
- `ReProccer Evolved`
- `Weapon Stat Synthesis Patcher` ([Nexus](#)) — Elder Wilds config: Damage Floor 4, Ceiling 22, Weight/Value Scale 1.0, Speed/Reach Normalization ON. Ignore List: Reliquary of Myth, ArteFakes, Unique Items Tweaks, Konahrik's Accoutrements. Run after `OWLLeveledListAddition`, `SpeedandReachFixes`, `AmmoTweaks`.
- `SpeedandReachFixes`
- `WeaponSpeedEffectsFix`
- `SynFixShieldData` / `SynFixWeaponAttackSpeed`
- `SynGlovesAddFirstsPerk`
- `ItemWeightCustomizer` / `SynPotionWeight`
- `SlotsSlotsSlots`
- `AmmoTweaks`
- `SynCGOStaves` — only if CGO staves in stack.

37.3.8. Weapon Stat Synthesis Patcher Configuration

Normalizes weapon stats across the entire load order when standalone weapon mods use different balance baselines. Runs in Stage 4 after `ReProccer Evolved`. Critical when adopting 16+ standalone weapon mods with varying stat philosophies.

What it does: Recalculates weapon damage, weight, value, speed, and reach for all weapons based on a unified scaling formula. Ensures every weapon — vanilla, Heavy Armory, Immersive Weapons, and standalone mods — fits a consistent power curve.

Elder Wilds config:

Option	Value	Notes
Damage Floor	4	Lowest damage any weapon can have
Damage Ceiling	22	Highest damage any weapon can have
Weight/Value Scale	1.0	Linear scaling between weight and value
Speed Normalization	ON	Standardizes attack speed ranges per weapon type
Reach Normalization	ON	Standardizes reach ranges per weapon type
Ignore List	<code>Reliquary of Myth.esp</code> , <code>ArteFakes.esp</code> , <code>Unique Items Tweaks.esp</code> , <code>Konahrik's Accoutrements.esp</code>	Do not rebalance unique/artifact items

When to run: After any weapon mod is added, removed, or updated. Run after `ReProccer Evolved` , `SpeedandReachFixes` , and `AmmoTweaks` . Output after `OWLLeveledListAddition` .

Key relationship: This patcher owns weapon stat balance. Do NOT allow Wrye Bash or other Synthesis patchers to override its damage/speed/reach output. The Ignore List prevents artifact

overhauls (Reliquary of Myth , ArteFakes , Unique Items Tweaks) from being re-scaled — those mods own their own balance.

37.3.9. Stage 5 — Magic, Spells, Enchanting

- Engarde_Synthesis
- DisenchantEverything / SmartDisenchantEverything
- Passive Enchanting Experience
- Spell Research Synthesizer — only if Spell Research added.
- StaffGenerator / StaffDuplicator
- SorcererPatcher
- ReadingIsGoodLegacy
- AllBooksHavePerks
- SpellTomePriceFixPatcher / SynSpellTomeNameExtender / DisplaySpellTomeLevelPatcher-Local / spelltome_yeeter — pick one.
- SynESLify — **must be last** in this stage.

37.3.10. Stage 6 — Snow, Weather, Region

- BDSPatcher — **required**. Use BDSPatcher entry (not Northpoint variant). Run with both BetterDynamicSnow and BetterDynamicAsh keywords active.
- SSBGPatcher
- Nights Adjuster / Darker Weather Nights — pick one.
- SkyVRaan Weather Patcher — only if installed.
- Unique Region Names / Unique Region Names Patcher
- RealisticWaterTwoPatcher
- Synthesis-BlendedShorelinesUniversal
- LandscapePatcher — only enable if landscape texture stack replaced.
- DestructibleSkyrimPatcher

37.3.11. Stage 7 — Trees, Flora, Mesh

- Pine Tree Rescaler
- Tree Scaler , Tree 3D Editor , TreeWindEditor
- FloraFixer
- ImmersiveEquipmentMeshGen / ALLGUDMeshGen
- Harvest Those Mammoth Tusks

37.3.12. Stage 8 — Visuals And Lighting

- ELE Patcher — skip (CS-only stack).
- NoShadowsPatch
- Radial Blur Remover
- **Remove Edge Glow**
- RemoveInteriorFogPatcher — disable if AOS/Sounds of Skyrim present.
- RemoveLandscapeVertexColor

37.3.13. Stage 9 — Audio

- `AOSISCSoundPatcher` — **required**. AOS and ISC both must be active when run.
- `TUDMFootstepPatcher`
- `Sounds of Seasons`
- `SFCOPaintingRemover`

37.3.14. Stage 10 — Performance And Cleanup

- `Grass FPS`
- `Configurable Grass Remover` — enable only if dense forest-floor route adopted.
- `NavmeshCollector-Updated` — **do not** enable if Bruma or Falskaar in stack.
- `Generic Synthesis Patcher` — keep ruleset empty until specific conflict identified.

37.3.15. Stage 11 — Utility, Compatibility, ESLification

- `Patchifier` — generates SkyPatcher/INI patches.
- `SynEDIDFixer`
- `SynStringMerger` — disable unless shipping translation layer.
- `CellEditorIDFixer` — required (Bruma, Falskaar, Wyrmsooth, Beyond Reach).
- `SynPEXPatcher`
- `SynPerkCOBJPatcher` / `SynESLify` — **keep at end**, ESLify output.

37.3.16. Build Order And Rebuild Triggers

The full rebuild order for the generated pipeline. Stages after `SSEEdit` can be skipped if their inputs have not changed — rebuild only the affected stages when their triggers fire.

1. **SSEEdit** — quick-conflict mode after every new plugin.
2. **Bashed Patch** — triggers: any plugin with Bash Tag change, any tweak setting change, any new tagged plugin. A Bash Tag change also triggers Synthesis (stage 3) — both must rebuild.
3. **Synthesis** — triggers: any patcher list change, any upstream mod change, any Bash Tag change.
4. **Pandora** — triggers: behavior/animation/skeleton change. Independent of patcher and LOD stages — can be run separately at any point.
5. **BodySlide** — triggers: body/skin/armor mesh change. Independent of patcher and LOD stages.
6. **xLODGen** — triggers: landscape/terrain/heightmap change (does not need TexGen).
7. **TexGen** — triggers: object/tree/building texture or mesh change producing LOD billboards. Requires xLODGen output when both are rebuilt together.
8. **Grass Cache** — triggers: grass mod change, worldspace change (Grass SSEEdit Script), or grass density INI tweak. Run after final worldspace load order is established.
9. **DynDOLOD** — triggers: worldspace/tree/object change, Bashed Patch or Synthesis change (form IDs affect reference records), or any xLODGen/TexGen output change. Includes Occlusion generation. Run twice: Medium to verify, final preset.
10. **SSE Display Tweaks + BethINI Pie** — final review after lighting/weather/graphics stack change.

Dependency note: DynDOLOD reads the final load order including Bashed Patch and Synthesis. If Synthesis changes form IDs or ESL flags, DynDOLOD must be rebuilt. Pandora and BodySlide do not affect LOD or patchers and can be run independently at any point without cascading into later stages.

37.3.17. Patcher Load Order

- `Bashed Patch` and `Synthesis` both write into same load-order range. Place Synthesis patch last unless a known conflict needs Bashed Patch to win.
- `SynESLify` **must** be last in its stage.
- `World: Timescale = 12` bakes into saves — never change on a long-running test save.
- `Skyrim-LeveledLoot` and `OWL` overlap: run `OWLLeveledListAddition` first.
- `AOSISCSoundPatcher` requires both AOS and ISC active.
- `HP\_NPC\_WIGS\_TO\_HEADPART` before `KS Hairs Bald Helmets Fixer`.
- `NavmeshCollector-Updated` dangerous with Bruma and Falskaar.
- Record Bashed Patch CRC after every rebuild. Verify by disabling source mods and confirming patch shrinks.

37.3.18. Key Principles

- The patch pipeline is the single source of truth for conflict resolution.
- A patcher or tag exists because a specific mod in the list requires it — never pre-install speculative coverage.
- The rebuild order is designed so a category change only triggers the affected stages.
- **Form ID compaction is dangerous.** Compacting form IDs (e.g., via xEdit “Compact FormIDs for ESL”) changes records that Synthesis patchers reference by FormID. If you compact, rebuild EVERYTHING from scratch — Bashed Patch, Synthesis, Pandora, BodySlide, DynDOLOD. Better: compact only after finalizing a section, before running the pipeline.
- **SynESLify must be last.** ESL-flagging changes plugin headers and can shift load order indices. Run it as the final Stage 11 patcher and never before DynDOLOD (DynDOLOD reads FormIDs directly).
- **Verify the Bashed Patch after rebuild.** Open in xEdit, expand Leveled Items and Leveled NPCs trees. Confirm the expected entries from tagged plugins are present. Record the CRC and file size in your build notes — compare on the next rebuild to ensure it didn’t silently shrink (common when mod plugins are disabled or removed).
- **Grass Cache must be last worldspace step before DynDOLOD.** If you rebuild the grass cache after DynDOLOD, grass LOD will be stale. Always: Bashed Patch → Synthesis → xLODGen → TexGen → Grass Cache → DynDOLOD.
- **Run DynDOLOD twice.** First pass at Medium to verify output is clean and no error messages appear. Second pass at the target preset (High/Ultra) for the final LOD.

38. Testing & Maintenance

38.1. Performance Testing And Tuning

The goal is consistent frametimes — not just a high average FPS — across the scenarios you actually play.

38.1.1. Tools

Tool	Purpose
NVIDIA FrameView	Lightweight overlay/logging: FPS, frametime, GPU power, thermals. No CPU per-core but lower overhead.
MSI Afterburner + RivaTuner	Full overlay/logging: FPS, frametime, GPU usage, CPU per-core, VRAM, draw calls, temps.
GPU-Z	Sensor logging for GPU clock, voltage, thermals, power limits.
Community Shaders debug overlay	Press F11 for render-time breakdown, draw-call count, VRAM usage.
SSE Display Tweaks OSD	Set <code>ShowOSD=true</code> in <code>SSEDisplayTweaks.ini</code> for basic overlay.
Skyrim console tools	<code>showstats</code> for real-time overlay; <code>sgtm 0.3</code> slows to 30% for inspecting stutter; <code>tfc 1</code> freezes camera.
Cathedral Assets Optimizer	Downscale uncompressed/oversized textures for VRAM relief.

38.1.2. Benchmark Methodology

1. Create a stable reference save with fixed weather (`fw 10a240`), fixed time (`set gamehour to 12`), `tfc 1` for reproducible screenshots. Save as `benchmark\_ref.ess`.
2. Use fixed routes — same save, same path for 60 seconds.
3. Log everything via MSI Afterburner or NVIDIA FrameView CSV.
4. Test one change at a time.

38.1.3. Benchmark Scenarios

Scenario	Location	Teleport	What It Tests
Open world	Whiterun plains west	<code>cow WhiterunWorld -10 0</code>	GPU/CPU balance, grass density, LOD load
Dense forest	Falkreath woods	<code>cow FalkreathWorld -30 -20</code>	Tree rendering, shadow complexity

Pine/marsh transition	Morthal swamp edge	cow HjaalmarchWorld -15 -10	Wetland shaders, mist, mixed flora
City centre (canals)	Riften marketplace	cow RiftenWorld 0 0	NPC count, water reflections, transitions
City centre (castles)	Solitude exterior	cow SolitudeWorld 0 0	Draw calls, shadow cascades, occlusion
Snowy tundra	Dawnstar coast	cow WinterholdWorld 10 5	Snow shaders, volumetric fog
Waterfall coast	Coast near Solitude	cow Tamriel -22 -4	Water displacement, foam, coastal LOD
Large interior	Blackreach	cow Blackreach 0 0	Particle lights, mesh density, alpha sorting
Large combat	Fort Neugrad assault	cow WhiterunWorld 15 -5 then run	AI packages, magic FX, simultaneous actors
Dragon encounter	Open world	player.placeatme 000FEA9F	Skeleton, shout VFX, cell transition
Magic VFX stress	Dense combat + spells	cow WhiterunWorld 0 0 then spawn dragon + 3 fire mages	Simultaneous particles, glow shaders, shadows
Heavy weather	Solstheim ash storm	cow SolstheimWorld 0 0 with fw 10A1E8	Particles, alpha-heavy weather, fog

38.1.4. Record Per Scenario

- Average FPS
- Frametime 99th percentile (stutter floor — >50ms means visible hitches)
- GPU usage % (below 90% = CPU bottleneck)
- VRAM usage (within 500MB of card total = stutter risk)
- Draw calls (above 10,000 = CPU struggling)

38.1.5. Tuning Knobs (Ordered By Impact)

Stop once worst-case scenario is playable.

1. **Grass density** (BethINI > Environment > Grass Density). 40-60 range.
2. **Grass render distance** (BethINI > View Distance > Grass LOD Fade). Drop to 50-70 for forest gains.
3. **Shadow resolution** (BethINI > Visuals > Shadow Resolution). 2048 is sweet spot.
4. **Particle count** (BethINI > Visuals > Max Particle Render Count). 7500 baseline; drop to 5000 if fire/frost causes framedrops.
5. **Tree LOD distance** — keep `fTreeLoadDistance=0` (DynDOLOD manages).
6. **Object LOD fade** — drop `fMeshLODLevel2FadeTreeDistance` to 6144, `fMeshLODLevel1FadeTreeDistance` to 8192.
7. **VSync / framerate cap** — use SSE Display Tweaks `LockFrameRate=60`, `VSync=0`.
8. **Skyrim Upscaler** — if GPU is bottleneck, enable with Quality preset.

38.1.6. When To Accept

- Occasional microstutter on cell border crossing is normal (Engine Fixes already mitigates).
- Combat in tight spaces with 6+ actors + magic VFX: if avg FPS >40 with no hitch >100ms, call it stable.
- If a specific mod cuts FPS 30%+ vs its branch alternative, consider swapping.

38.1.7. Benchmark Reports

Export to CSV (NVIDIA FrameView Save CSV, MSI Afterburner *History* > *Log to file*). Keep a `benchmark-sheet.md`:

Pass	Scenario	Avg FPS	1% Low	GPU%	VRAM	Hitch Max	Delta vs Prev
1	Whiterun	58	42	92%	6.16	48ms	—

Every tuning attempt recorded in git alongside modlist changes.

—

38.2. Maintenance And Rebuild Epilogue

38.2.1. Patcher Revisit Triggers

Patcher	Trigger	Settings Worth Tweaking
LOOT	Every meaningful load-order change	No custom tweaks
Synthesis	Patcher list change, upstream mod added/removed/updated, Bash Tag change on any plugin	Weapon Stat Synth: raise/lower Damage Ceiling; add to Ignore List for new artifact mods
Wrye Bash	Bash Tag change on any plugin, tweak setting change, new tagged plugin, or leveled-list-heavy mod added/removed	No per-run settings unless Bashed Patch imports unintended items
Pandora	Animation mod added/removed/reordered; skeleton or behavior change	Direct output to Pandora Output
BodySlide	Body preset/skin texture/armor mod change	No per-run settings beyond chosen preset
Grass Cache	Grass/tree/landscape texture/worldspace change; density INI tweak	Rebuild from scratch
xLODGen	Landscape texture/heightmap/worldspace change; terrain LOD mod change	Quality slider per-run for coastline artefacts
TexGen	Object/tree/building texture changes producing LOD billboards; tree mod swaps	Revisit GrassModelHeightMultiplier and TreeMSAlphaThreshold

DynDOLOD	Worldspace/tree/large-reference/ LOD resource change; TexGen output change; Bashed Patch or Synthesis change (form IDs affect reference records)	Brightness/Contrast for LOD32; DoubleSidedTextureMask for new mountain types
-----------------	--	--

38.2.2. Generator Output Order (First-Time Build)

Run in this sequence when generating outputs for the first time. Independent outputs (Pandora, BodySlide) are generated earlier here than in the canonical → [Section 37](#) because they can be iterated on while patchers are still in flux. The final LOD sequence (xLODGen → TexGen → Grass Cache → DynDOLOD) matches the rebuild order exactly.

1. **SSEEdit** — quick-conflict check after every major mod addition
2. **Pandora** — after animation stack stabilizes
3. **BodySlide** — after body/armor changes stabilize
4. **Bashed Patch** — after load order is stable enough for tagging
5. **Synthesis** — patchers the final stack actually uses
6. **xLODGen** — terrain LOD first
7. **TexGen** — before DynDOLOD
8. **Grass Cache** — after grass/landscape/worldspace choices stabilize
9. **DynDOLOD** — last major output (includes Occlusion generation)
10. **SSE Display Tweaks + BethINI Pie** — final review after graphics baseline is set

After generation, verify: `DynDOLOD.esm` at end of ESM block, `DynDOLOD.esp` near end, `Occlusion.esp` after, `Synthesis.esp`